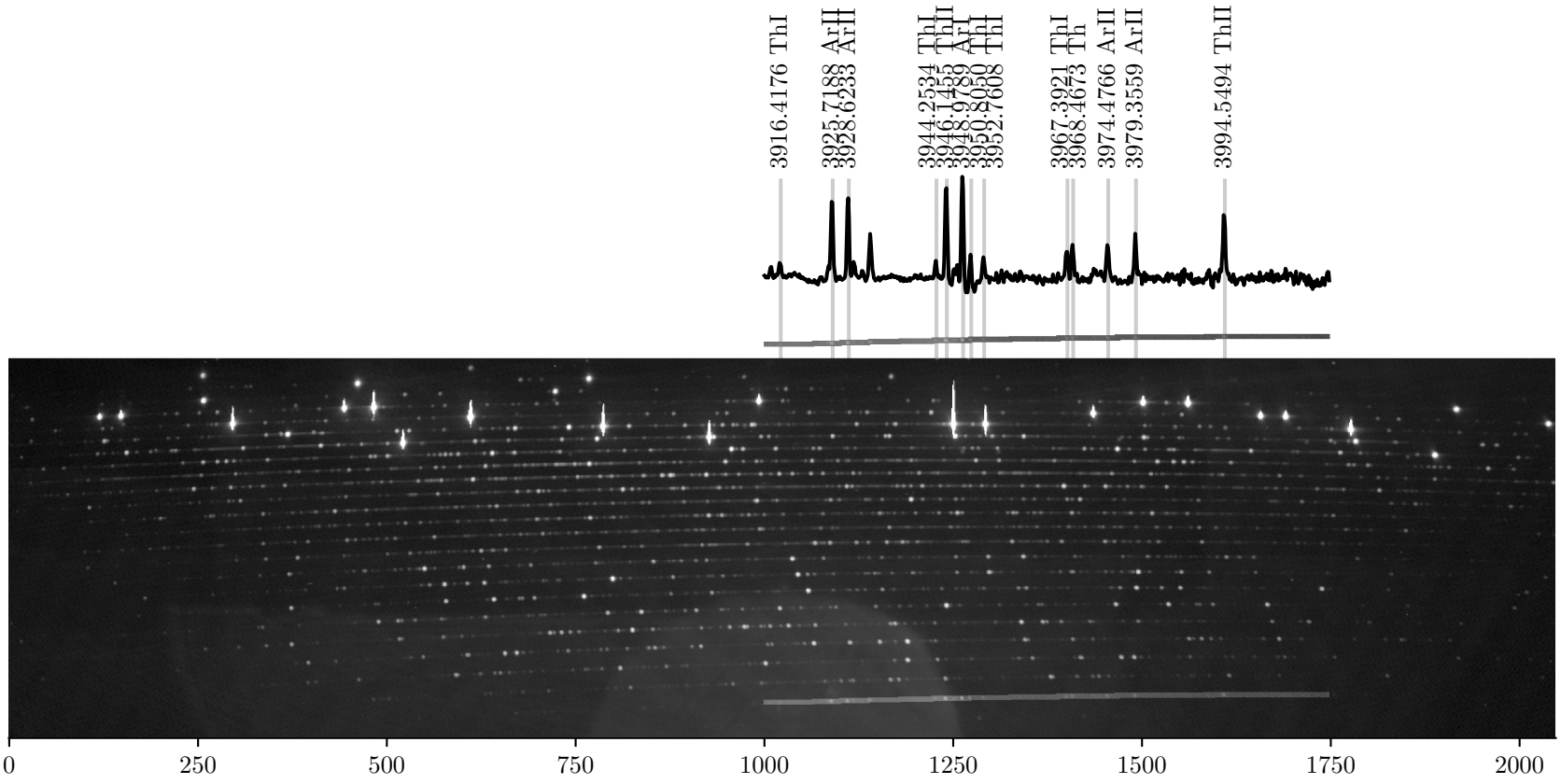
Guide to Medium Resolution ThAr Spectra

Echelle Spectrograph for the Nanshan Observatory

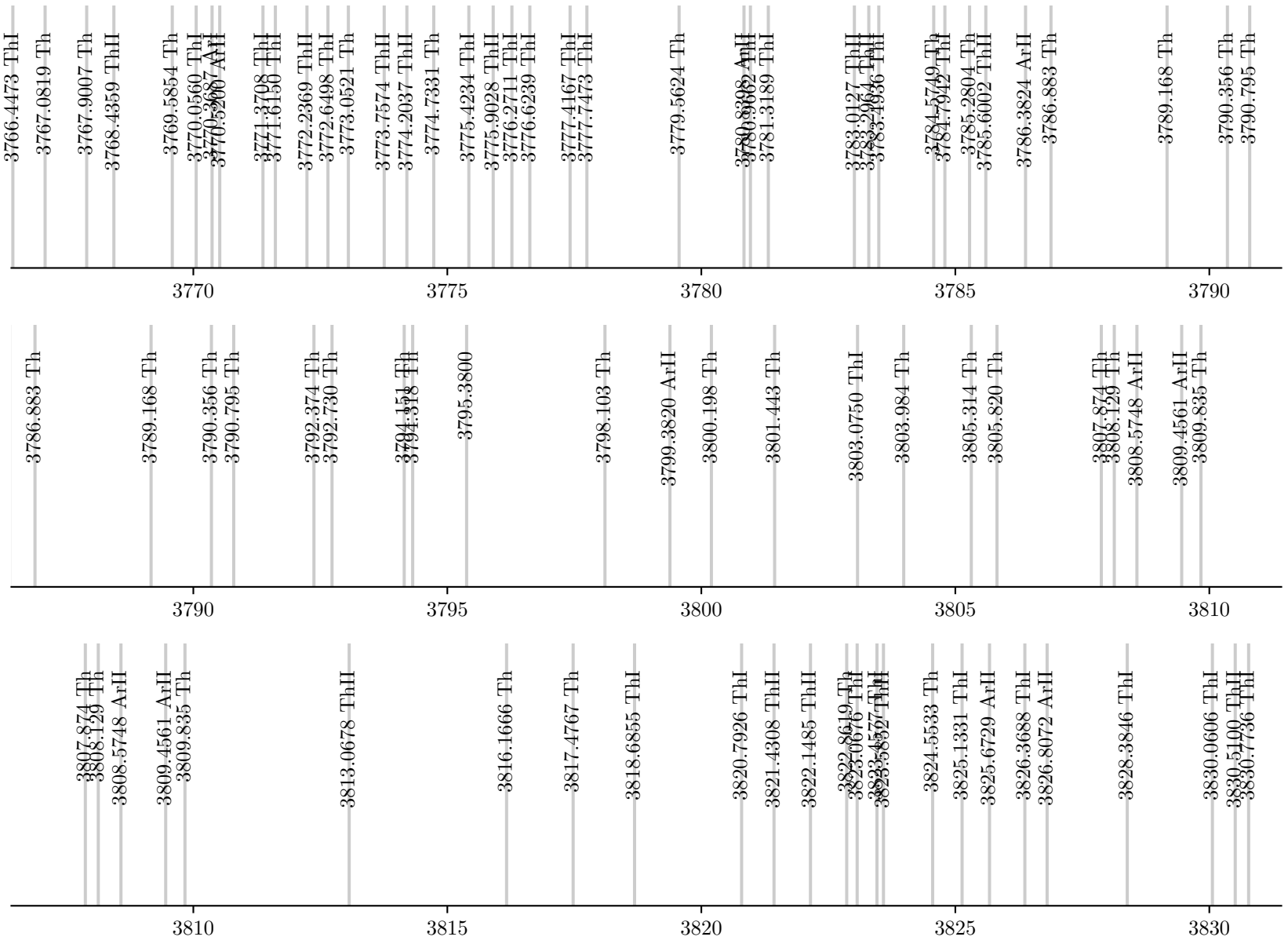
Astronomical Consultants & Equipment, Inc.

November 9, 2020

Order 38



Dispersion:	0.137 Å pix^{-1}
Central wavelength:	3917 Å
Wavelength range:	3769 Å – 4050 Å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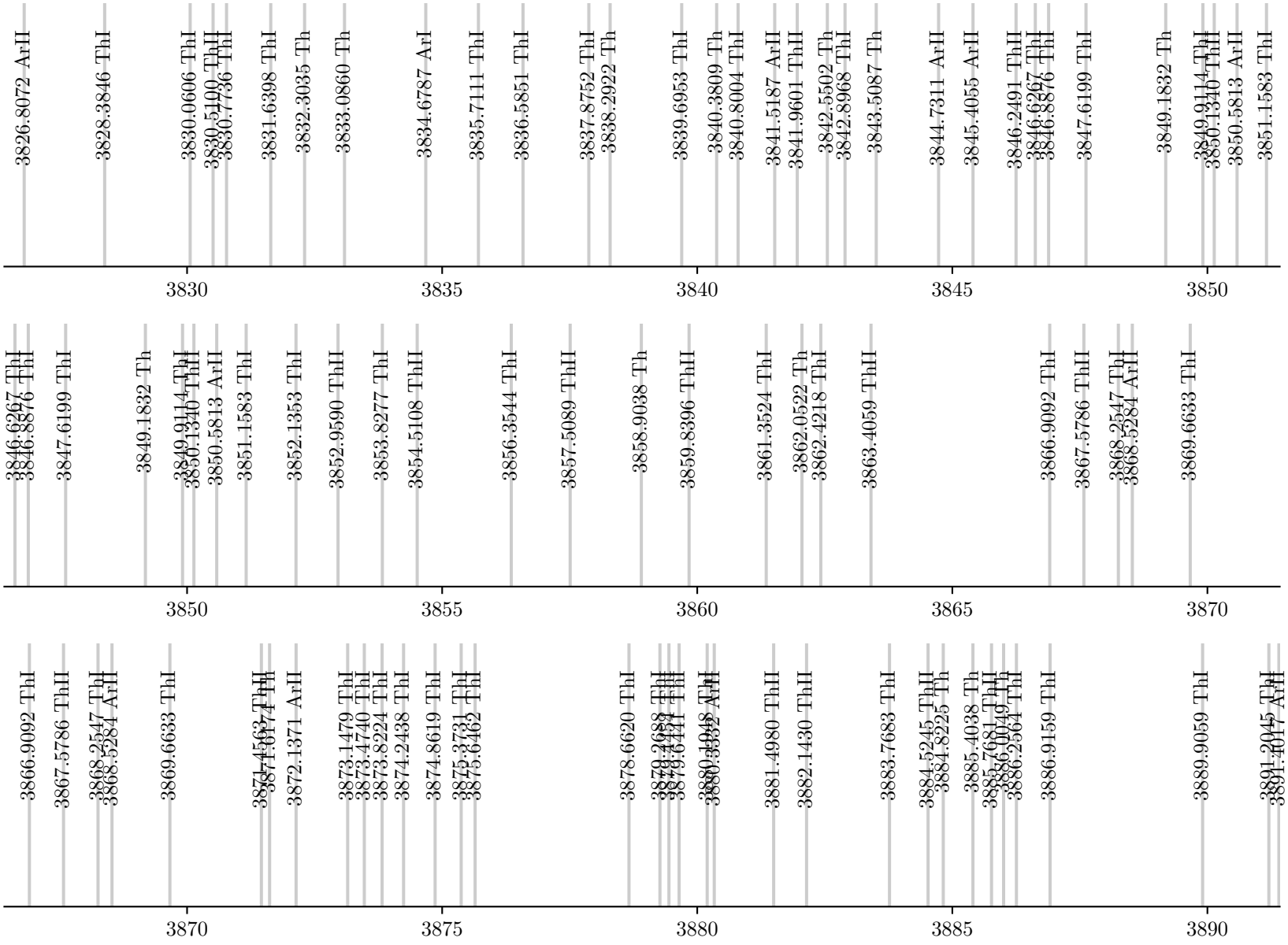
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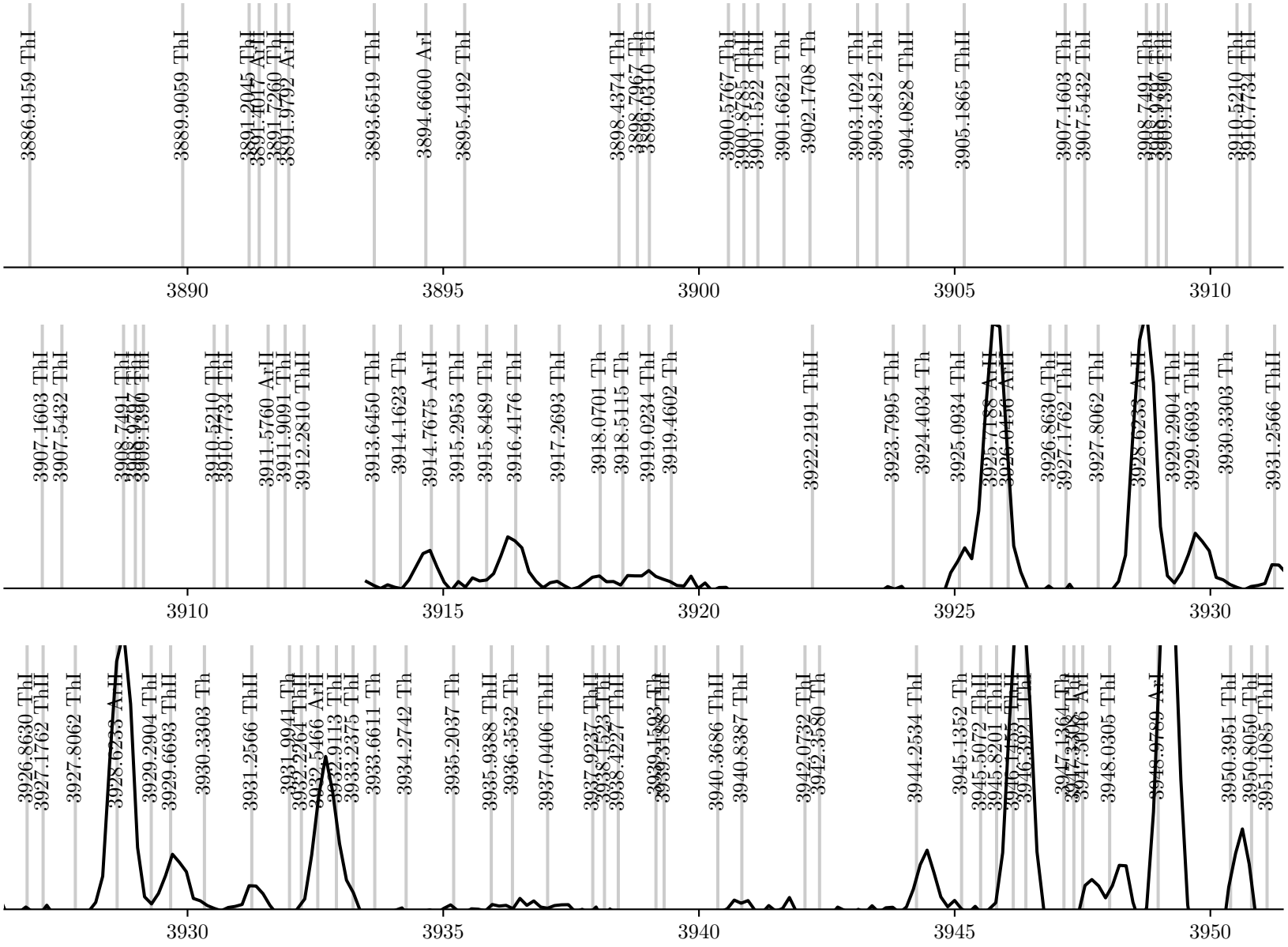
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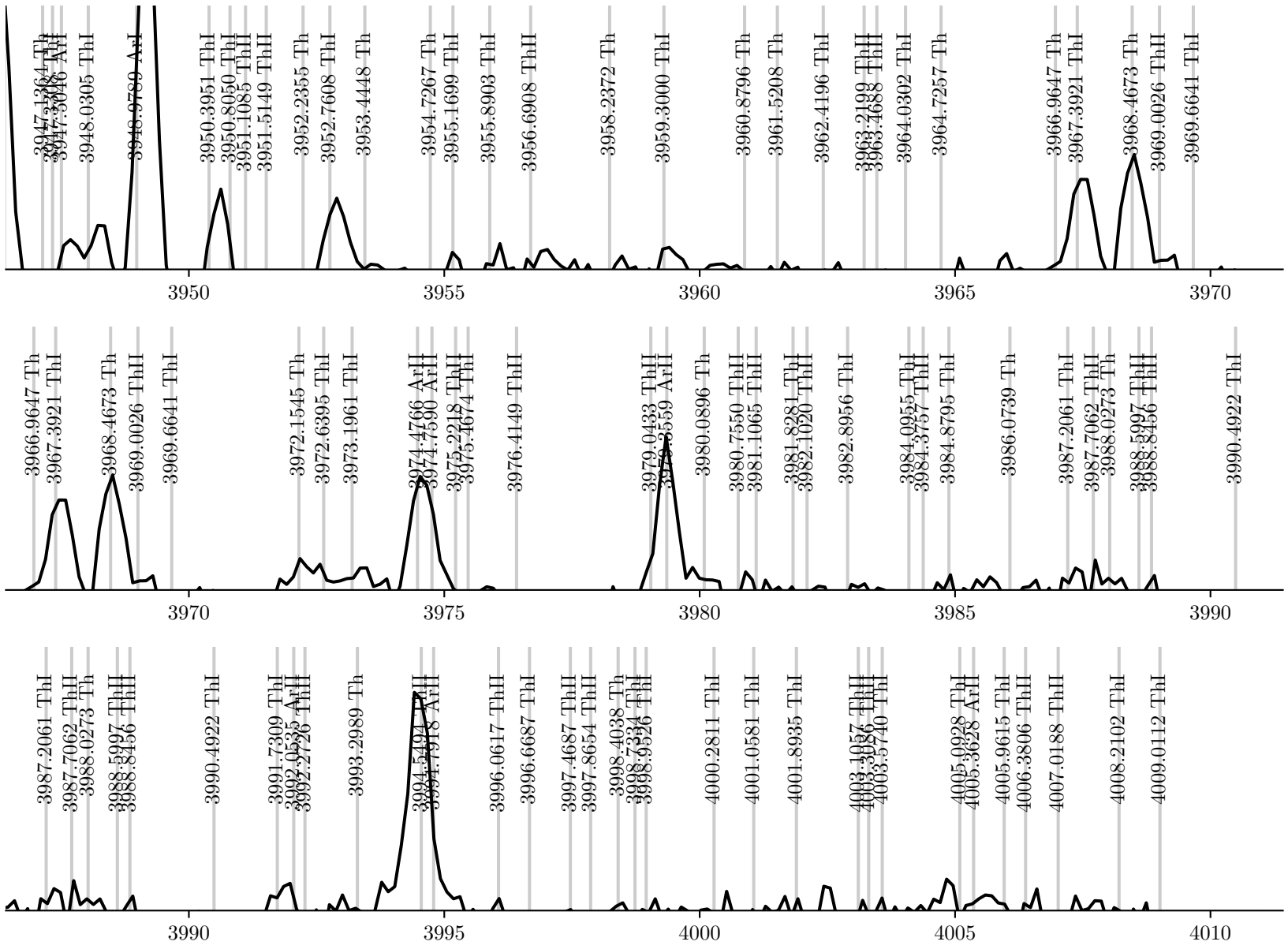
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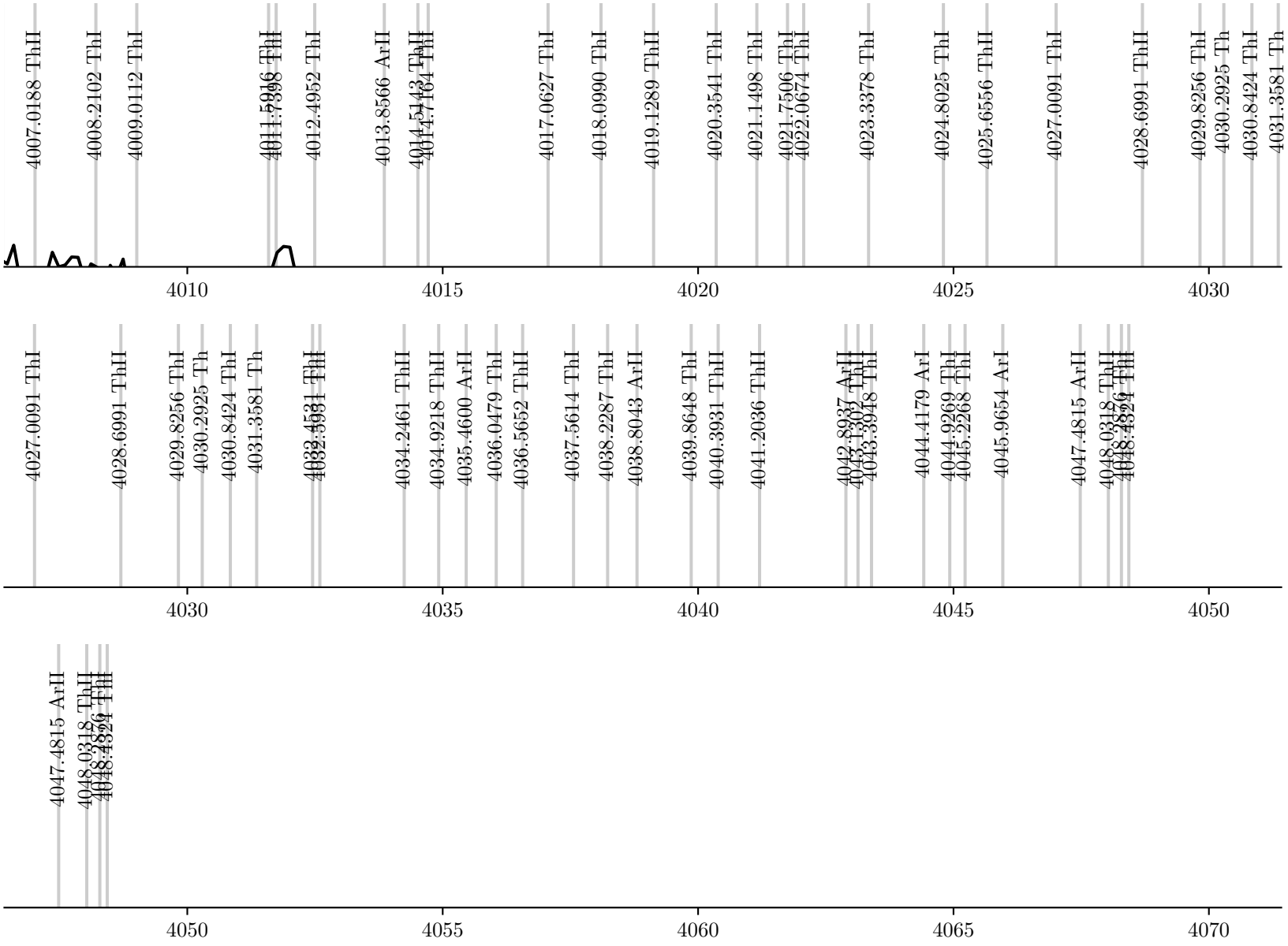
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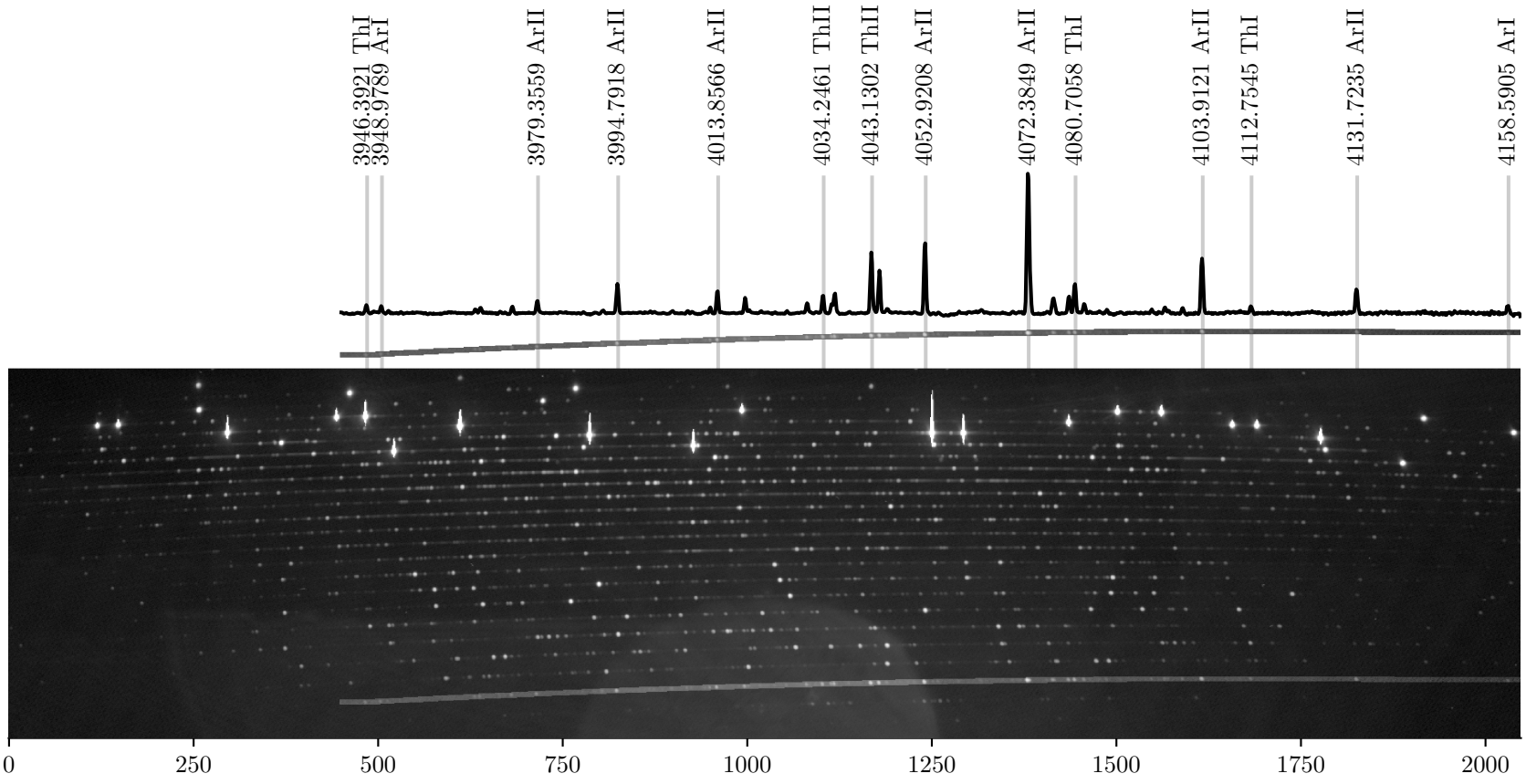

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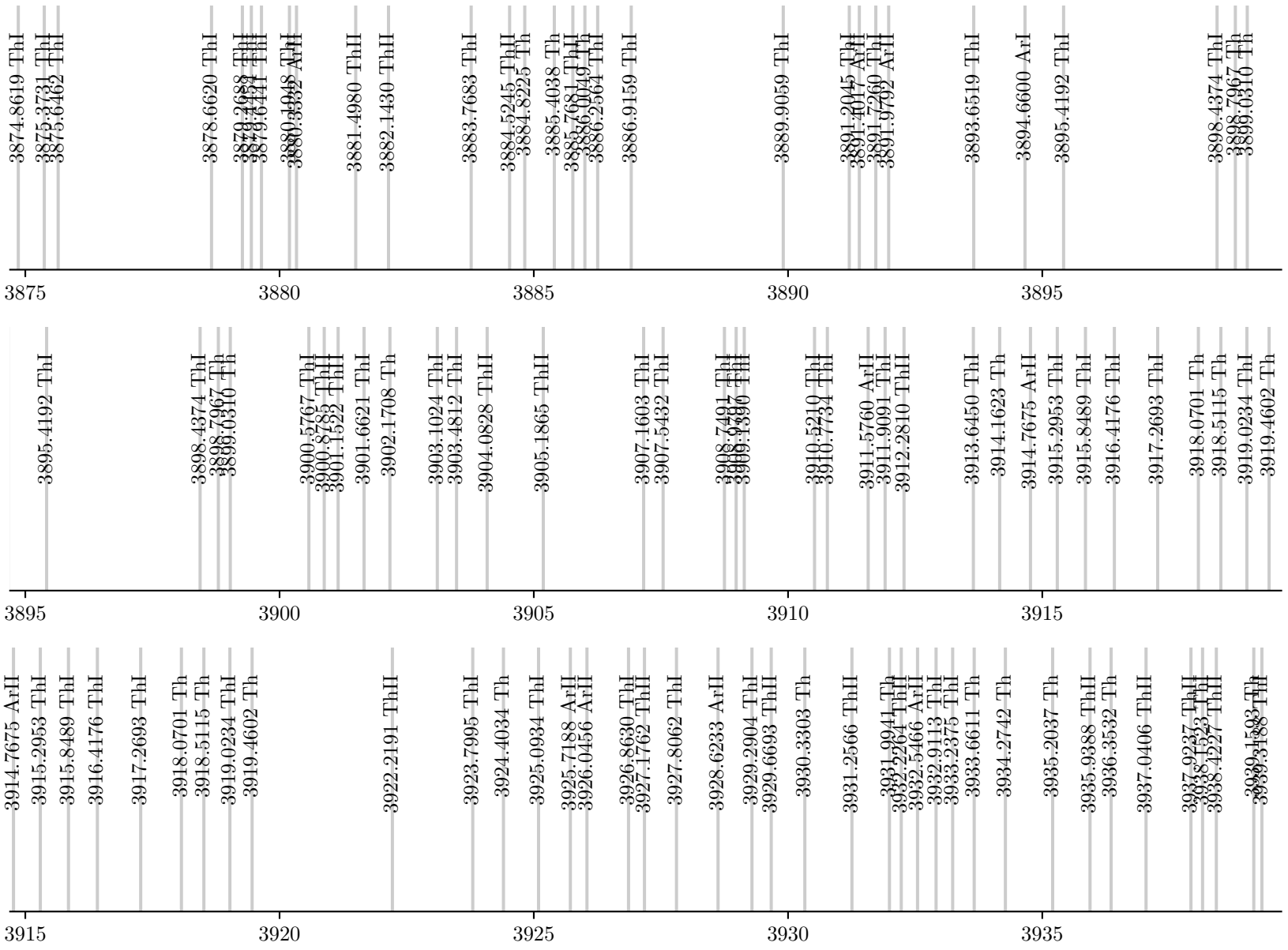
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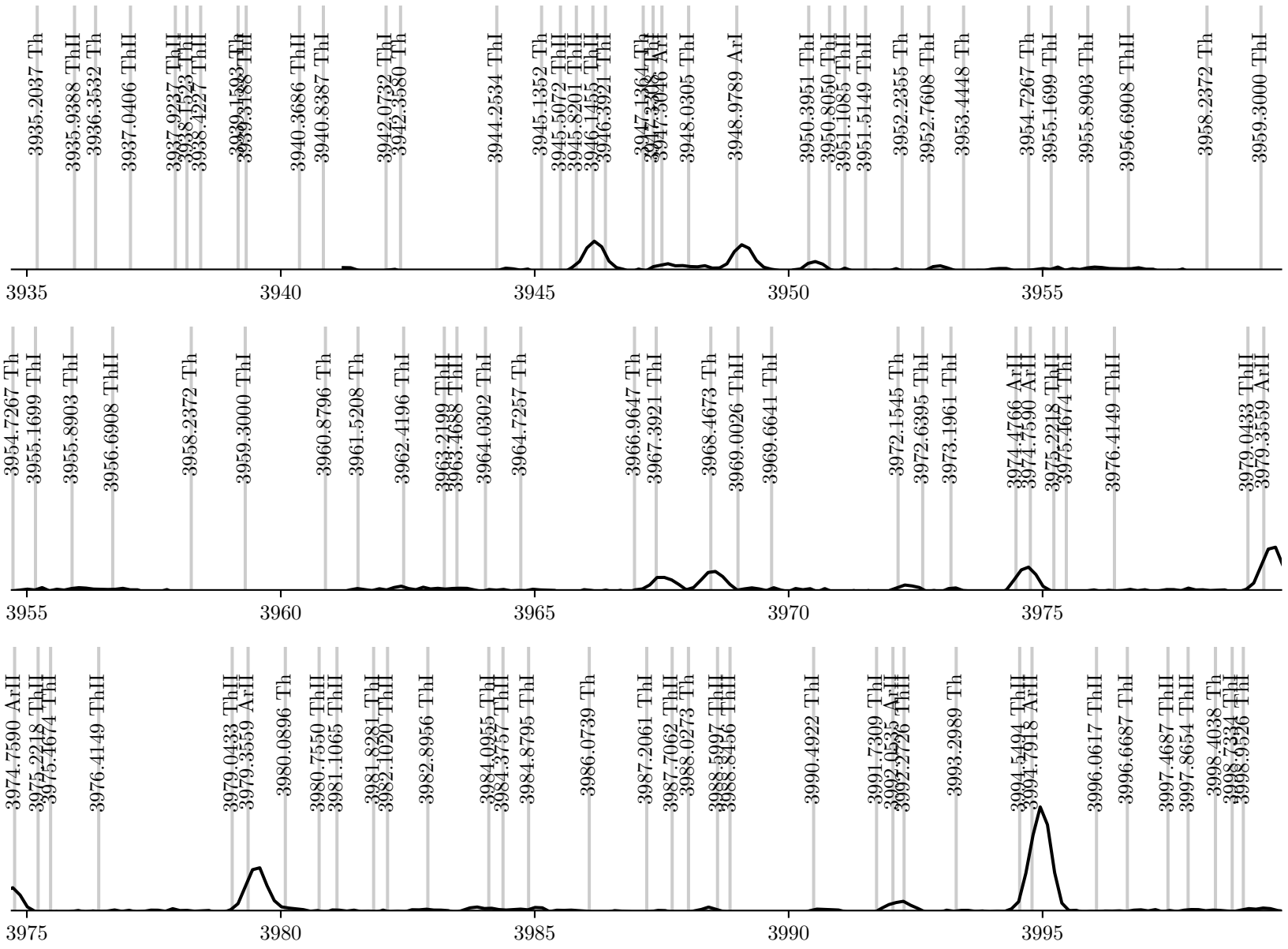
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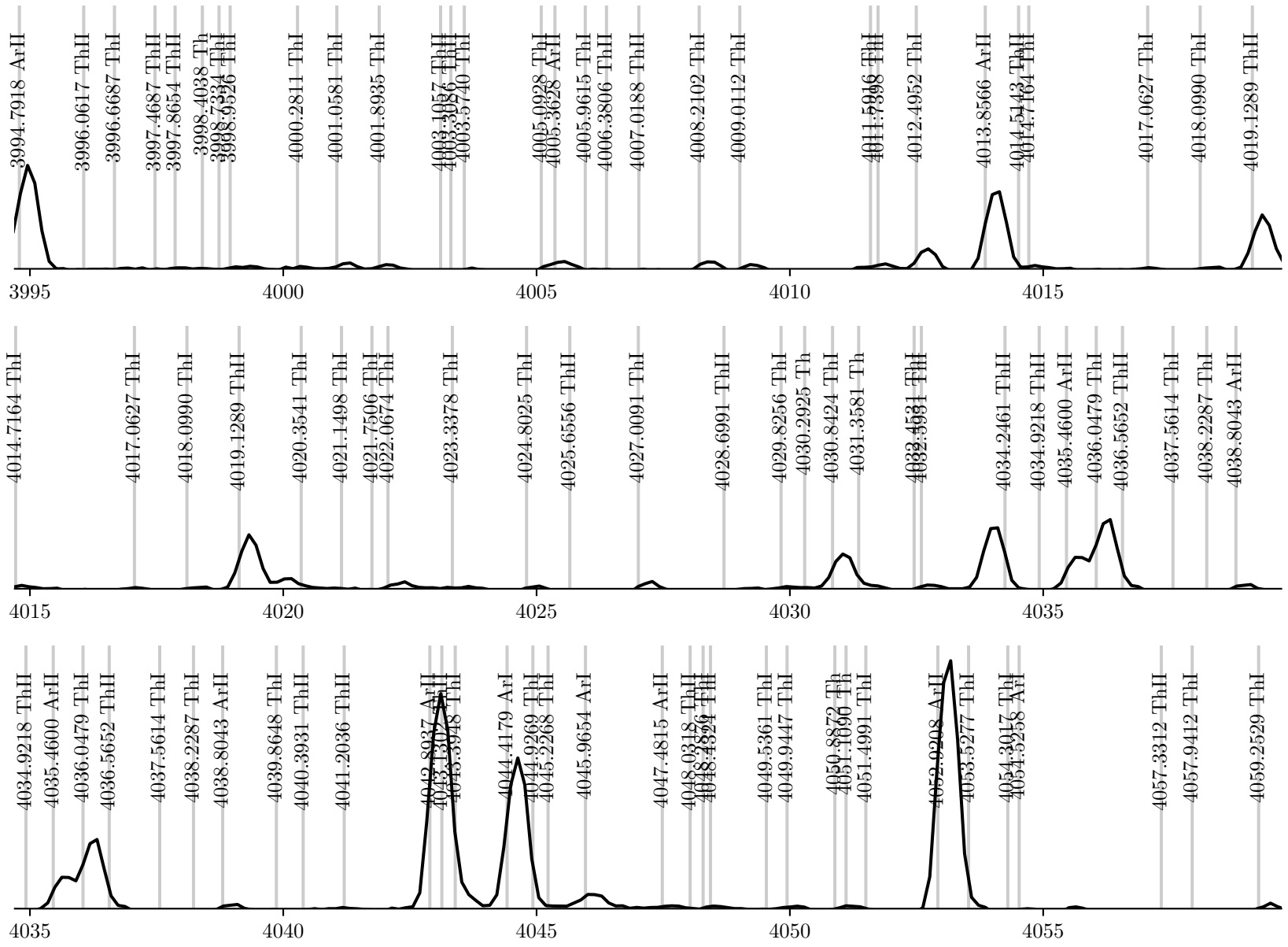
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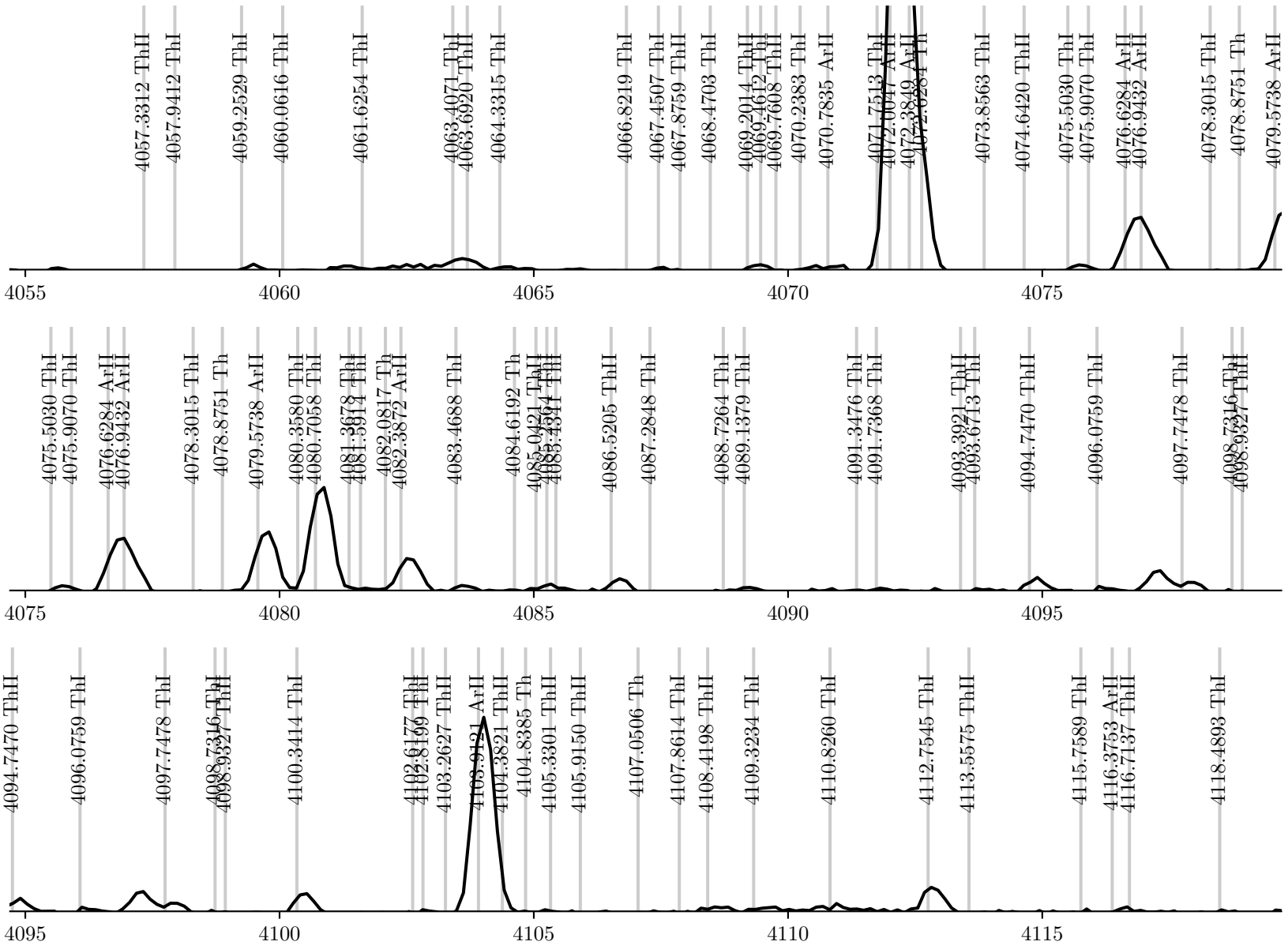
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4076.9432 ArII
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4070
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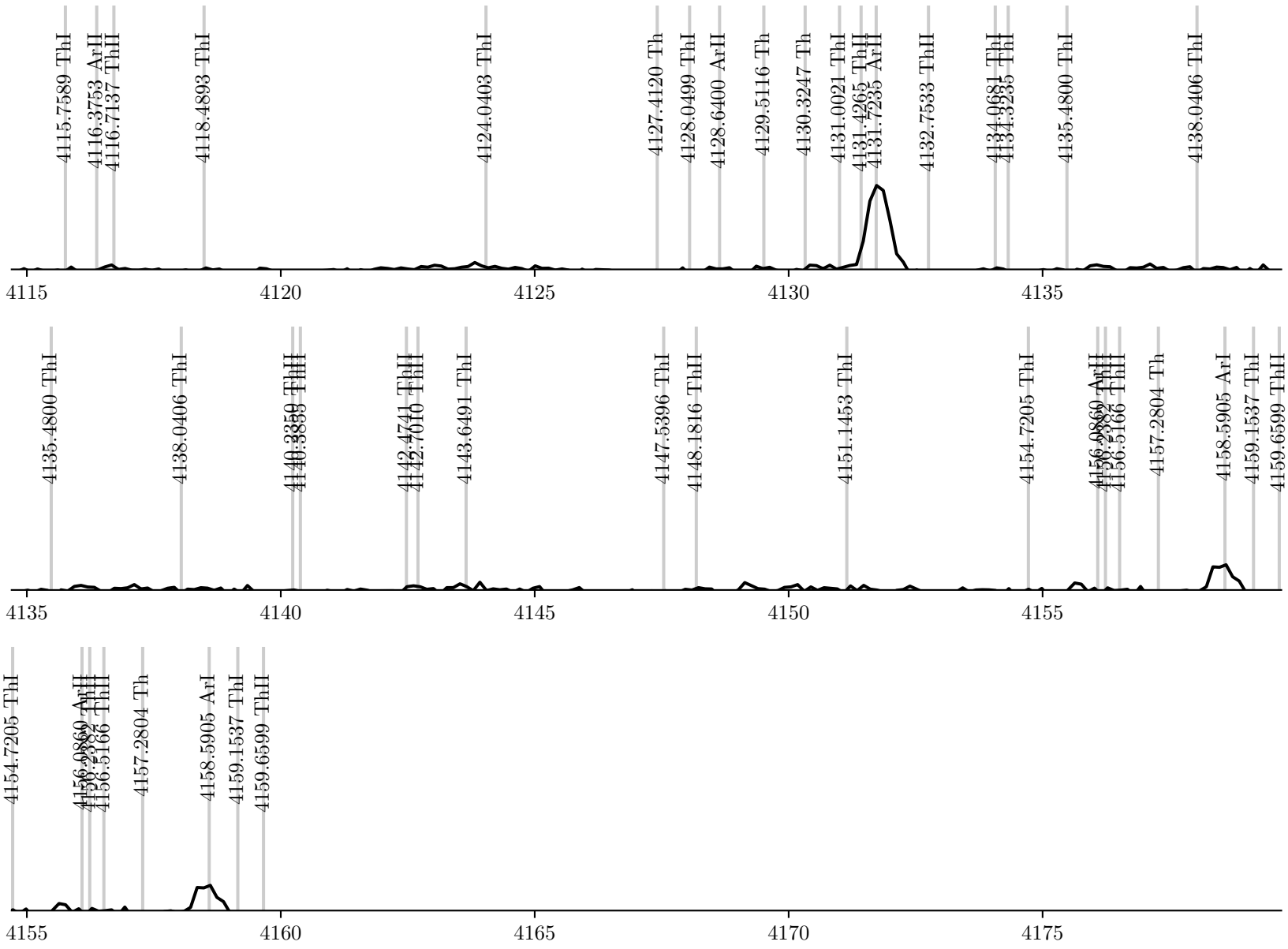
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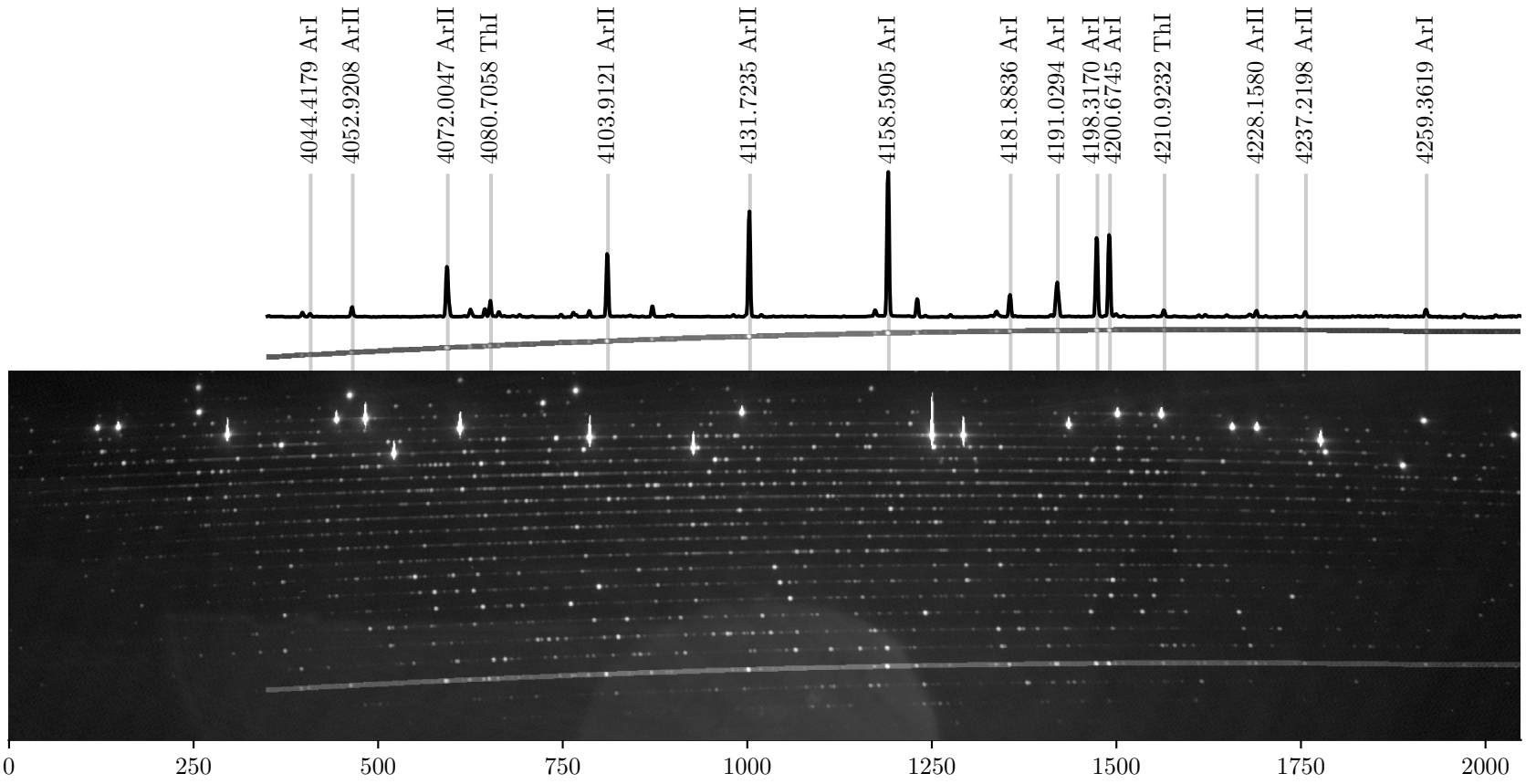
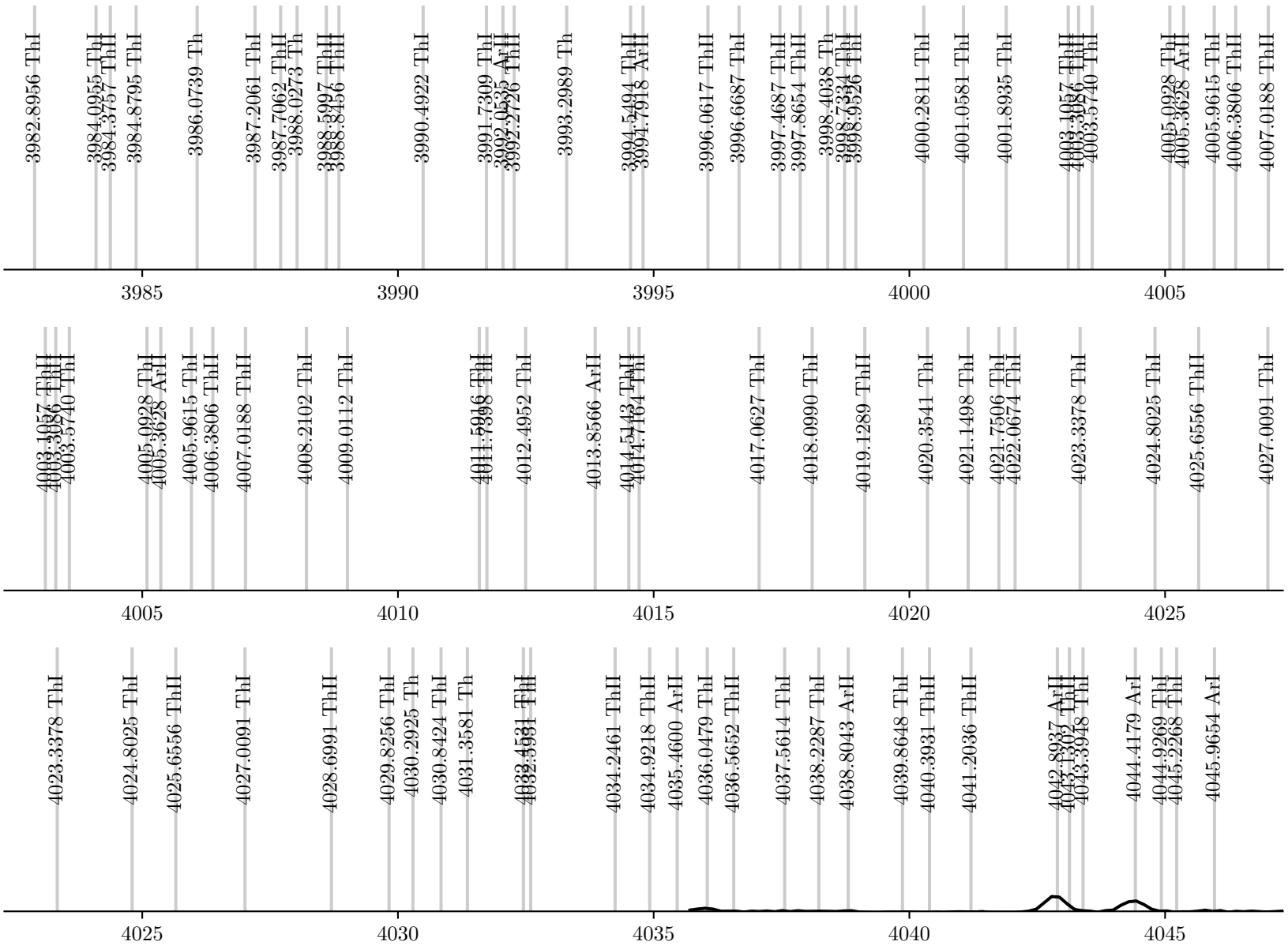
Order 36



Dispersion:	0.144 Å pix^{-1}
Central wavelength:	4135 Å
Wavelength range:	3983 Å – 4276 Å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3991.7309 ThI
3992.0535 ArII
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3998.9526 ThI
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4005.3628 ArII
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3990
3995
4000
4005
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4027.0091 ThI
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4010
4015
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4030.2925 Th
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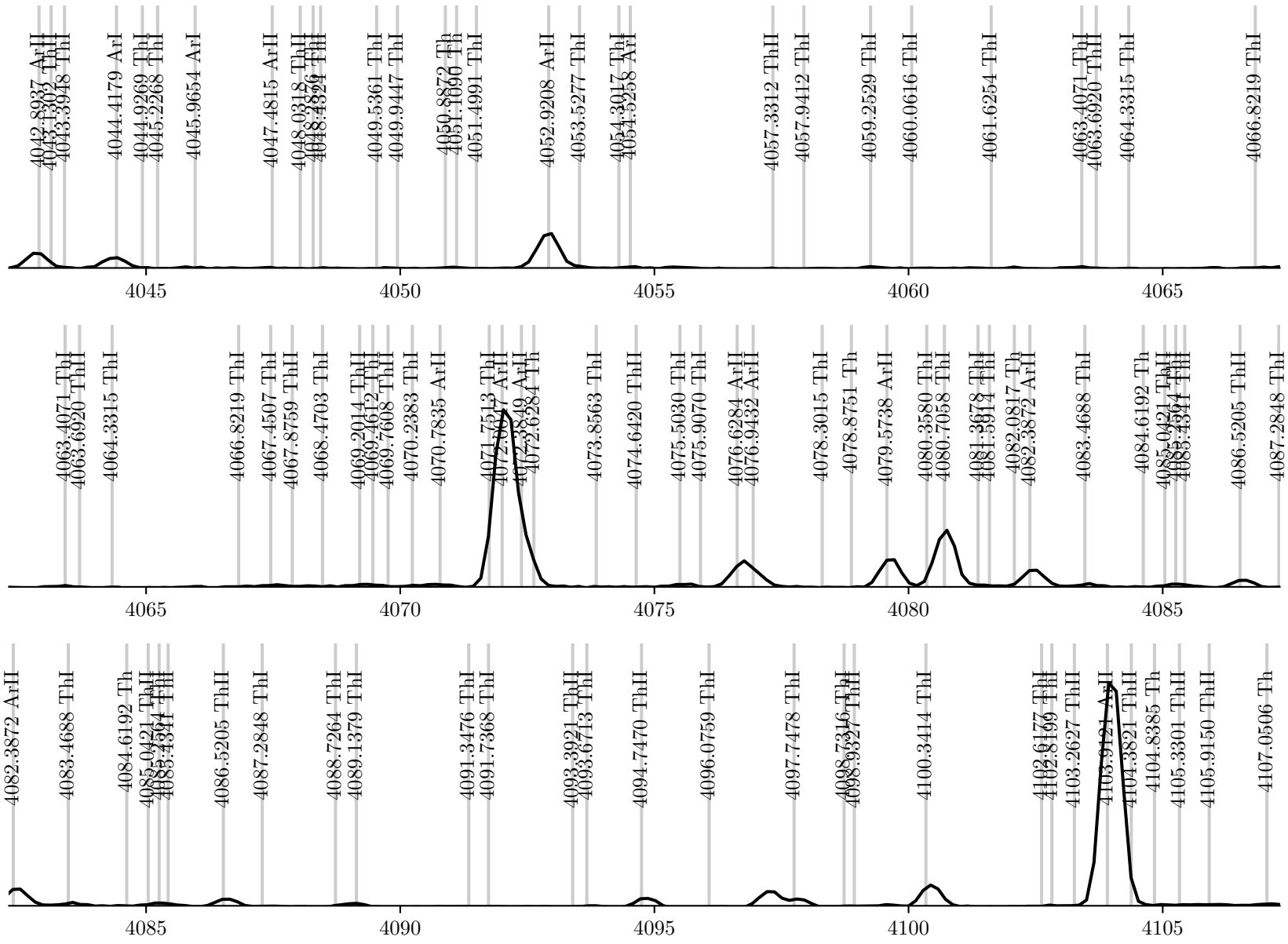
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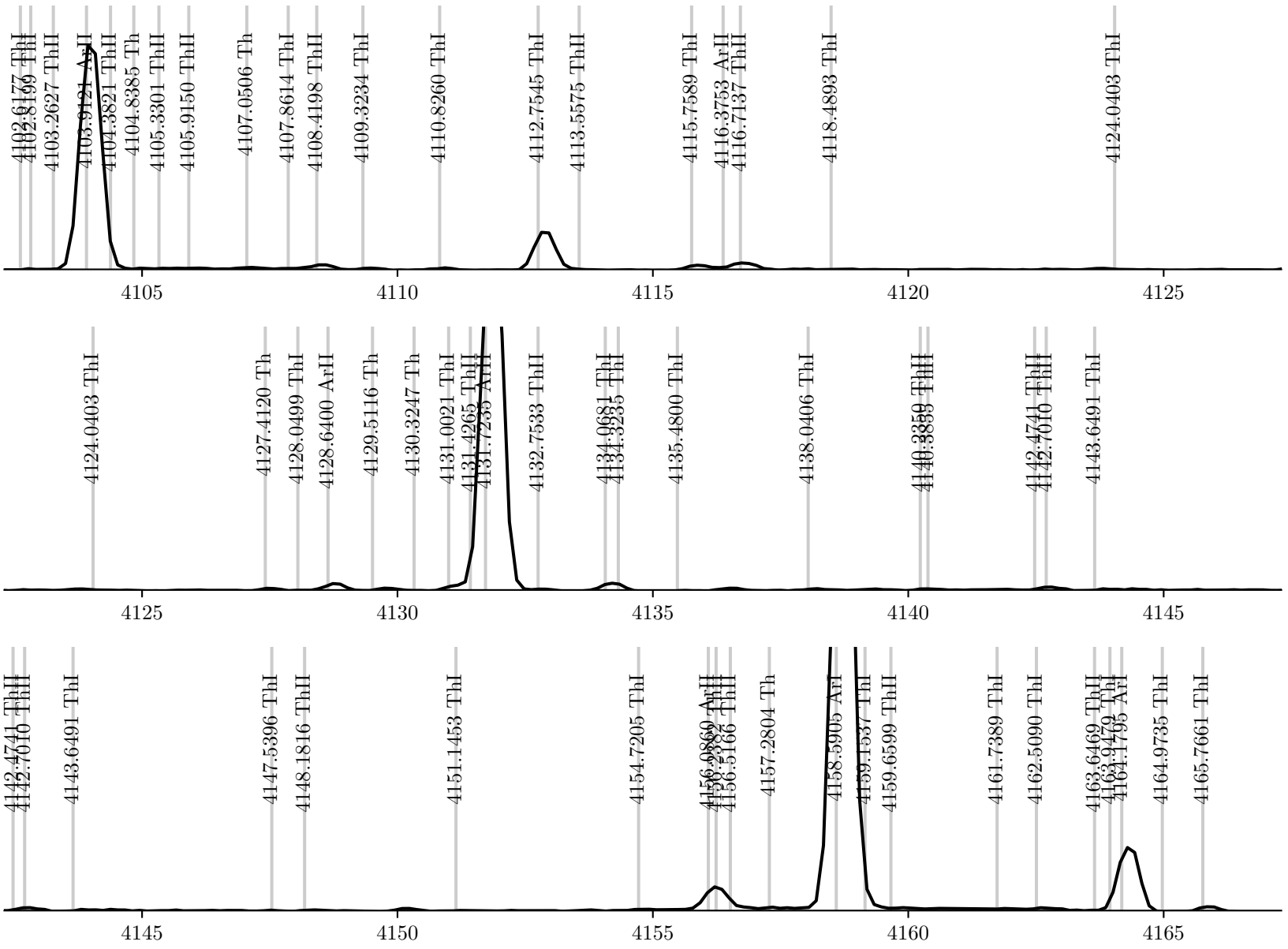
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4131.0021 ThI
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4132.7533 ThII
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4134.3235 ThI
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4156.2382 ThII
4156.5166 ThII
4157.2804 Th
4158.5905 ArI
4159.1537 ThI
4159.6599 ThII
4161.7389 ThI
4162.5090 ThI
4163.6469 ThII
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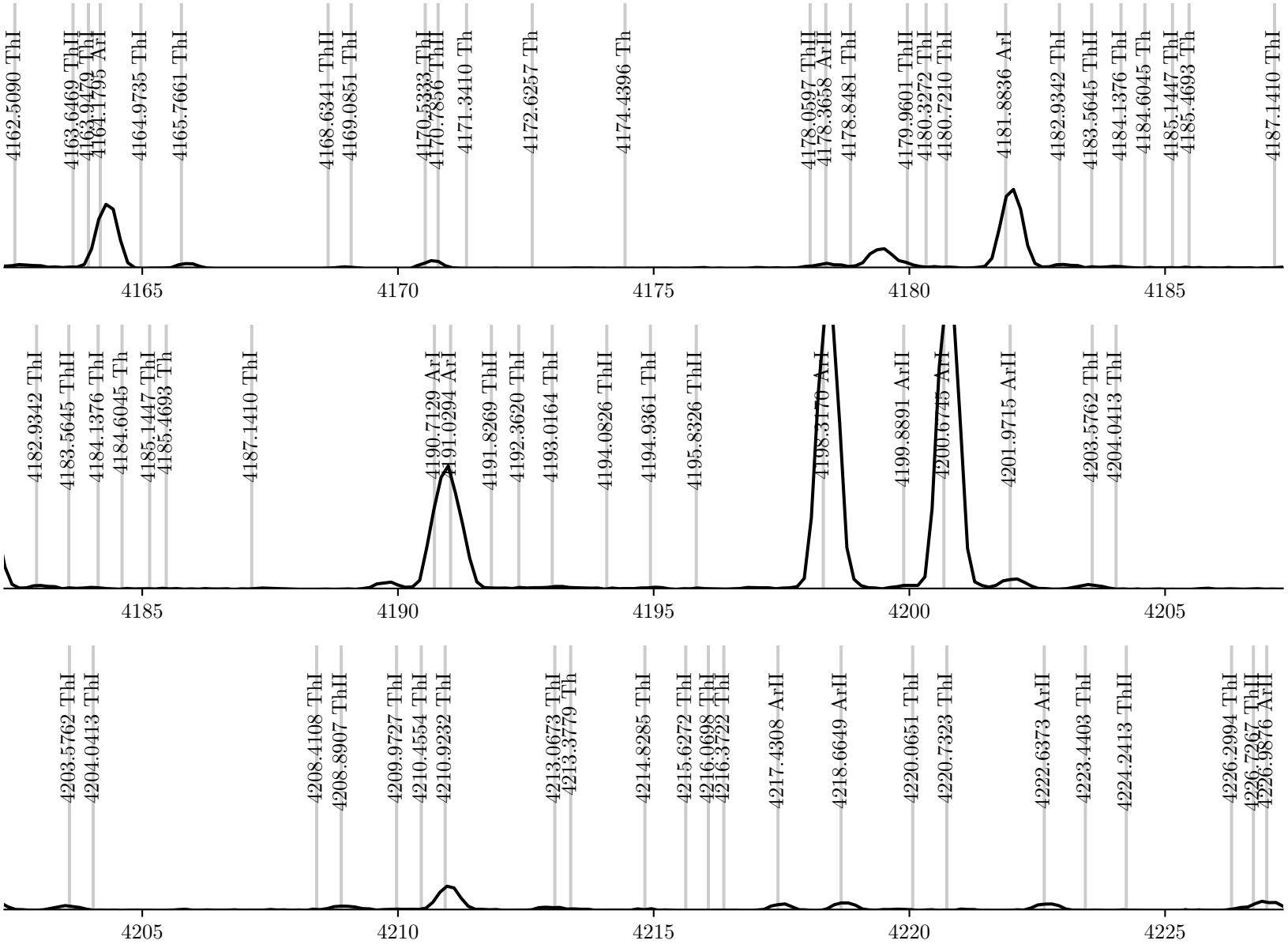
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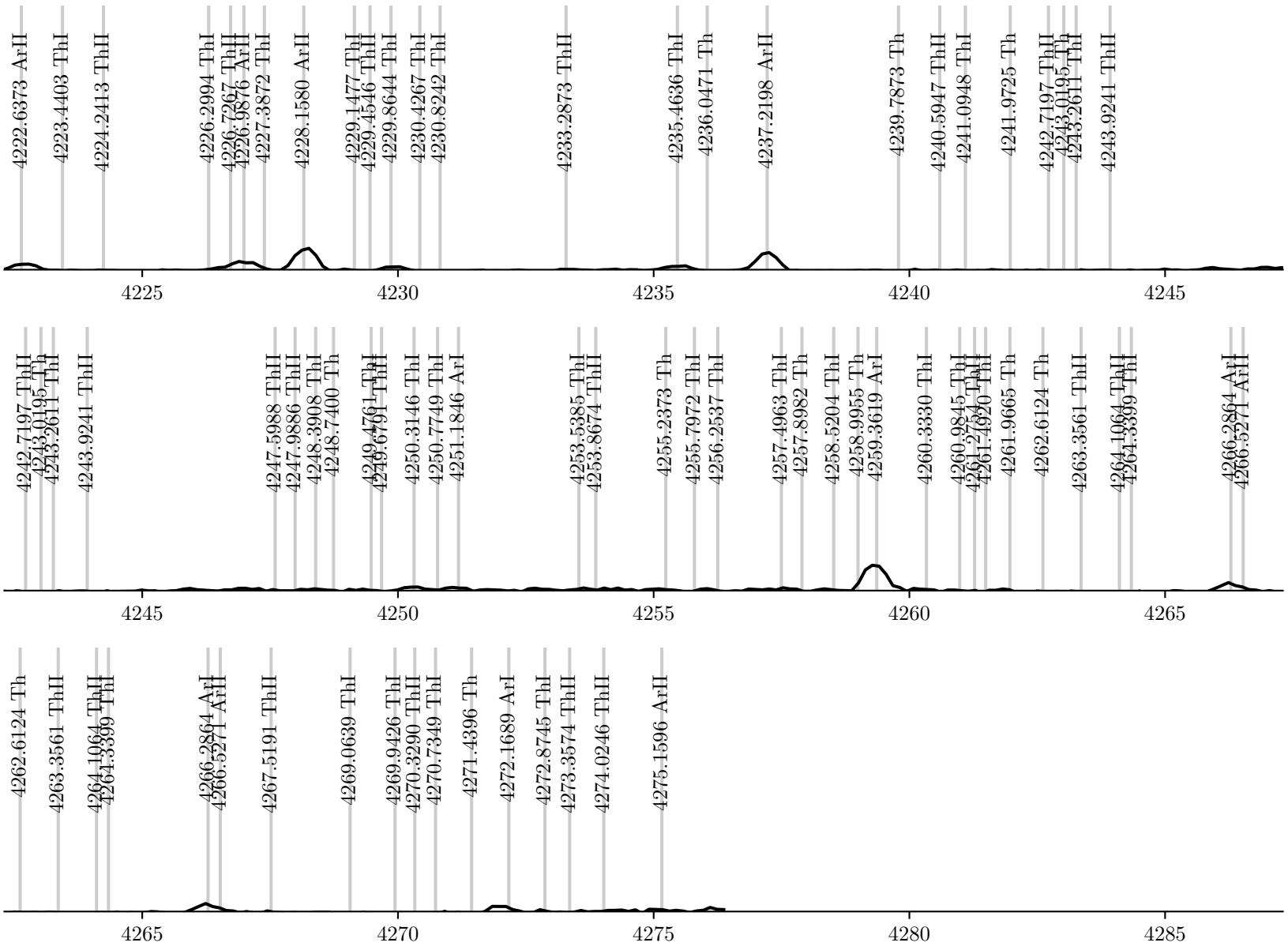
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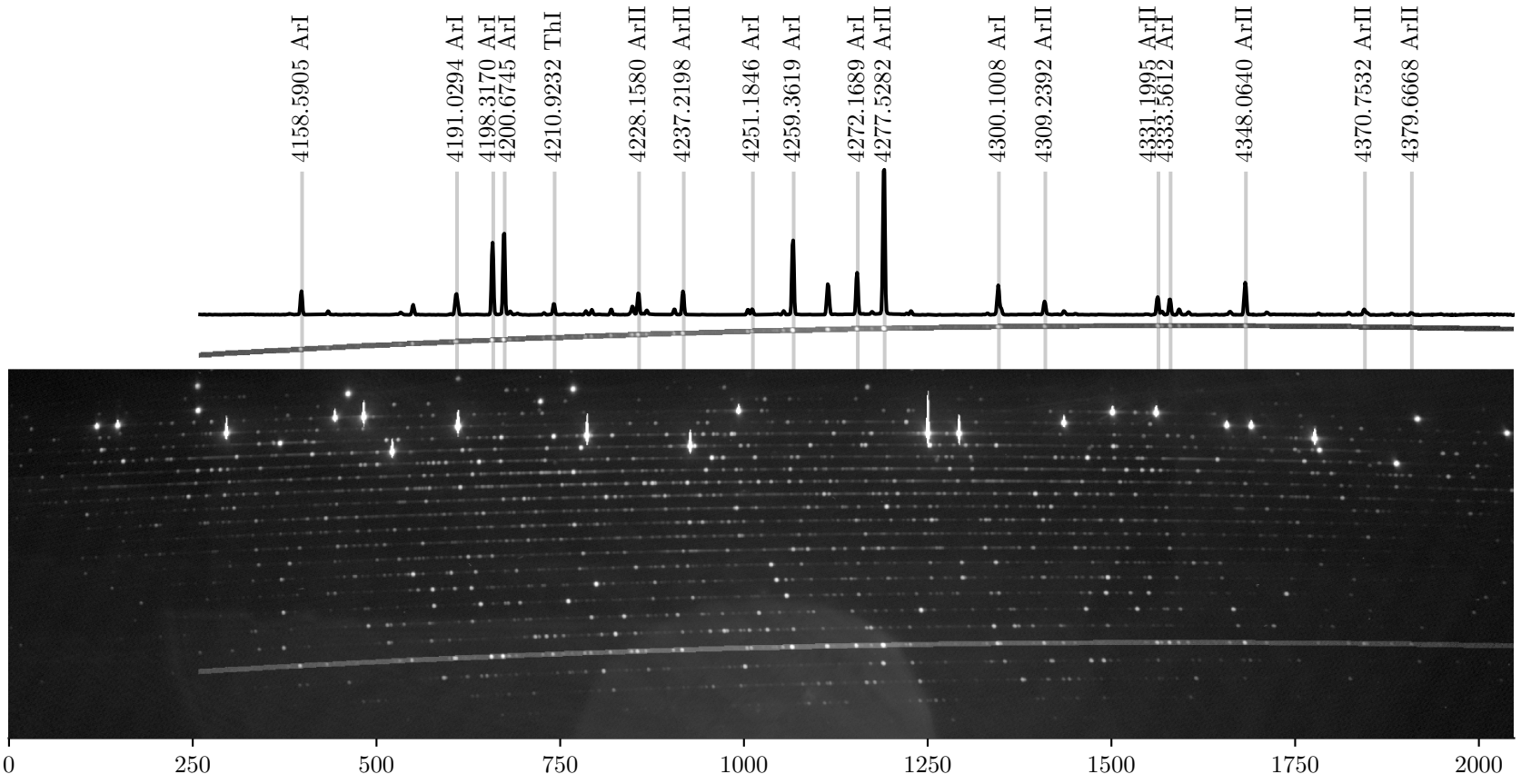
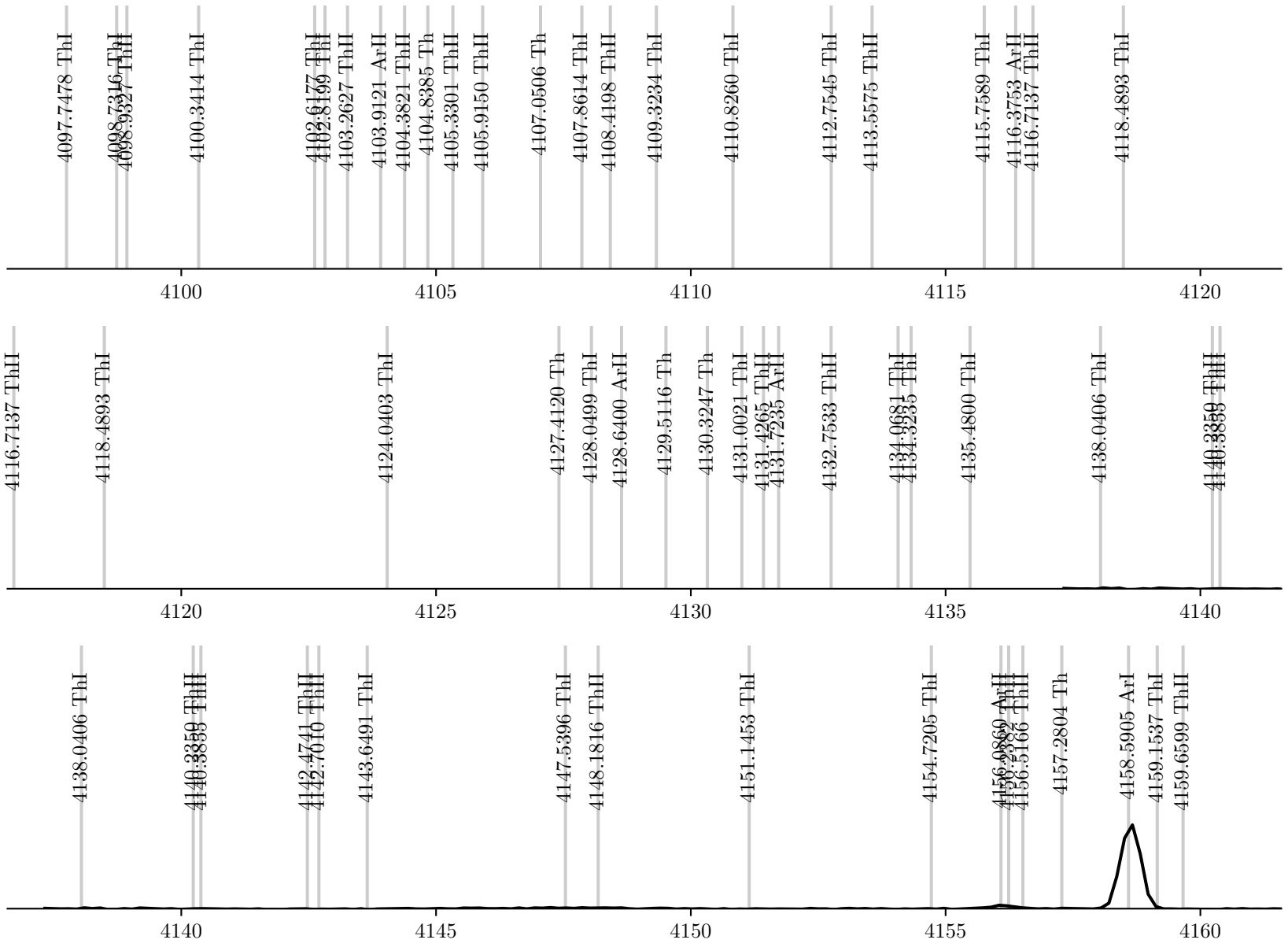
Order 35



Dispersion:	0.148 Å pix^{-1}
Central wavelength:	4253 Å
Wavelength range:	4097 Å – 4399 Å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4109.3234 ThI
4110.8260 ThI
4112.7545 ThI
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4116.3753 ArII
4116.7137 ThII
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4127.4120 Th
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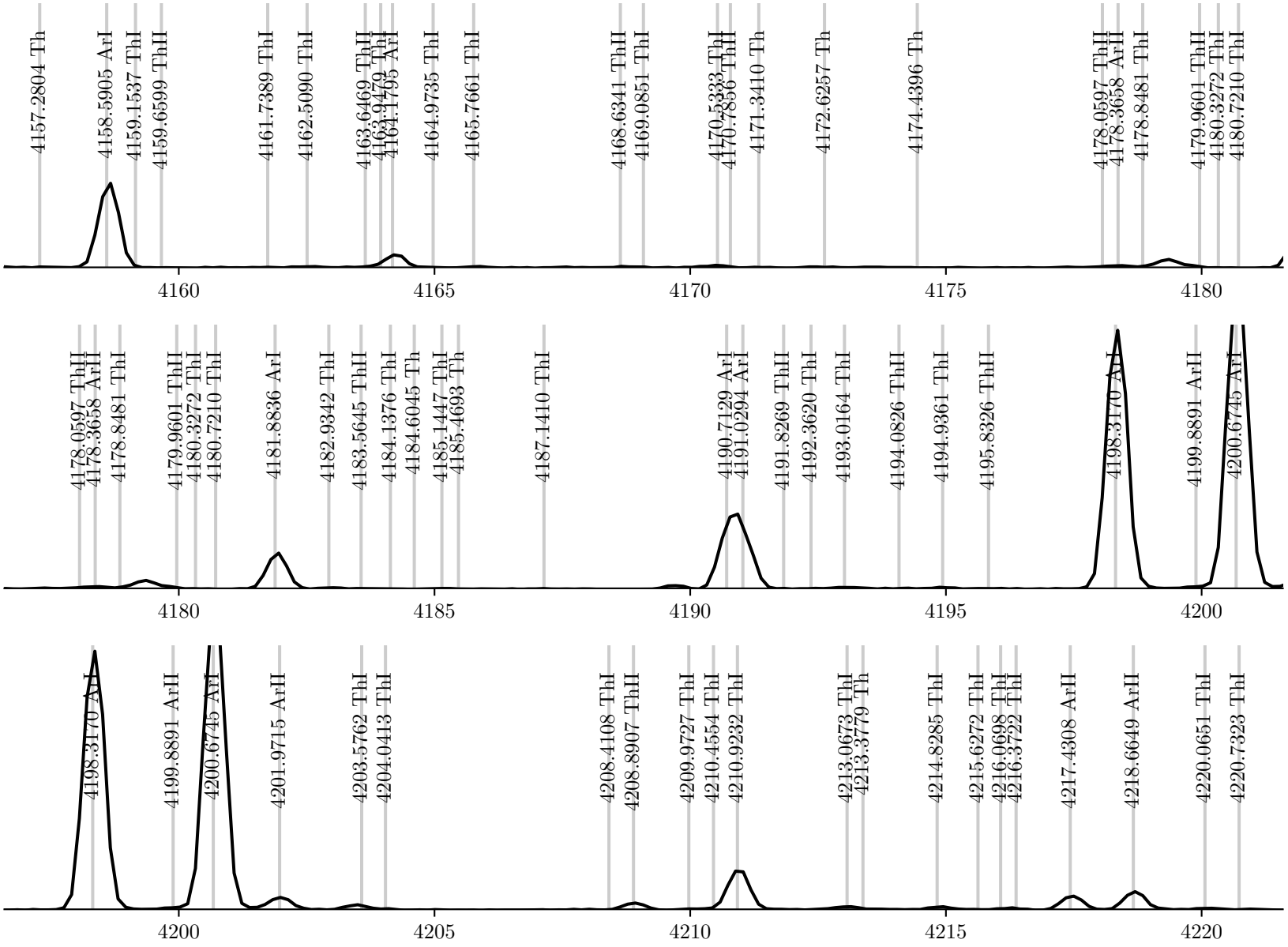
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4172.6257 Th
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4178.0597 ThII
4178.3658 ArII
4178.8481 ThI
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4180.3272 ThI
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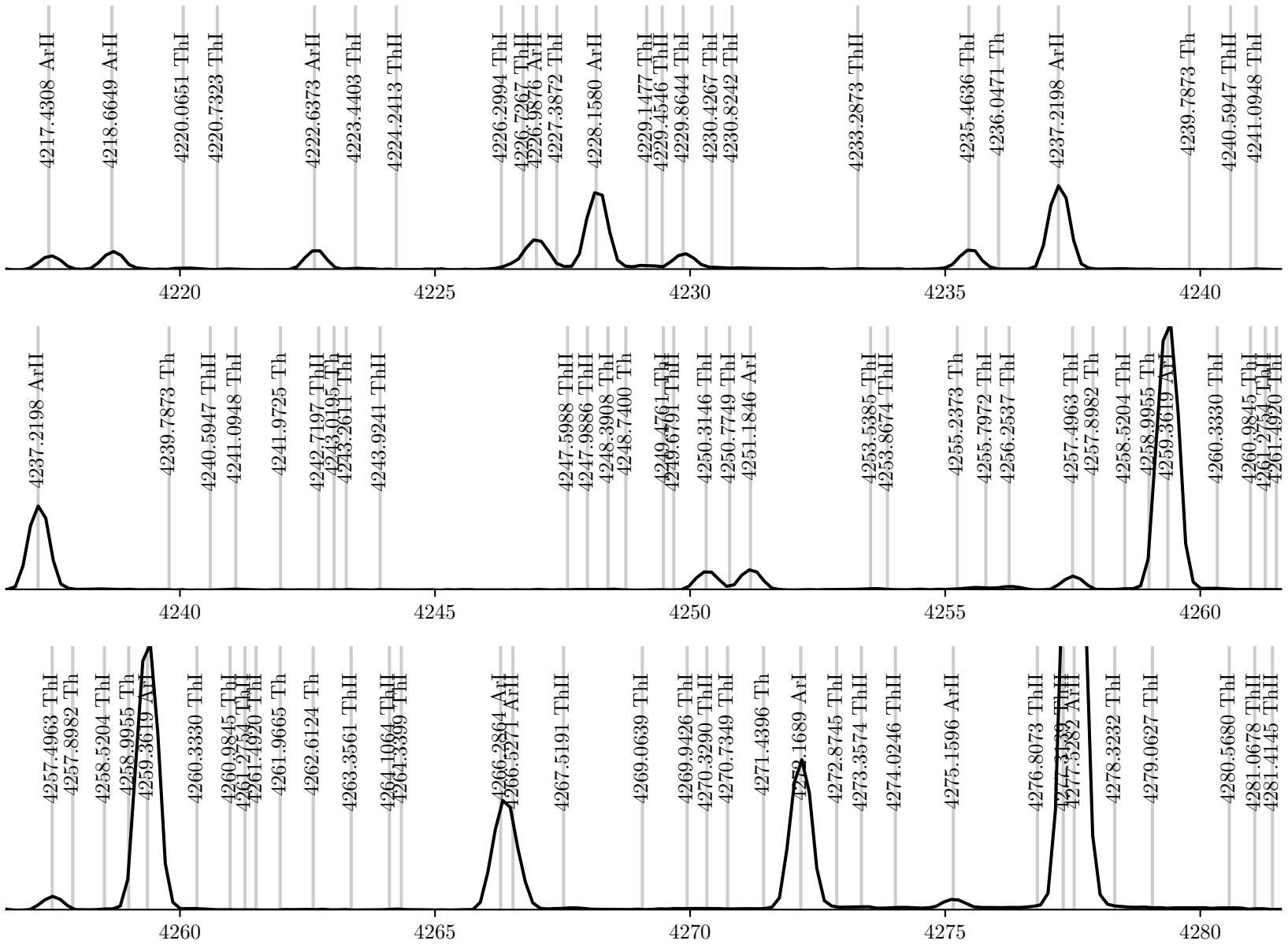
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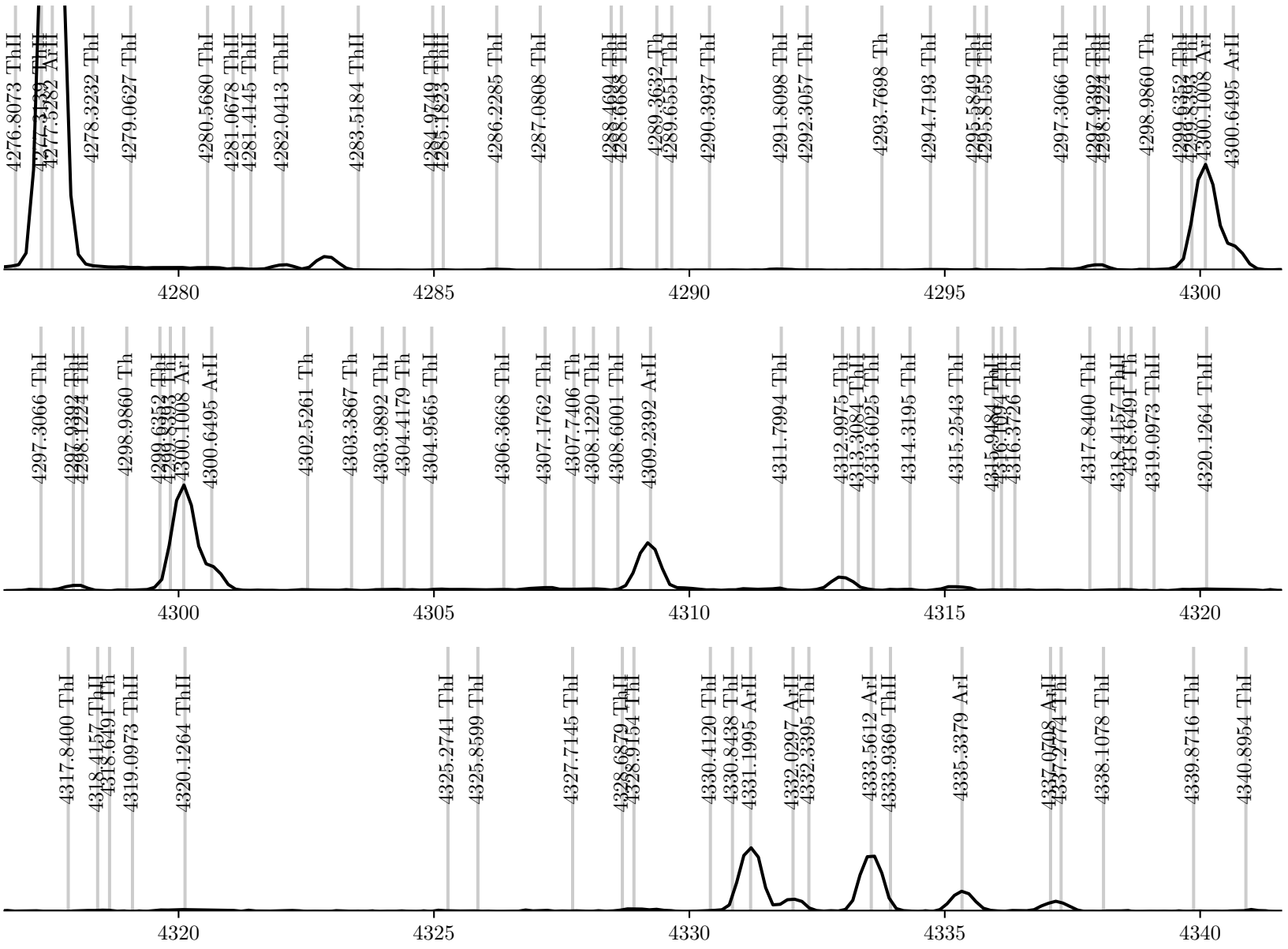
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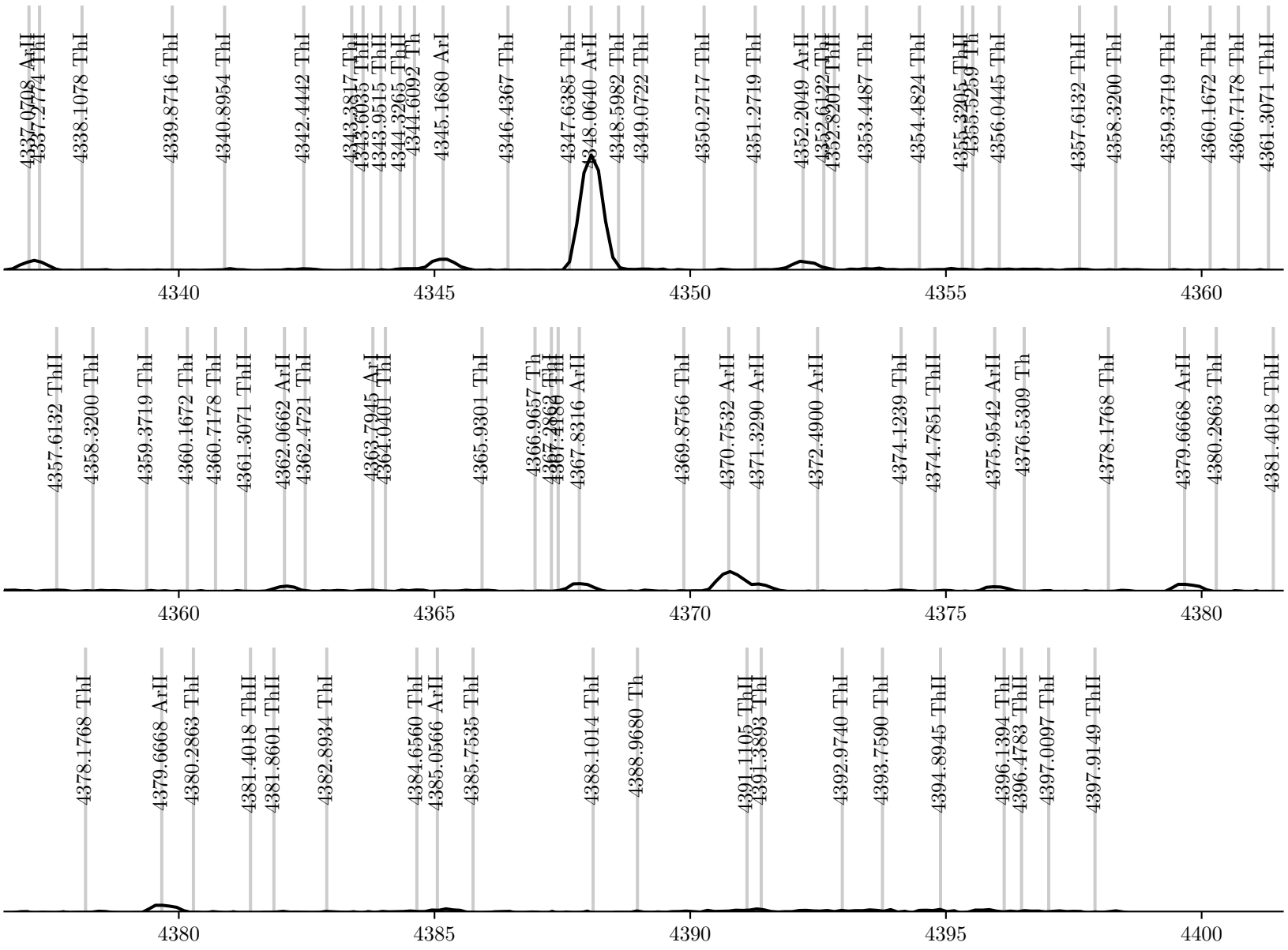
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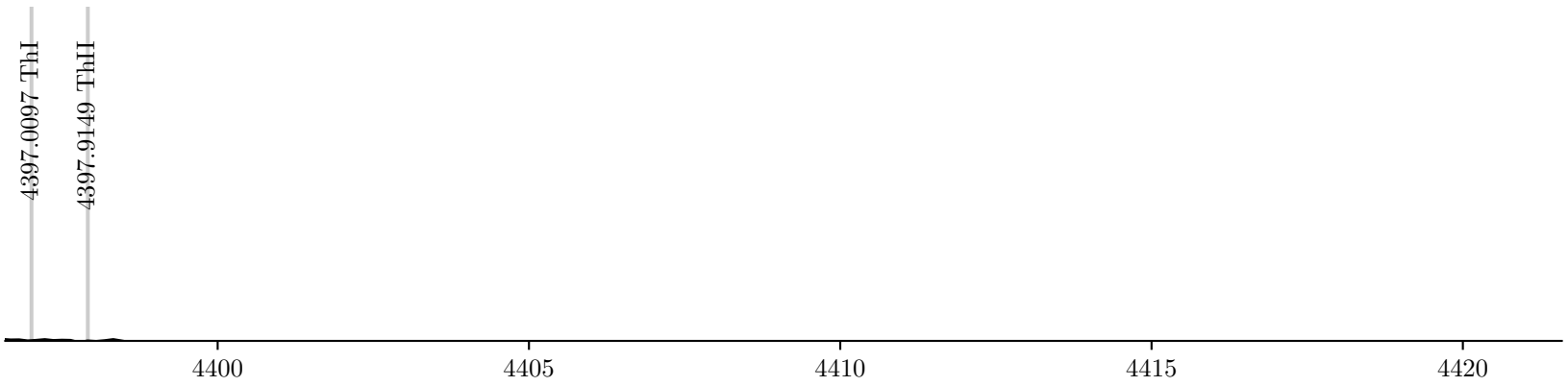
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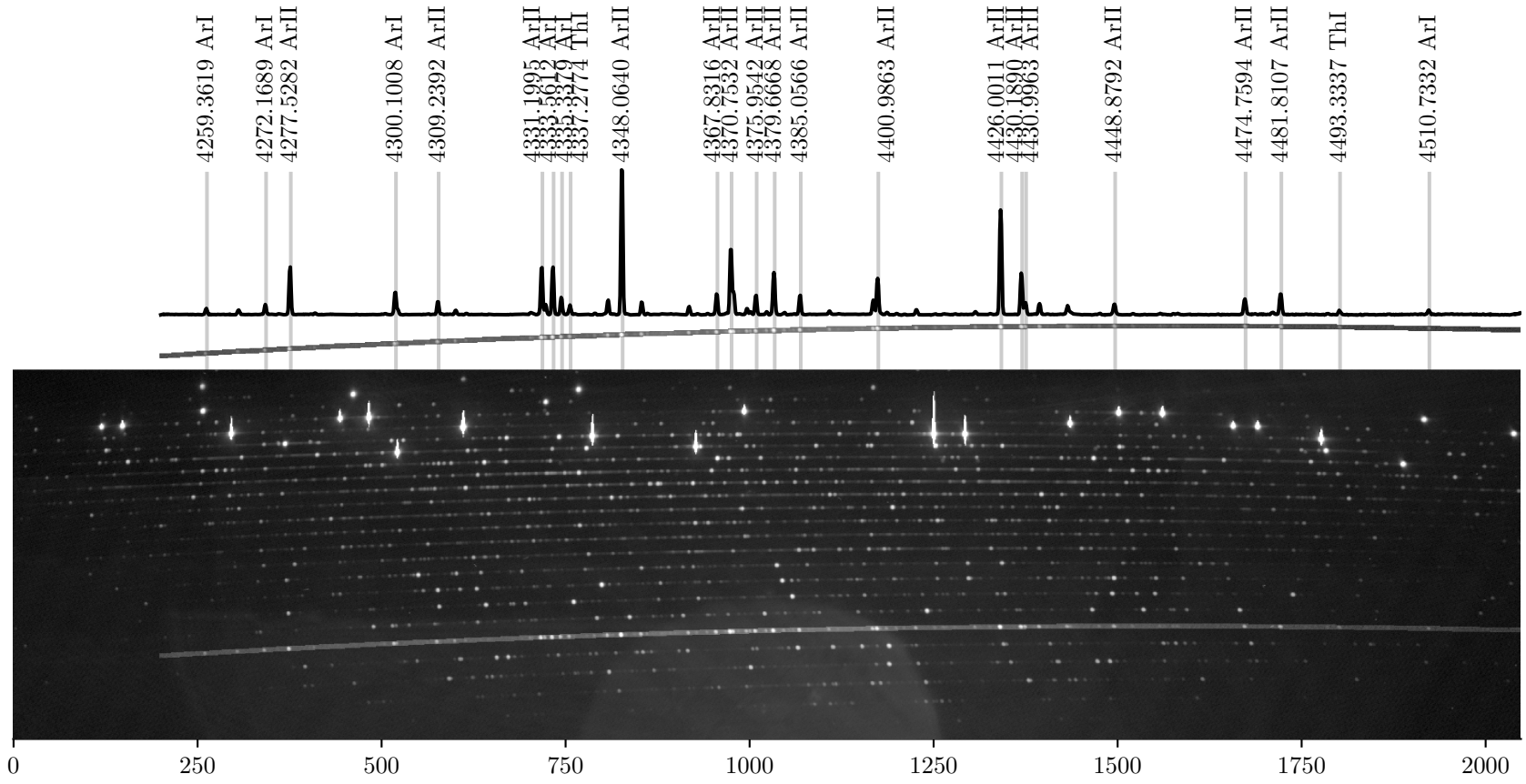
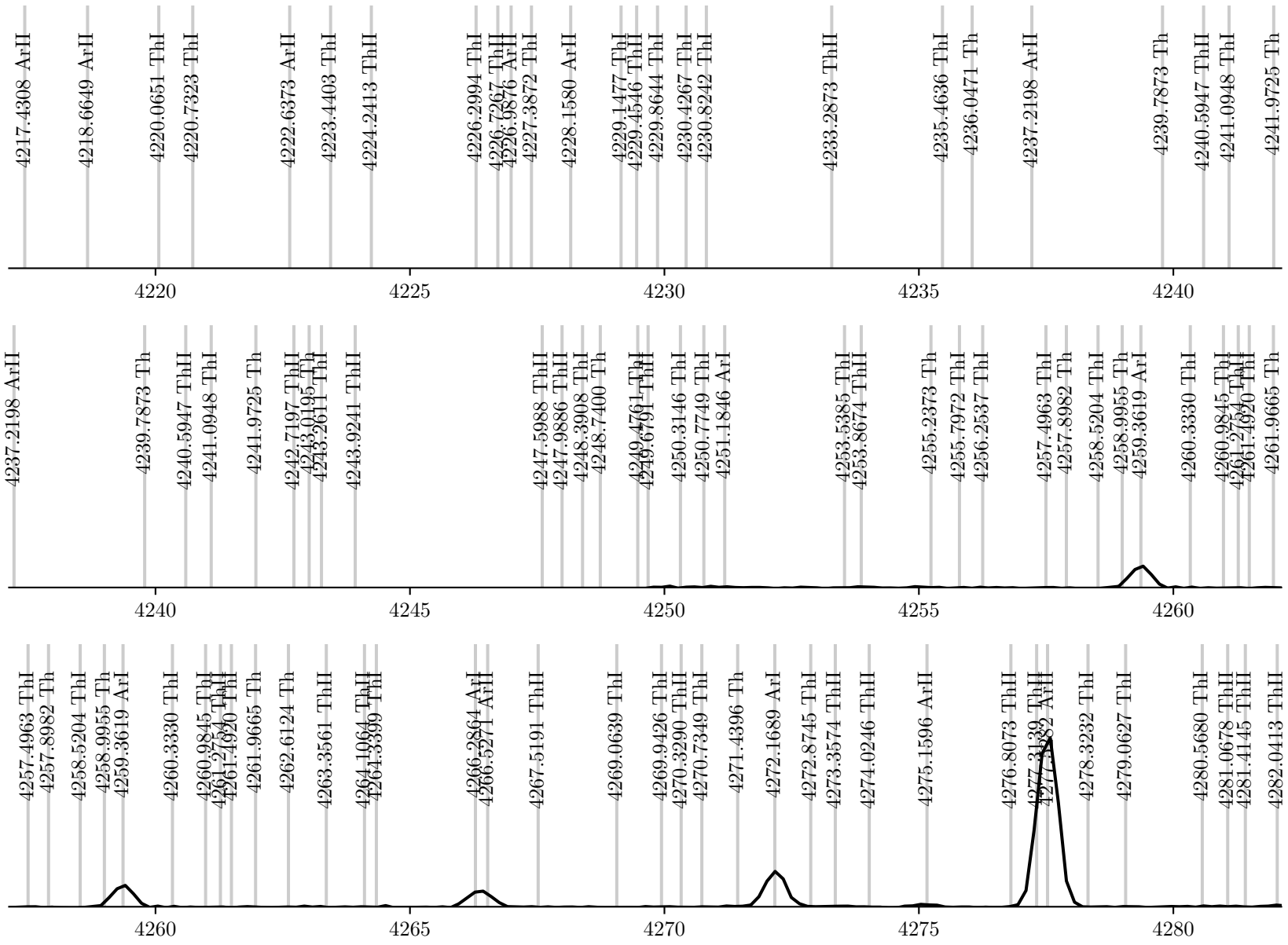


Dispersion:	0.152 Å pix^{-1}
Central wavelength:	4378 Å
Wavelength range:	4217 Å – 4528 Å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4257.8982 Th
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4258.9955 Th
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4260.9845 ThI
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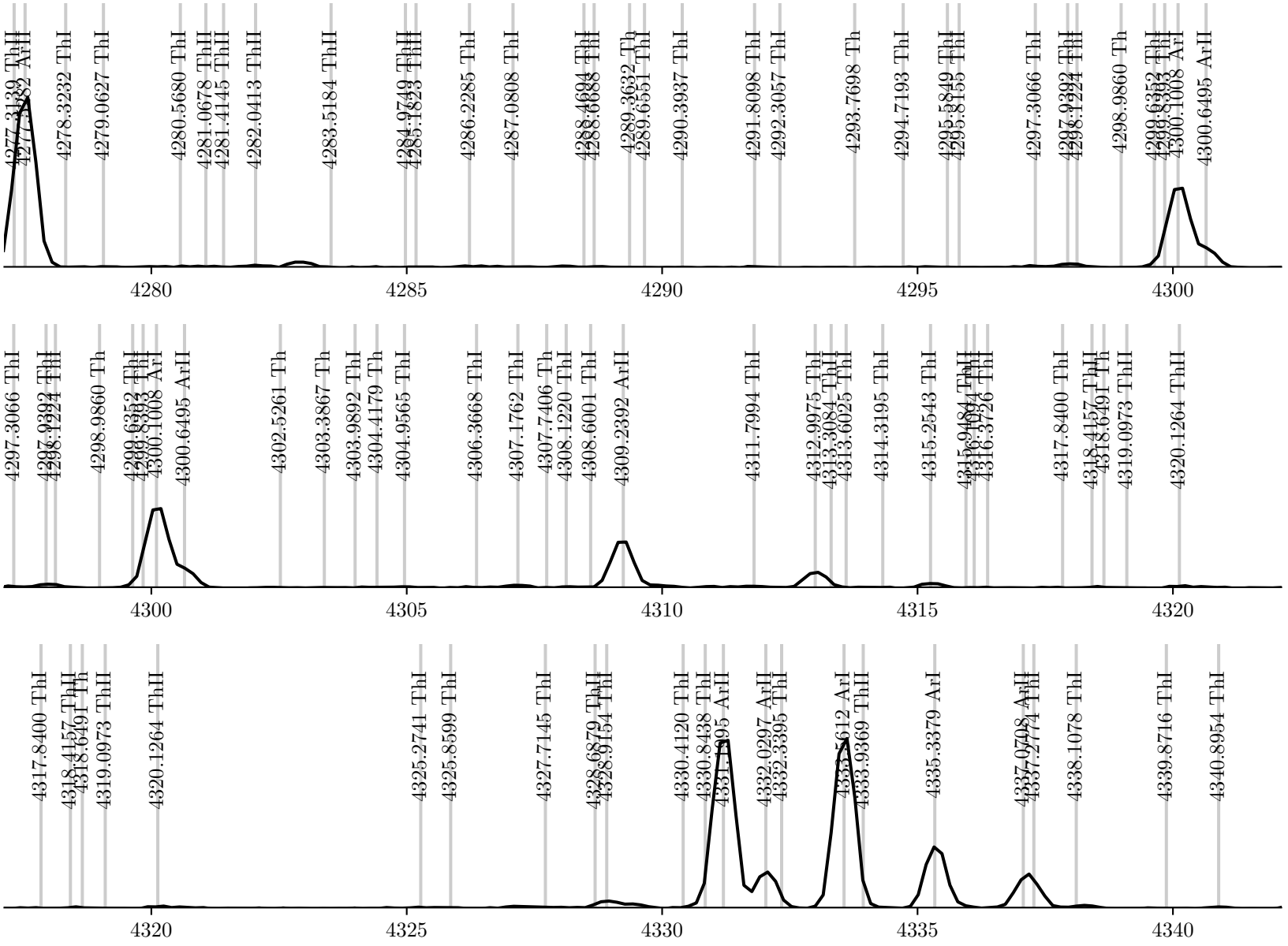
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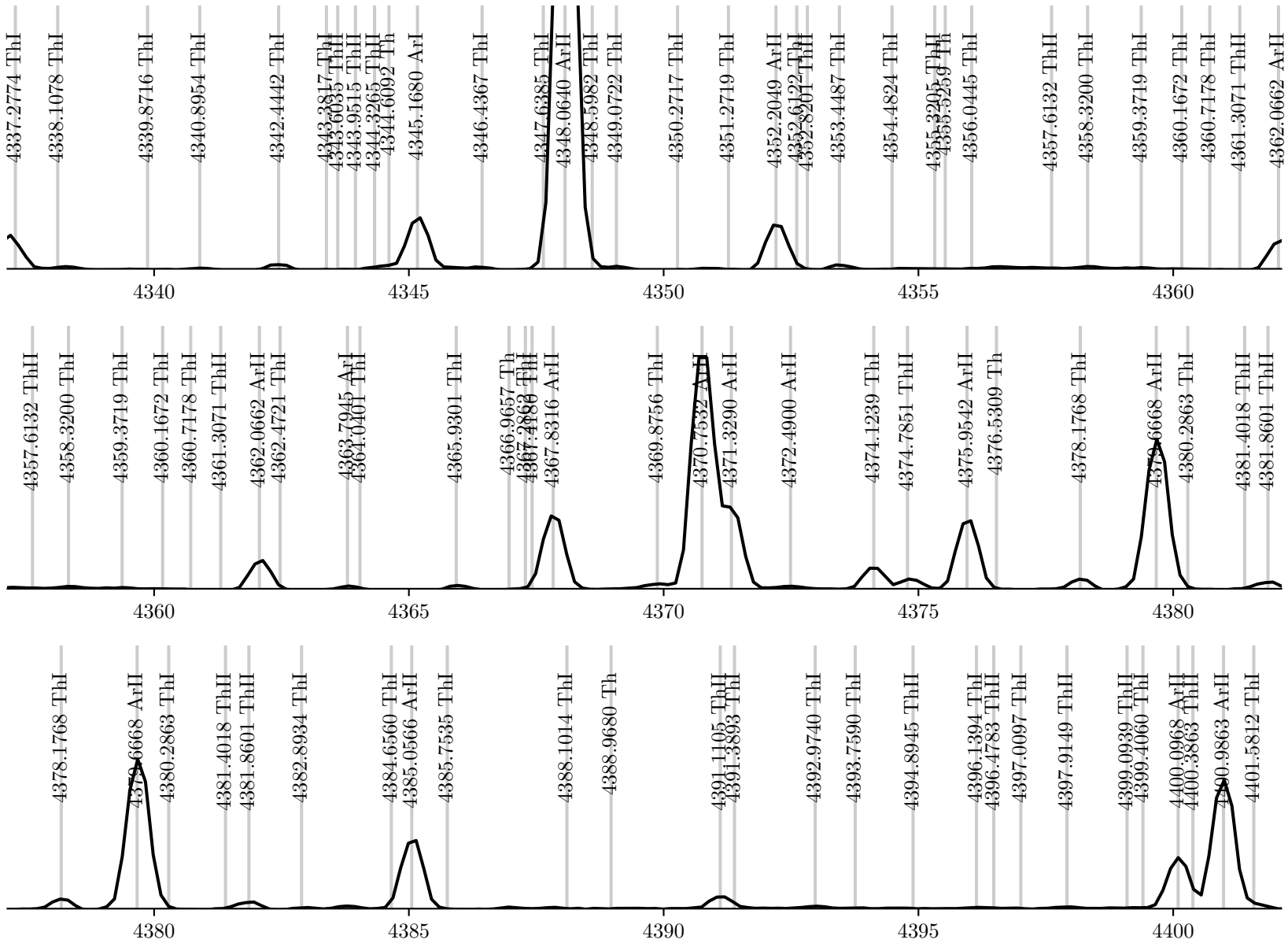
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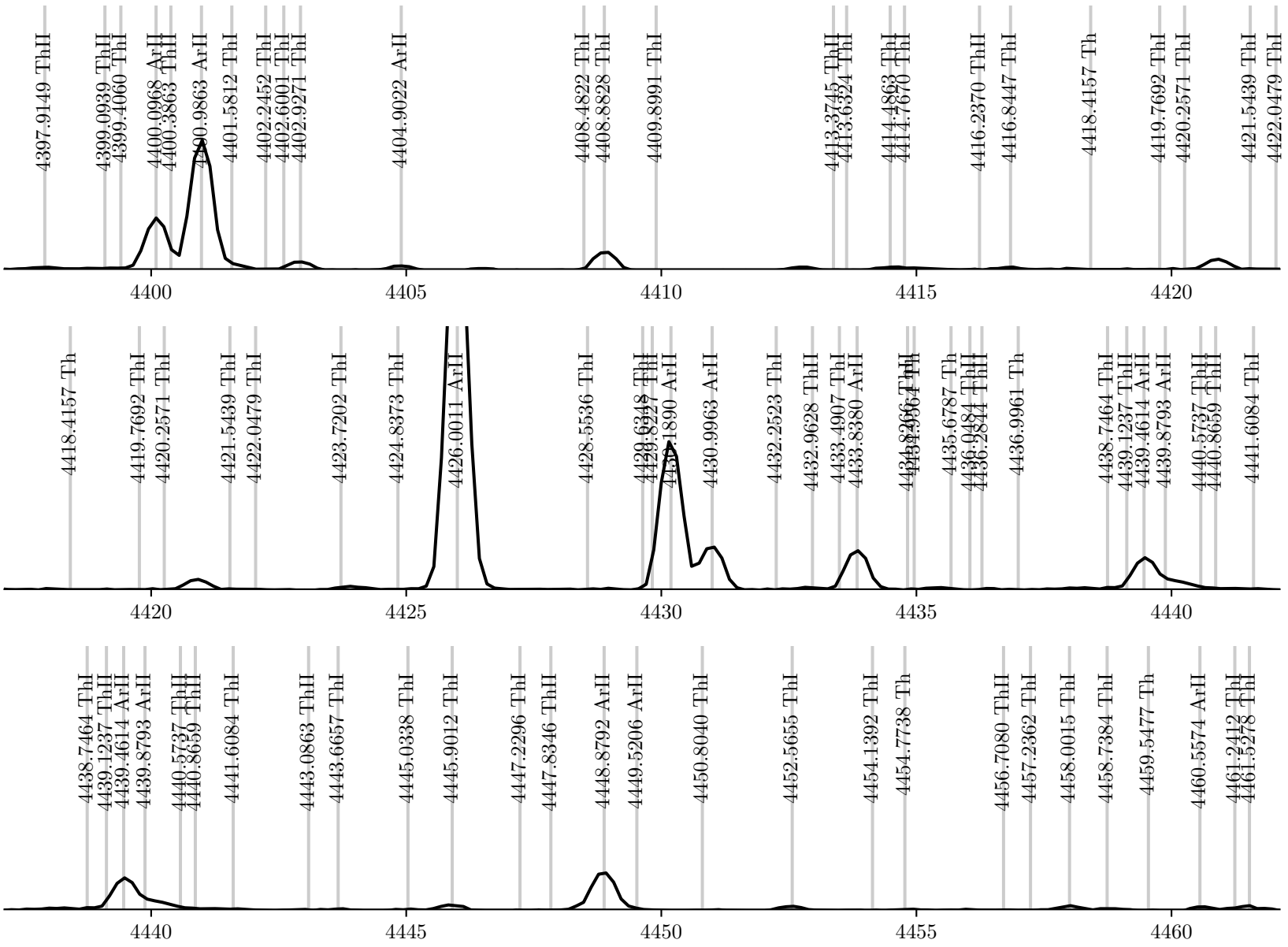
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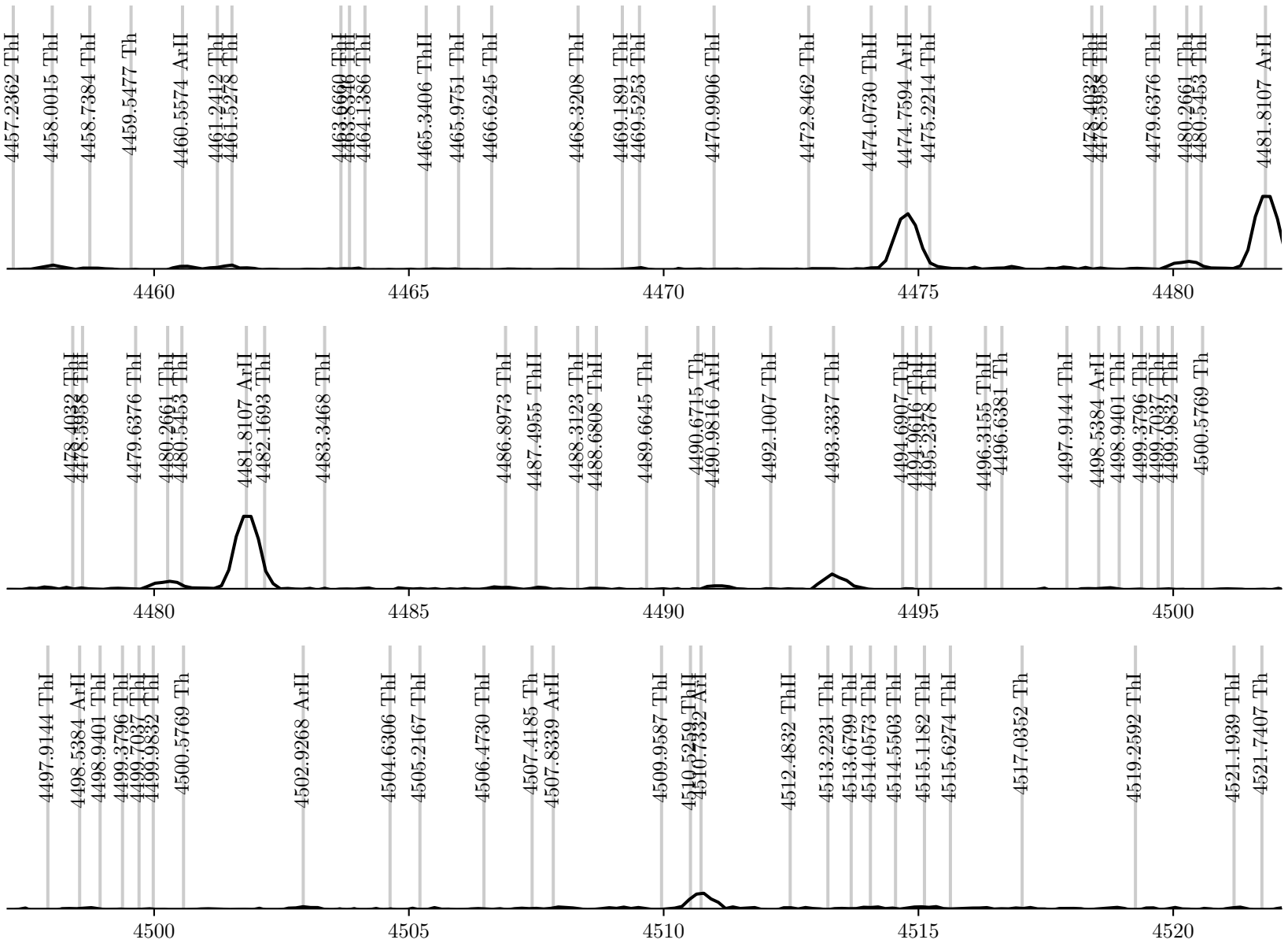
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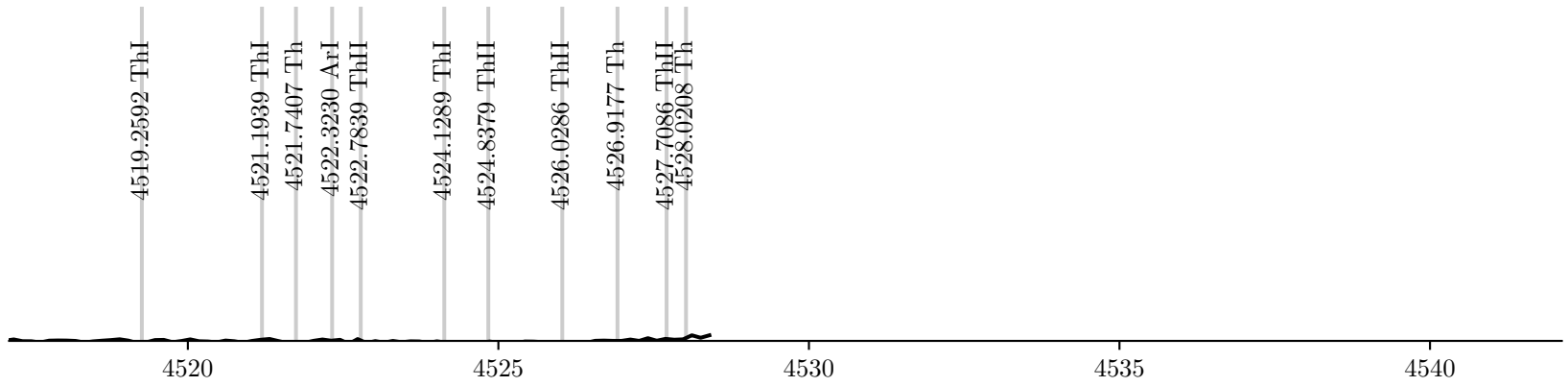
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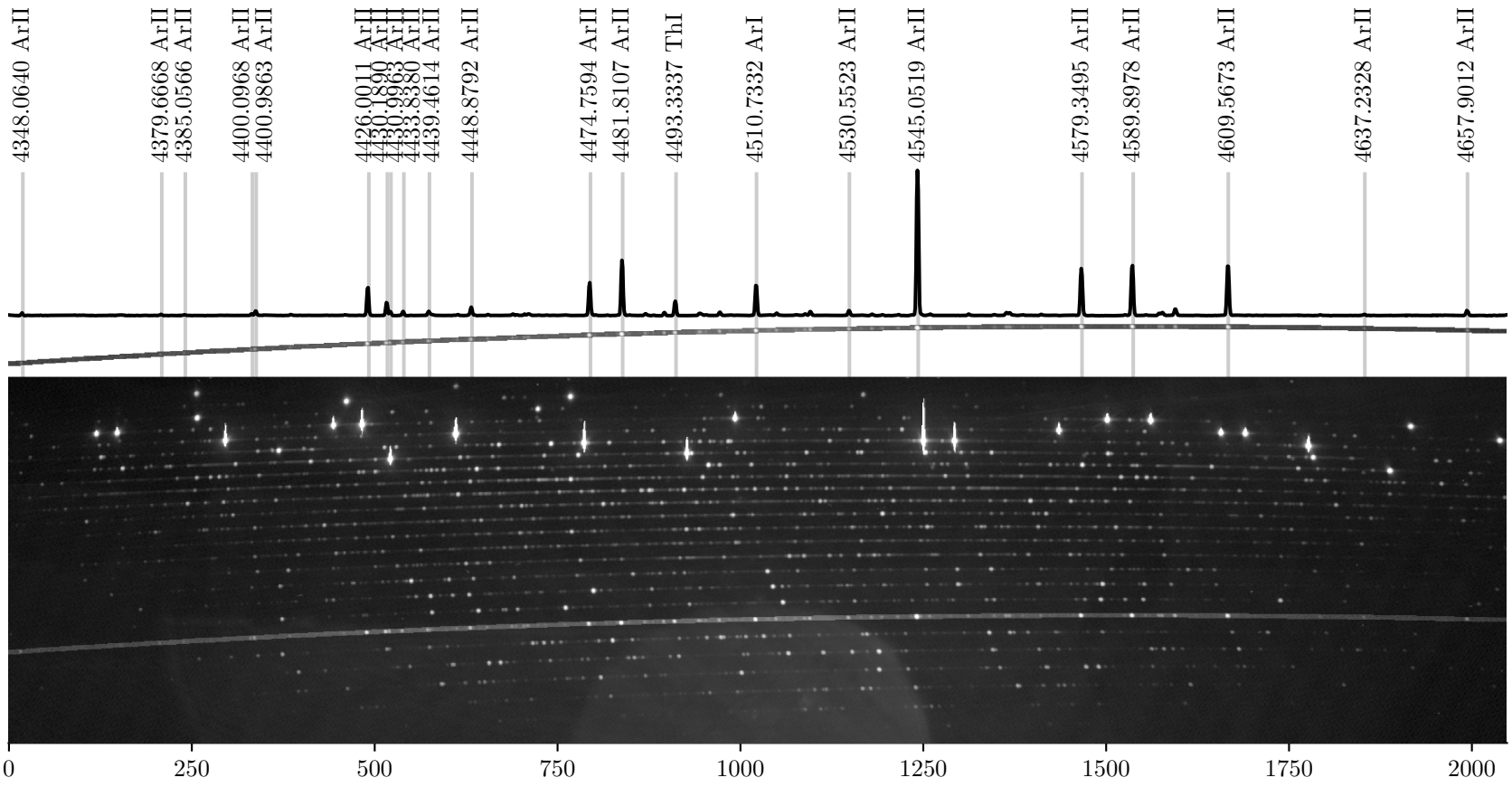
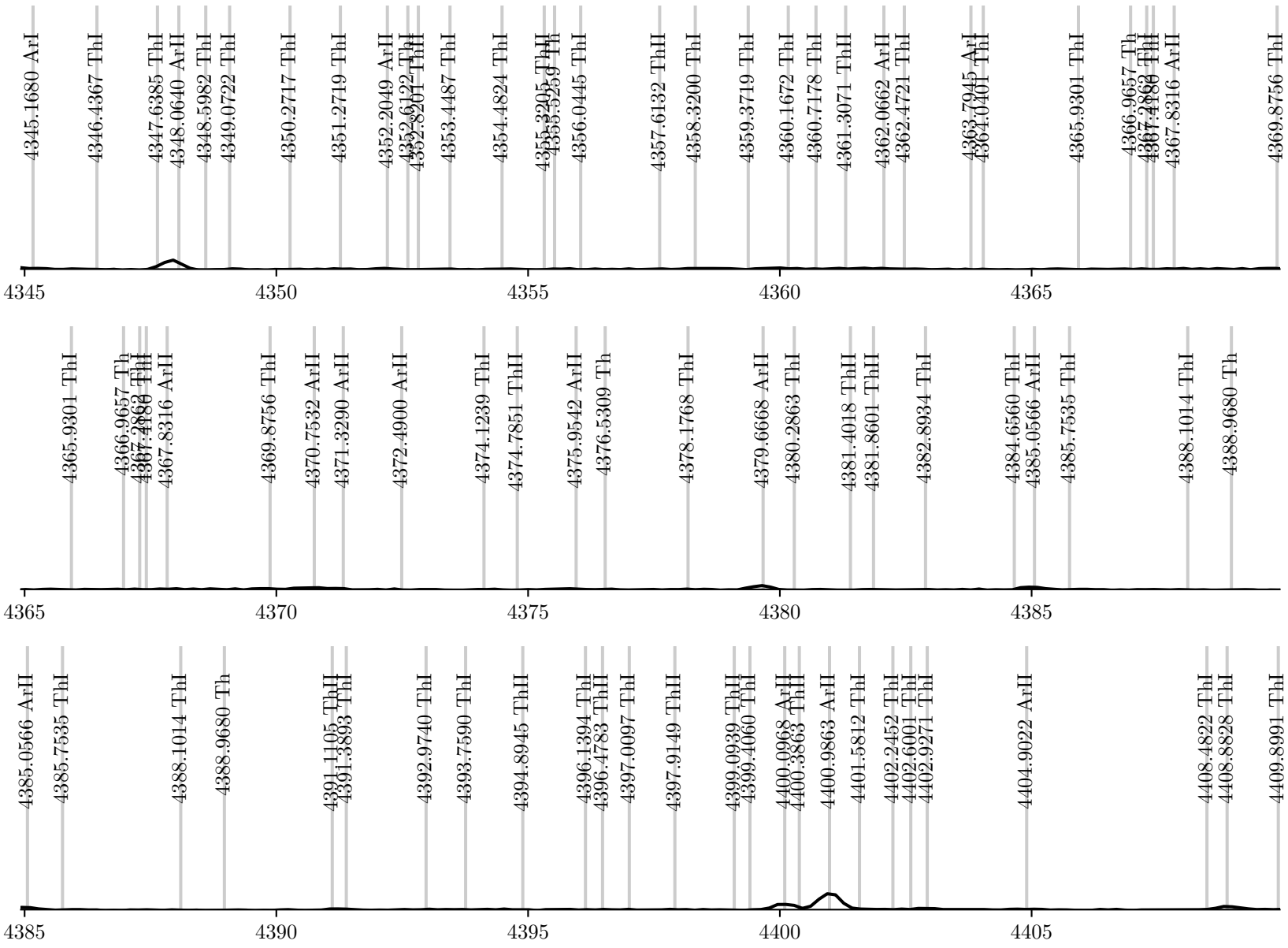
Order 33



Dispersion:	0.157 Å pix^{-1}
Central wavelength:	4511 Å
Wavelength range:	4345 Å – 4666 Å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4385.7535 ThI
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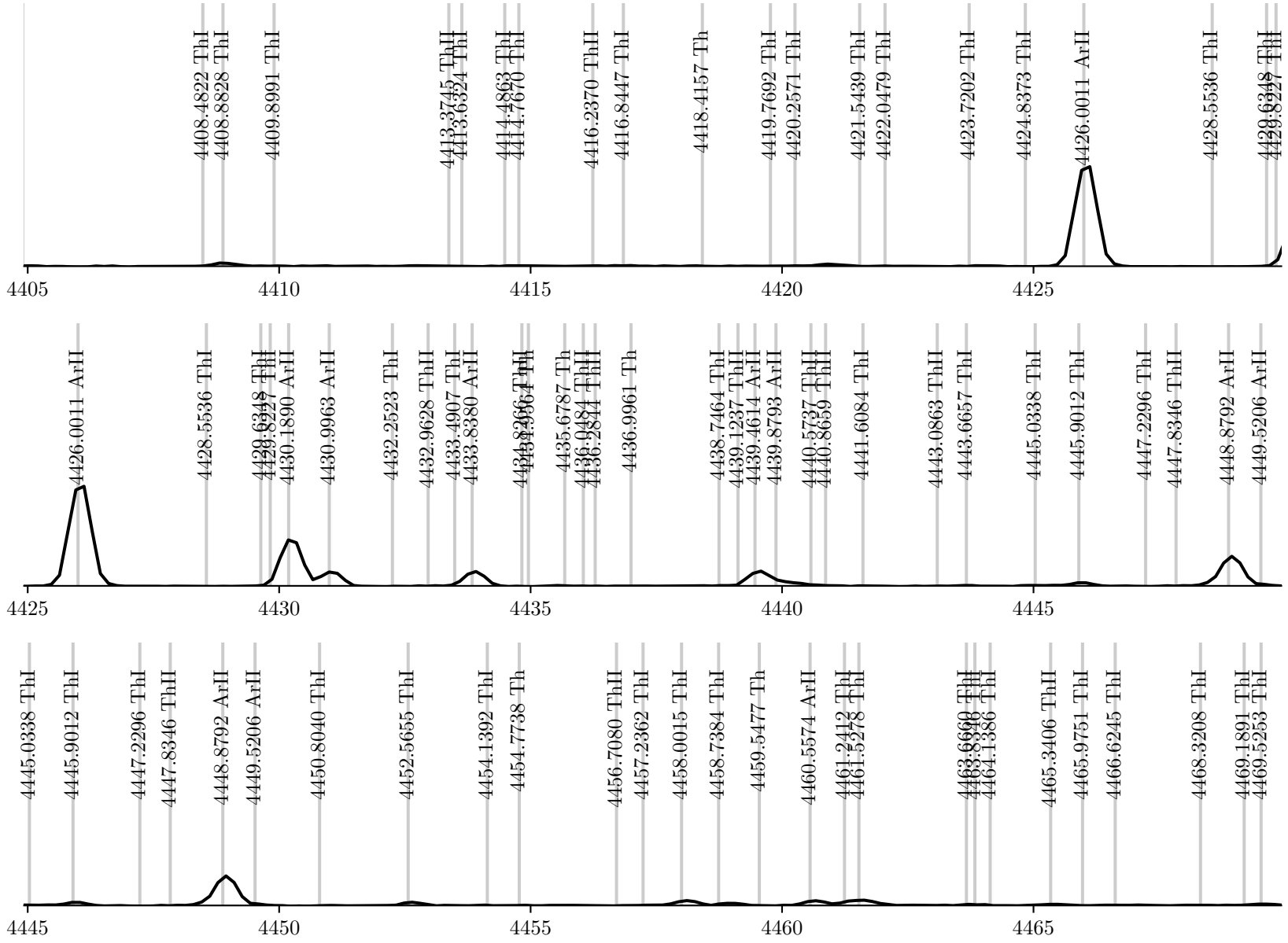
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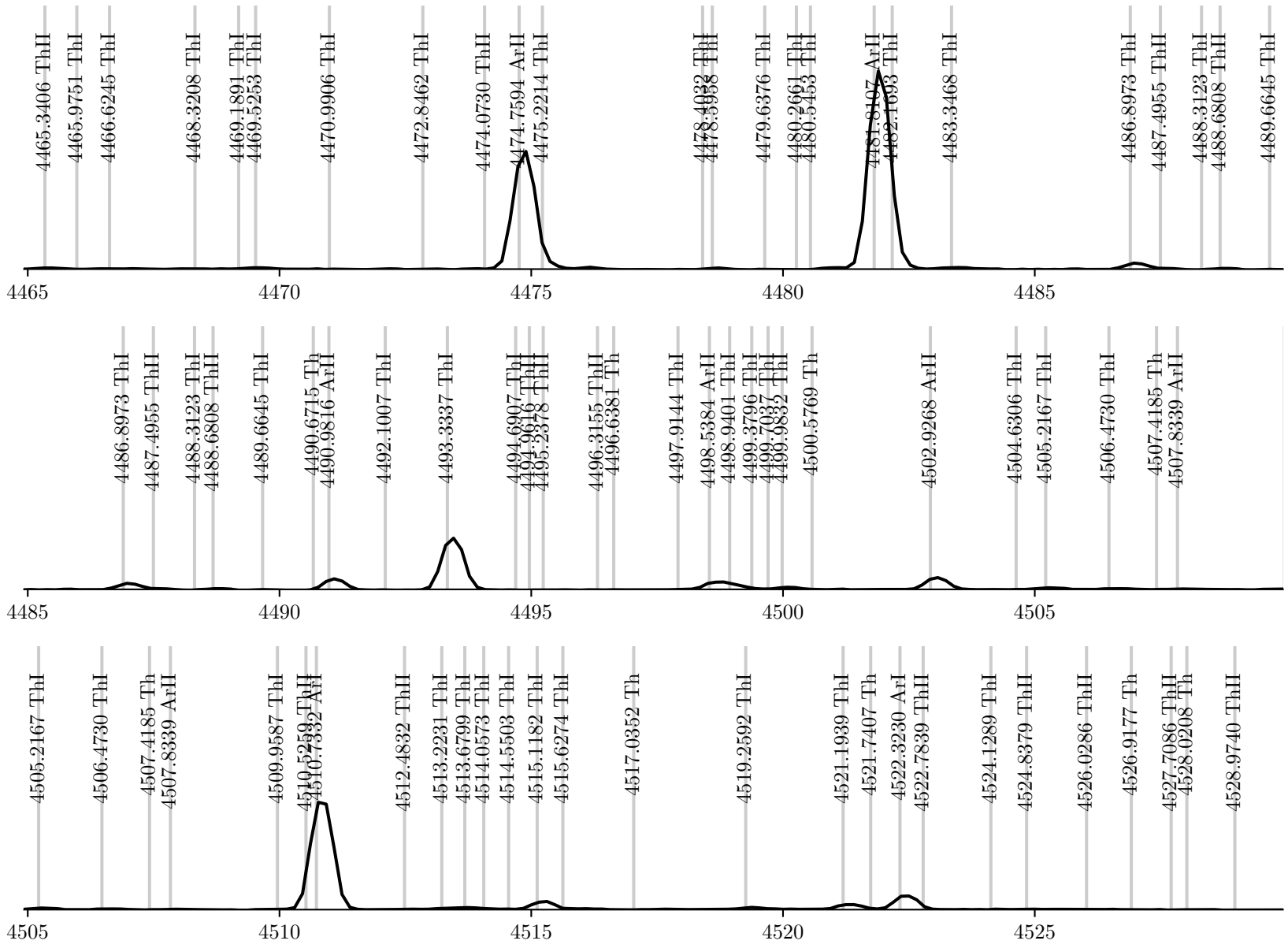
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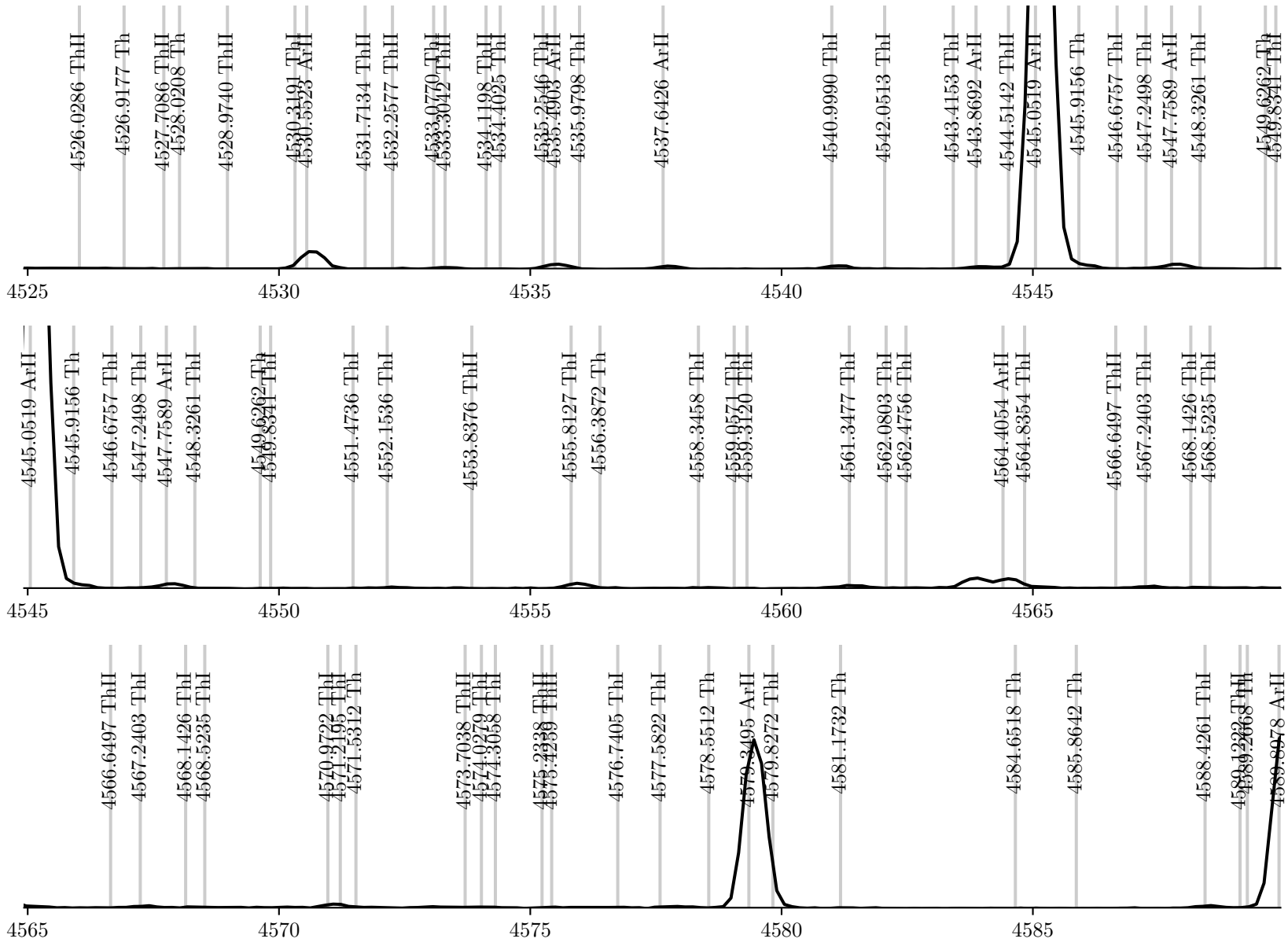
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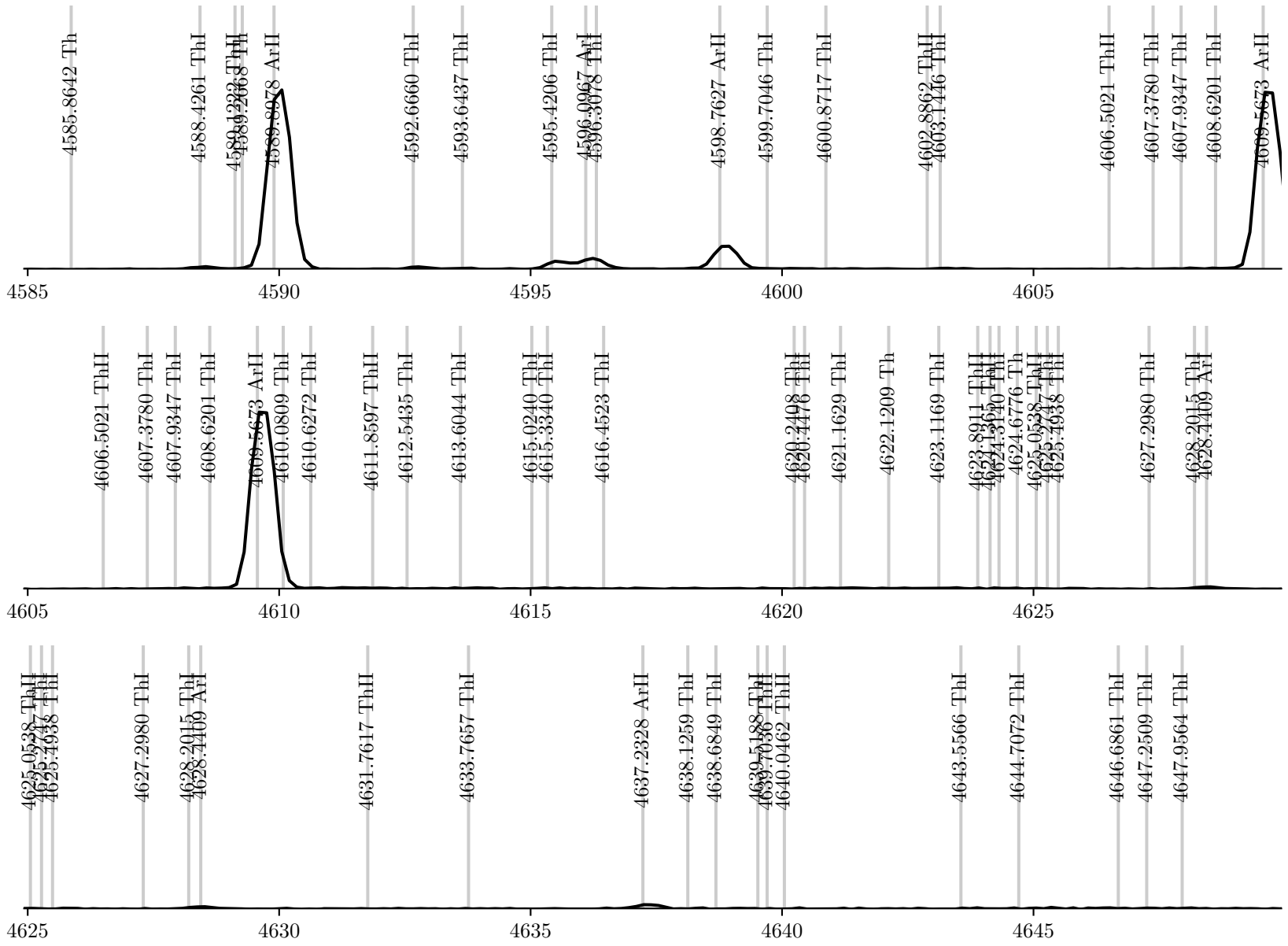
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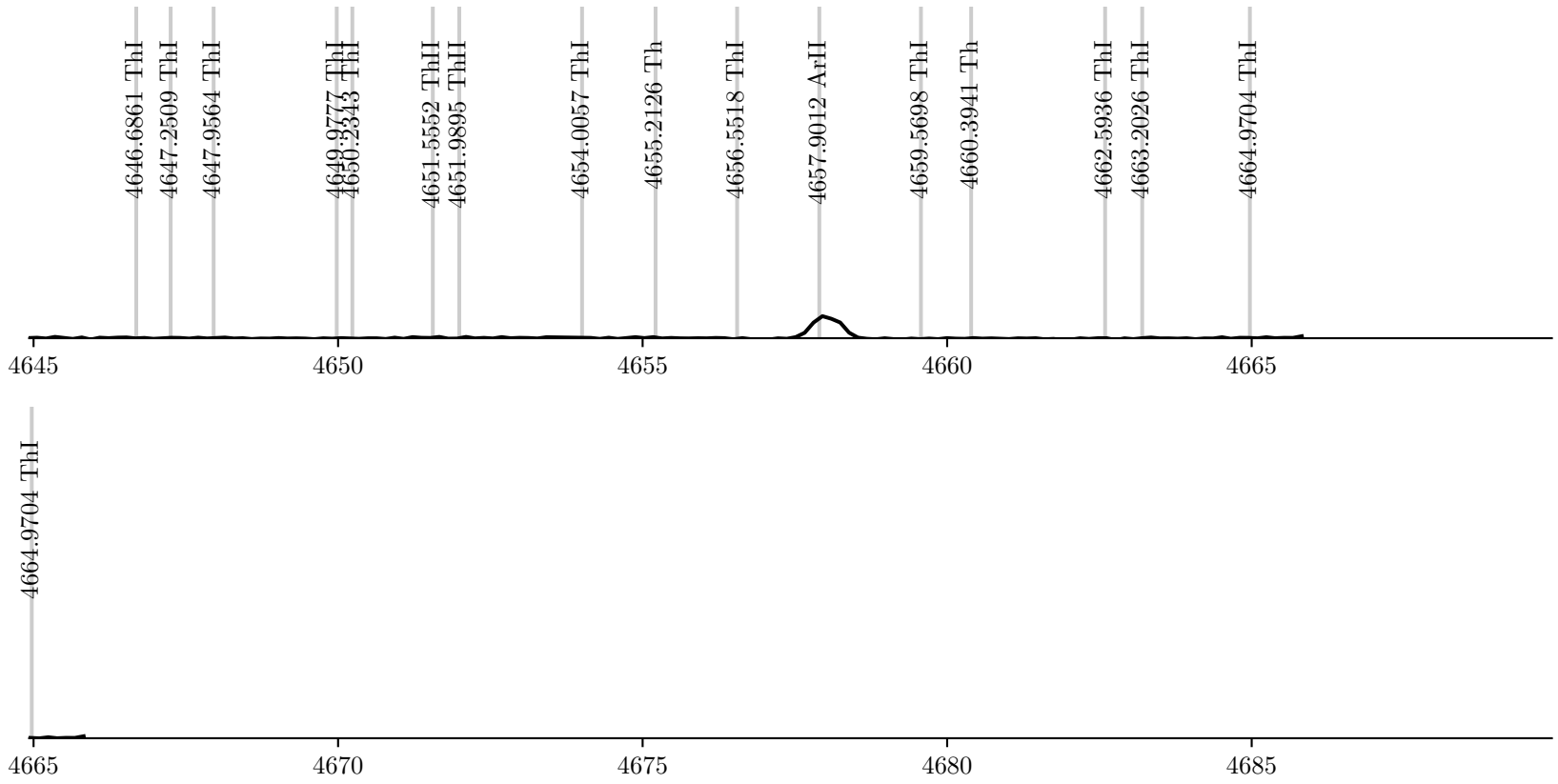
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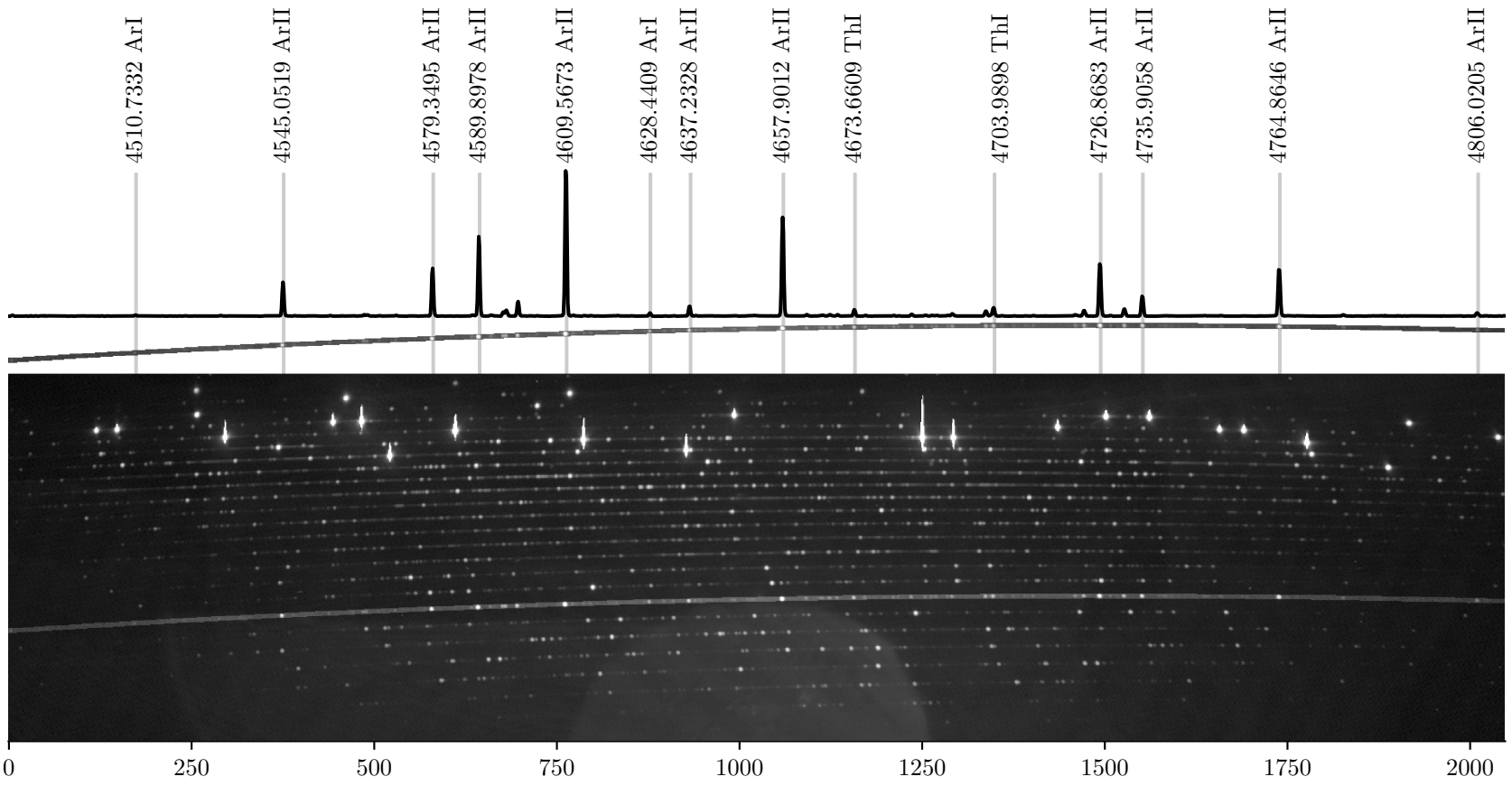
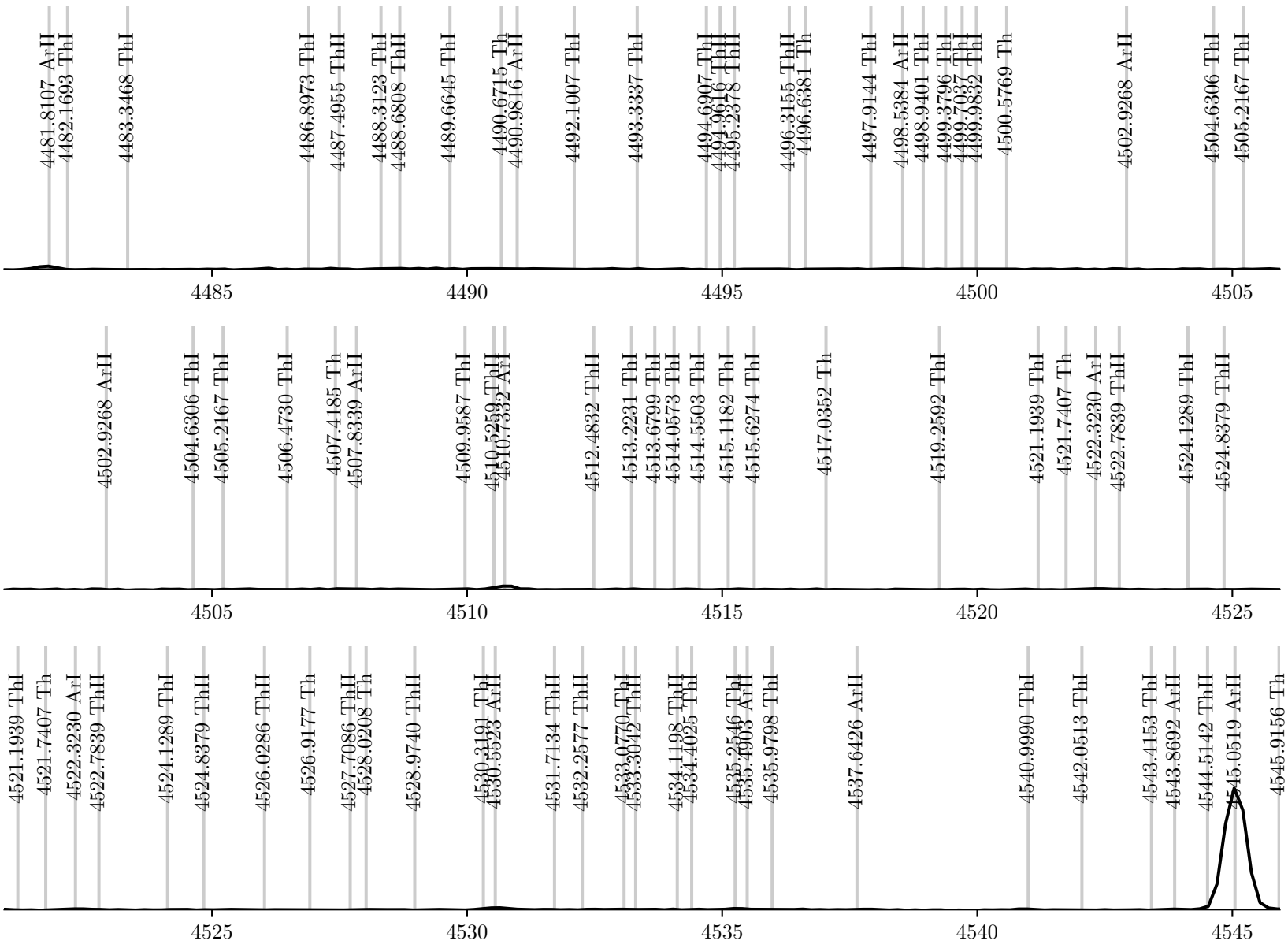
Order 32



Dispersion:	0.162 Å pix^{-1}
Central wavelength:	4652 Å
Wavelength range:	4481 Å – 4812 Å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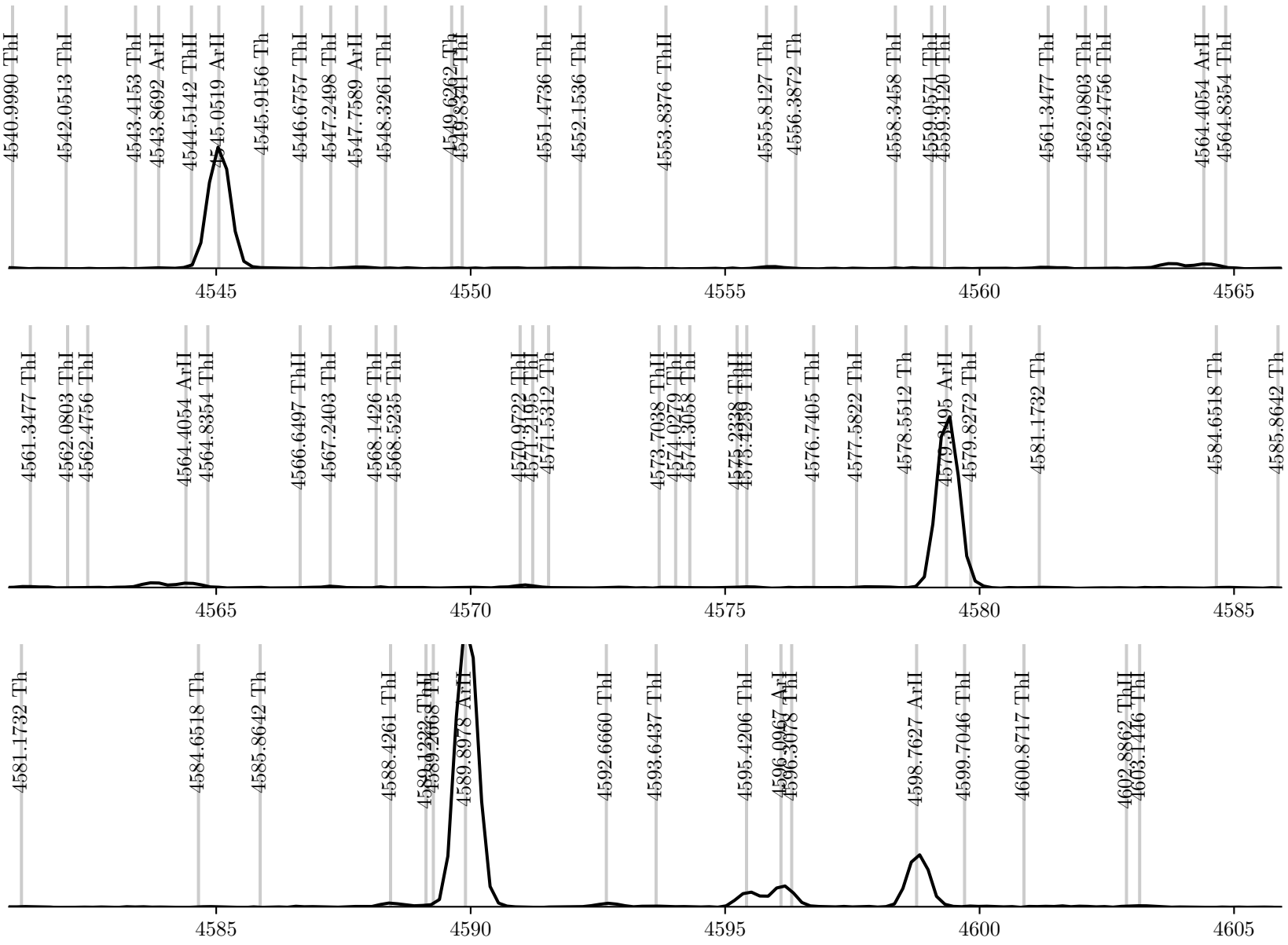
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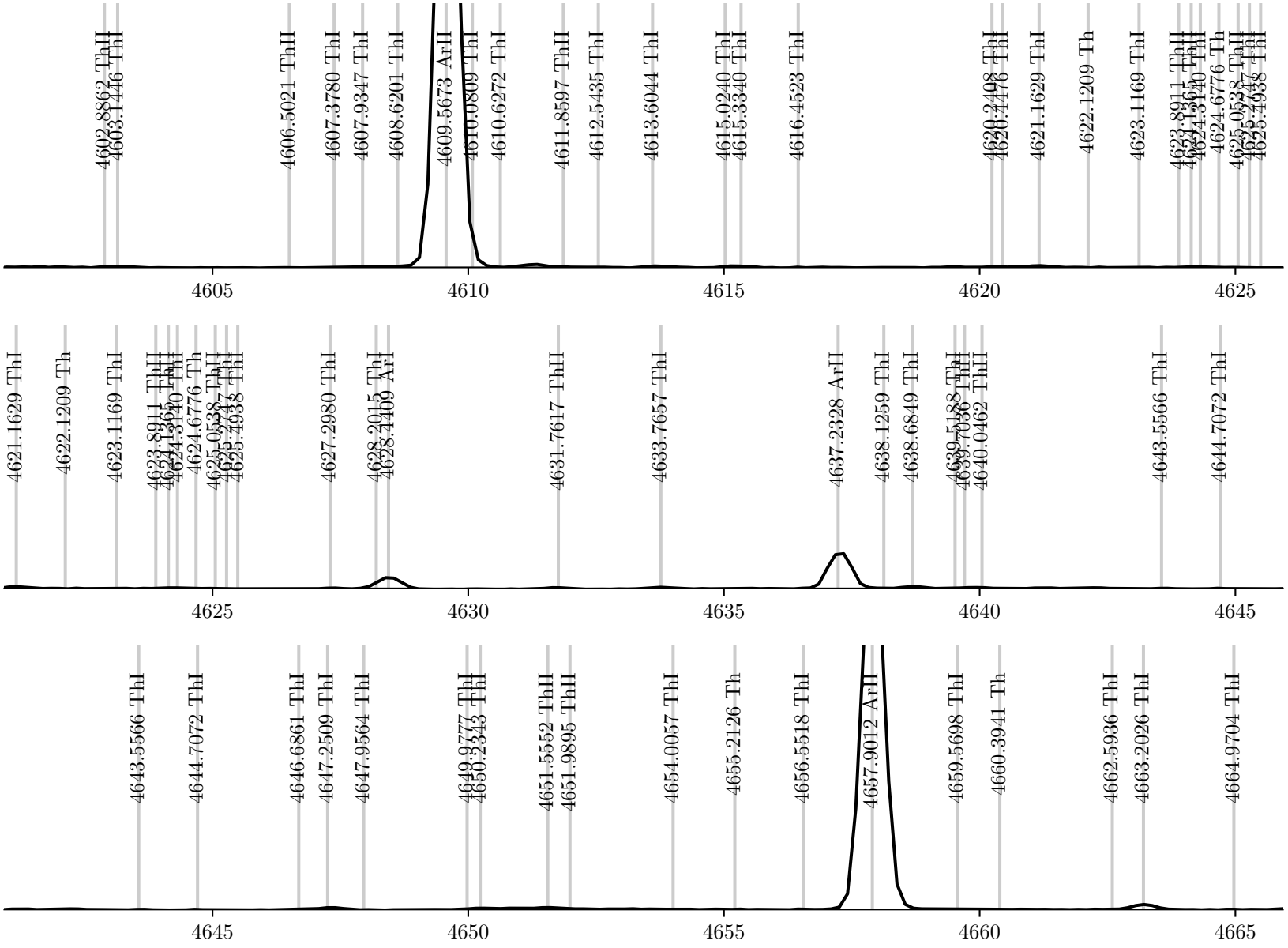
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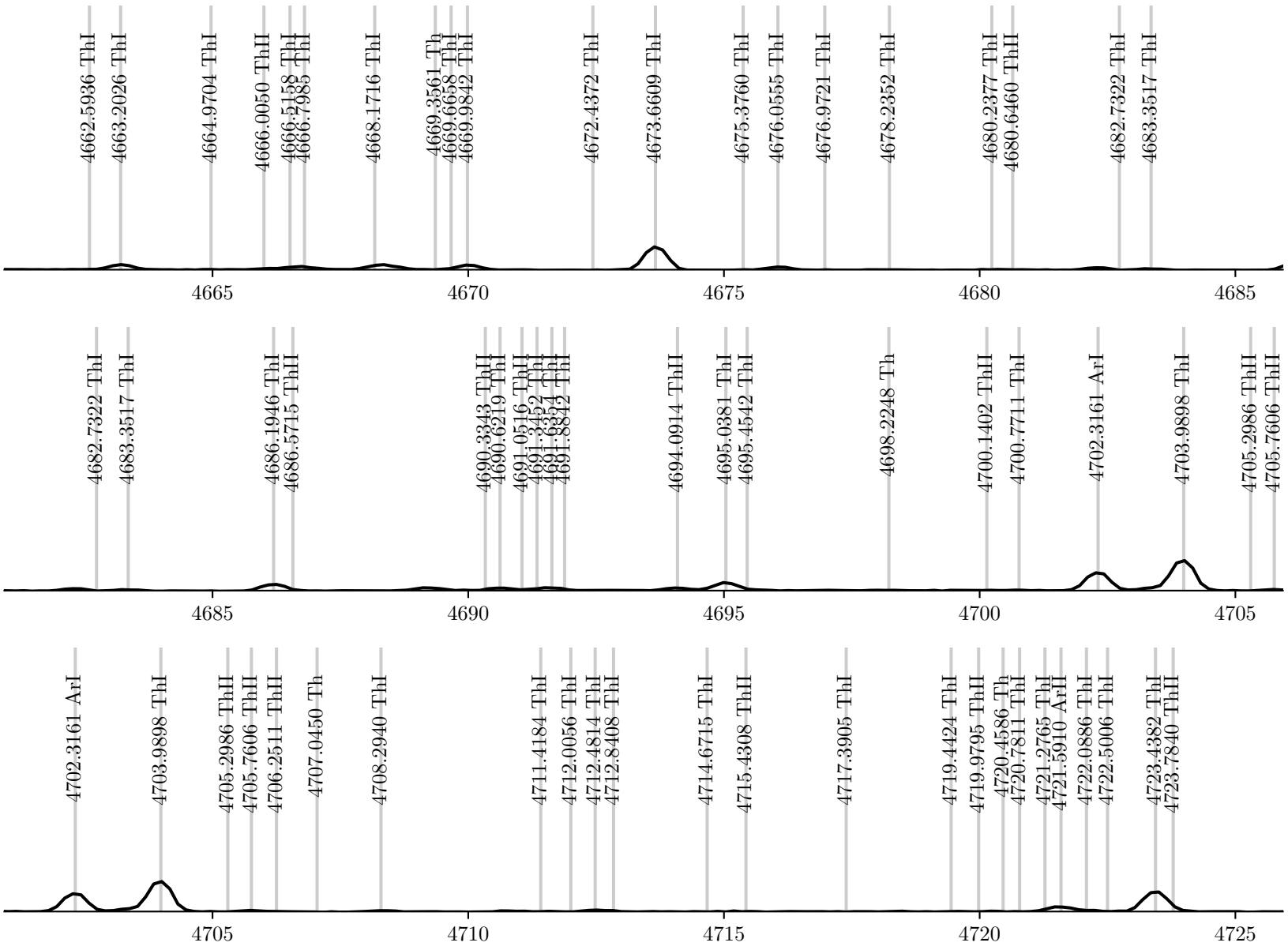
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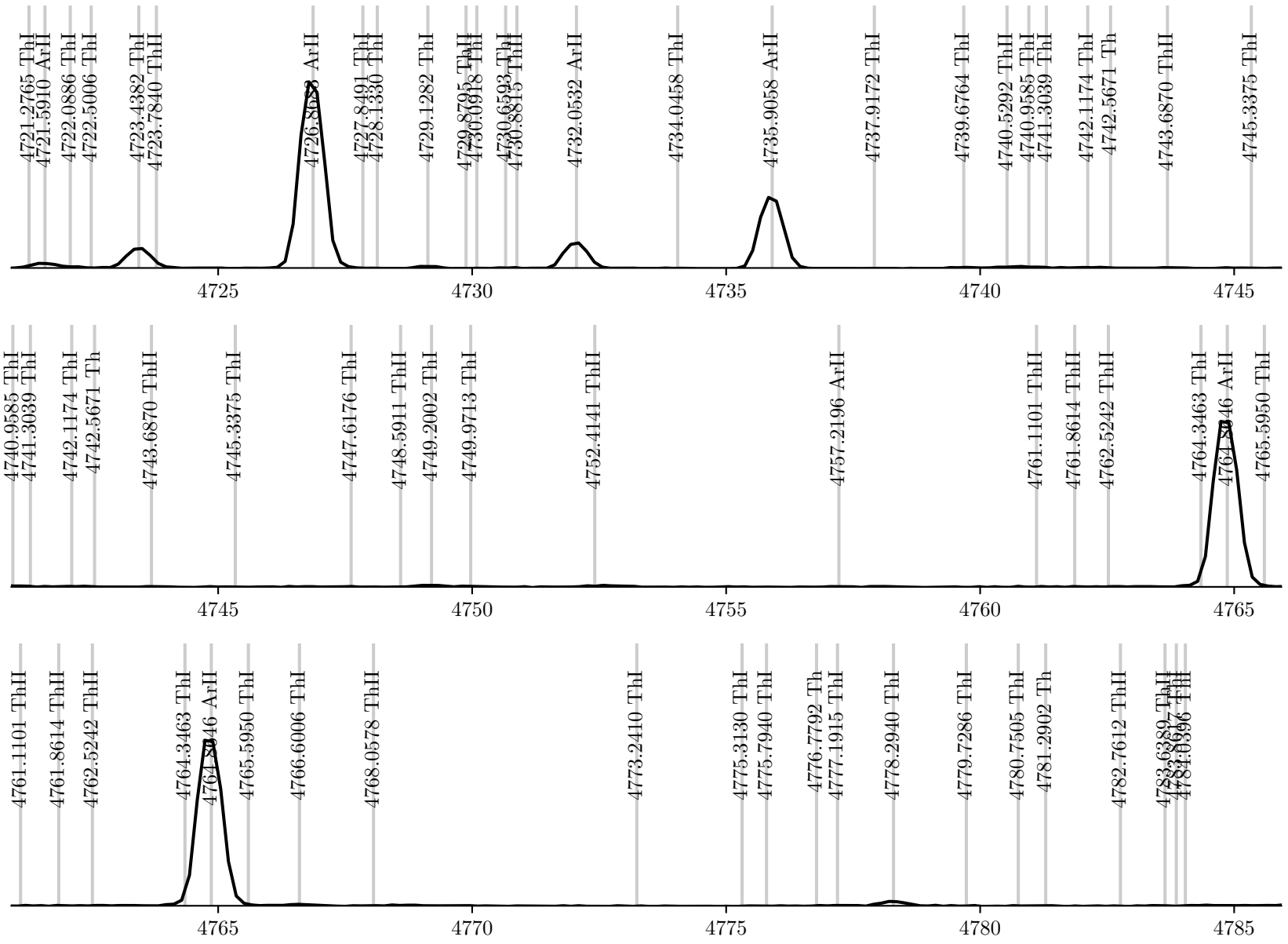
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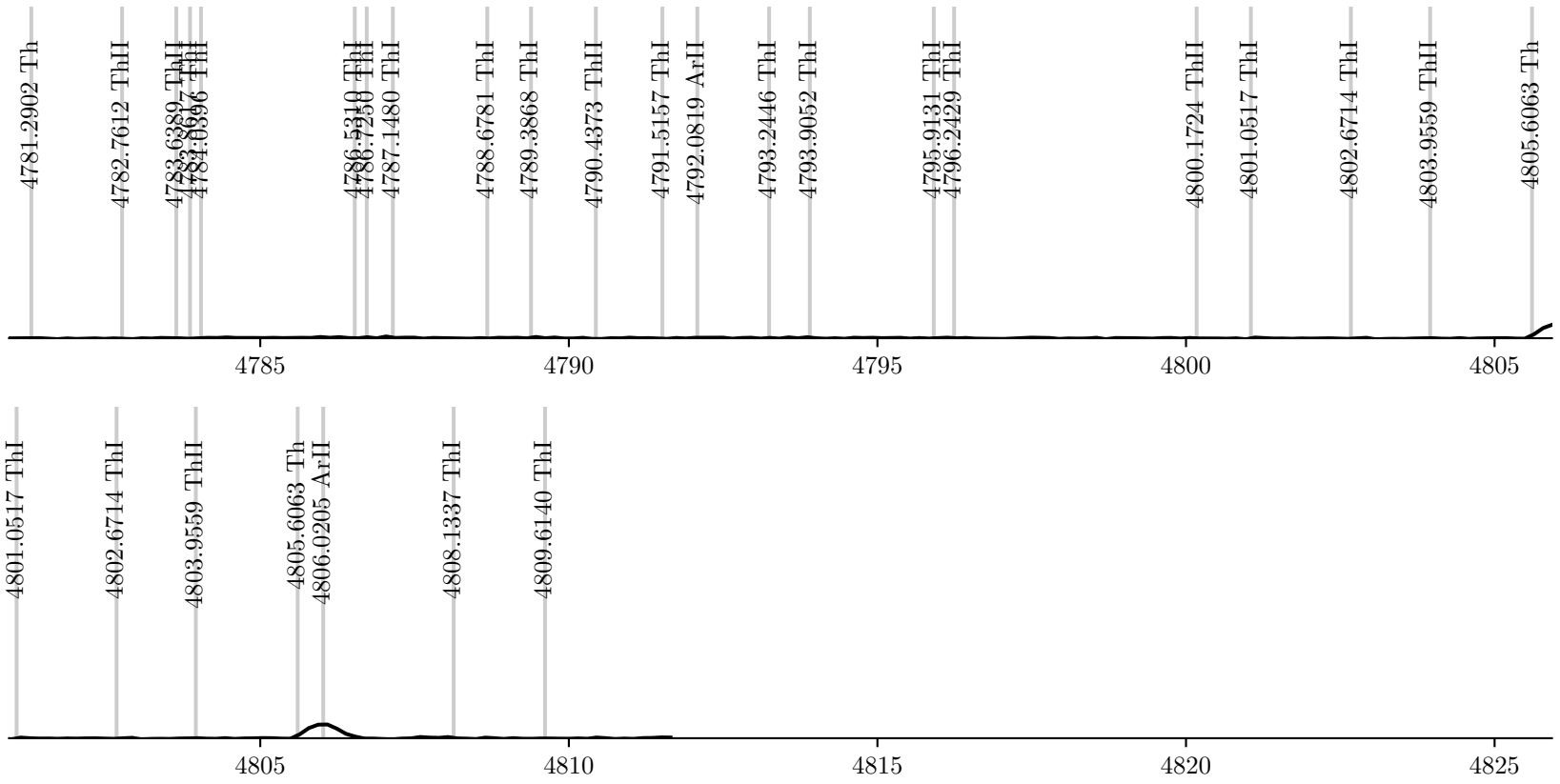
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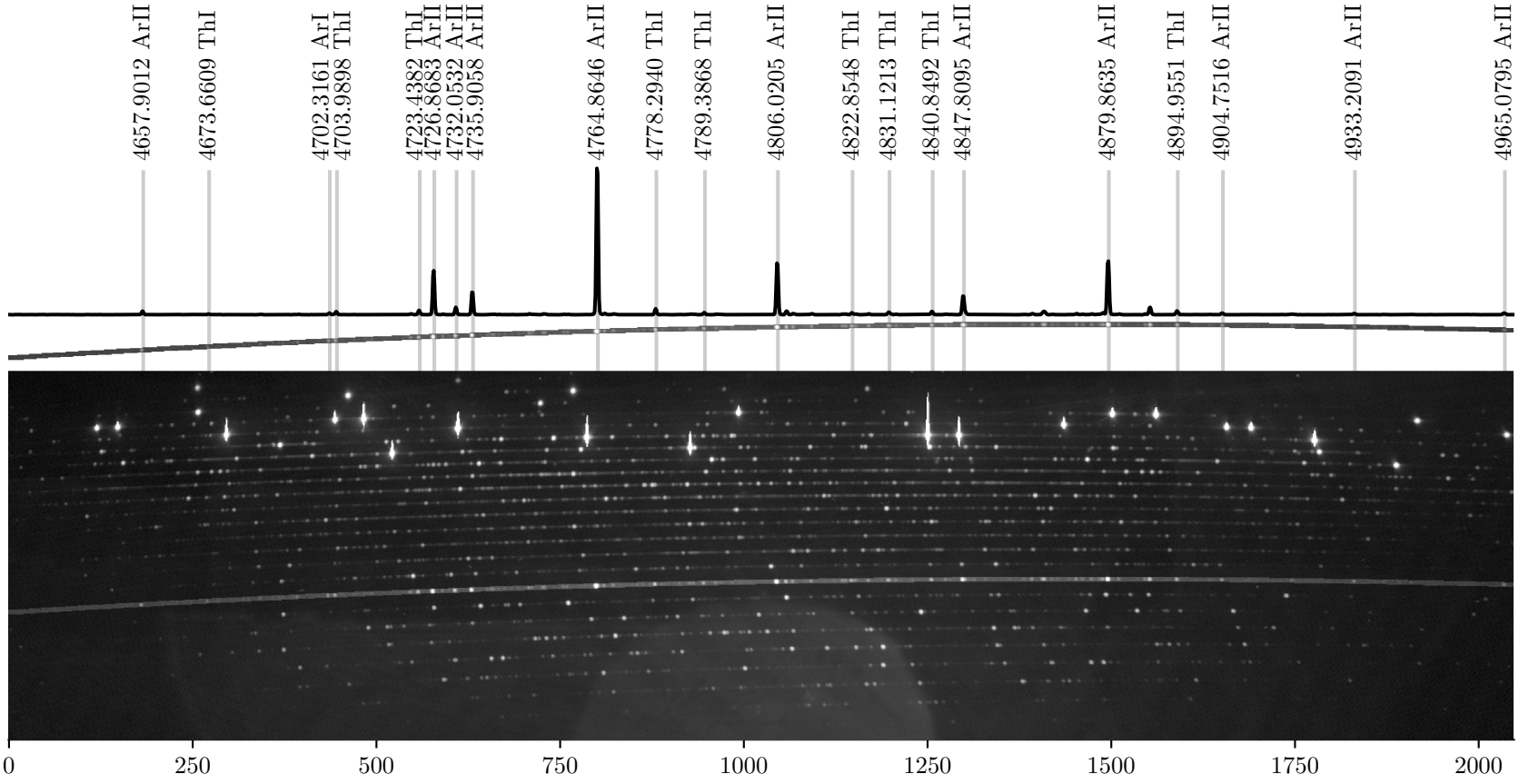
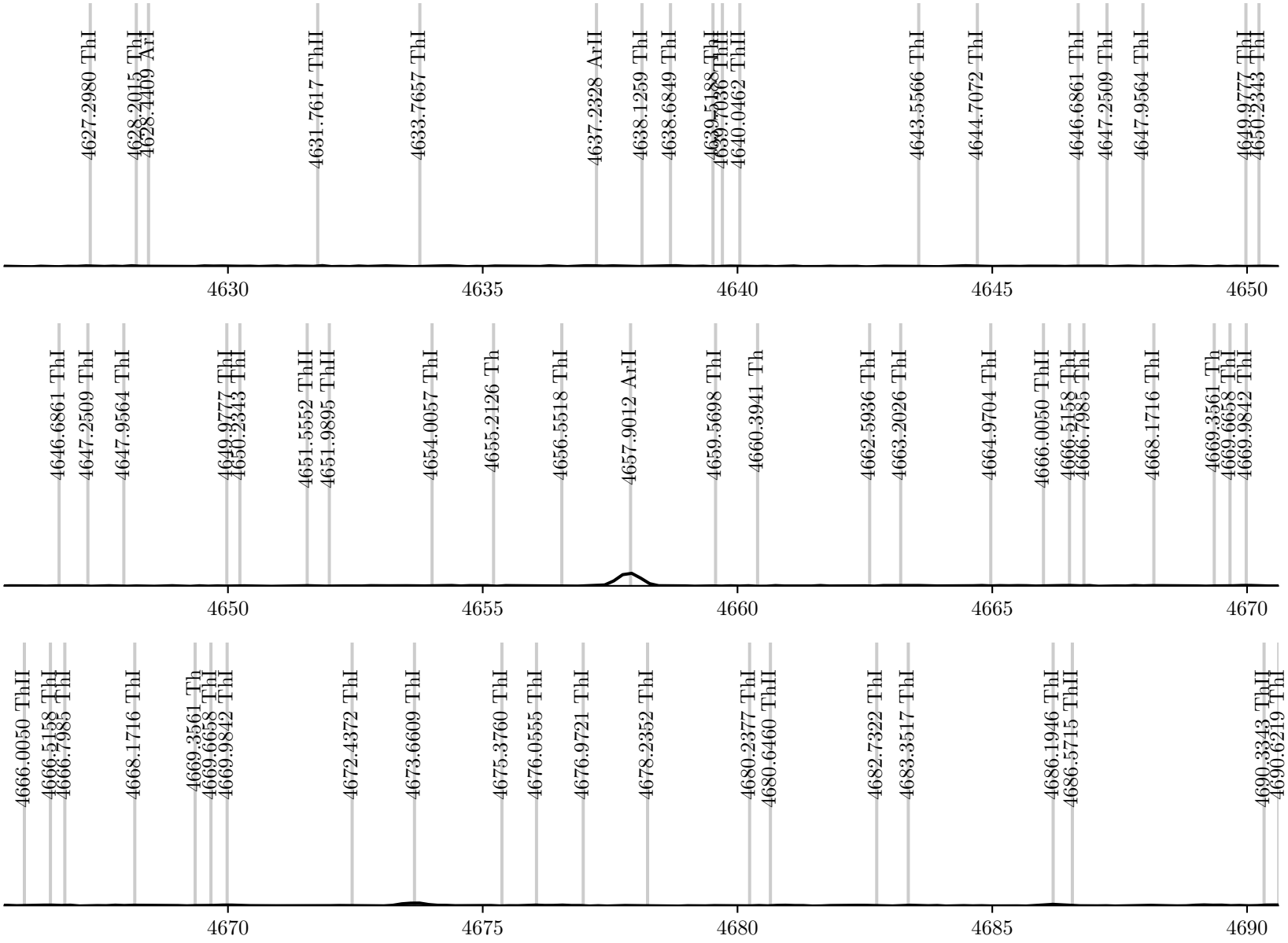
Order 31



Dispersion:	0.167 Å pix^{-1}
Central wavelength:	4802 Å
Wavelength range:	4626 Å – 4967 Å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4639.7036 ThII
4640.0462 ThII
4643.5566 ThI
4644.7072 ThI
4646.6861 ThI
4647.2509 ThI
4647.9564 ThI
4649.9777 ThI
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4645
4650
4646.6861 ThI
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4651.9895 ThII
4654.0057 ThI
4655.2126 Th
4656.5518 ThI
4657.9012 ArII
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4663.2026 ThI
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4675.3760 ThI
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4678.2352 ThI
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4680.6460 ThII
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4683.3517 ThI
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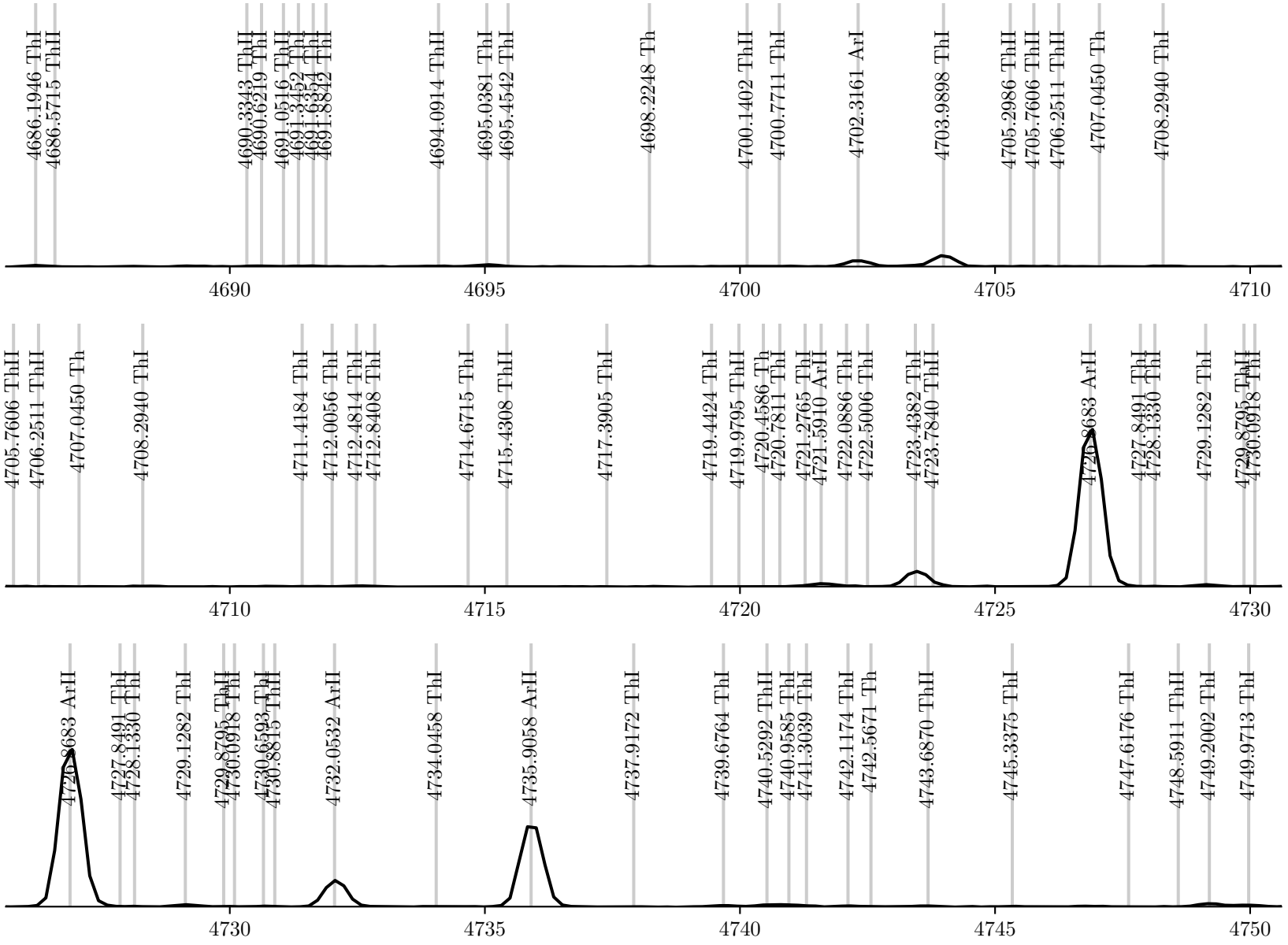
4690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4702.3161 ArI
4703.9898 ThI
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4695
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4715
4720
4725
4730
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4728.1330 ThI
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4742.5671 Th
4743.6870 ThII
4745.3375 ThI
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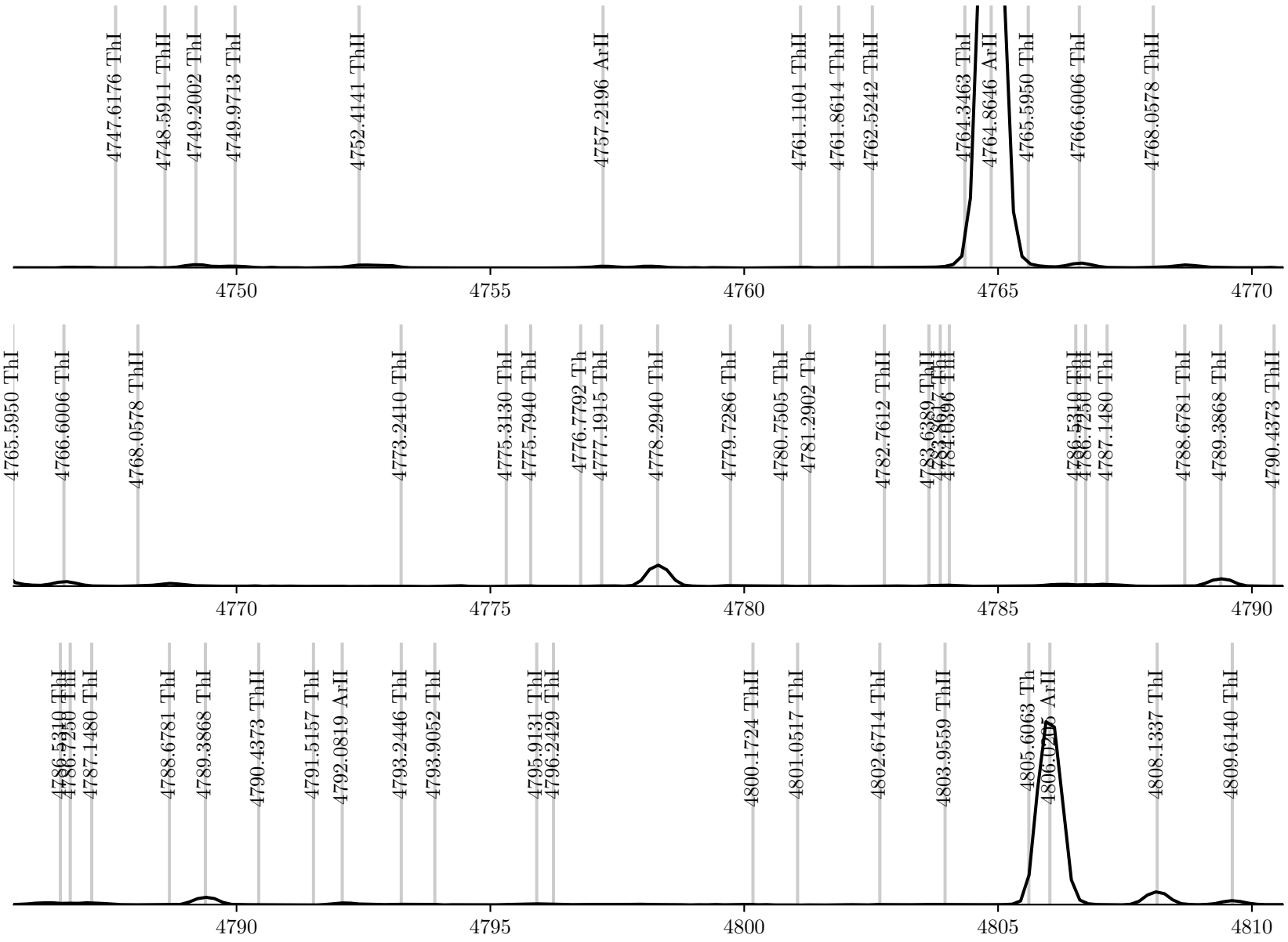
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4762.5242 ThII
4764.3463 ThI
4764.8646 ArII
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4750
4755
4760
4765
4770
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4766.6006 ThI
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4775.3130 ThI
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4777.1915 ThI
4778.2940 ThI
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4781.2902 Th
4782.7612 ThII
4786.5310 ThI
4787.1480 ThI
4788.6781 ThI
4789.3868 ThI
4790.4373 ThII
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4785
4790
4786.5310 ThI
4787.1480 ThI
4788.6781 ThI
4789.3868 ThI
4790.4373 ThII
4791.5157 ThI
4792.0819 ArII
4793.2446 ThI
4793.9052 ThI
4795.9131 ThI
4796.2429 ThI
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4801.0517 ThI
4802.6714 ThI
4803.9559 ThII
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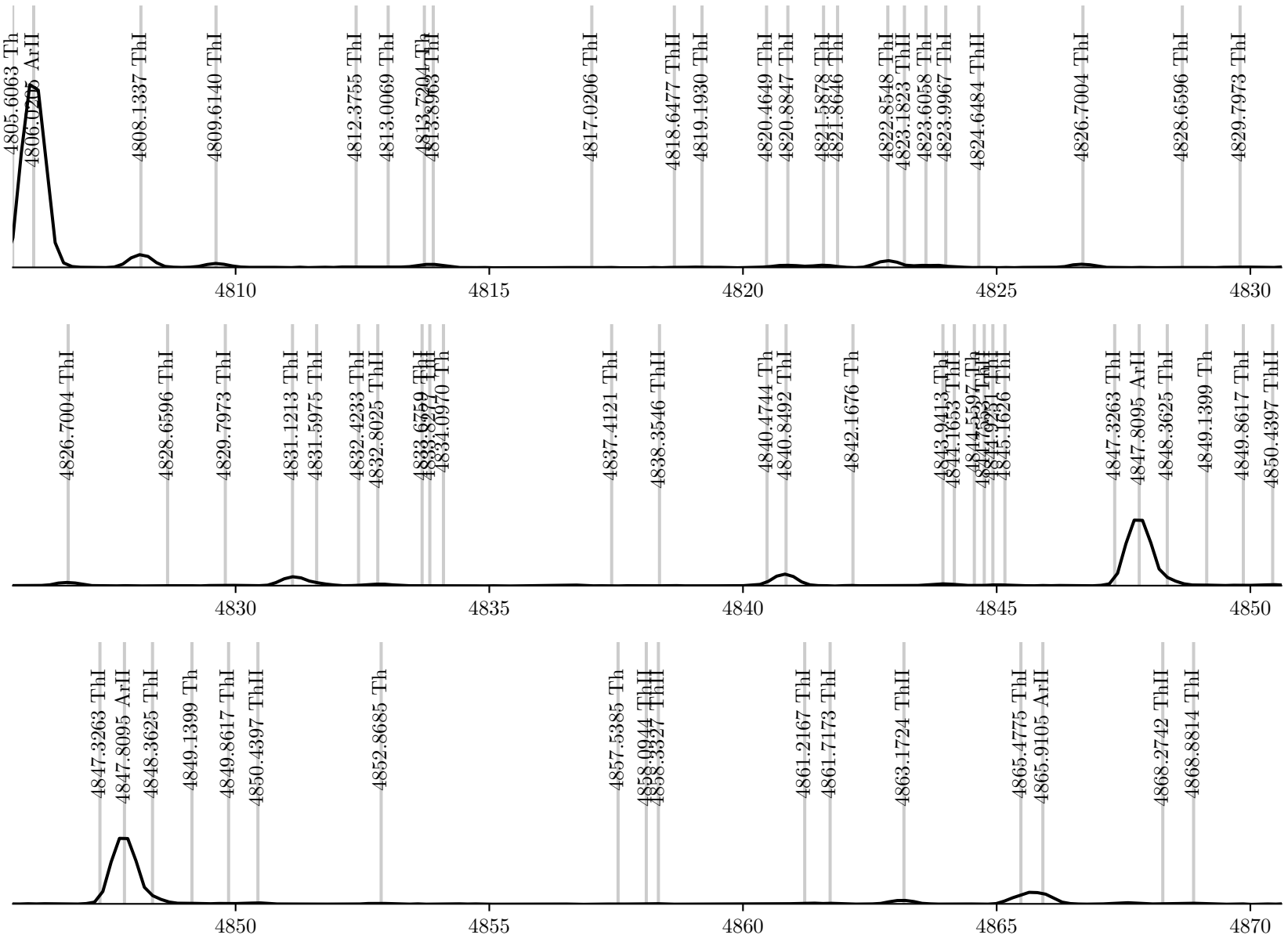
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4823.1823 ThII
4823.6058 ThI
4823.9967 ThI
4824.6484 ThII
4826.7004 ThI
4828.6596 ThI
4829.7973 ThI
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4815
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4825
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4829.7973 ThI
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4831.5975 ThI
4832.4233 ThI
4832.8025 ThII
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4838.3546 ThII
4840.4744 Th
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4844.5597 Th
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4847.3263 ThI
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4835
4840
4845
4850
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4858.3327 ThII
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4861.7173 ThI
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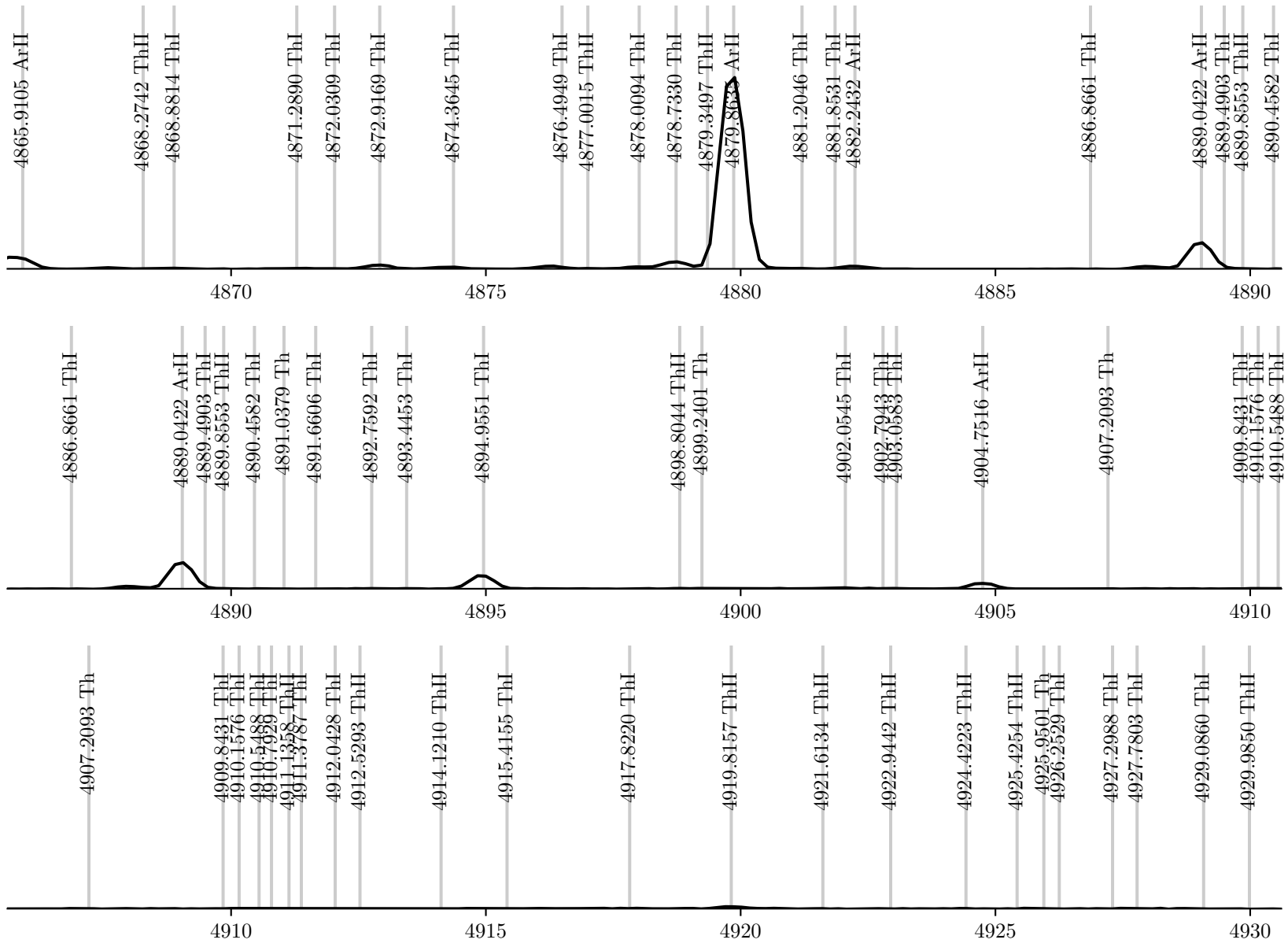
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4882.2432 ArII
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4892.7592 ThI
4893.4453 ThI
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4903.0583 ThI
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4909.8431 ThI
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4922.9442 ThII
4924.4223 ThII
4925.4254 ThII
4925.9501 Th
4926.2529 ThI
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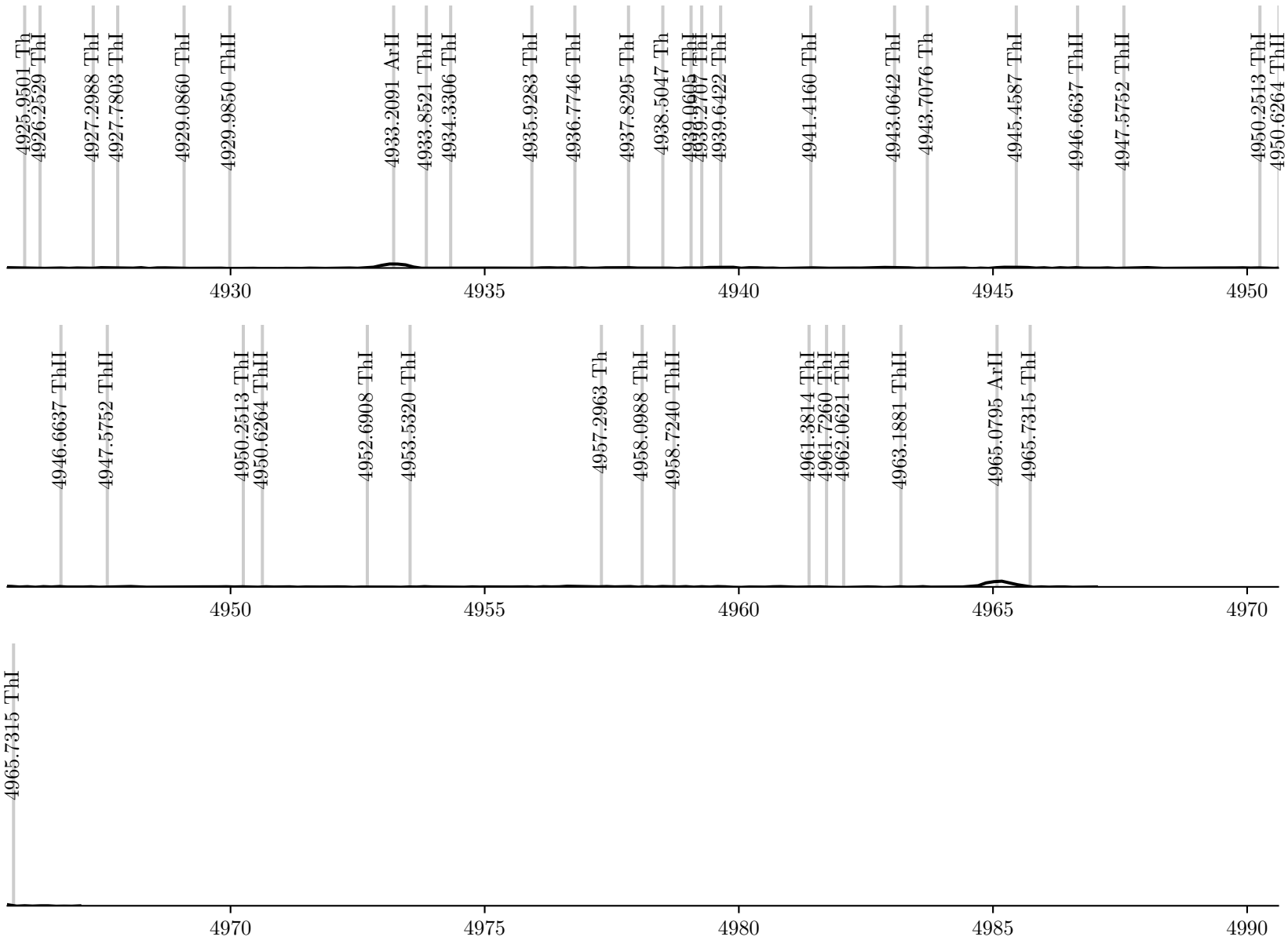
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4957.2963 Th
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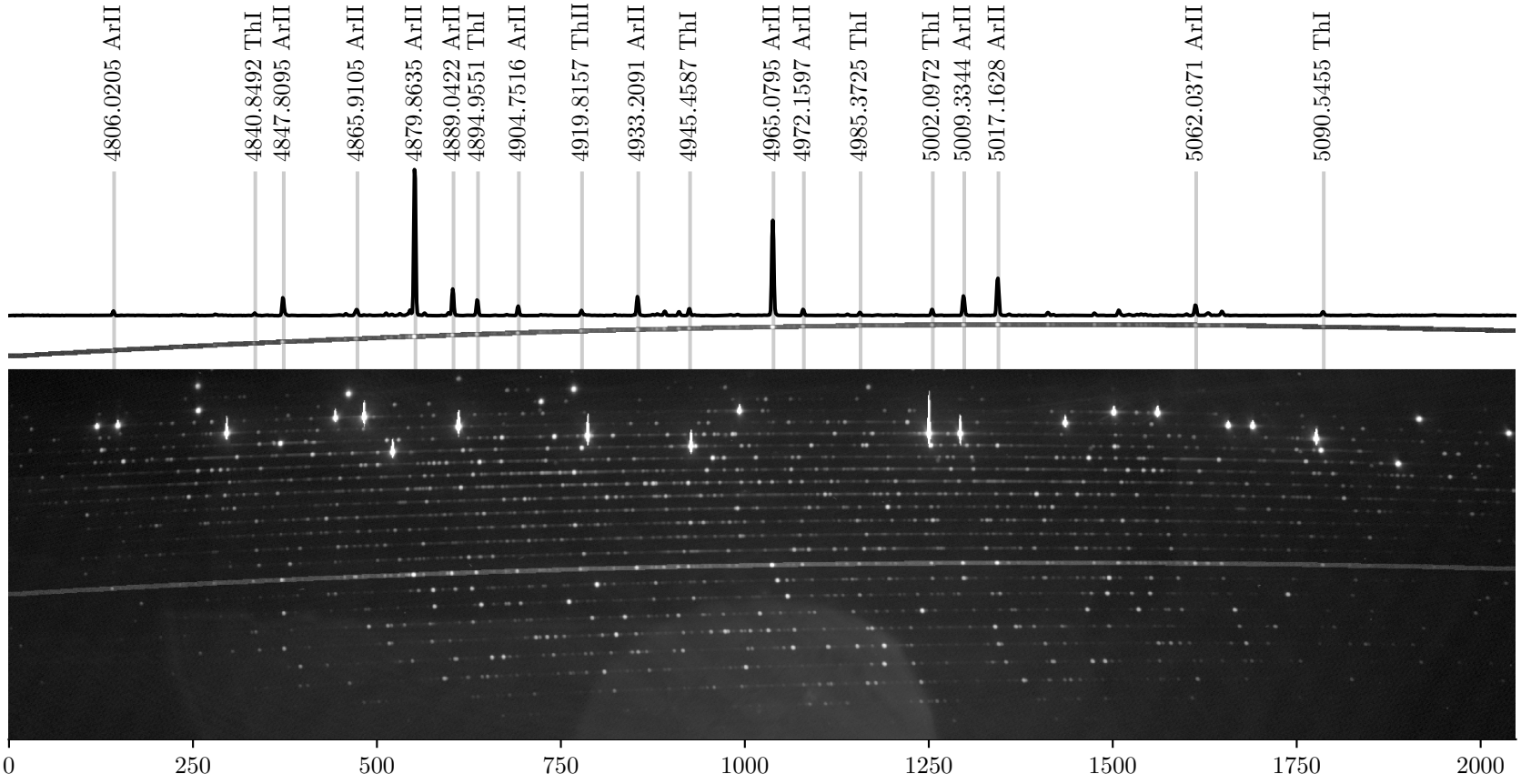
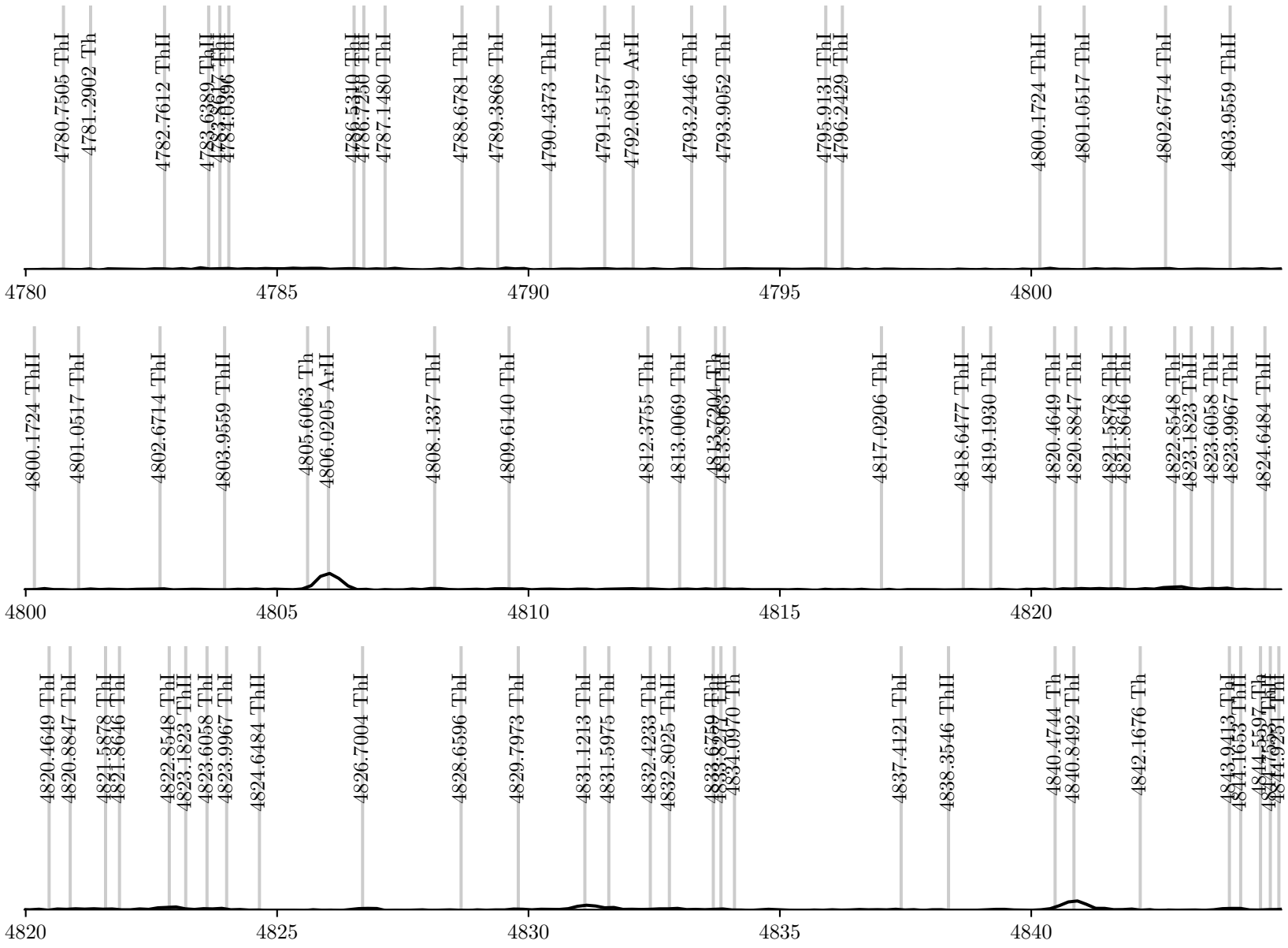
4980
4985
4990

Order 30



Dispersion:	0.172 Å pix^{-1}
Central wavelength:	4962 Å
Wavelength range:	4780 Å – 5133 Å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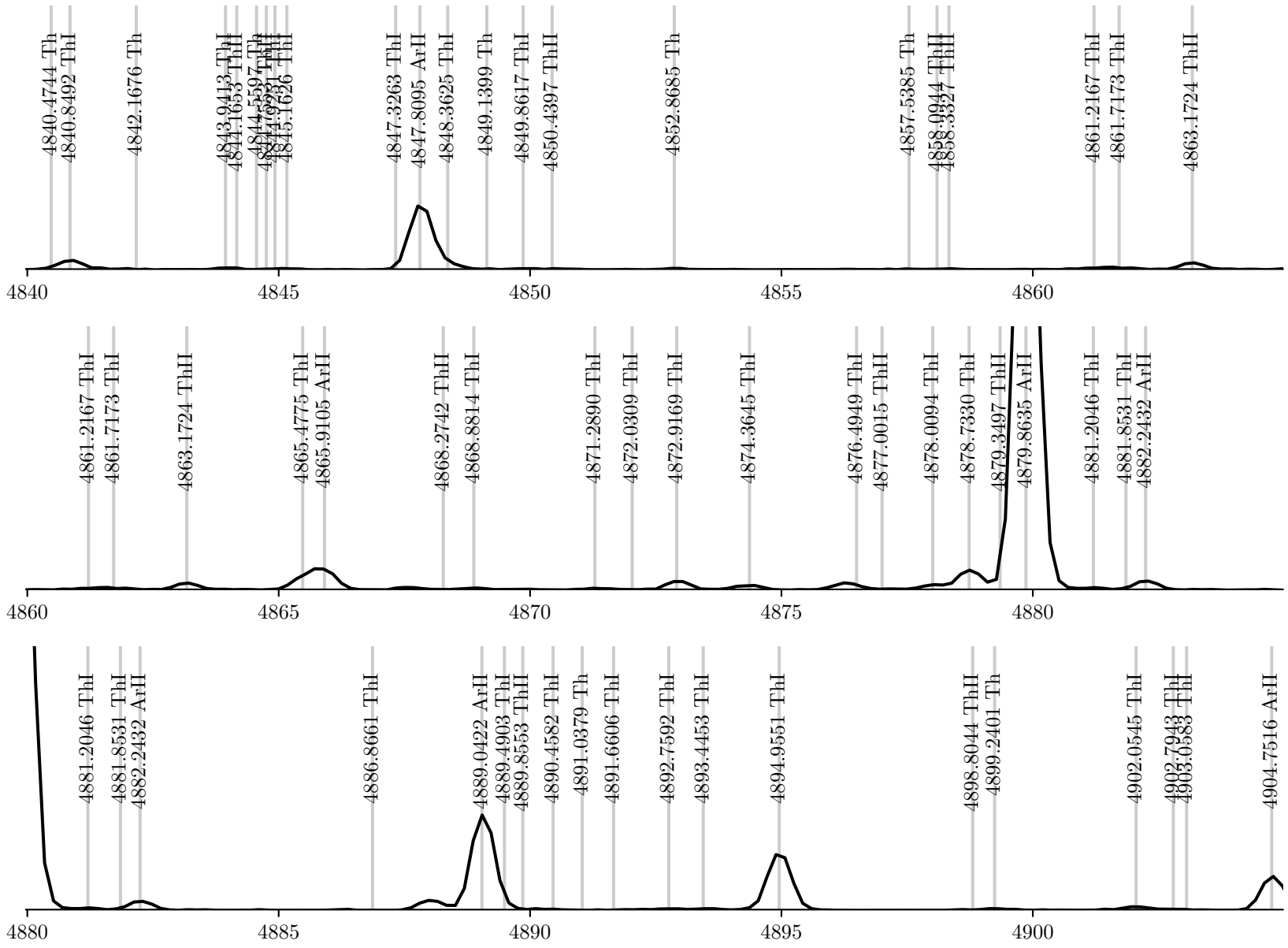
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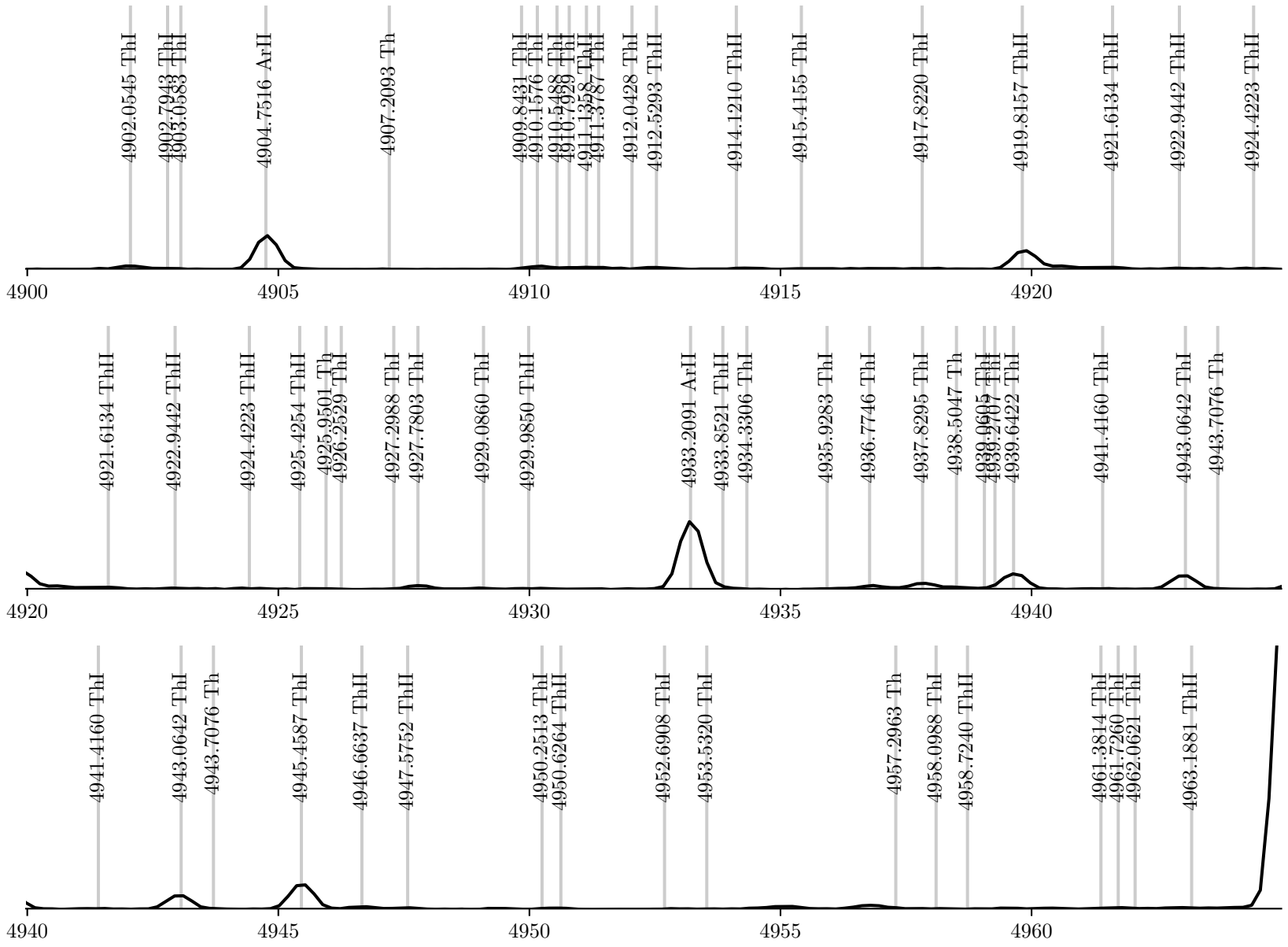
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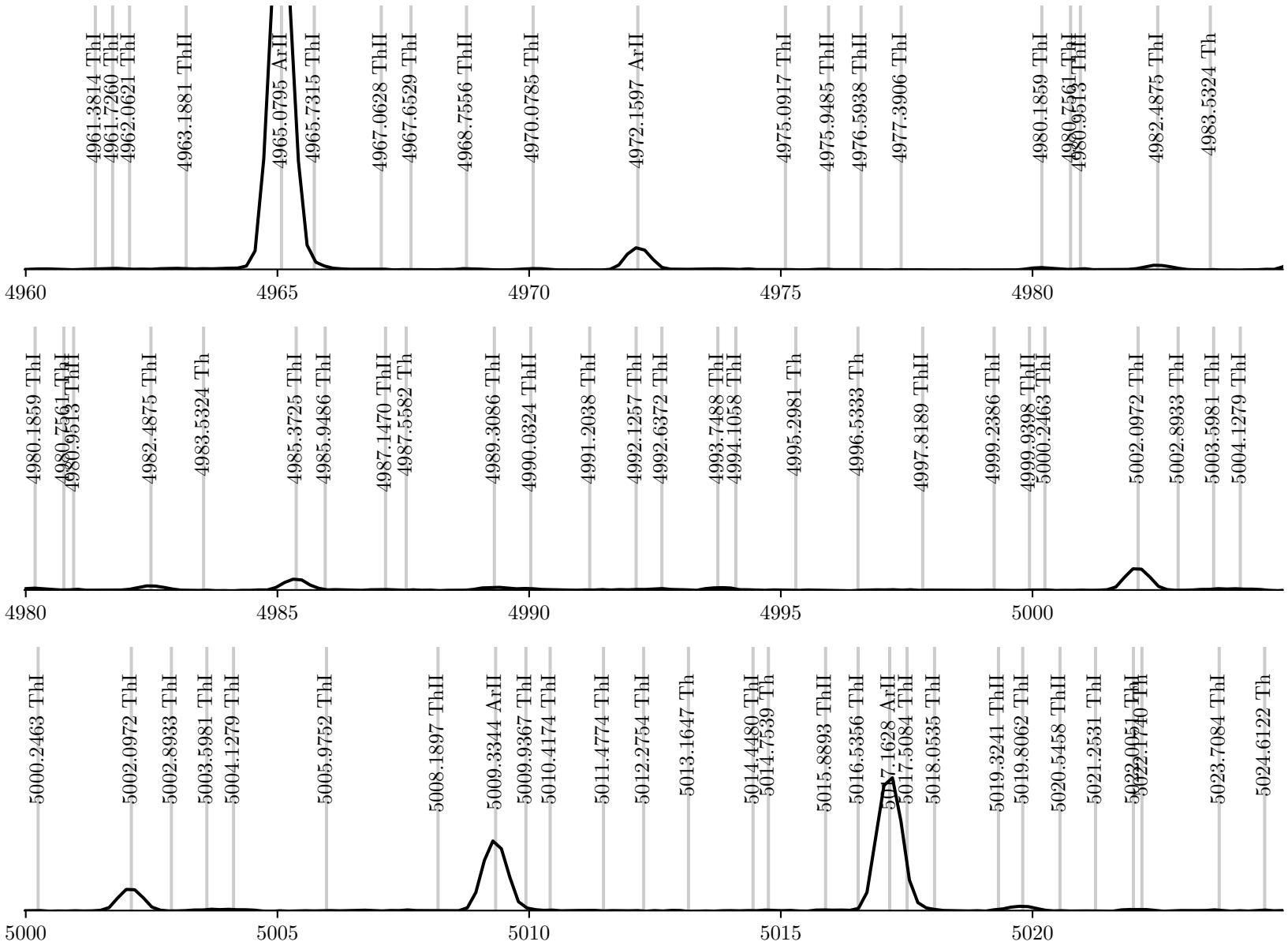
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4991.2038 ThI
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5003.5981 ThI
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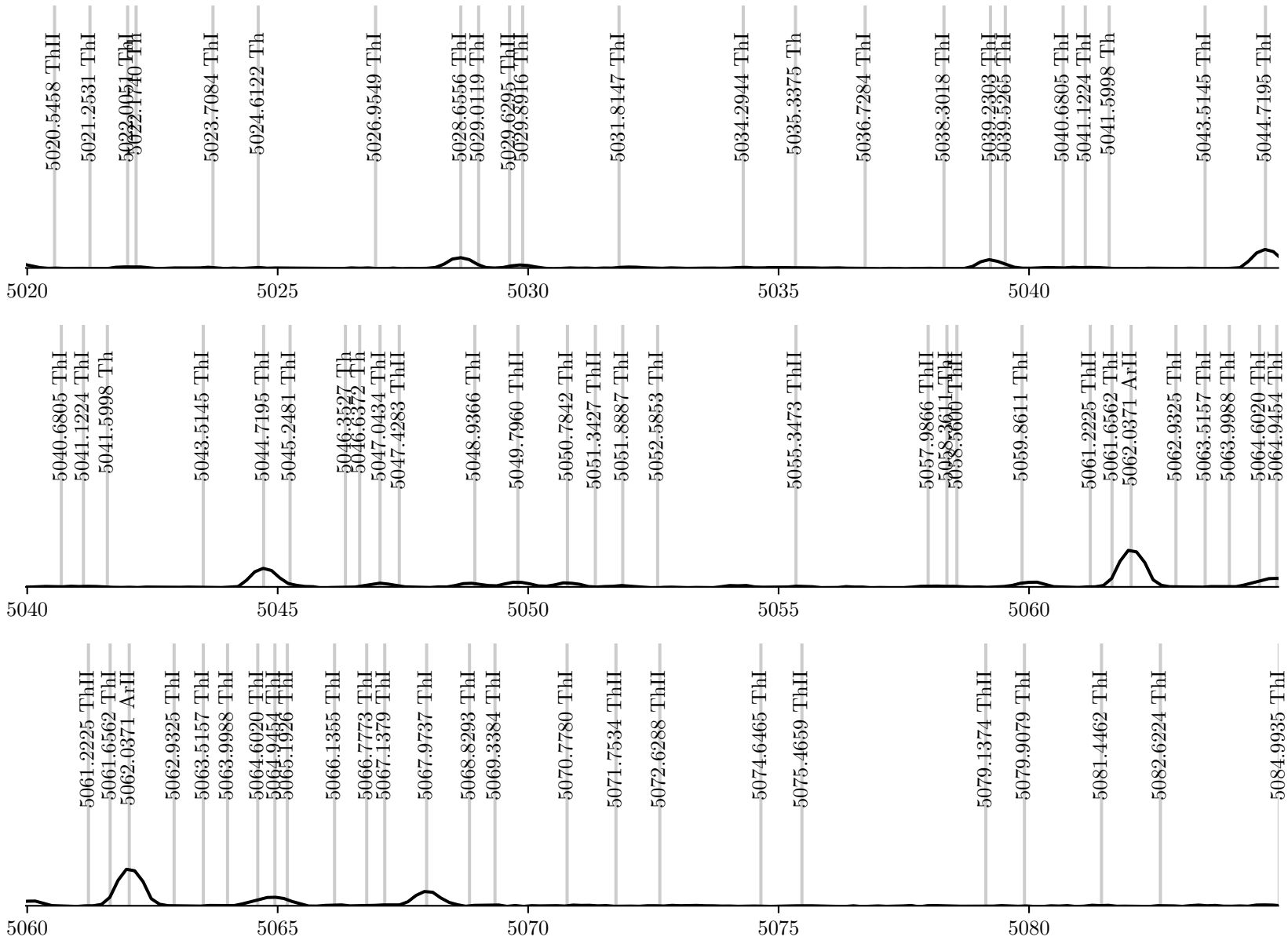
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5041.5998 Th
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5045.2481 ThI
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5064.9454 ThI
5065.1926 ThI
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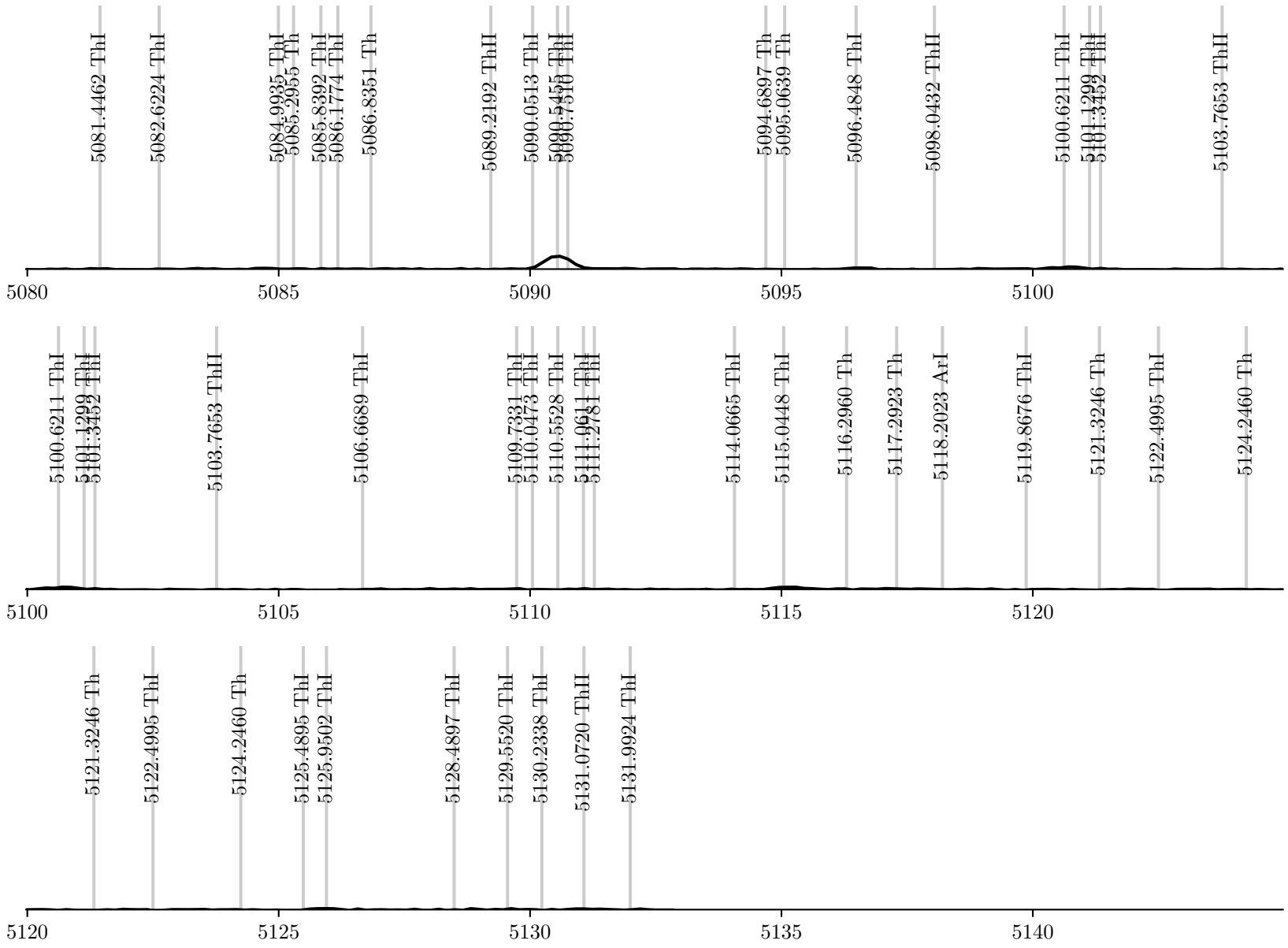
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5117.2923 Th
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5121.3246 Th
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5121.3246 Th
5122.4995 ThI
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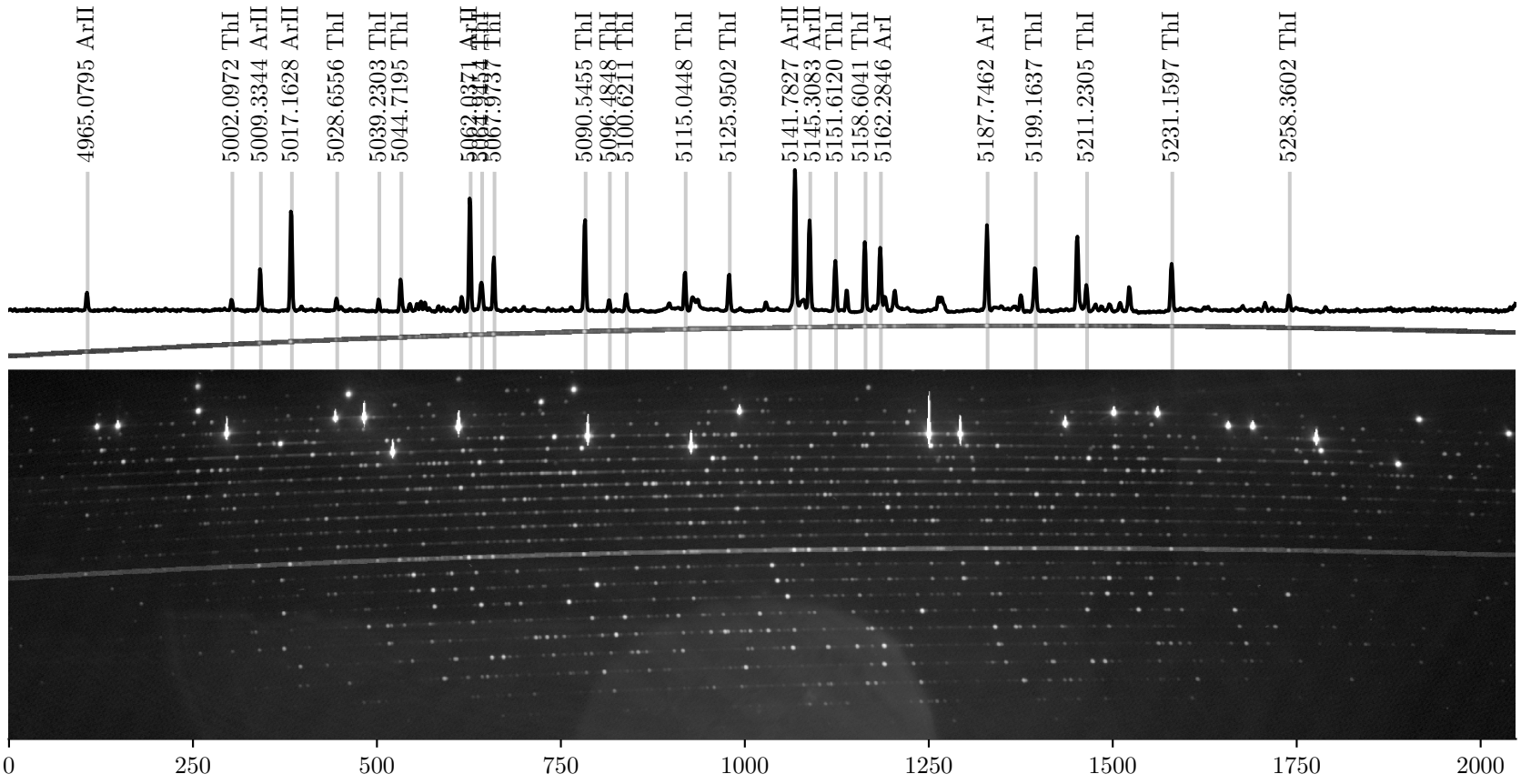
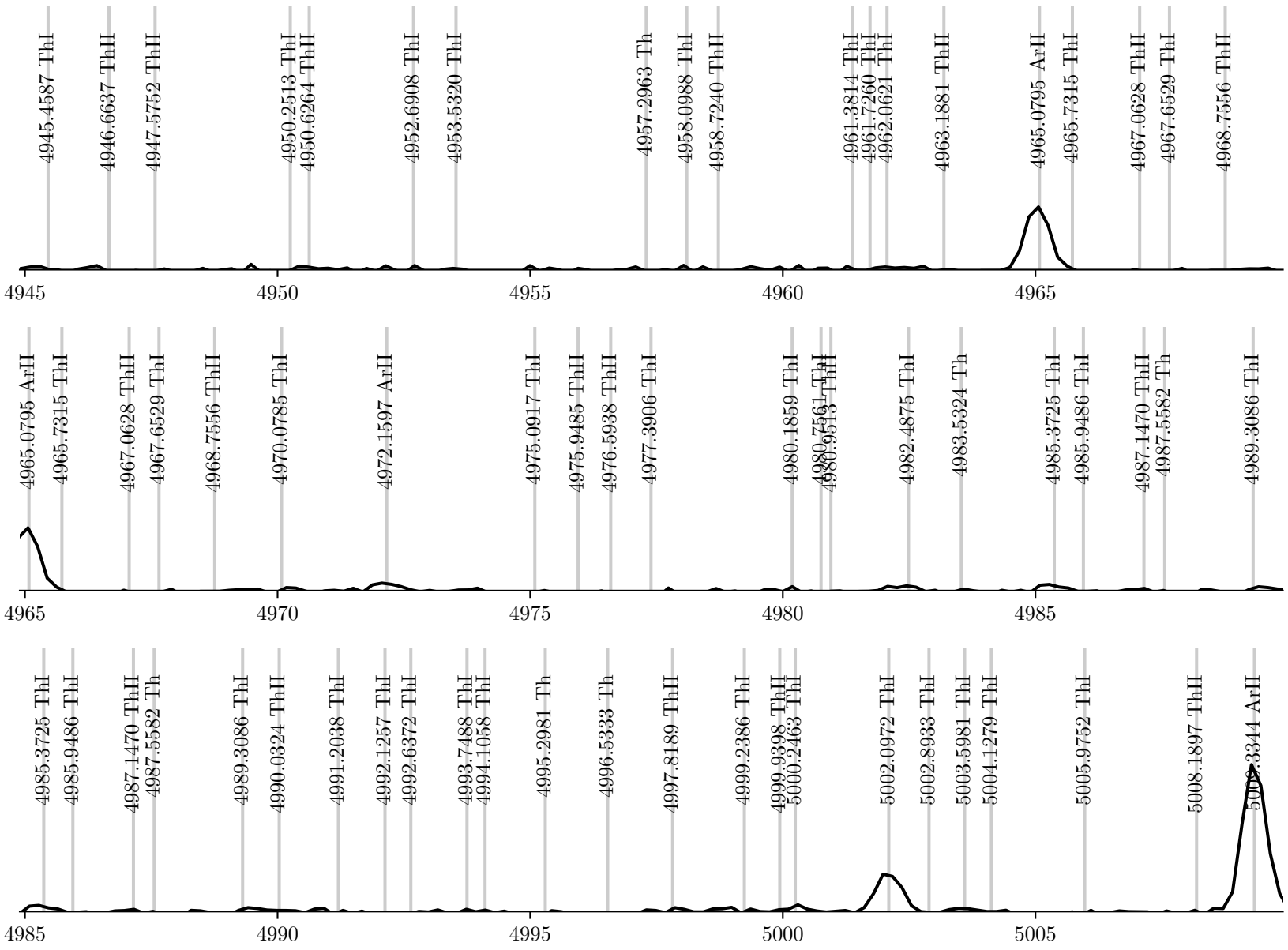
Order 29



Dispersion:	0.178 Å pix^{-1}
Central wavelength:	5134 Å
Wavelength range:	4945 Å – 5310 Å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4958.7240 ThII
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4965.0795 ArII
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4967.0628 ThII
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4968.7556 ThII
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4955
4960
4965
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4970
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4980
4985
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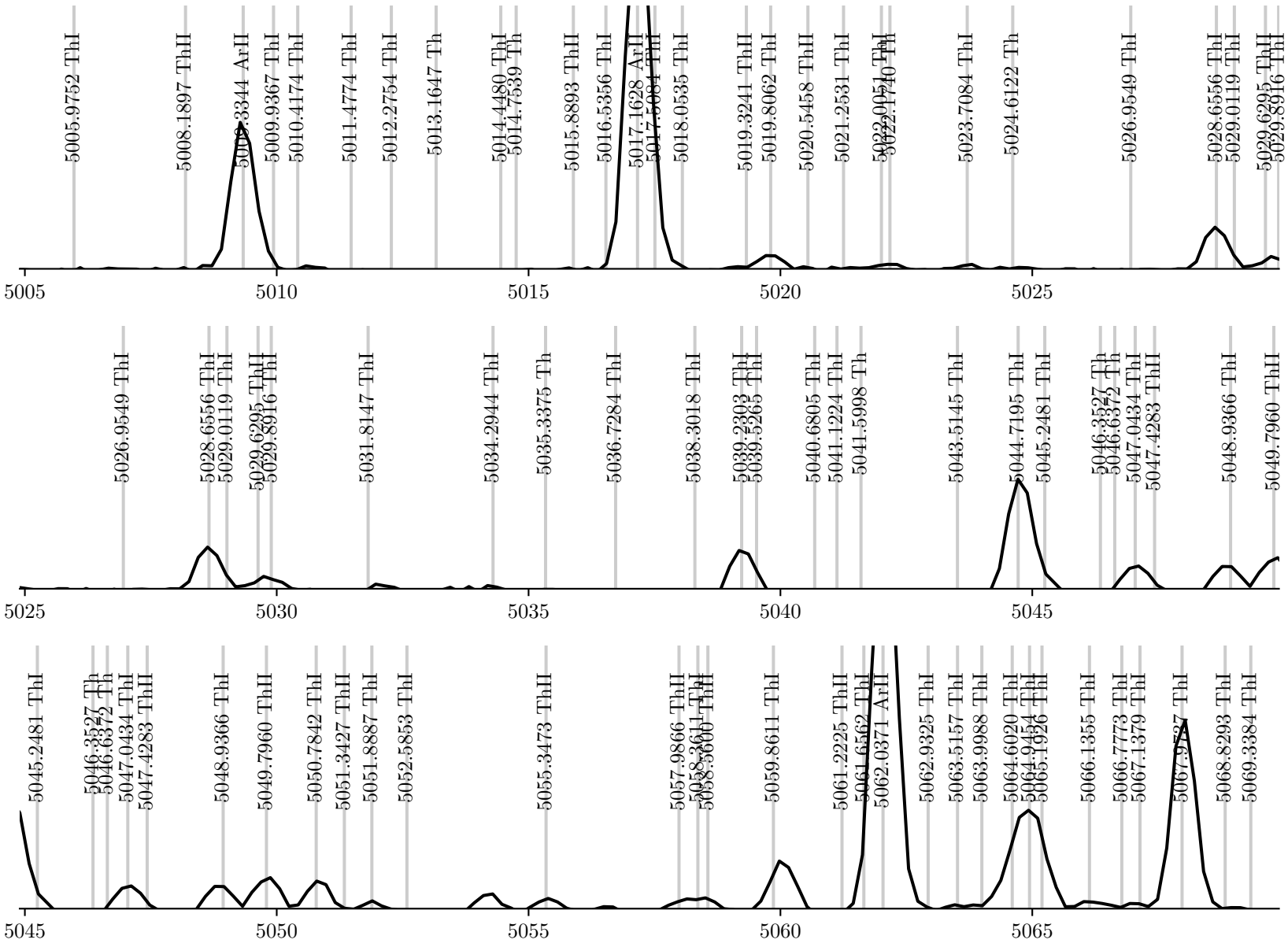
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5066.7773 ThI
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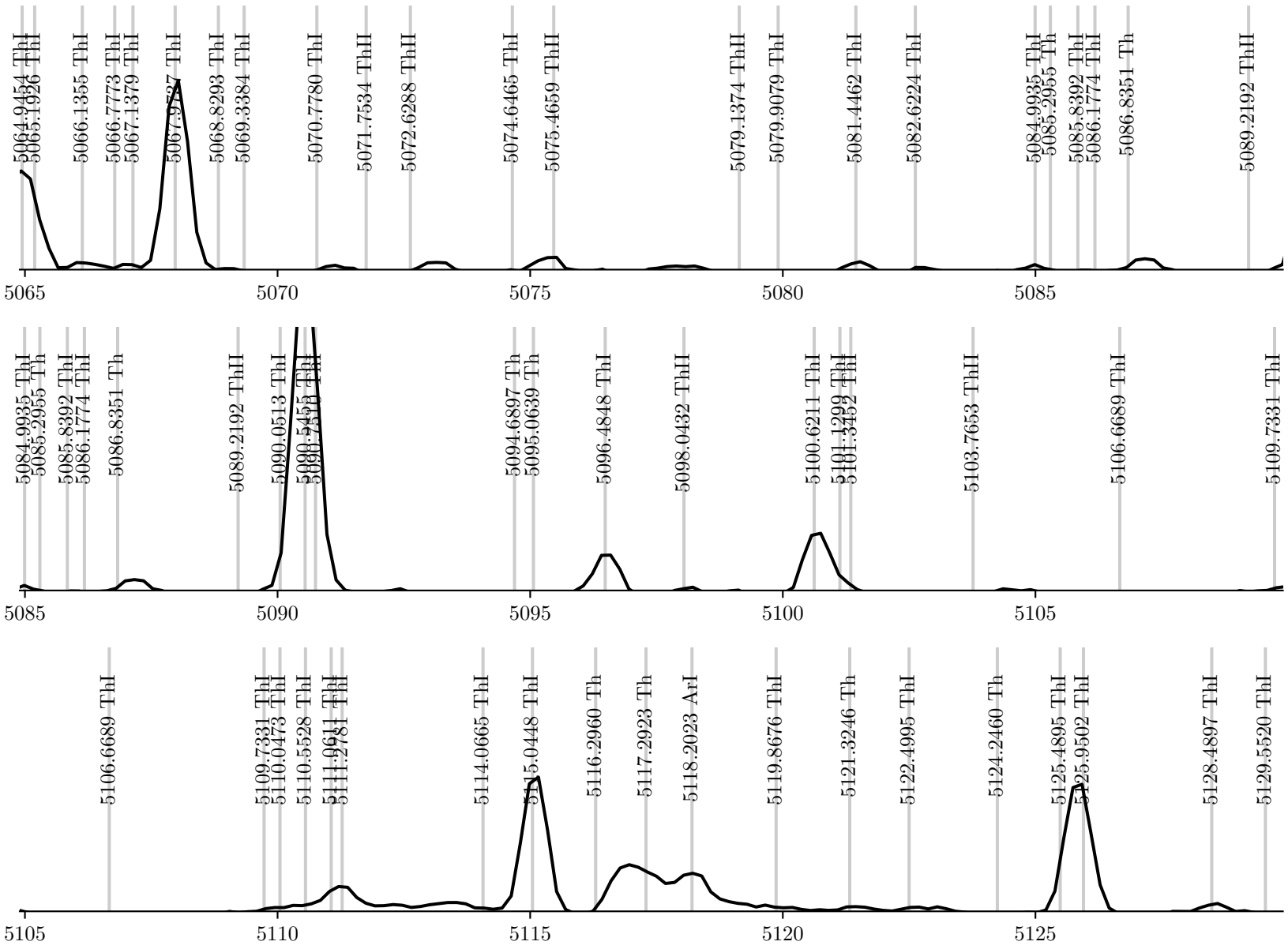
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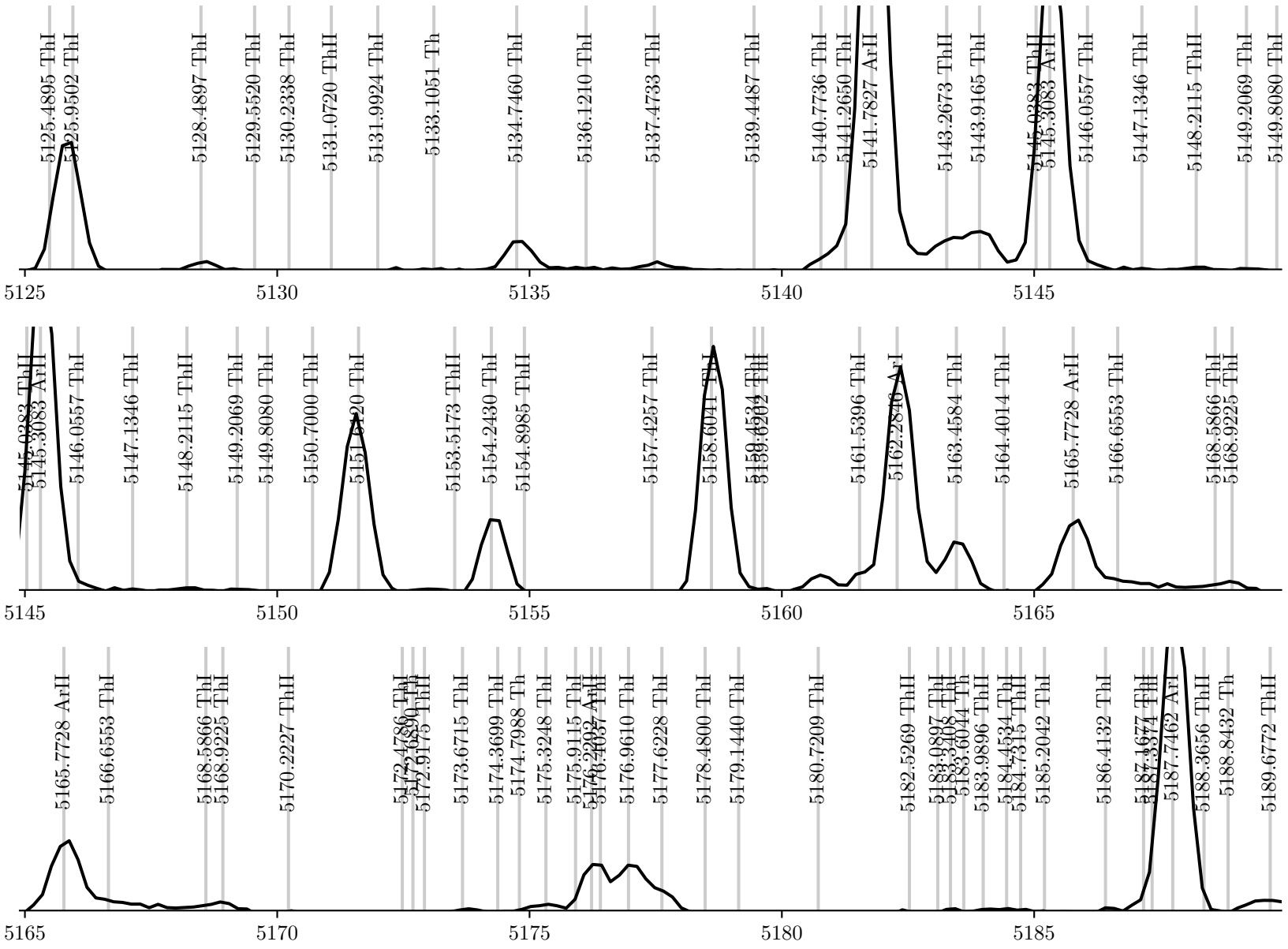
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5143.9165 ThI
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5145.3083 ArII
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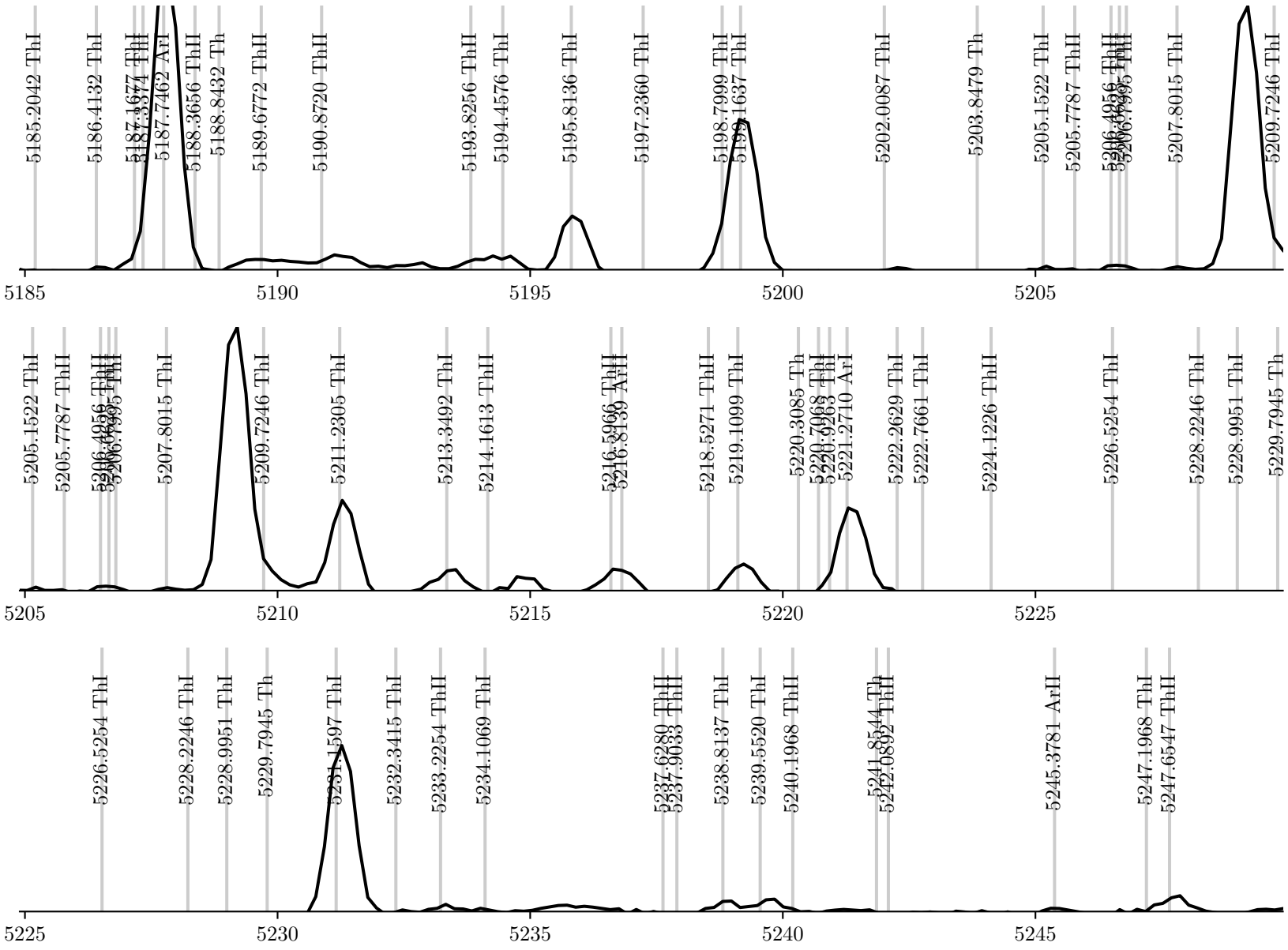
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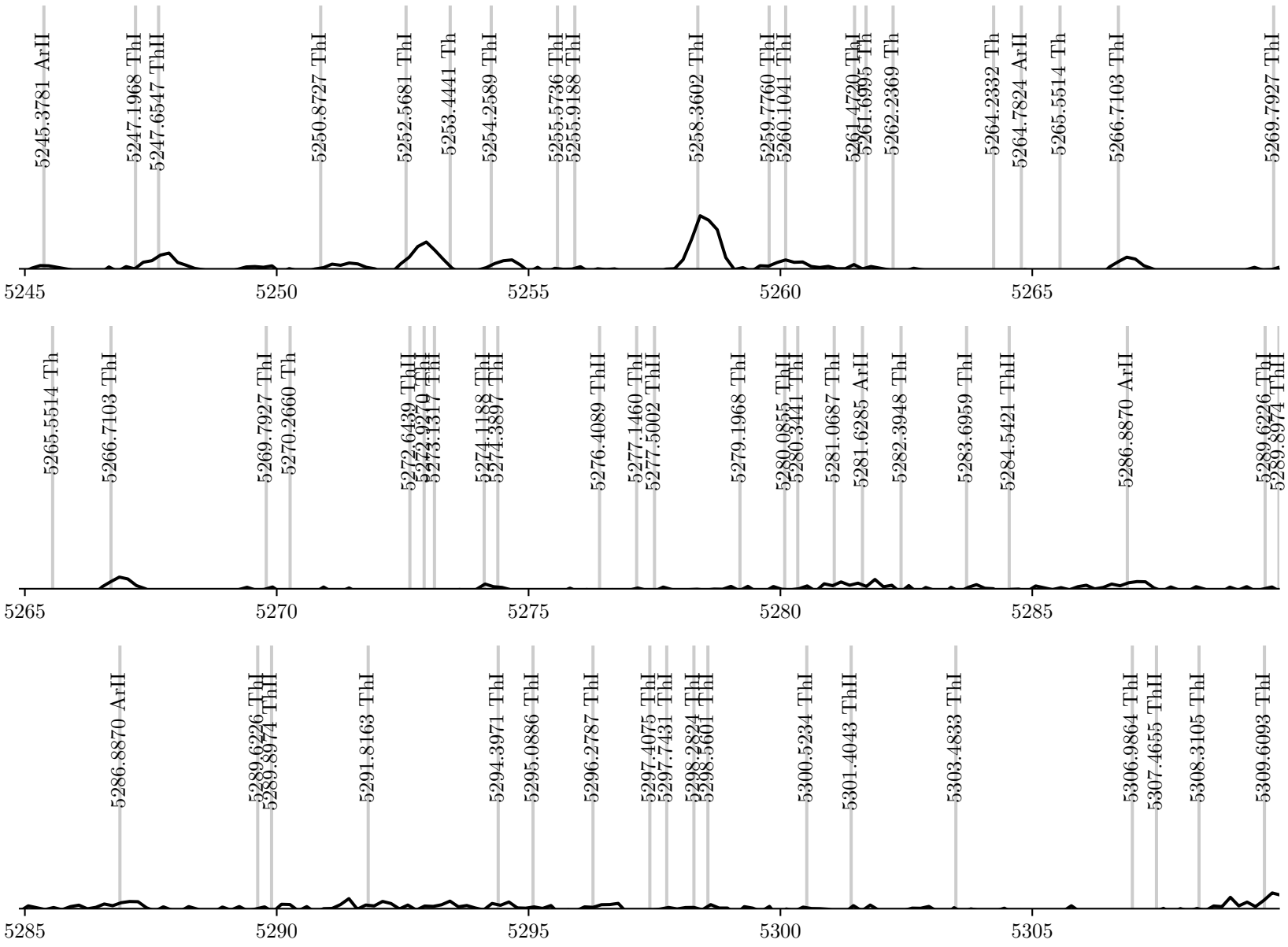
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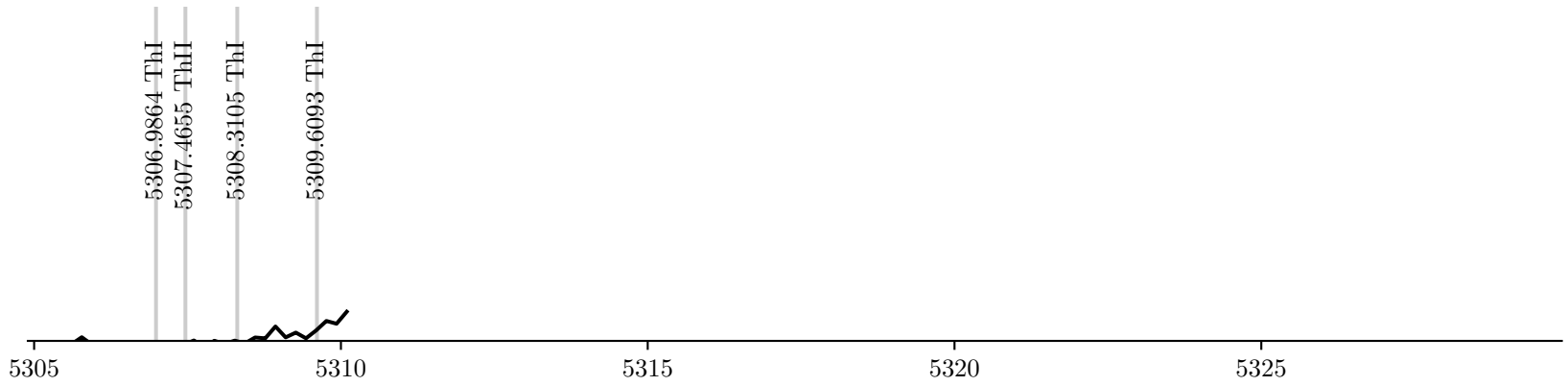
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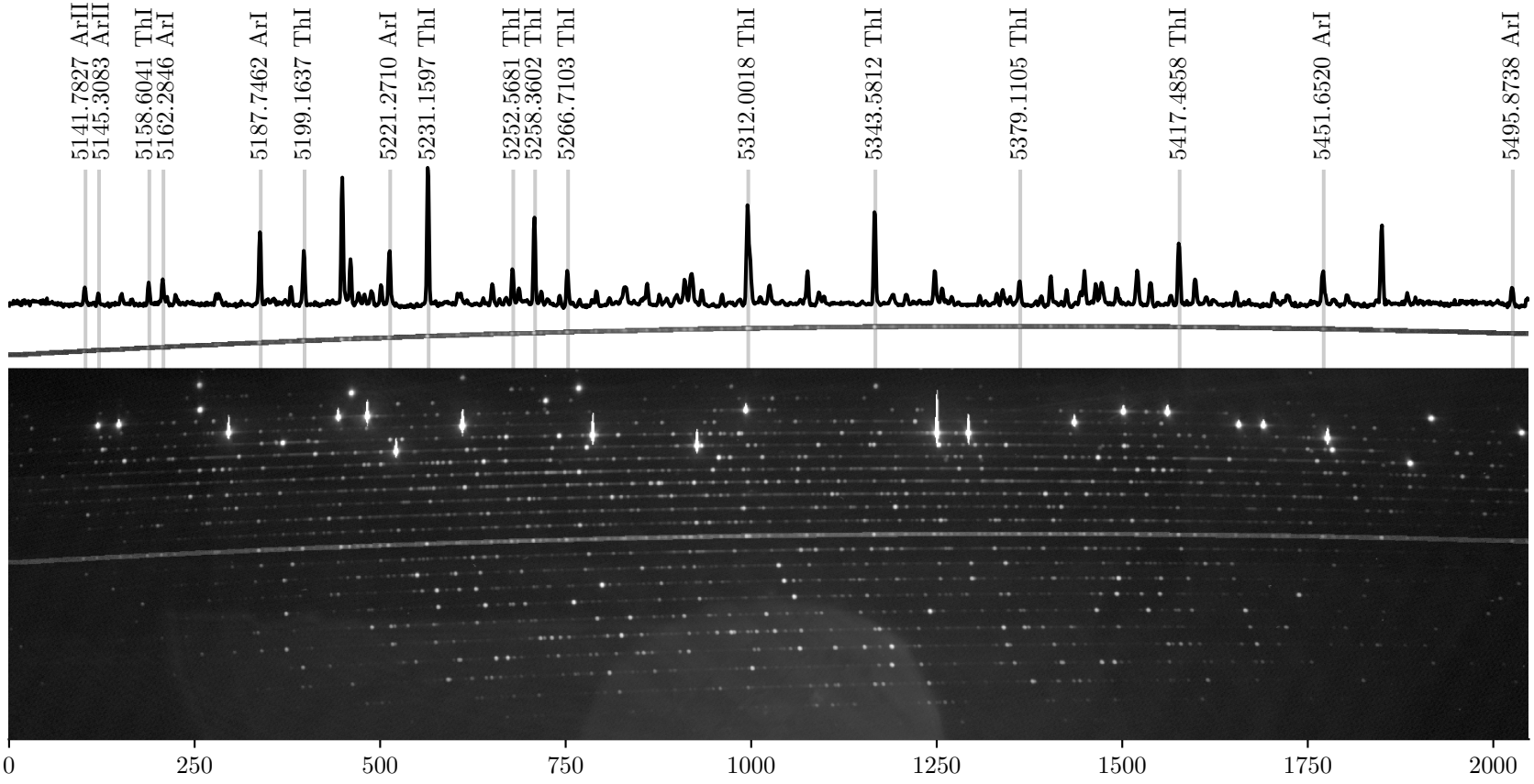
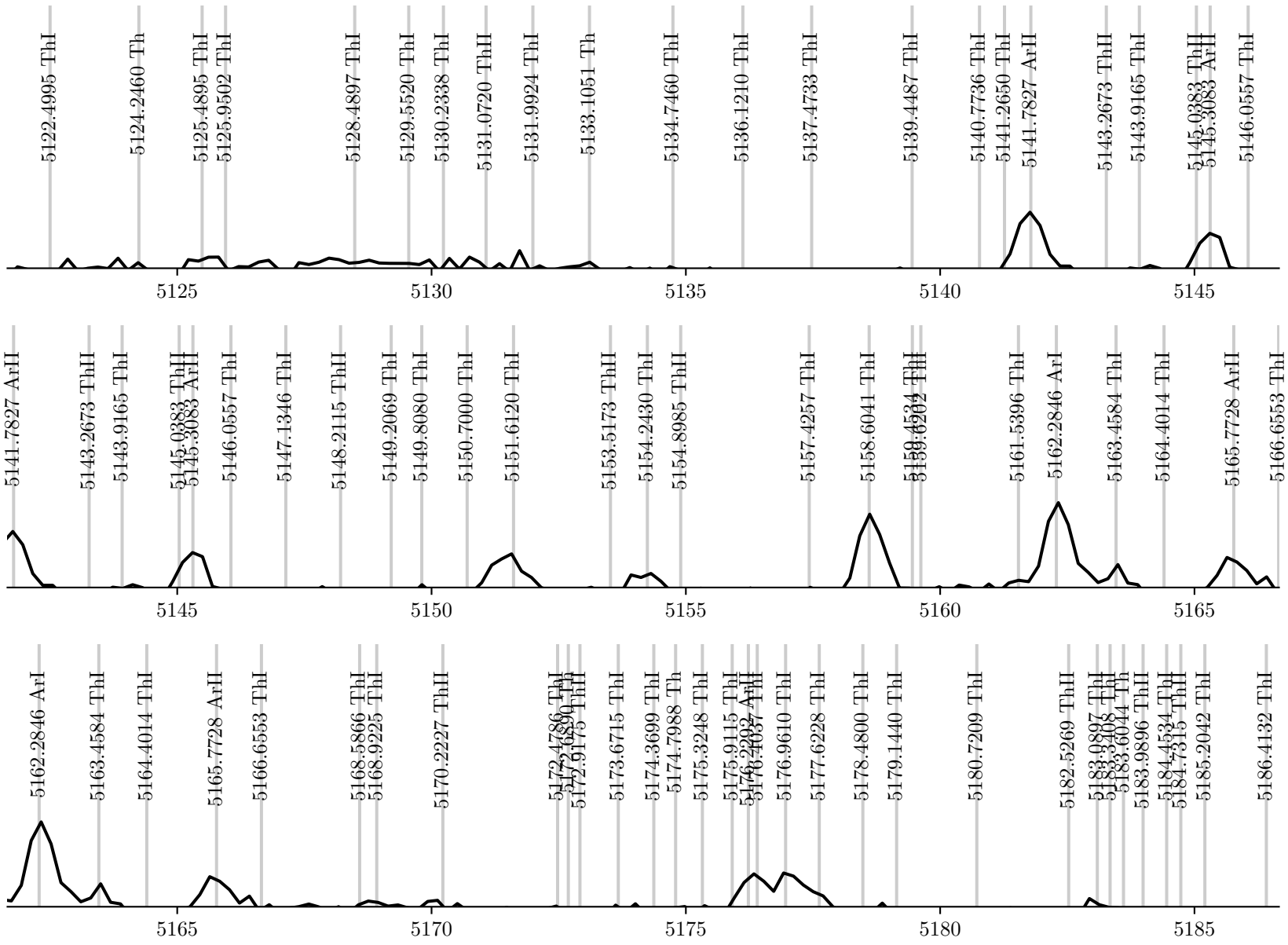
Order 28



Dispersion:	0.185 Å pix^{-1}
Central wavelength:	5317 Å
Wavelength range:	5122 Å – 5500 Å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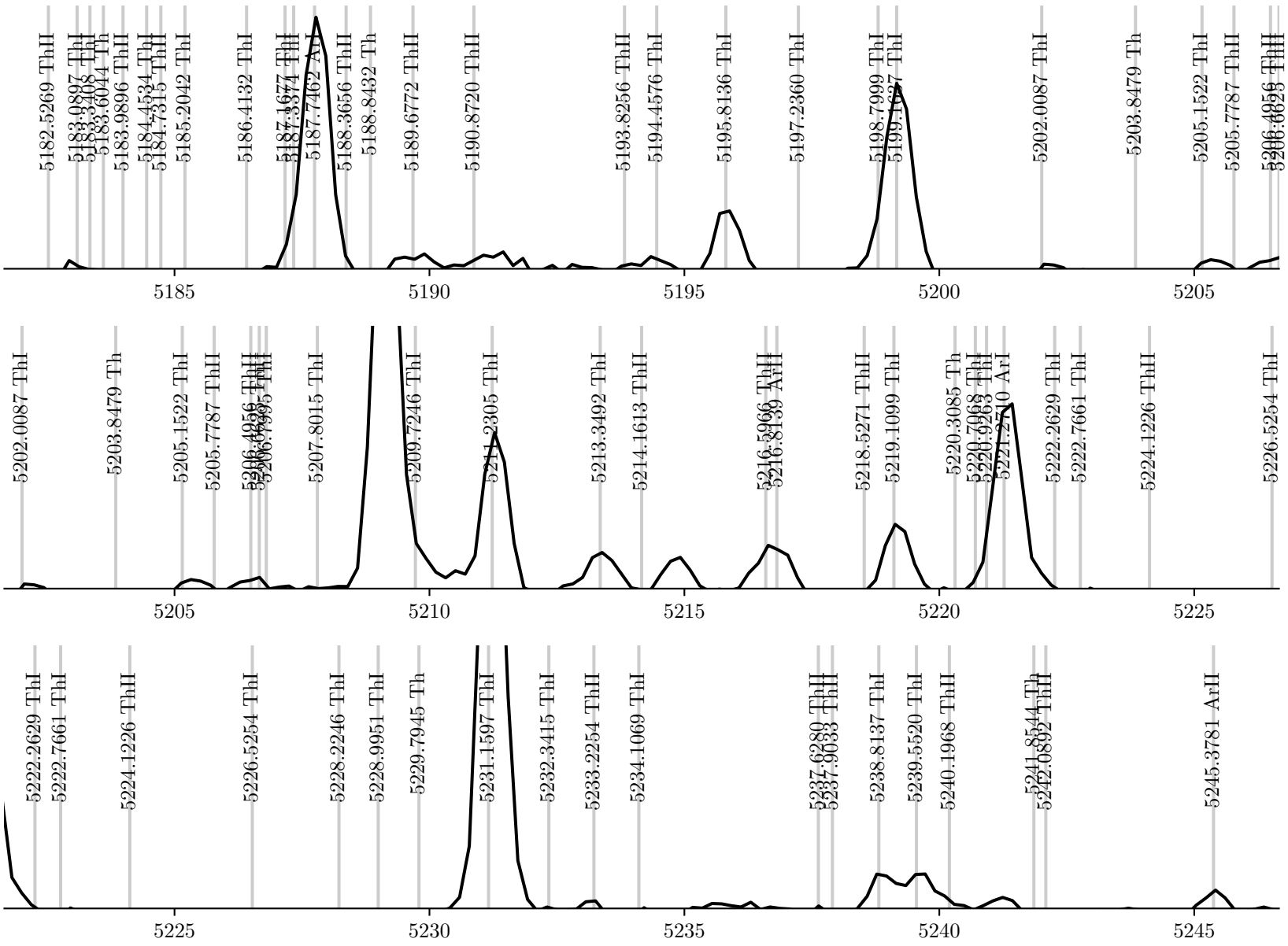
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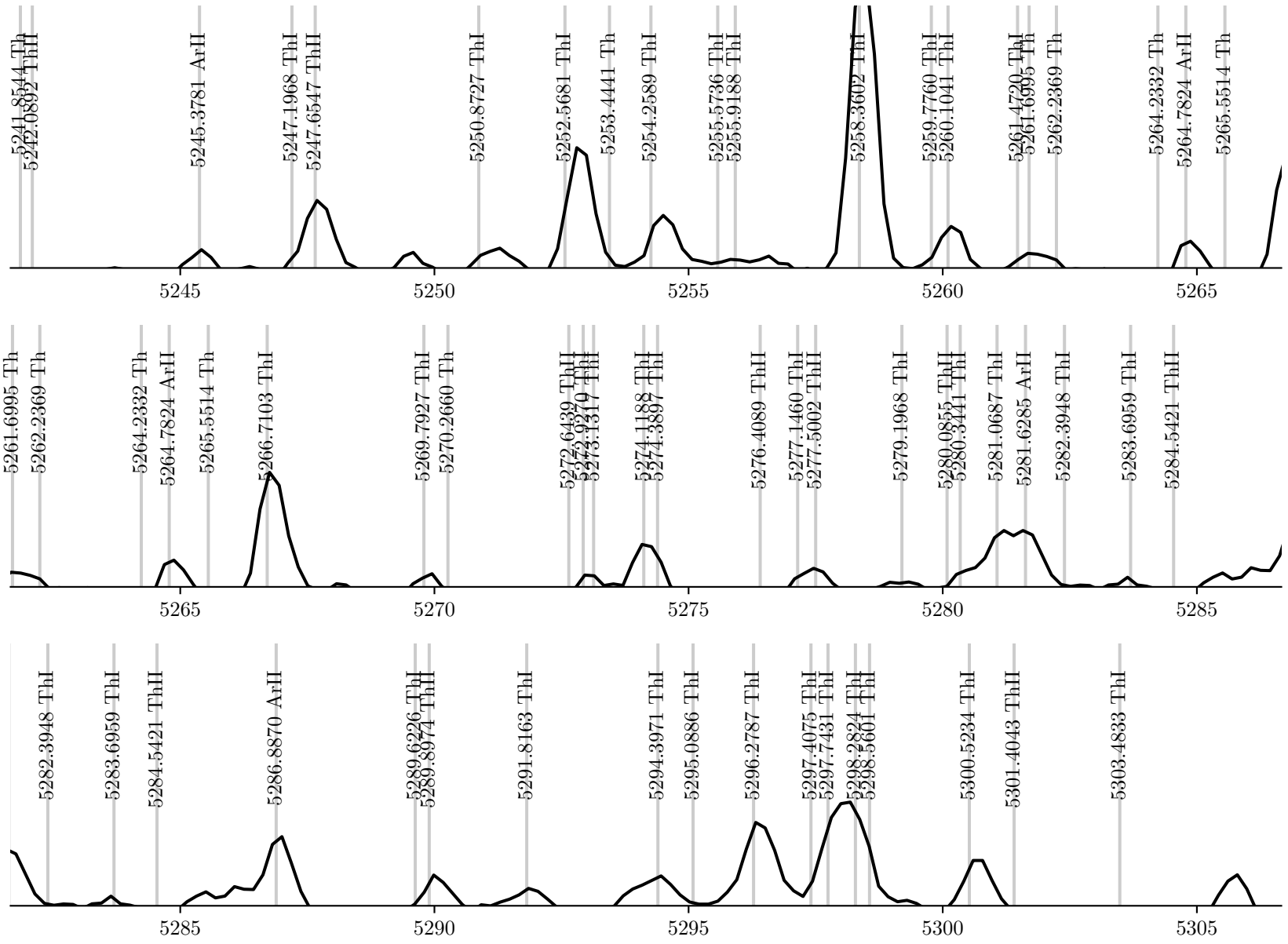
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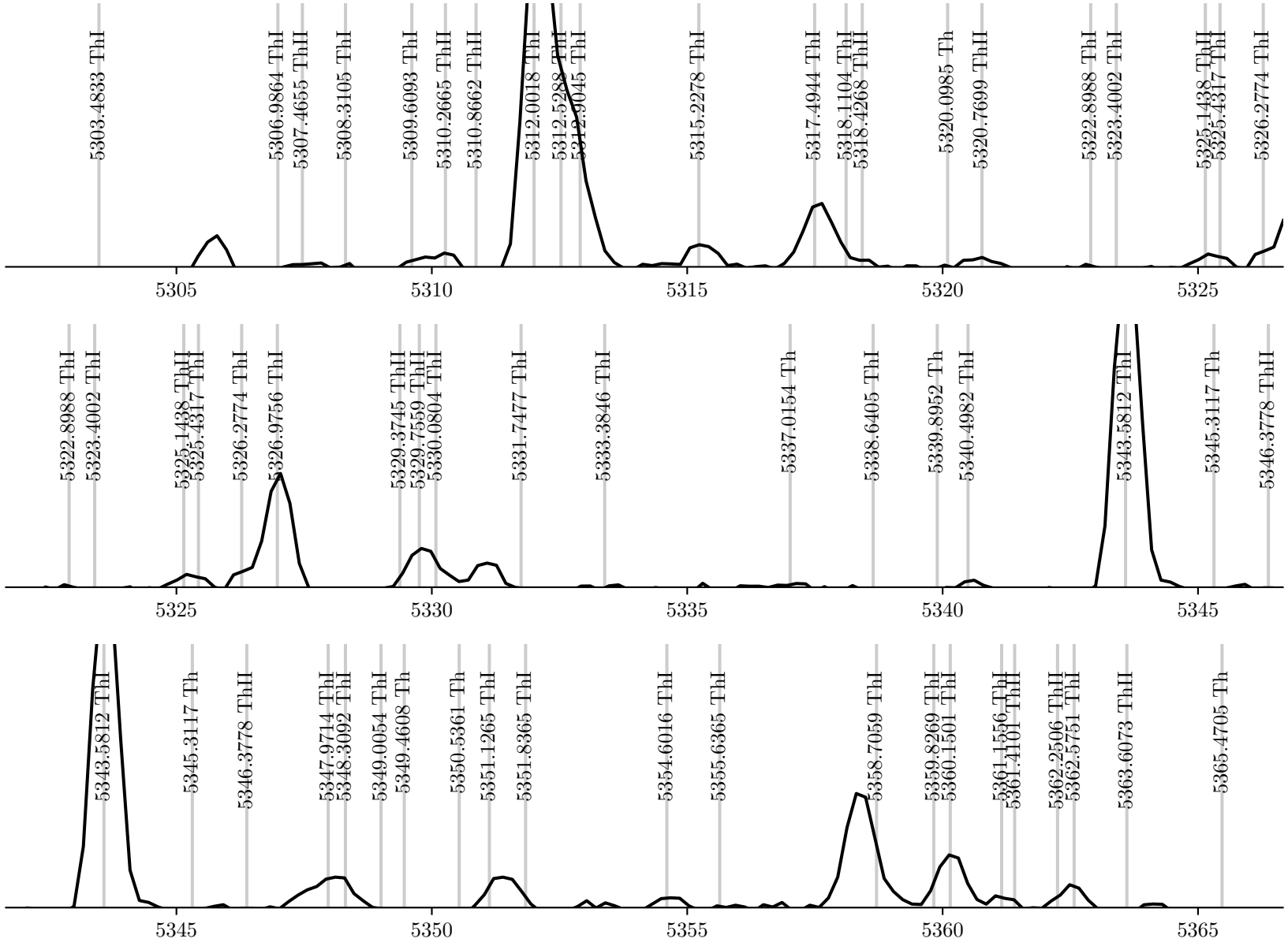
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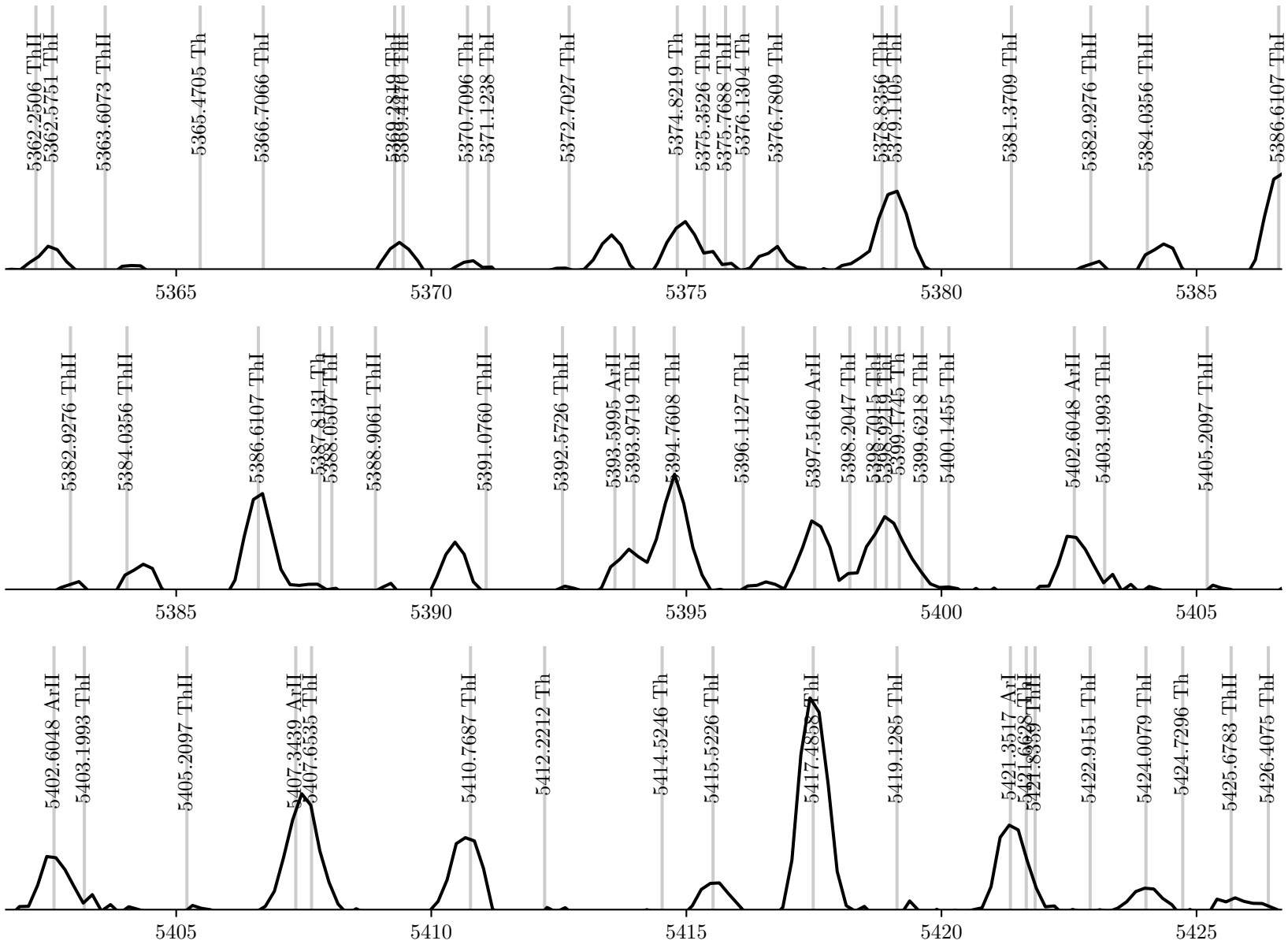
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5400
5405
5402.6048 ArII
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5414.5246 Th
5415.5226 ThI
5417.4858 ThI
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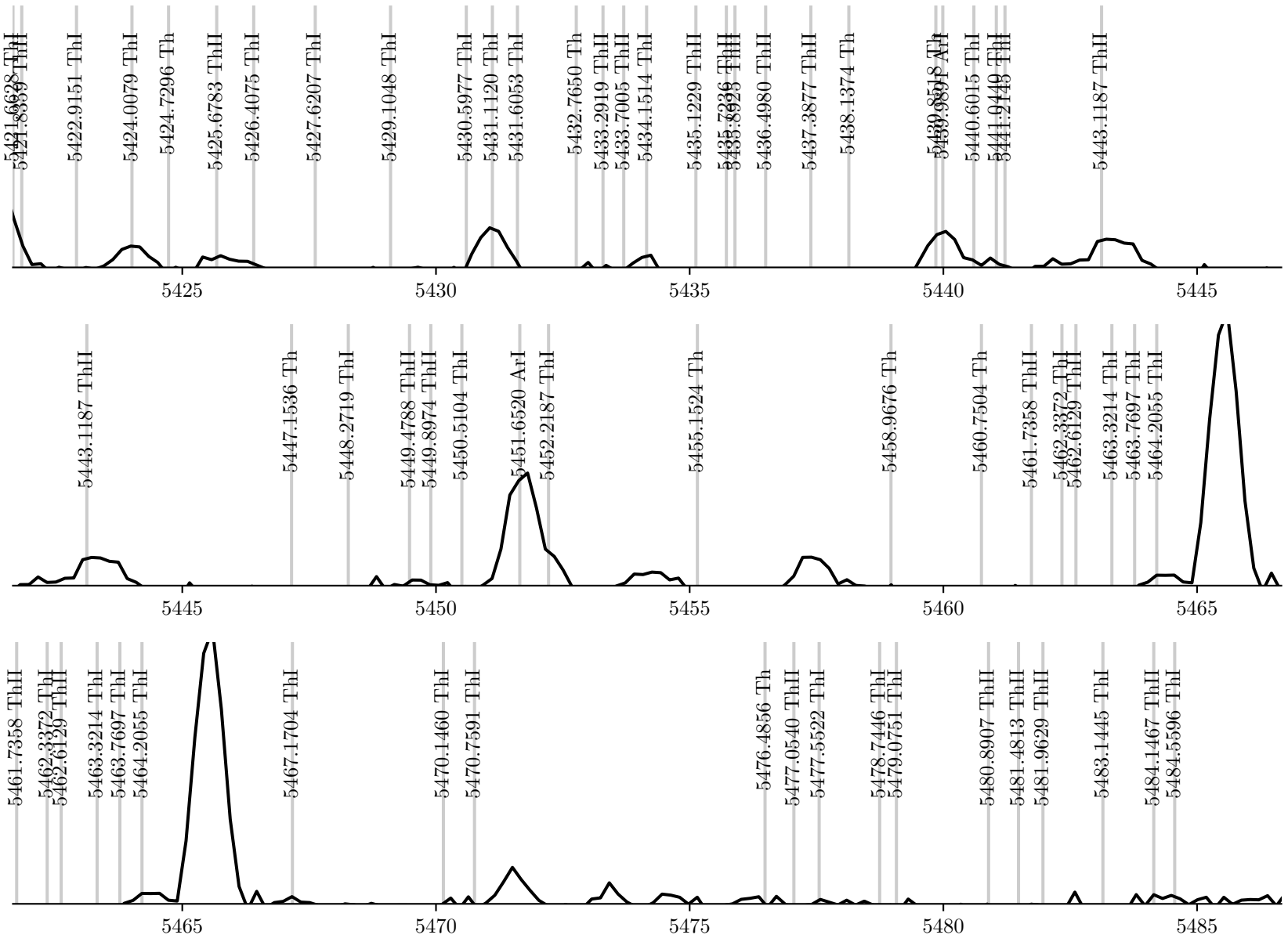
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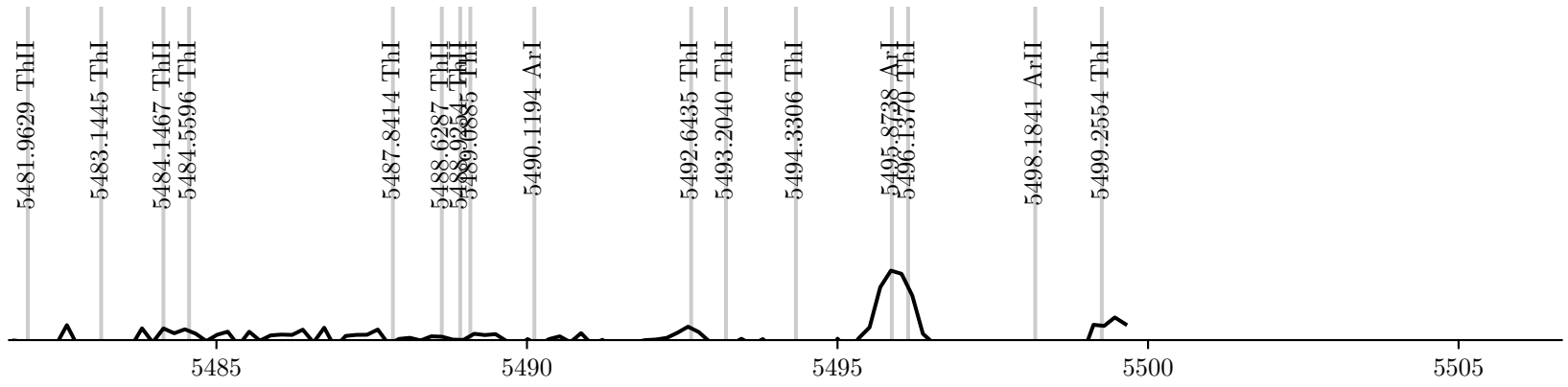
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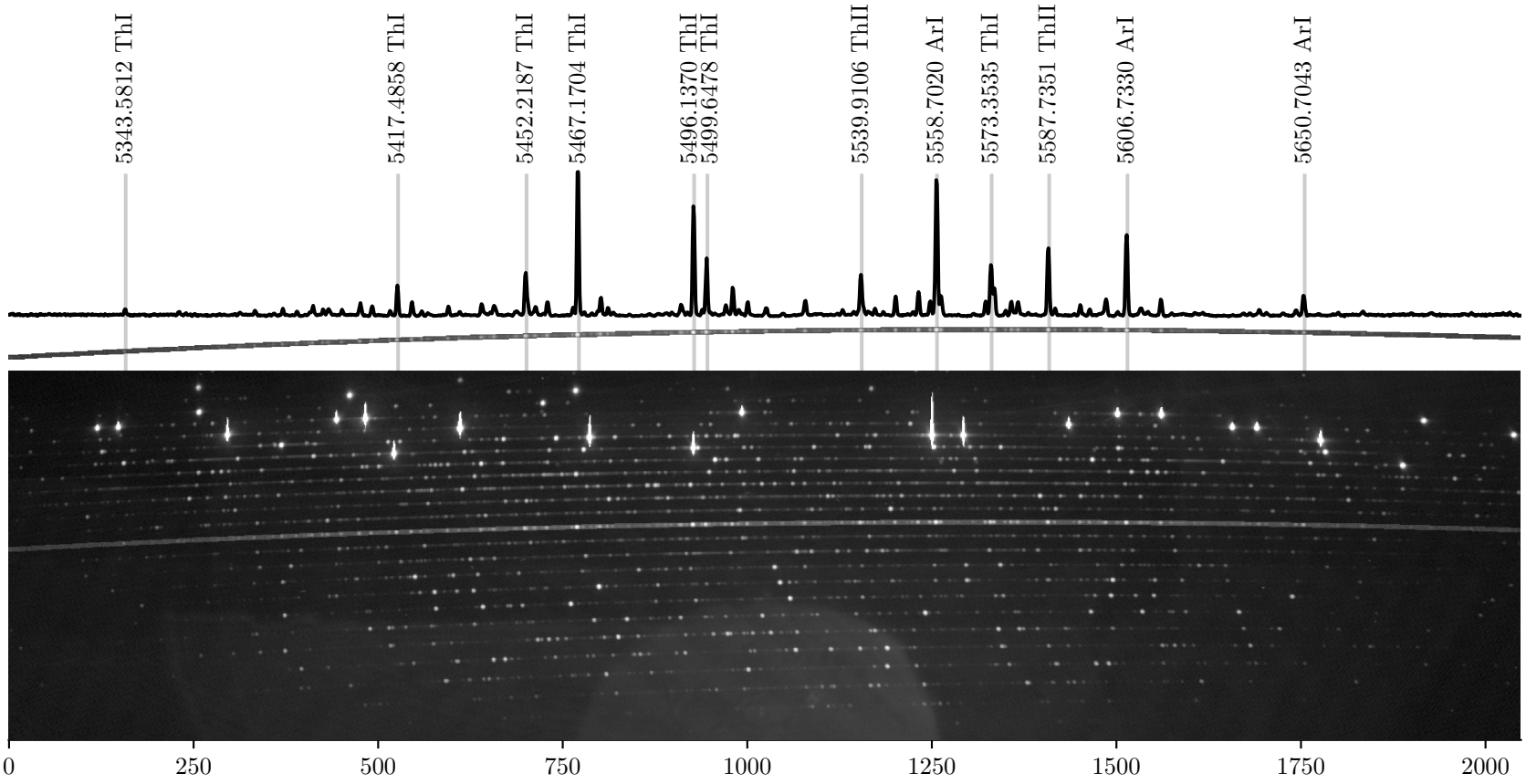
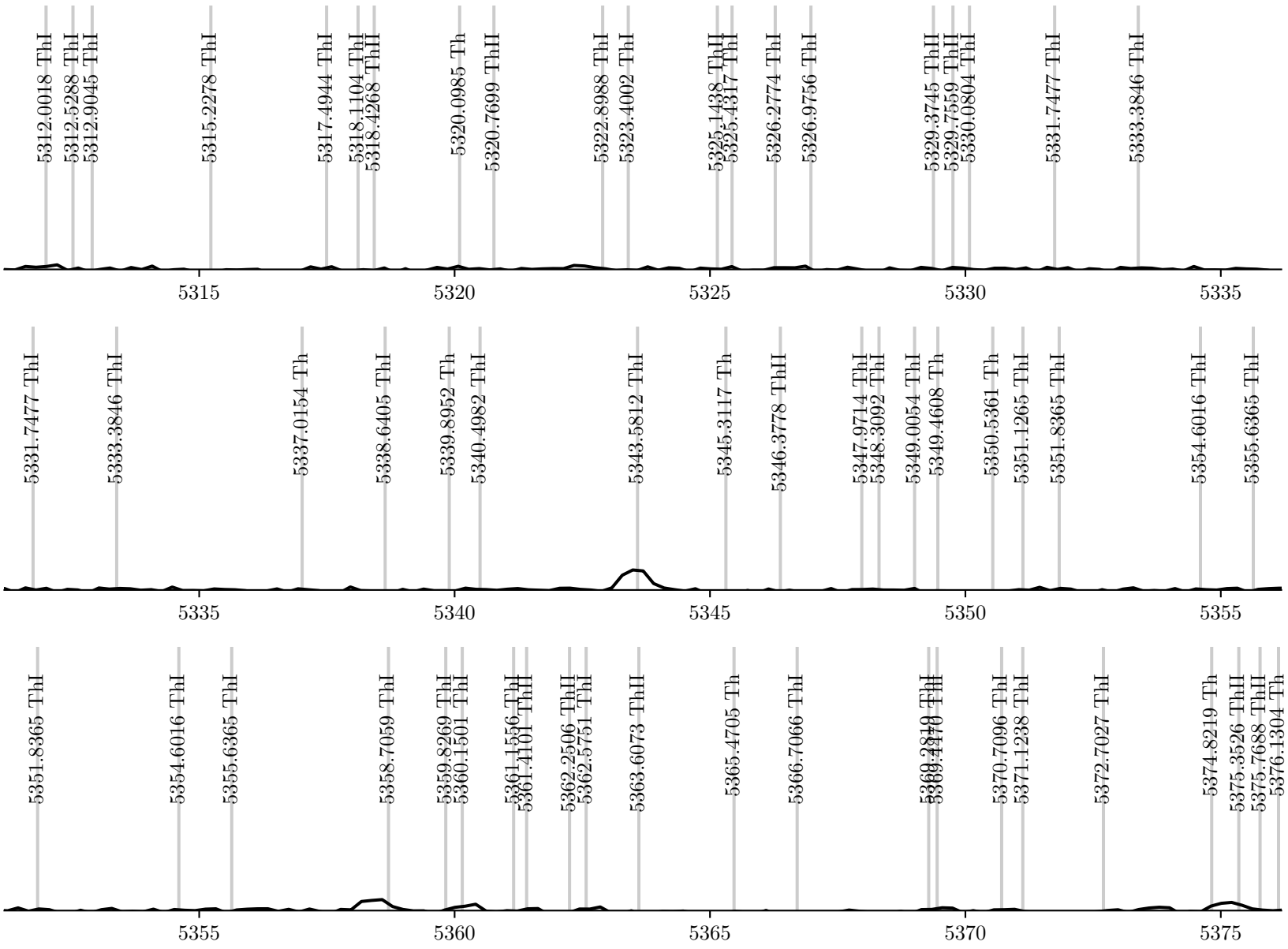
5500
5505

Order 27



Dispersion:	0.191 Å pix^{-1}
Central wavelength:	5515 Å
Wavelength range:	5311 Å – 5703 Å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5323.4002 ThI
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5325.4317 ThI
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5326.9756 ThI
5329.3745 ThII
5329.7559 ThII
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5333.3846 ThI
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5335
5331.7477 ThI
5333.3846 ThI
5337.0154 Th
5338.6405 ThI
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5340.4982 ThI
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5346.3778 ThII
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5348.3092 ThI
5349.0054 ThI
5349.4608 Th
5350.5361 Th
5351.1265 ThI
5351.8365 ThI
5354.6016 ThI
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5351.8365 ThI
5354.6016 ThI
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5362.5751 ThI
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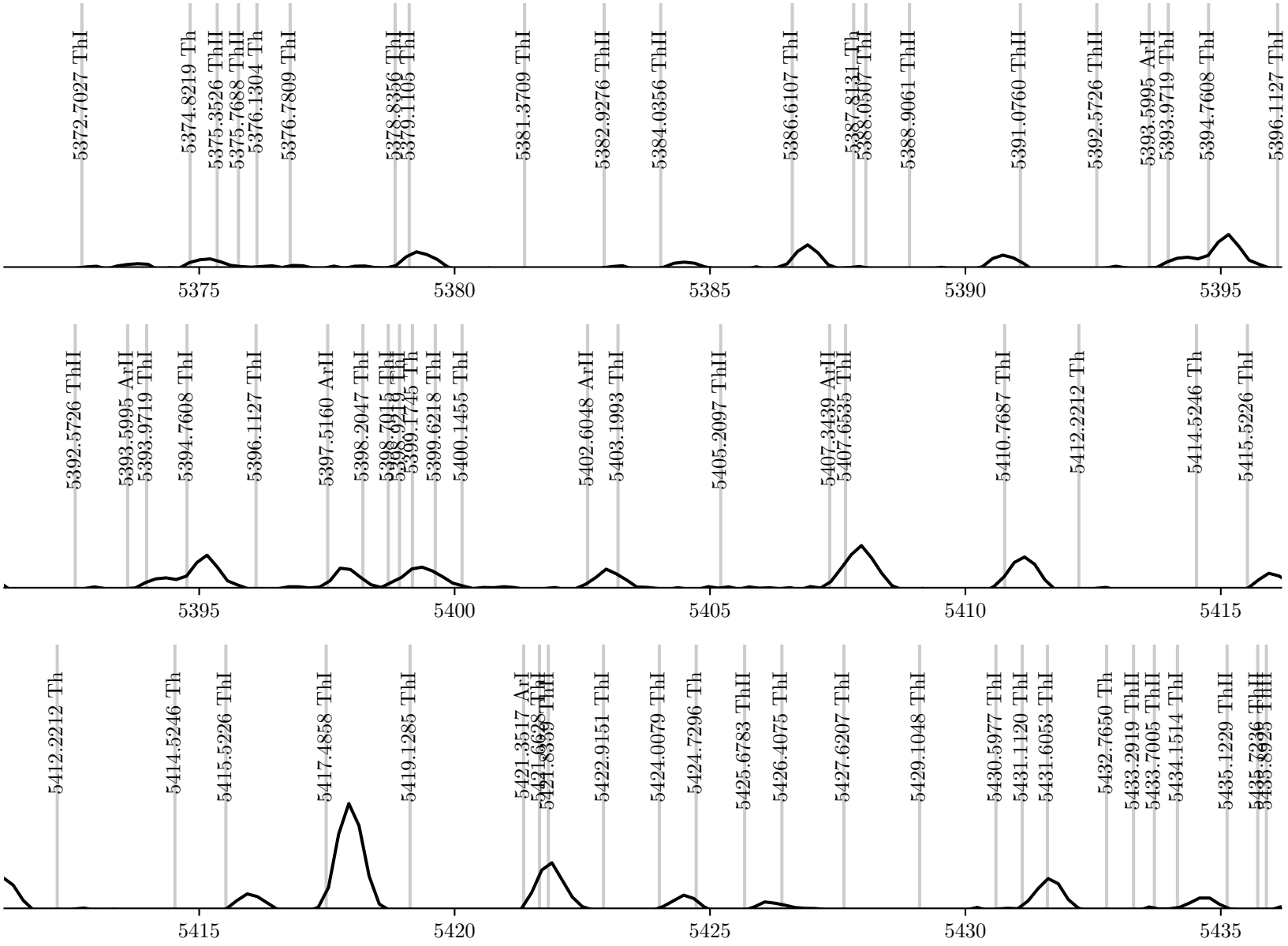
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5381.3709 ThI
5382.9276 ThII
5384.0356 ThII
5386.6107 ThI
5387.8131 Th
5388.0507 ThI
5388.9061 ThII
5391.0760 ThII
5392.5726 ThII
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5394.7608 ThI
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5432.7650 Th
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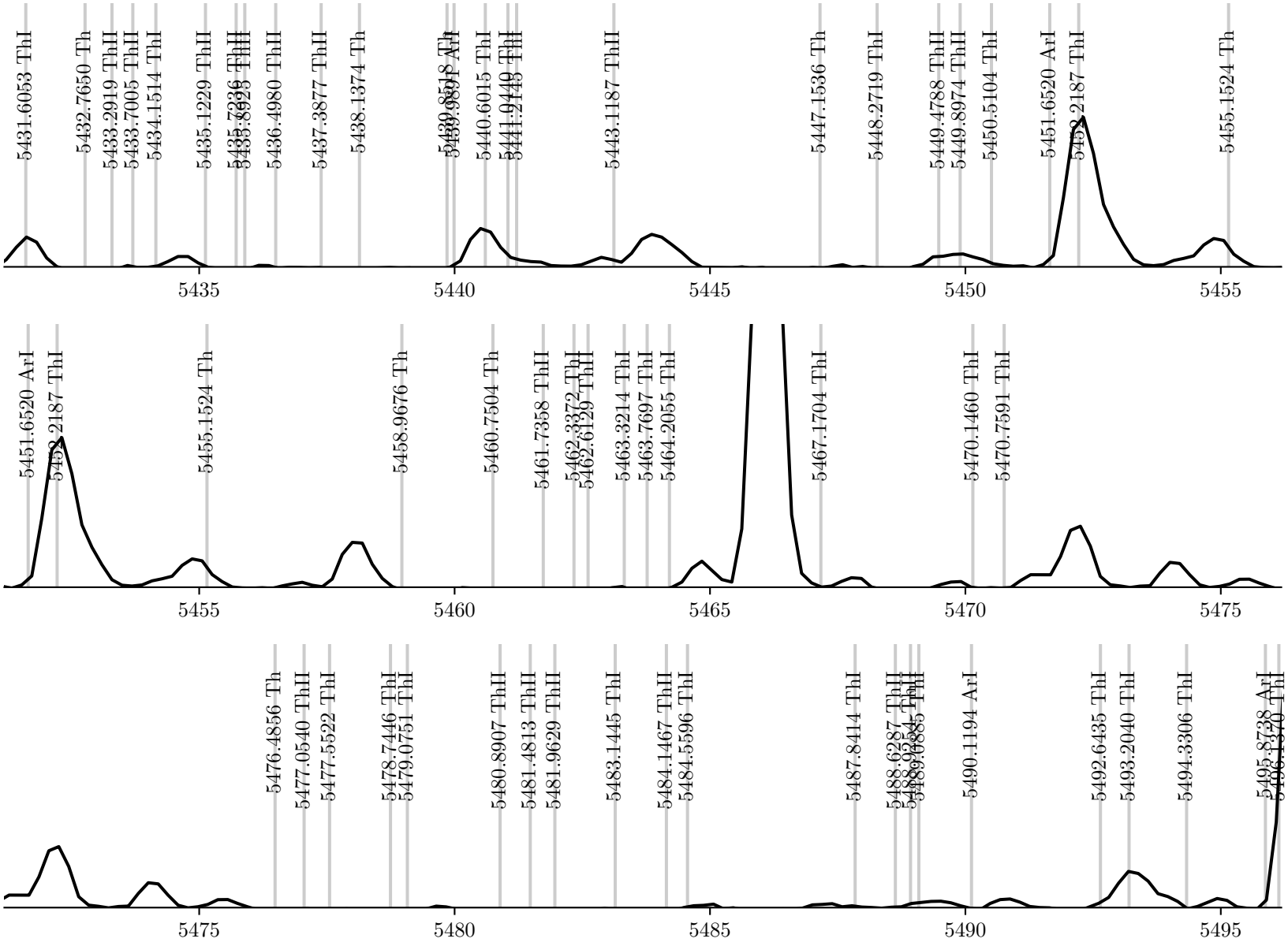
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5449.8974 ThII
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5452.2187 ThI
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5476.4856 Th
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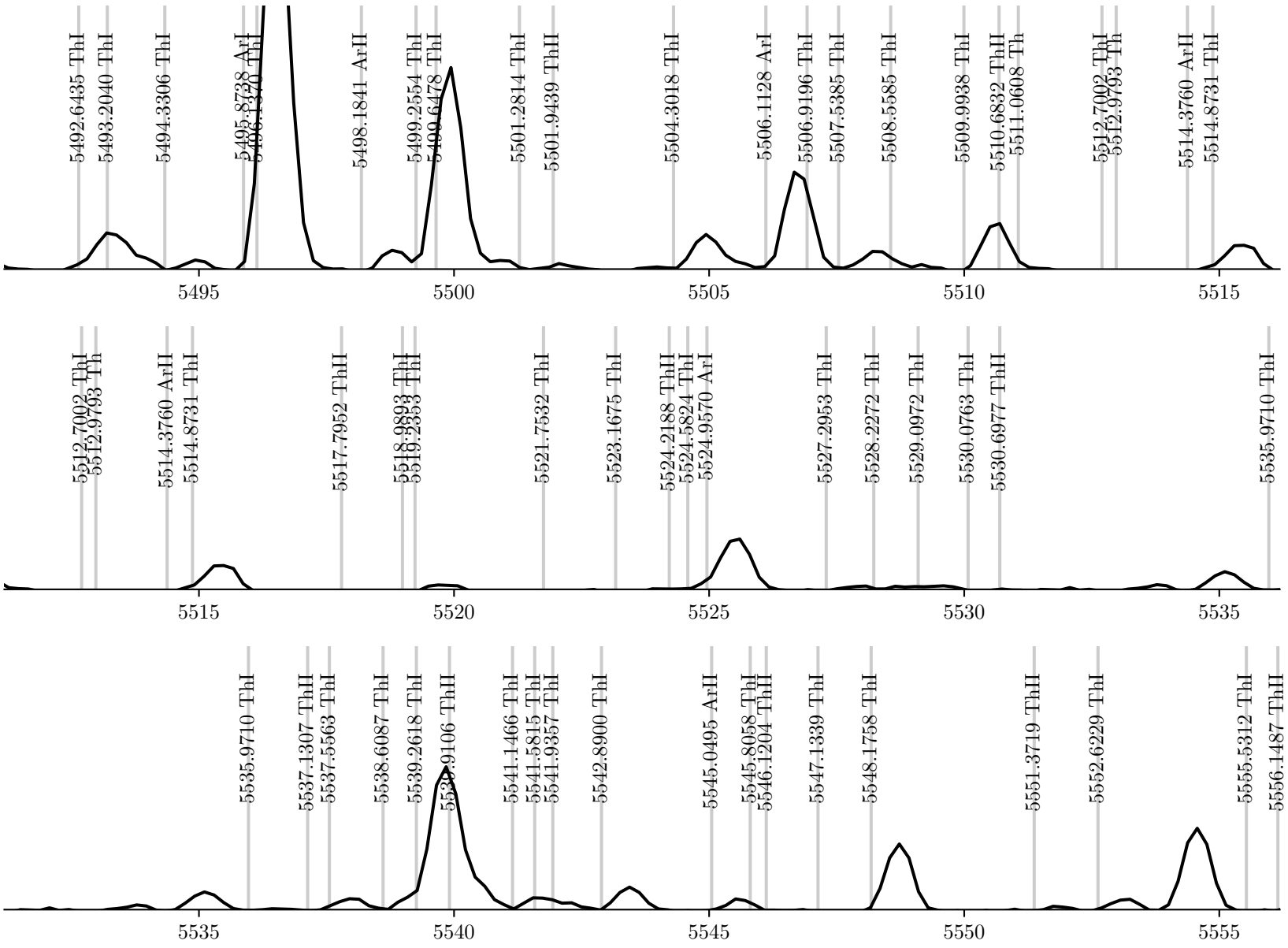
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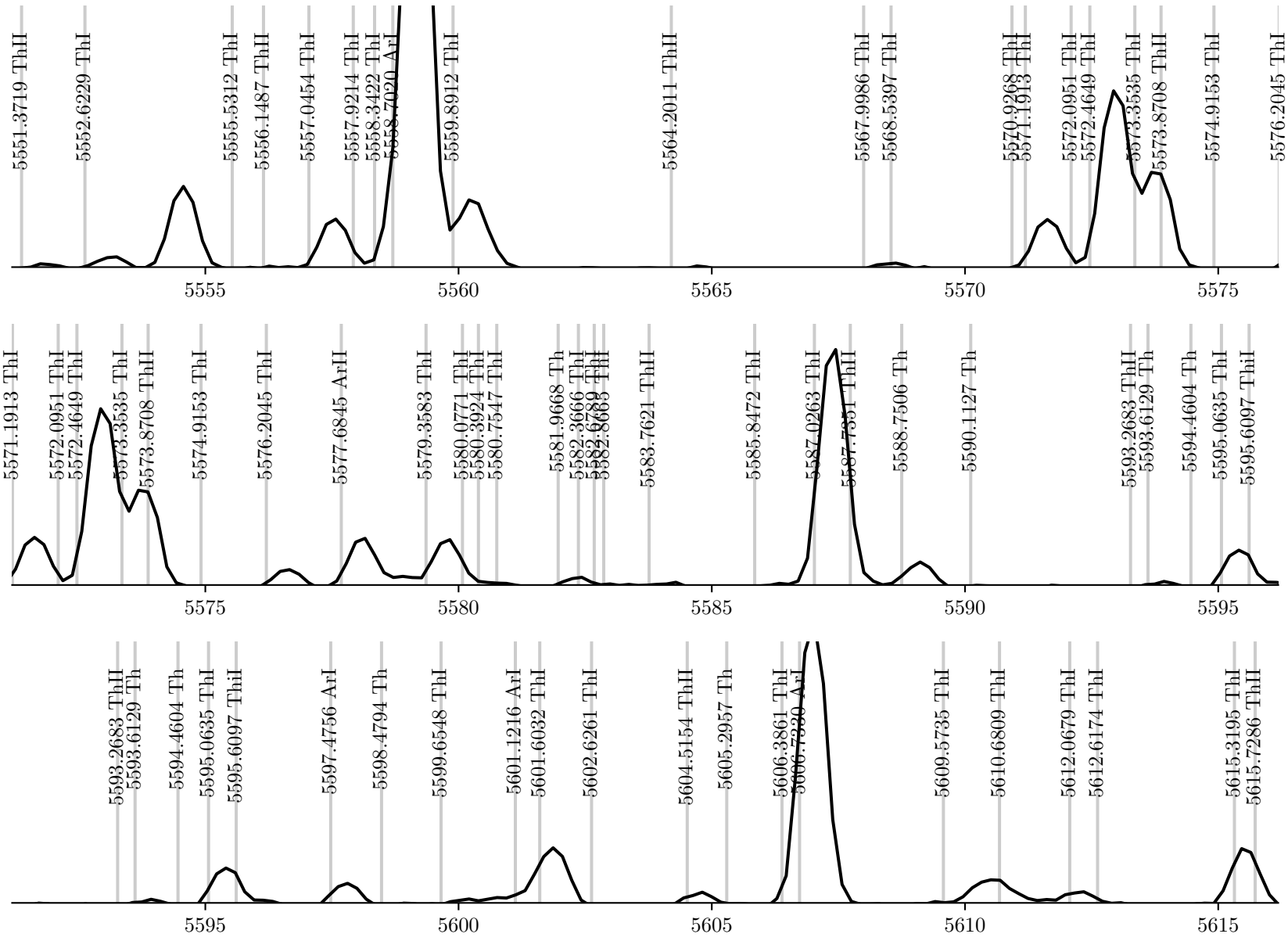
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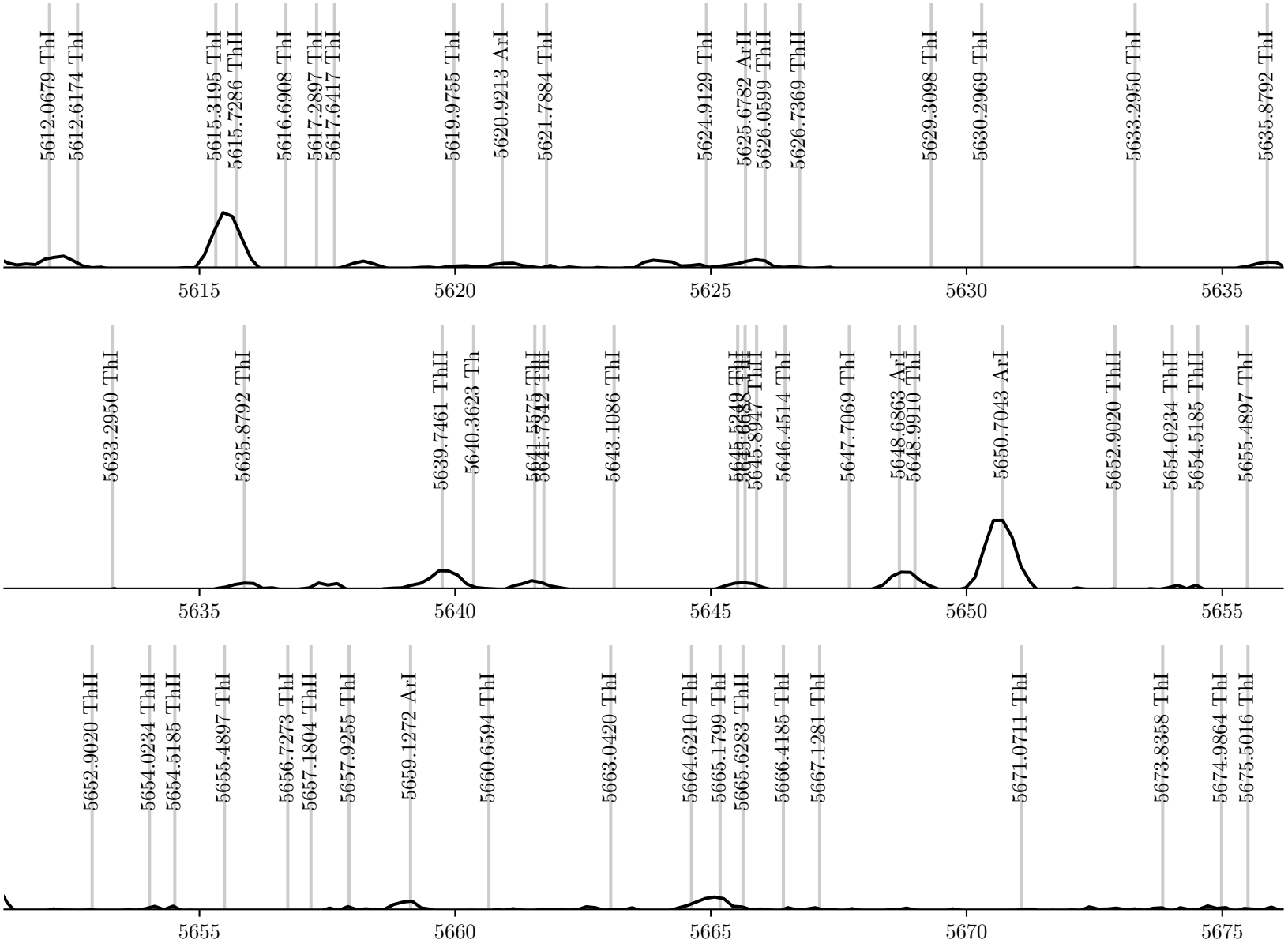
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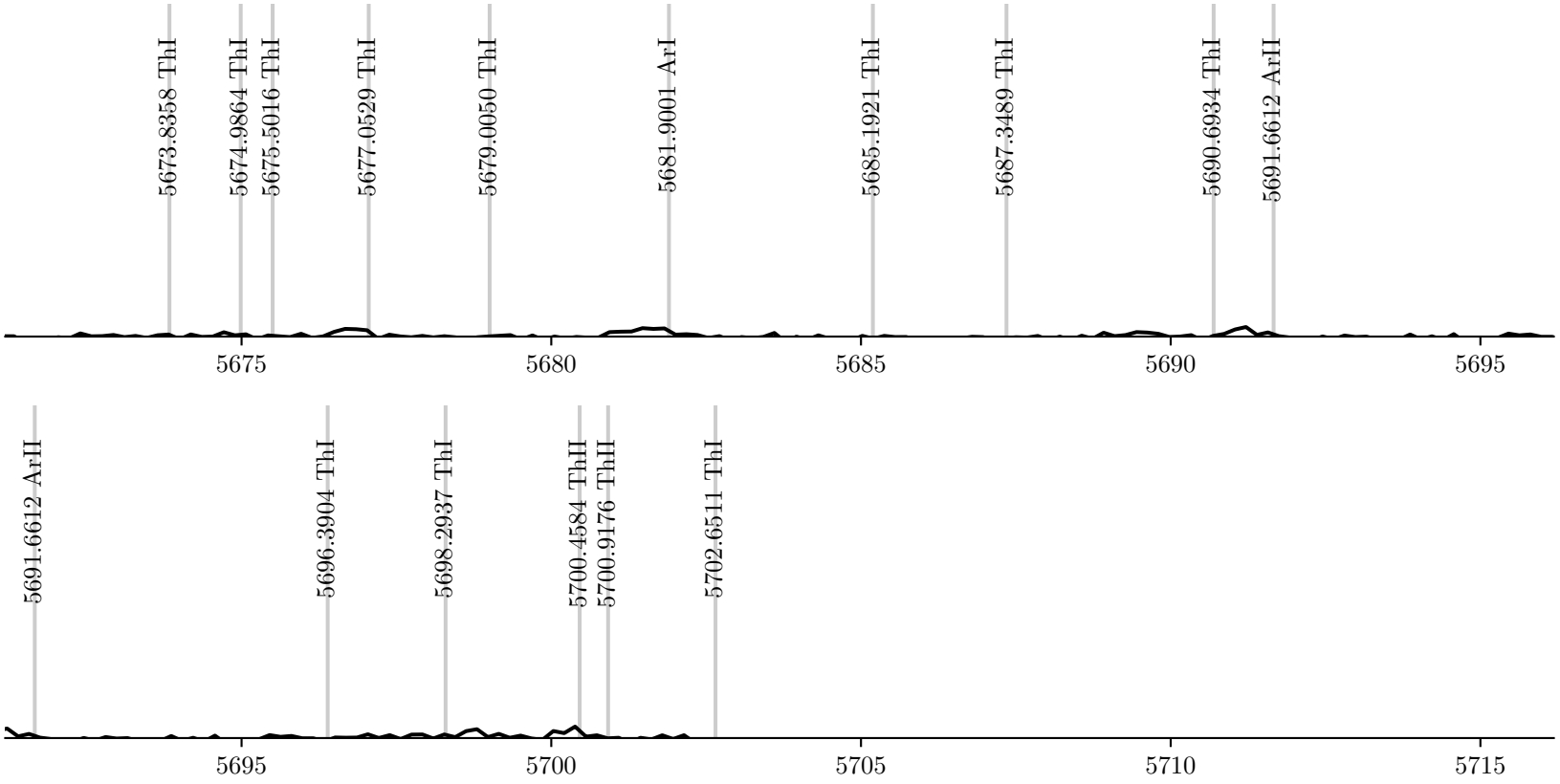
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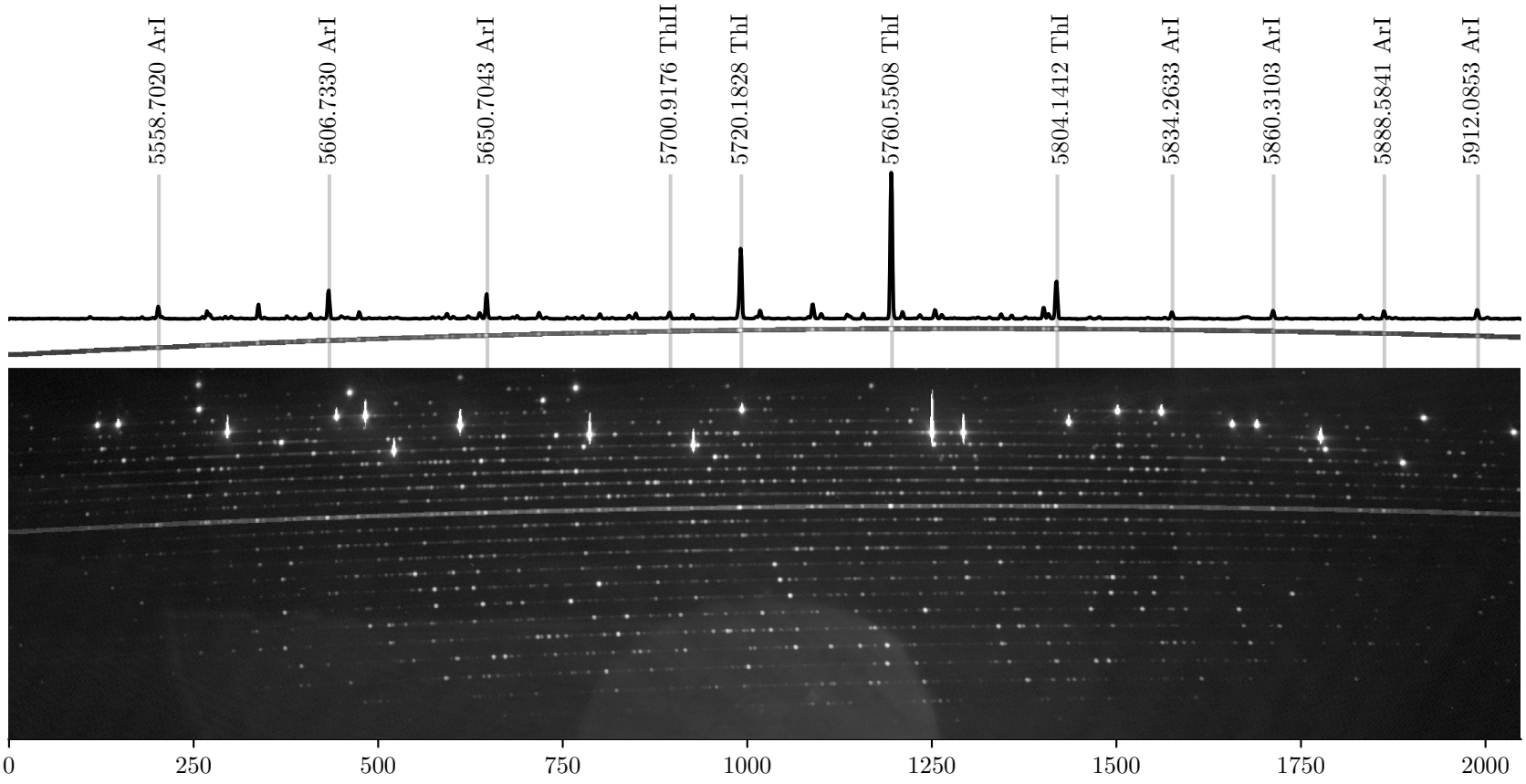
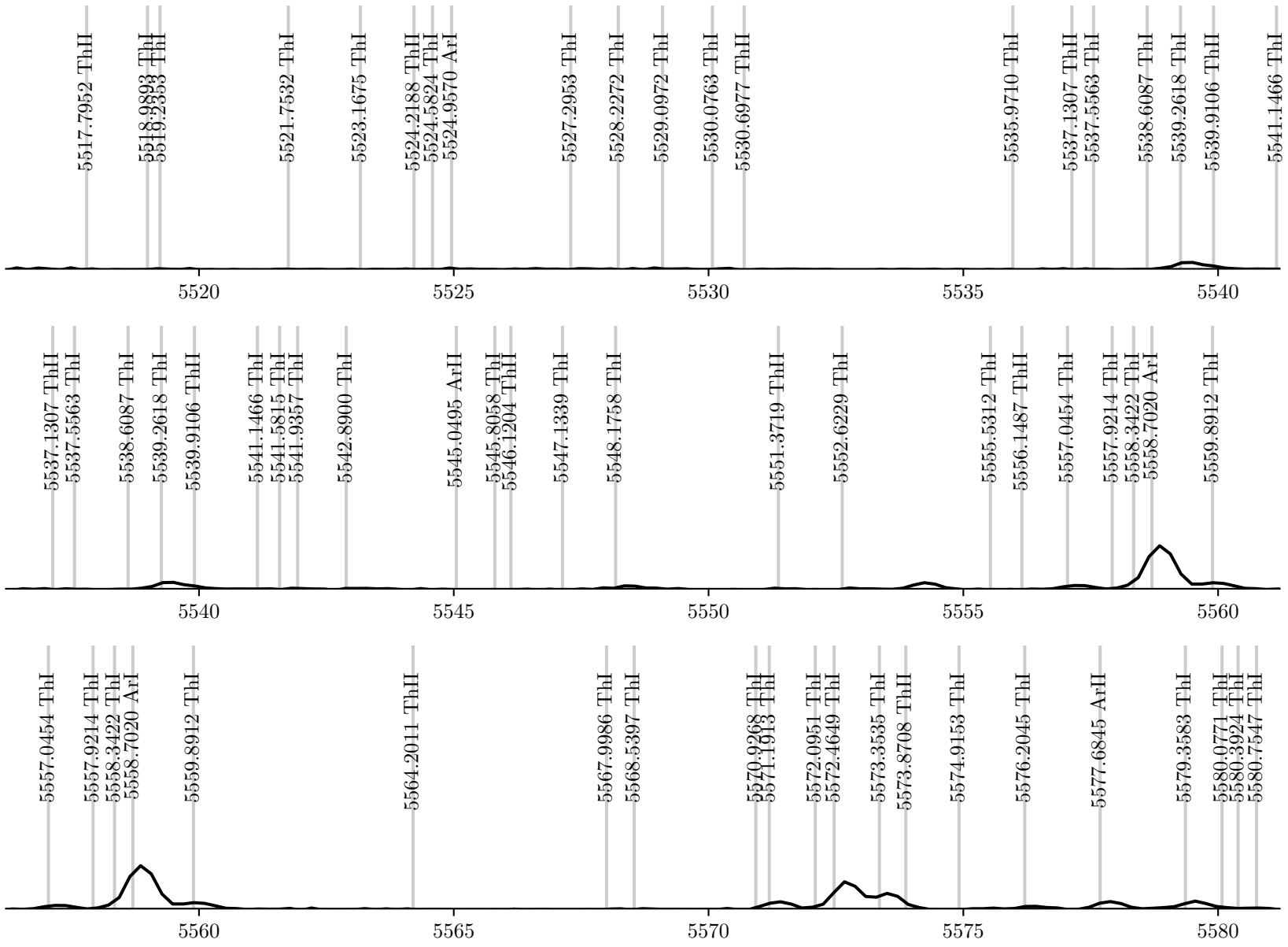
Order 26



Dispersion:	0.199 Å pix^{-1}
Central wavelength:	5727 Å
Wavelength range:	5516 Å – 5923 Å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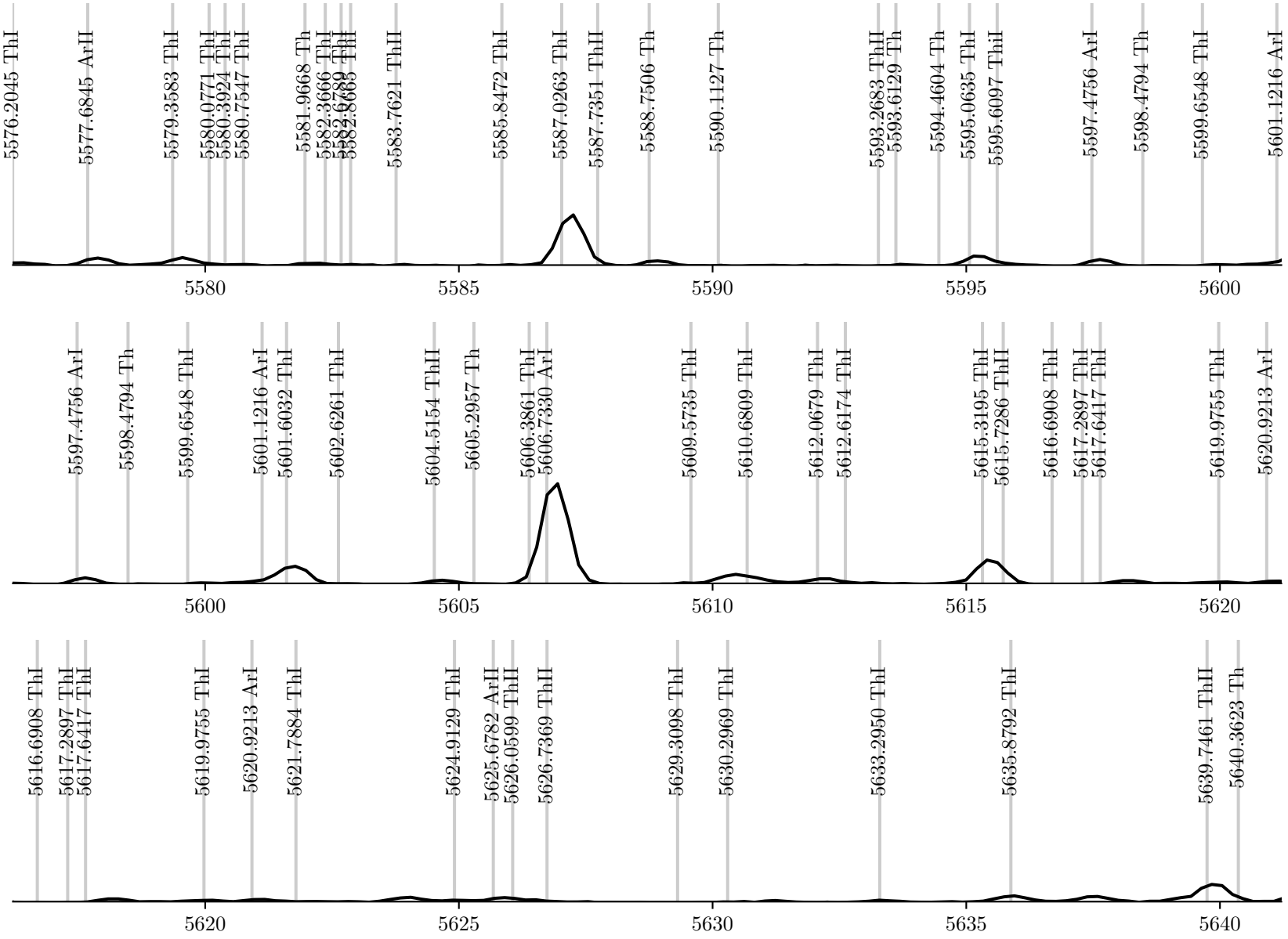
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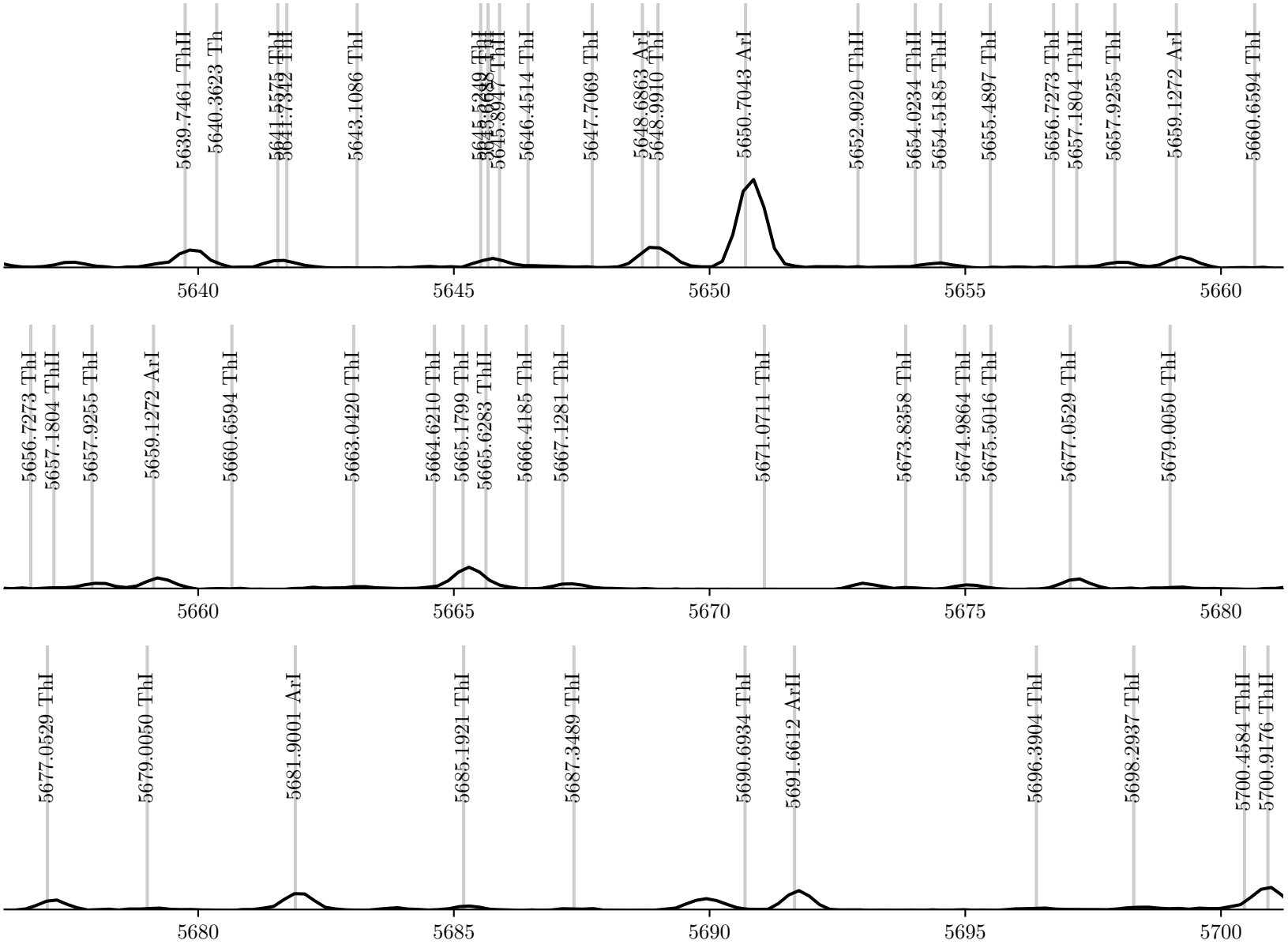
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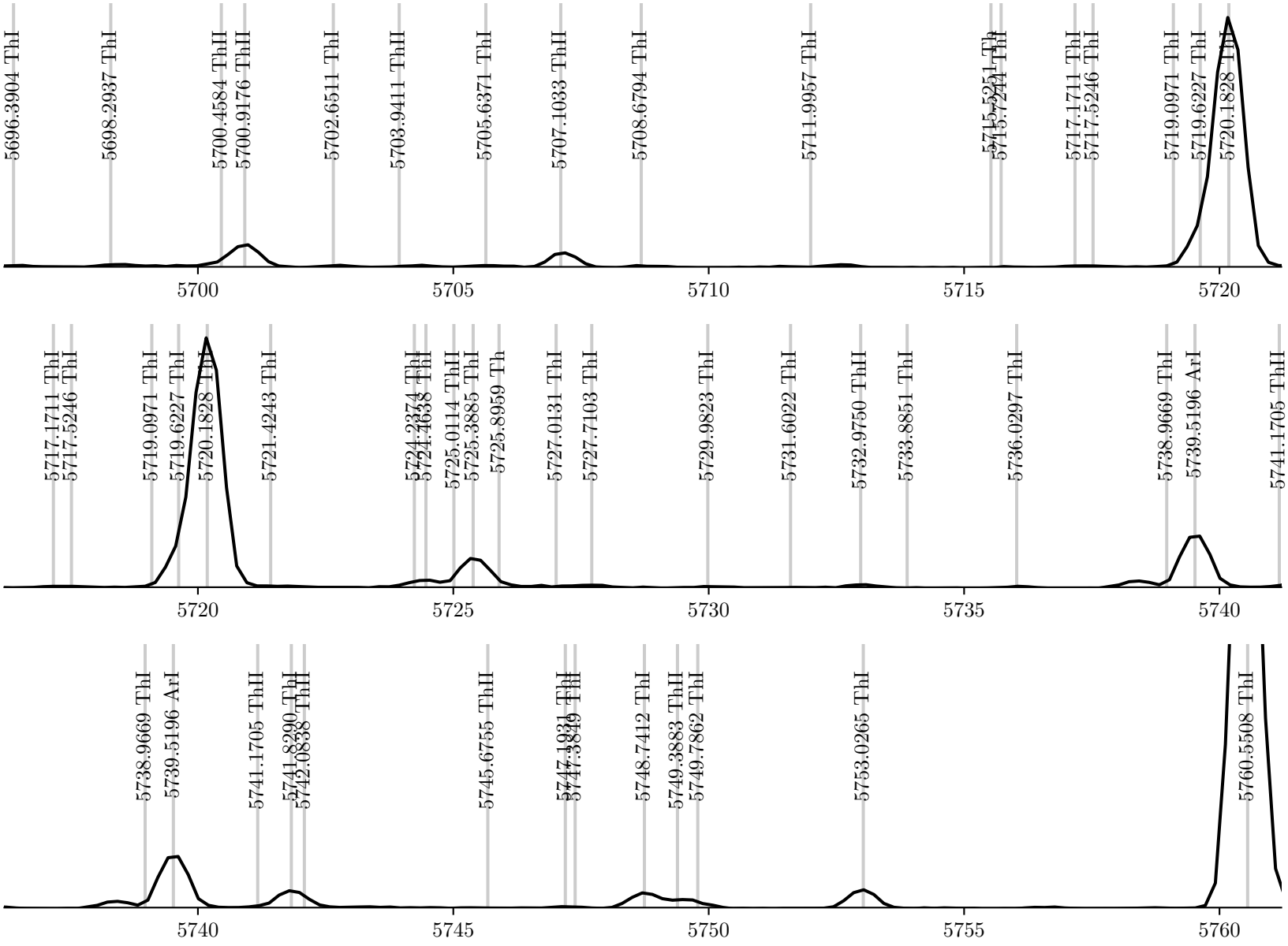
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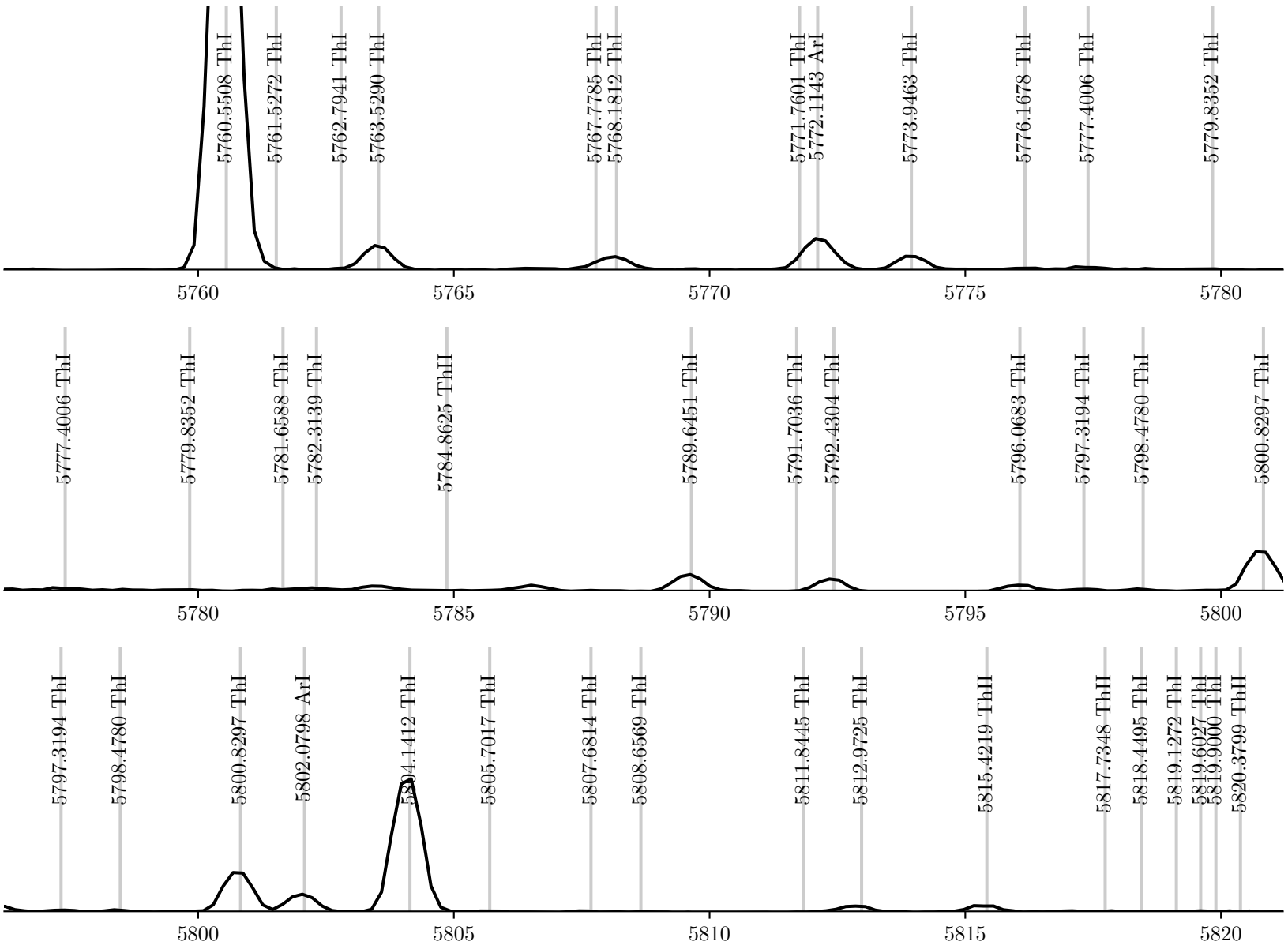
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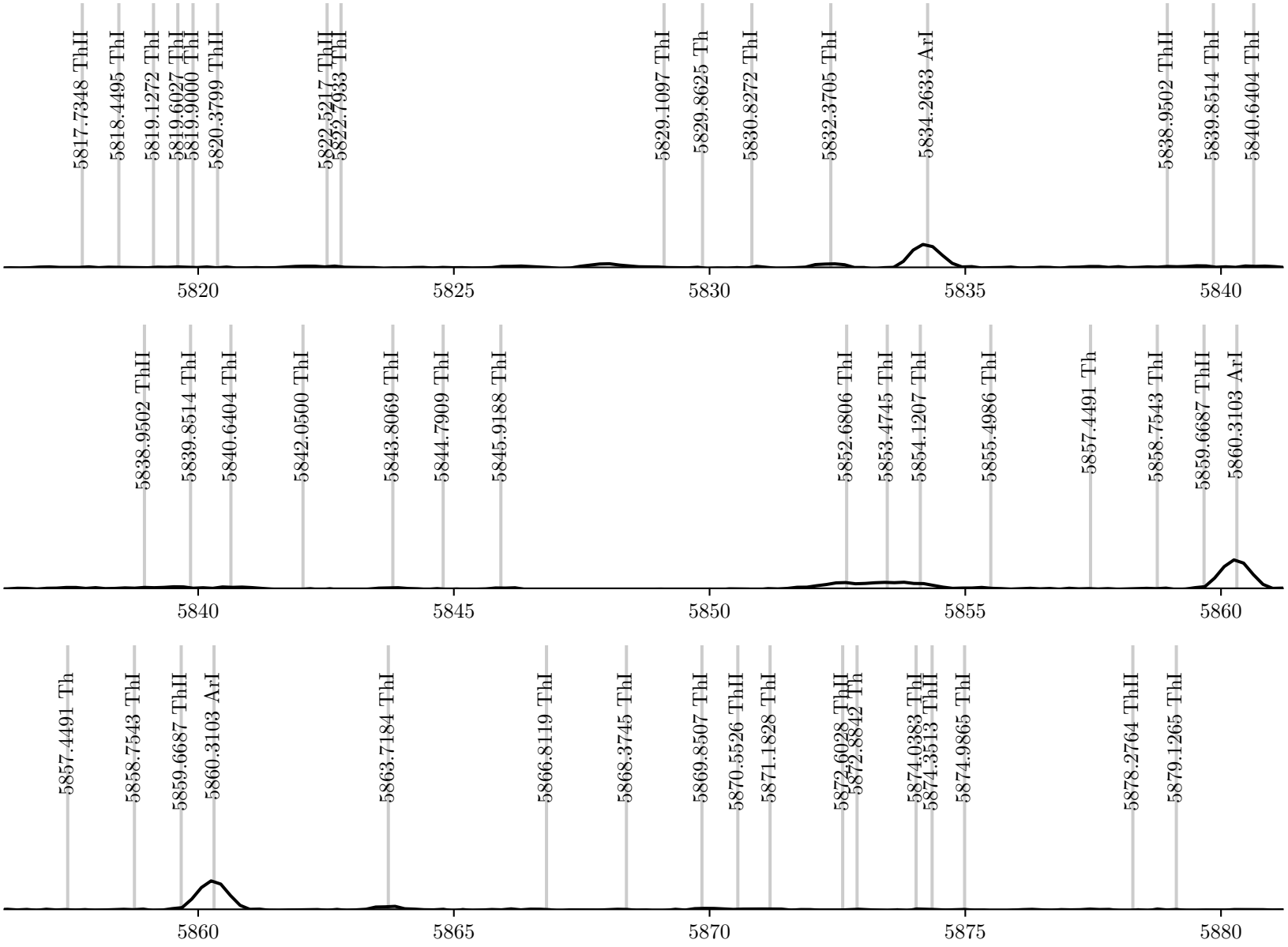
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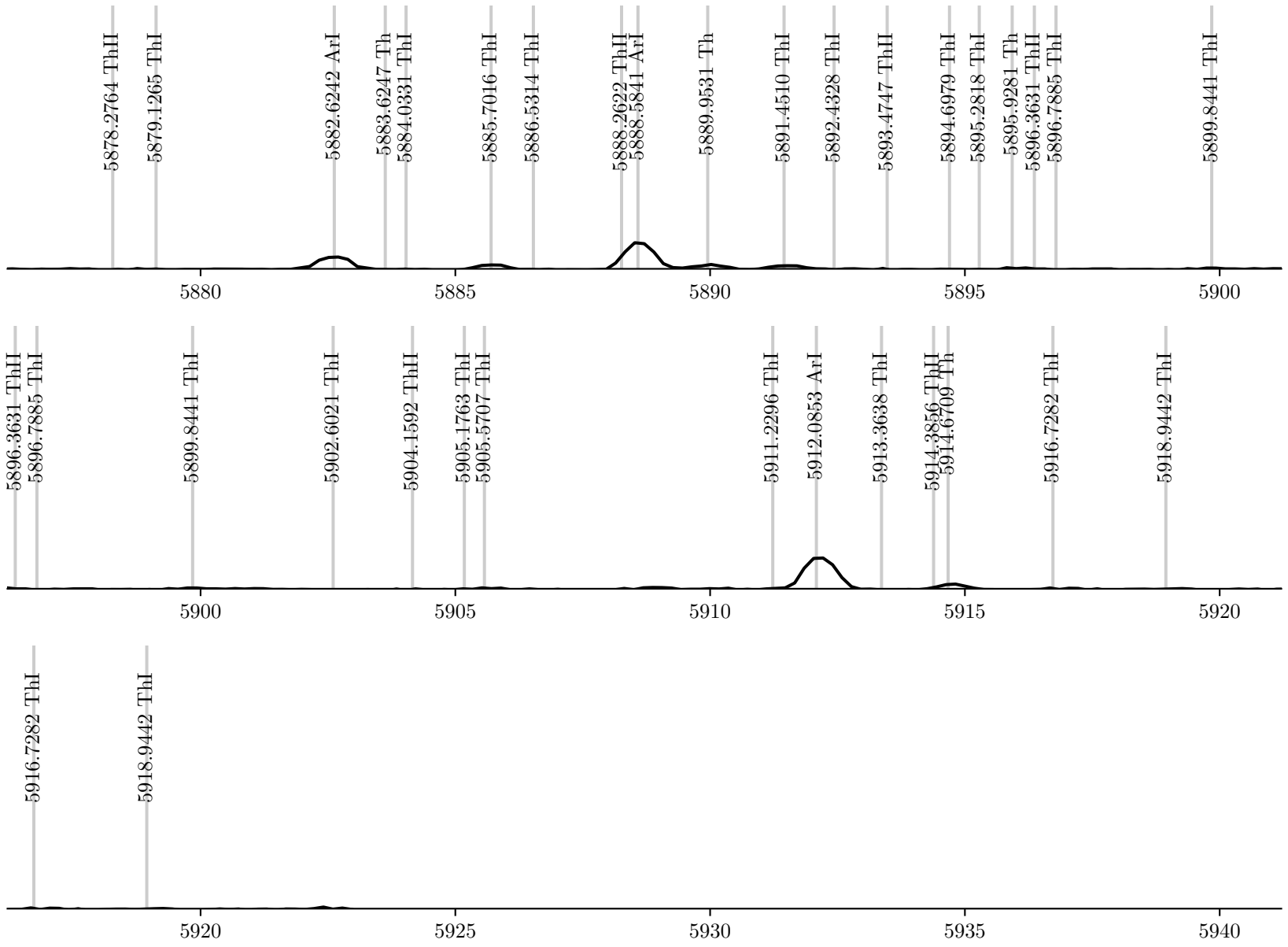
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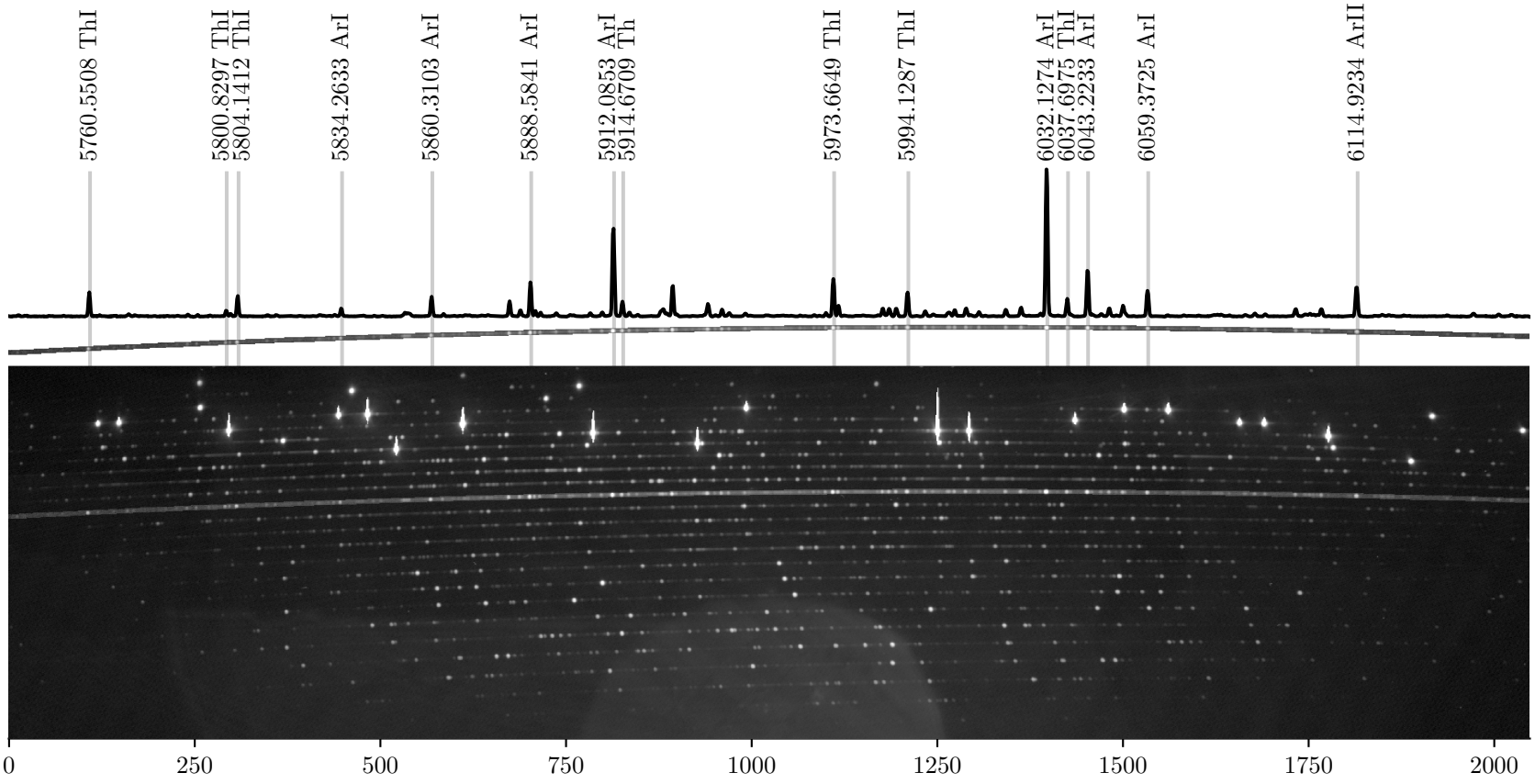

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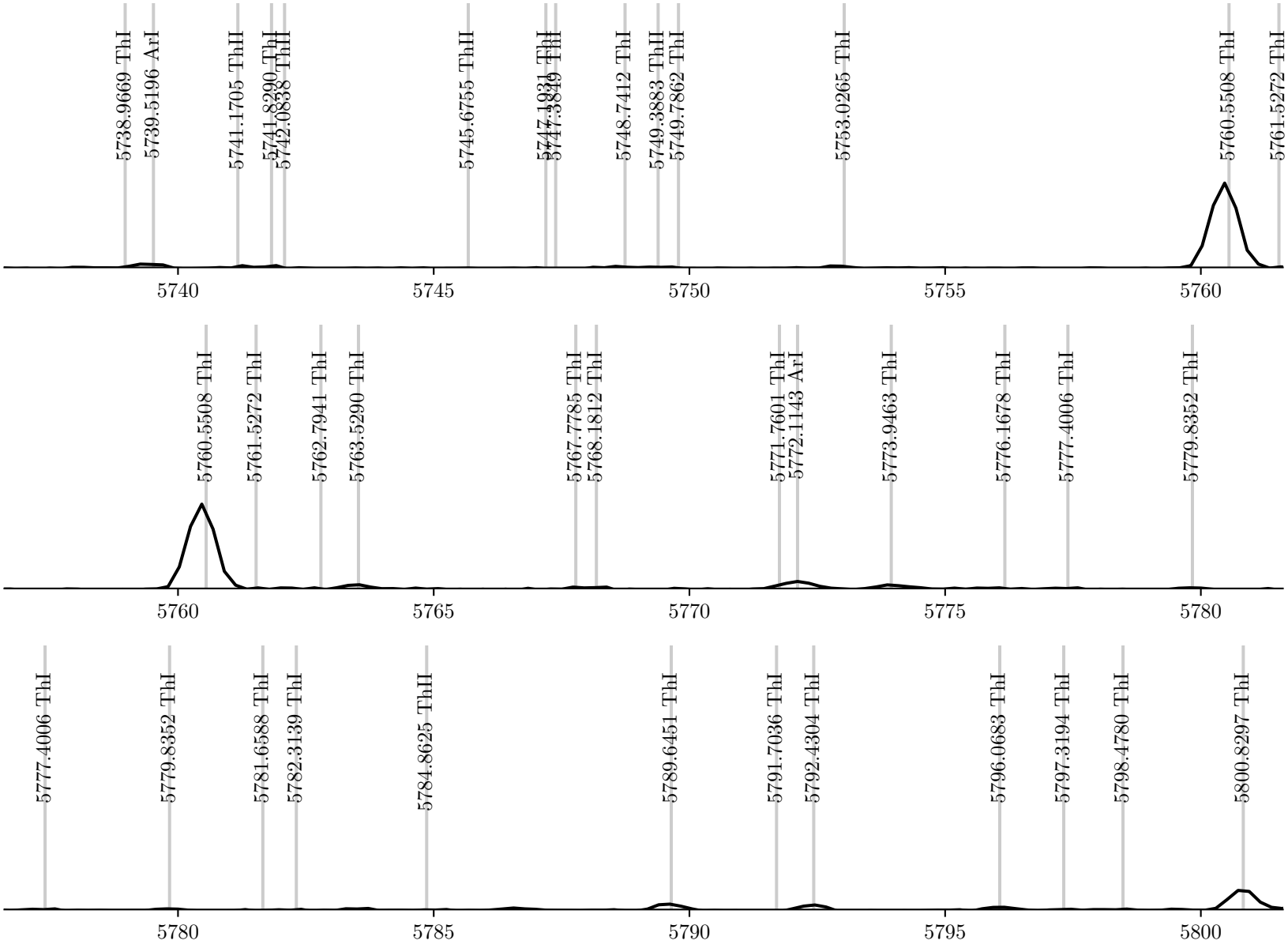
Central wavelength:	5956 Å
Wavelength range:	5737 Å – 6160 Å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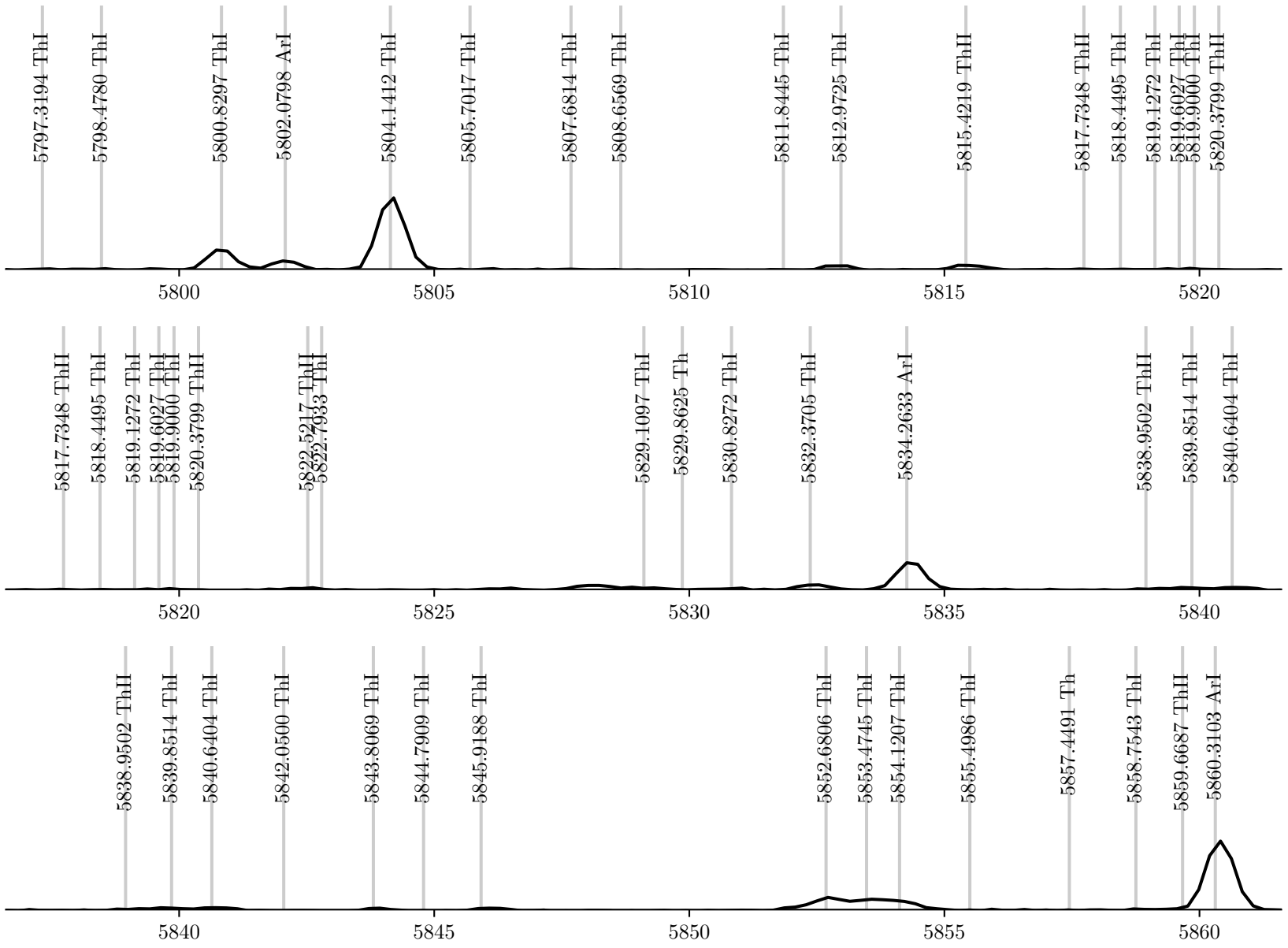
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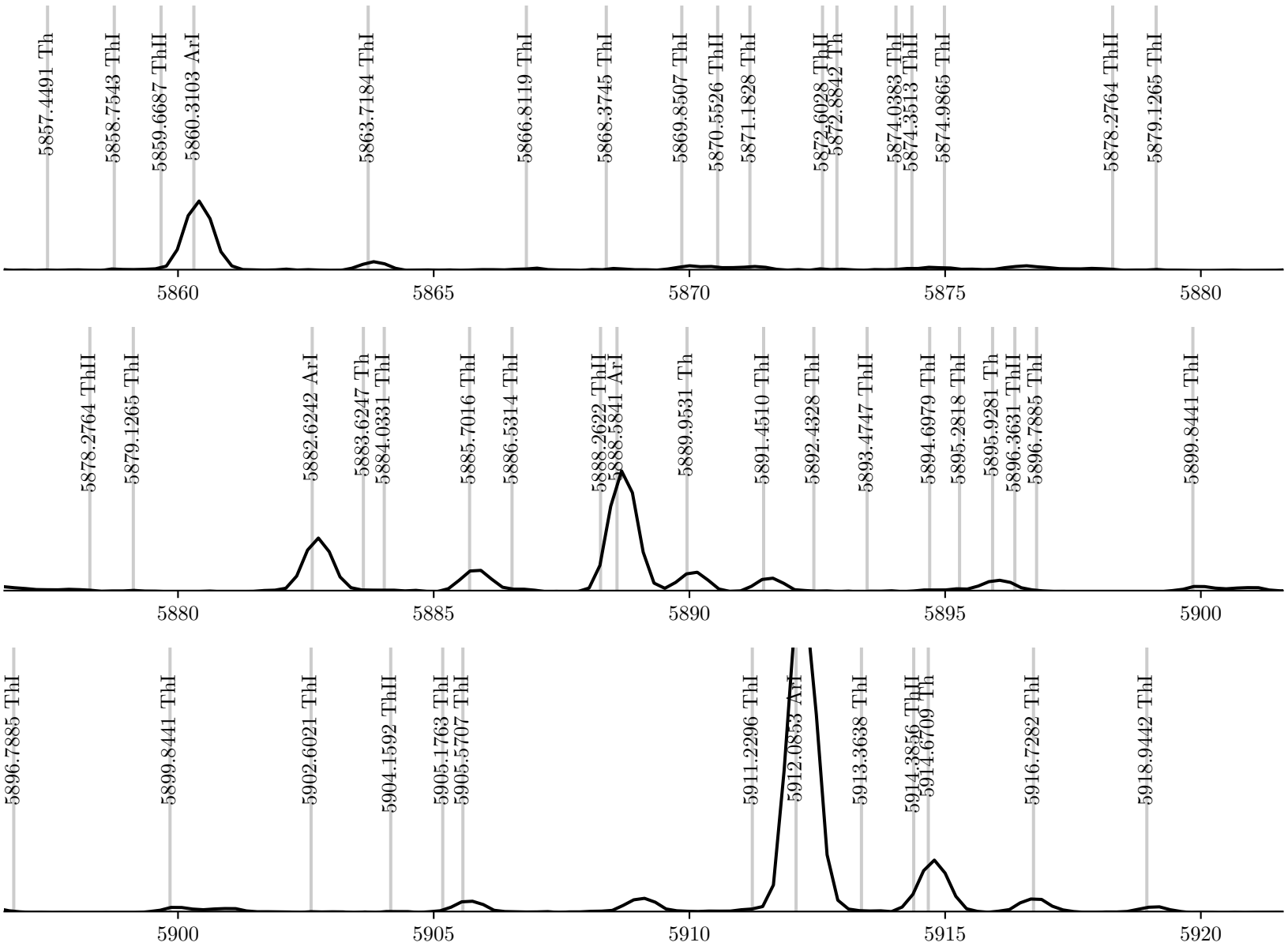
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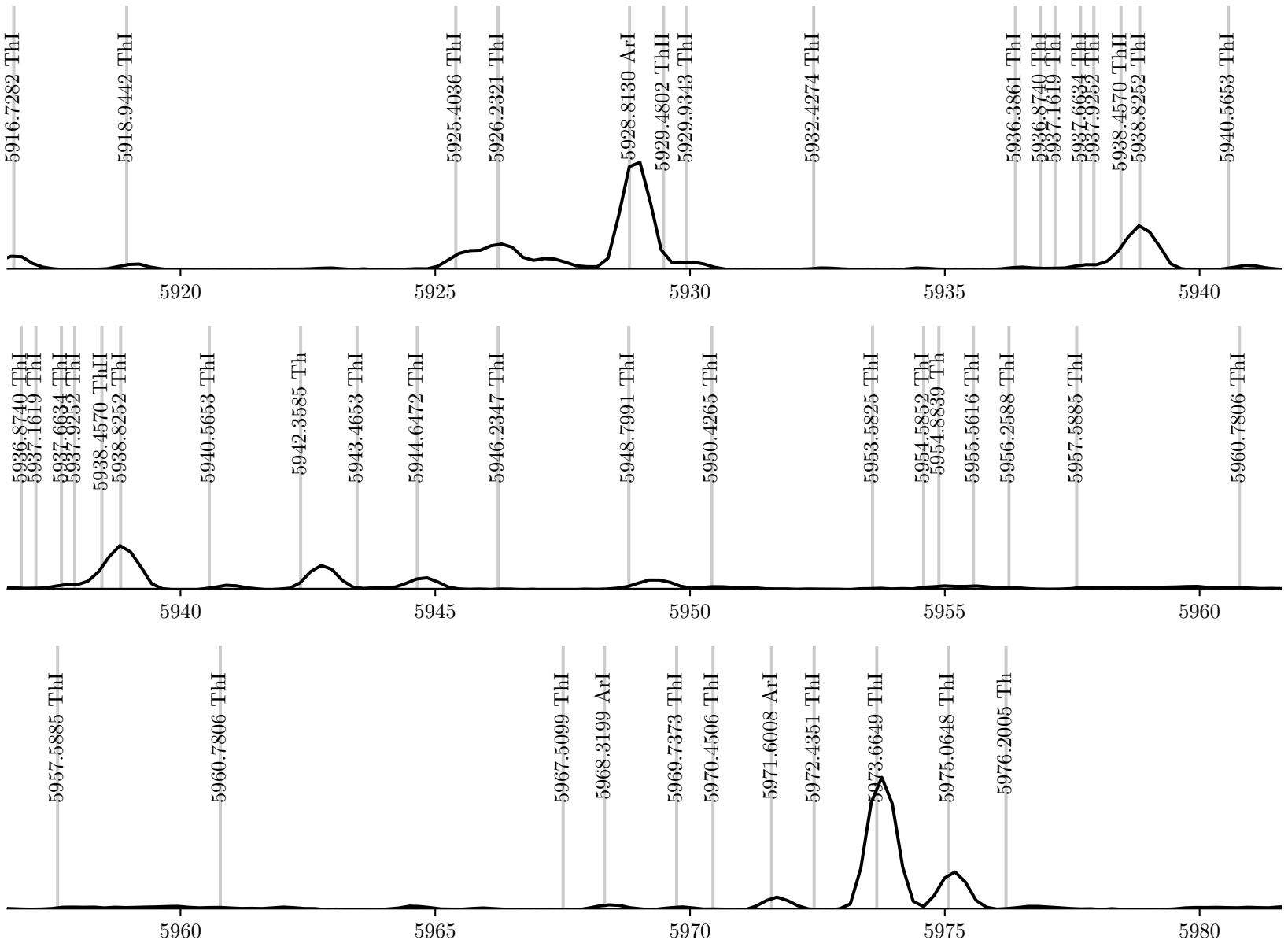
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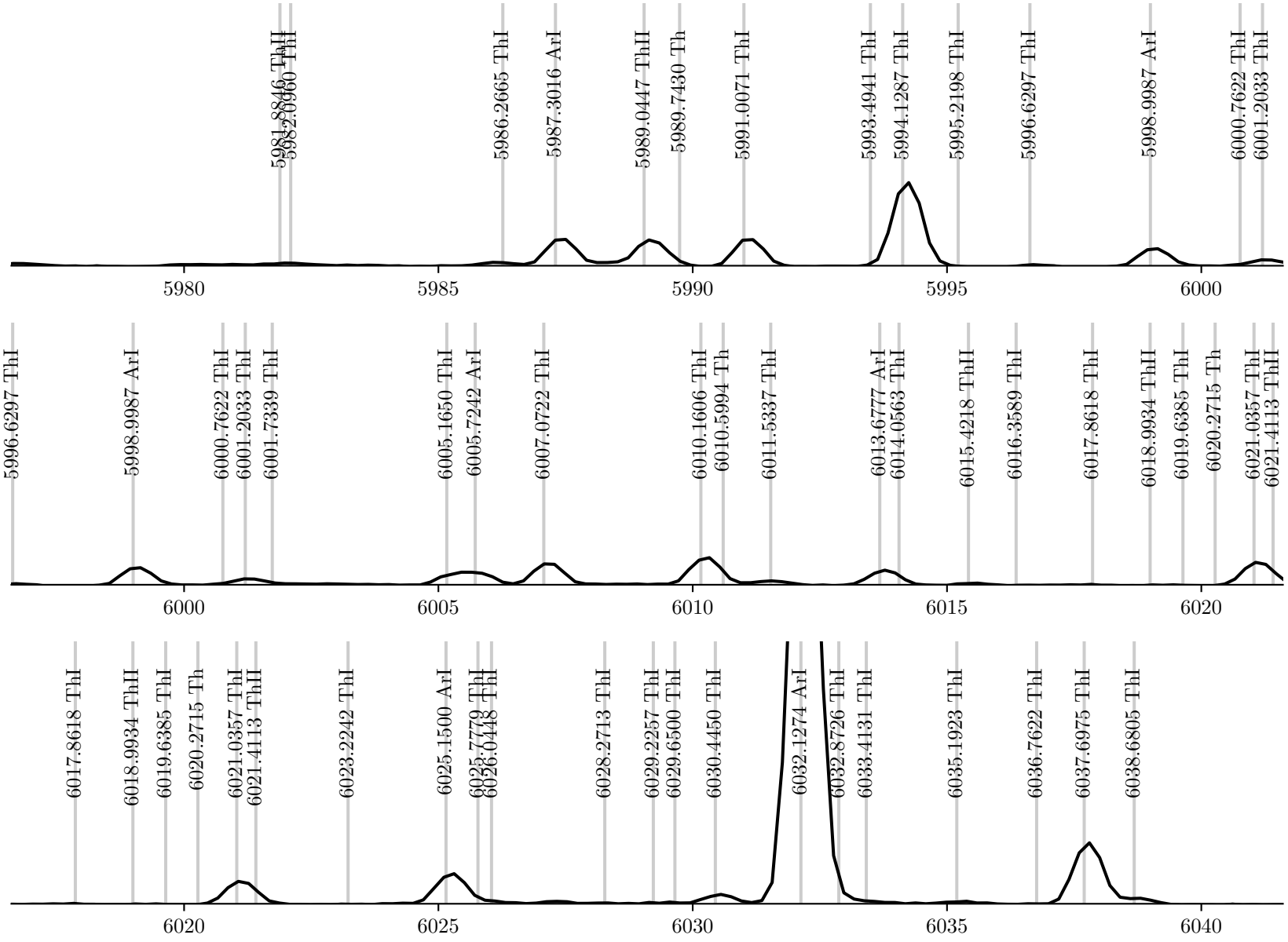
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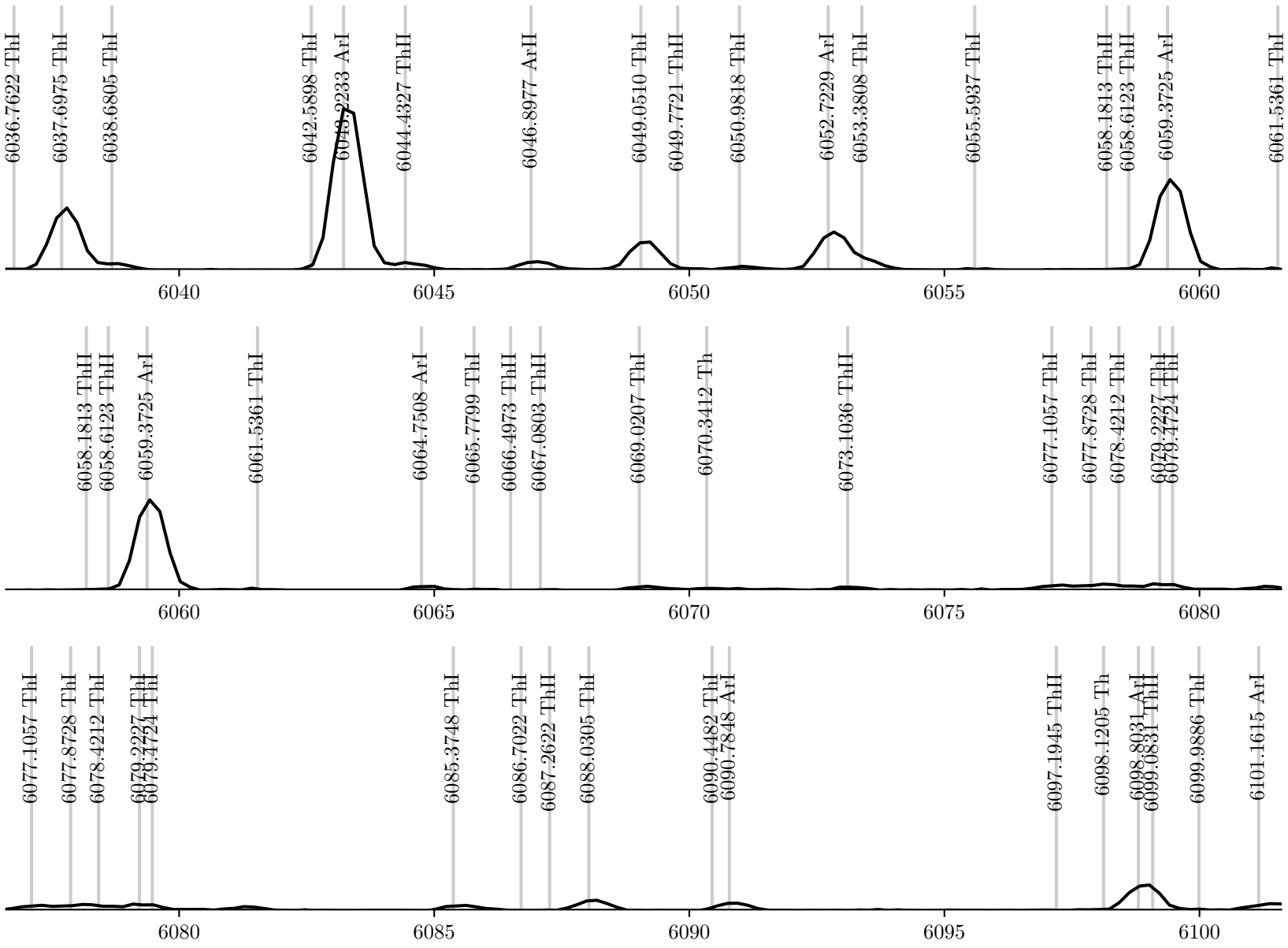
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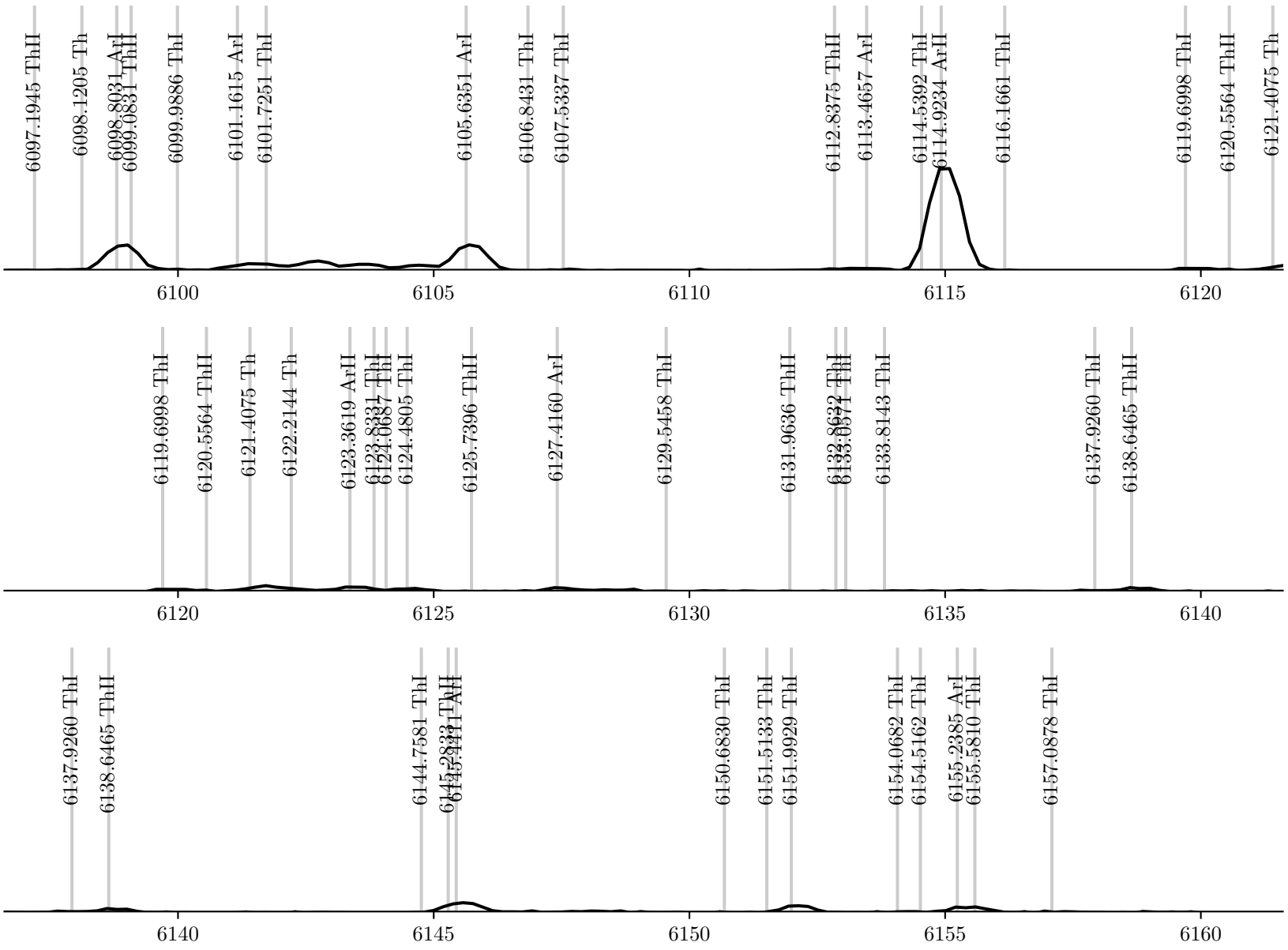
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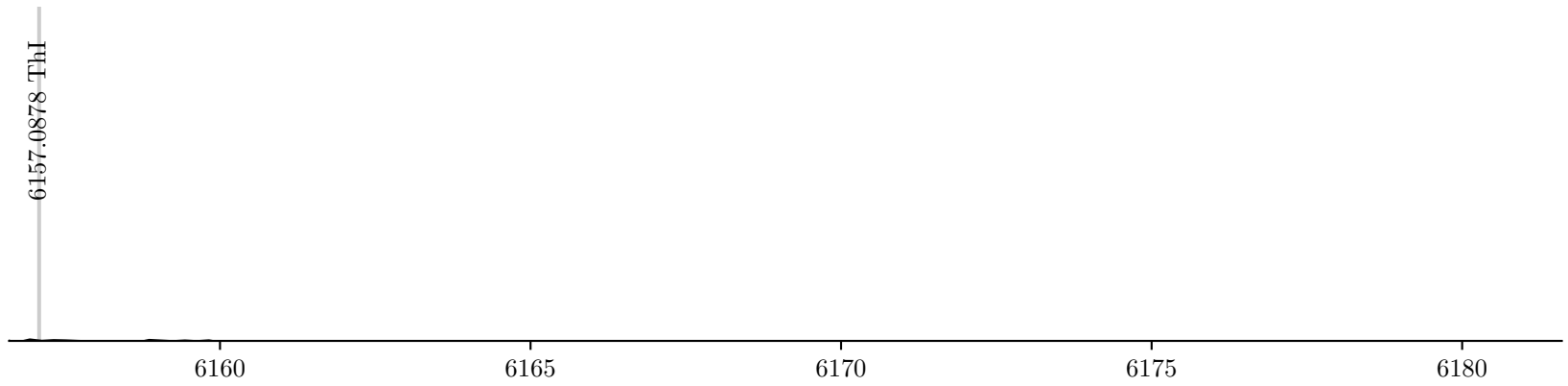
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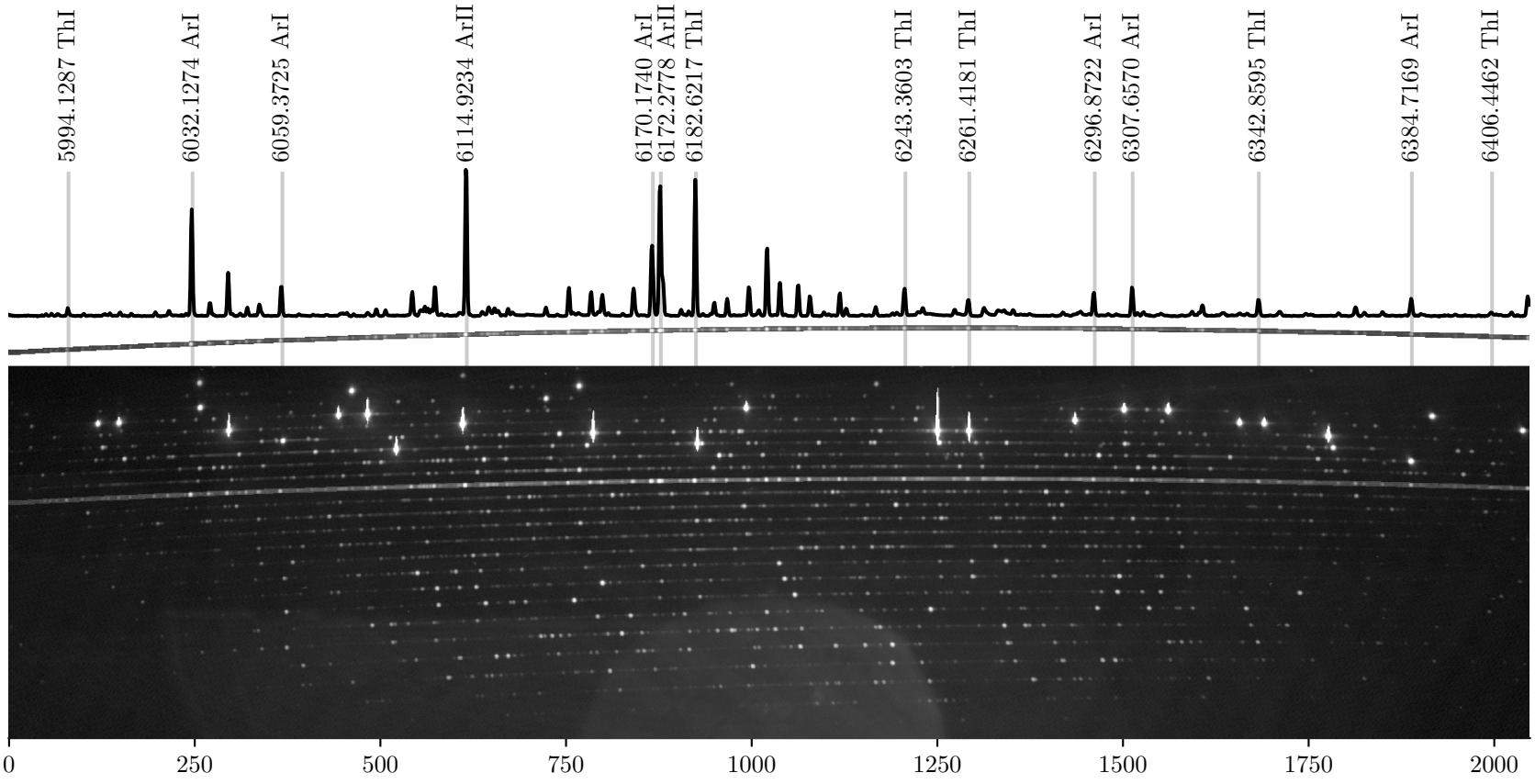
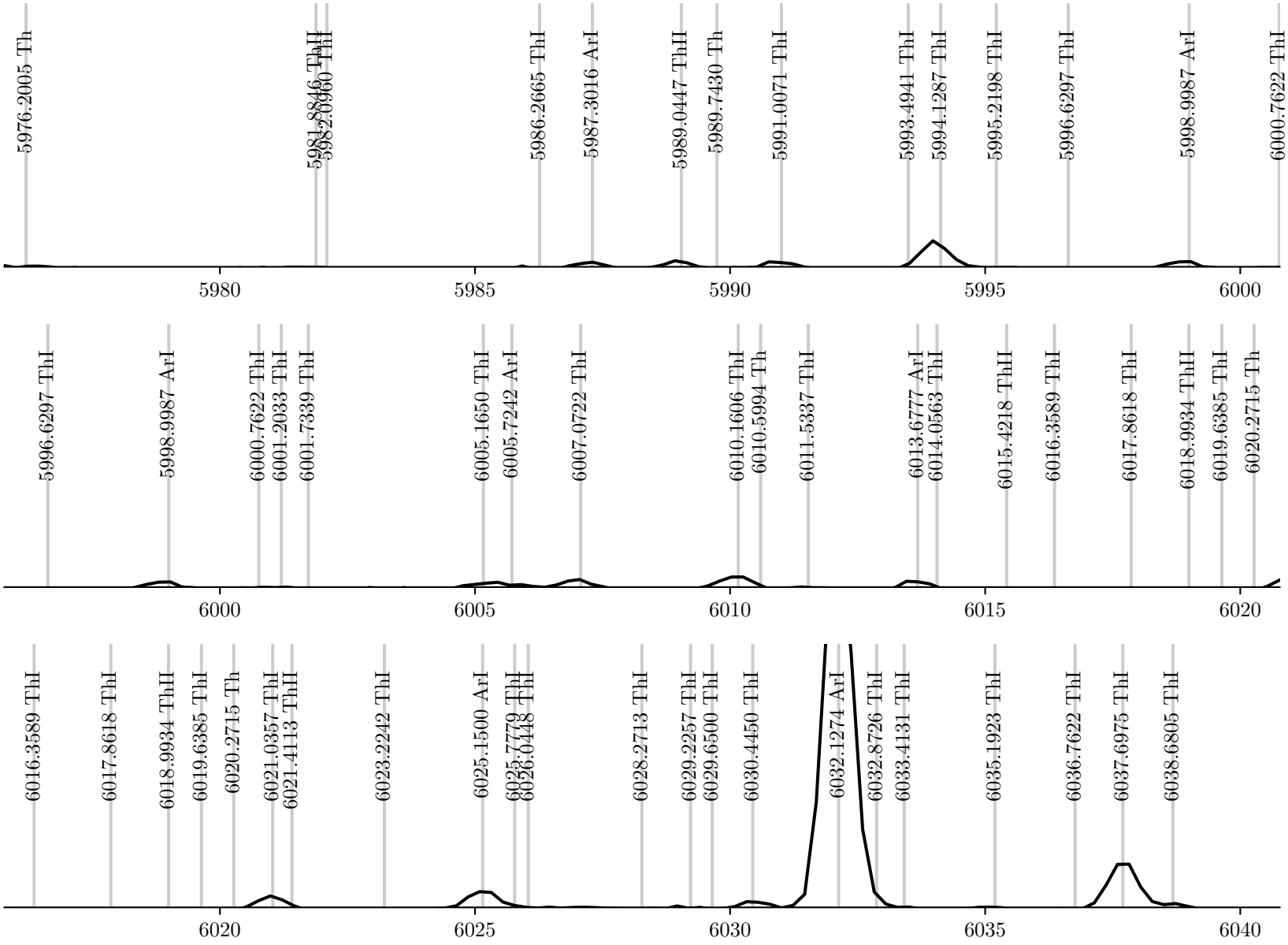
Order 24



Dispersion:	0.215 Å pix^{-1}
Central wavelength:	6204 Å
Wavelength range:	5976 Å – 6417 Å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5986.2665 ThI
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5989.0447 ThII
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5993.4941 ThI
5994.1287 ThI
5995.2198 ThI
5996.6297 ThI
5998.9987 ArI
6000.7622 ThI
5980
5985
5990
5995
6000
5996.6297 ThI
5998.9987 ArI
6000.7622 ThI
6001.2033 ThI
6001.7339 ThI
6005.1650 ThI
6005.7242 ArI
6007.0722 ThI
6010.1606 ThI
6010.5994 Th
6011.5337 ThI
6013.6777 ArI
6014.0563 ThI
6015.4218 ThII
6016.3589 ThI
6017.8618 ThI
6018.9934 ThII
6019.6385 ThI
6020.2715 Th
6000
6005
6010
6015
6020
6016.3589 ThI
6017.8618 ThI
6018.9934 ThII
6019.6385 ThI
6020.2715 Th
6021.0357 ThI
6021.4113 ThII
6023.2242 ThI
6025.1500 ArI
6025.7779 ThI
6026.0448 ThI
6028.2713 ThI
6029.2257 ThI
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6030.4450 ThI
6032.1274 ArI
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6033.4131 ThI
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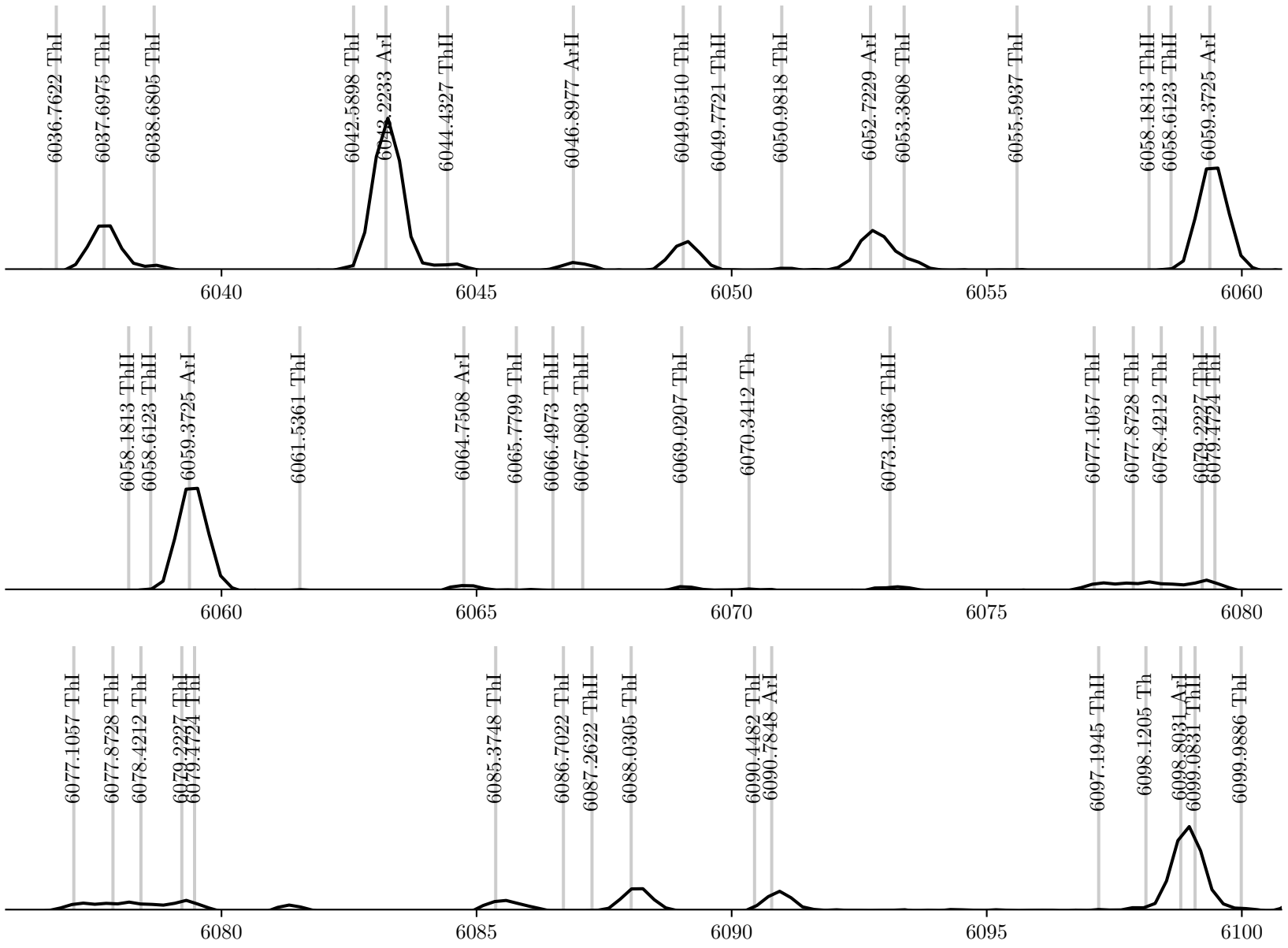
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6052.7229 ArI
6053.3808 ThI
6055.5937 ThI
6058.1813 ThII
6058.6123 ThII
6059.3725 ArI
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6059.3725 ArI
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6079.2227 ThI
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6085.3748 ThI
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6090.7848 ArI
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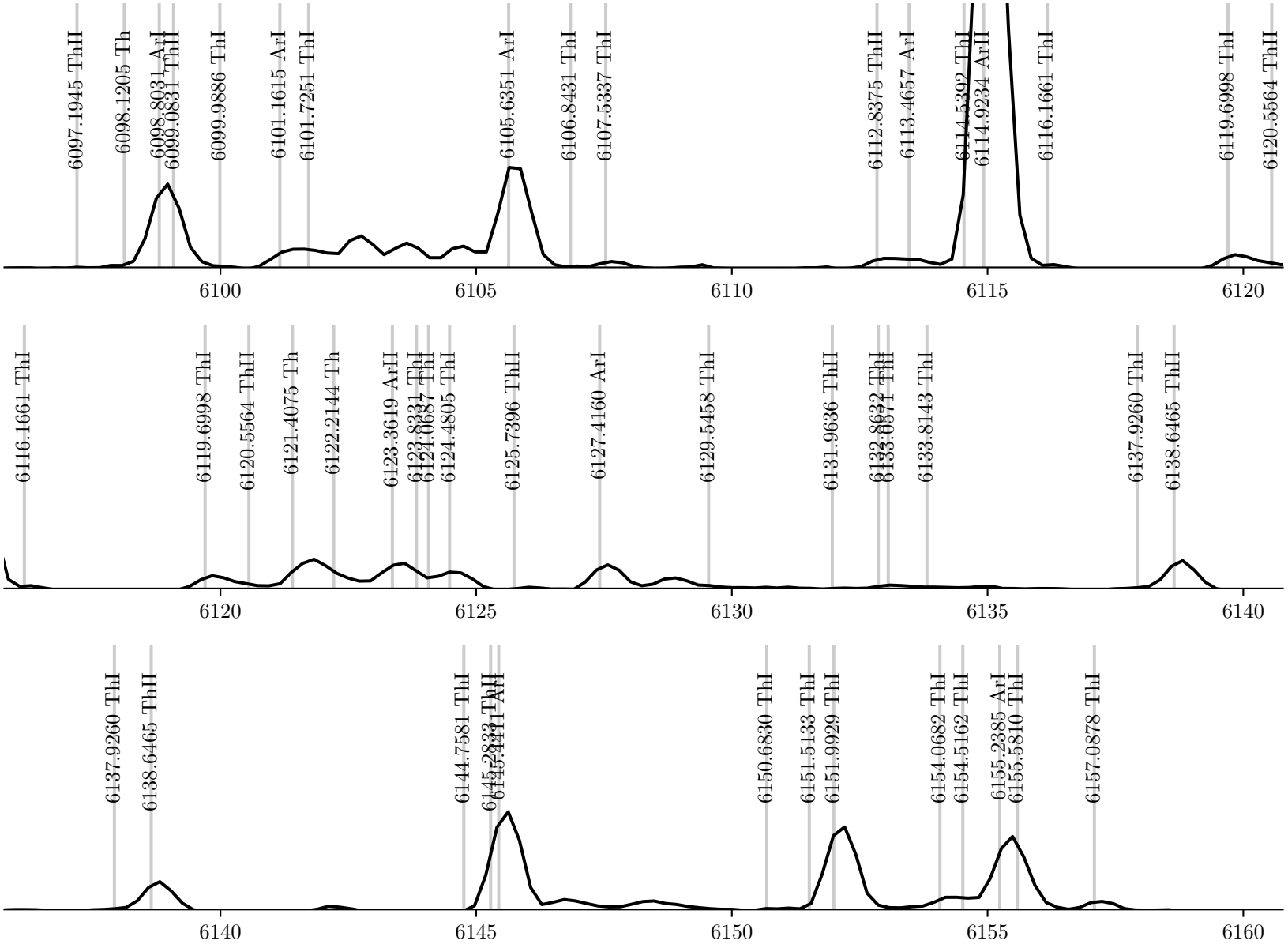
6100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6113.4657 ArI
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6122.2144 Th
6123.3619 ArII
6123.8331 ThI
6124.0687 ThI
6124.4805 ThI
6125.7396 ThII
6127.4160 ArI
6129.5458 ThI
6131.9636 ThII
6133.8143 ThI
6137.9260 ThI
6138.6465 ThII
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6125
6130
6135
6140
6137.9260 ThI
6138.6465 ThII
6144.7581 ThI
6150.6830 ThI
6151.5133 ThI
6151.9929 ThI
6154.0682 ThI
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6155.2385 ArI
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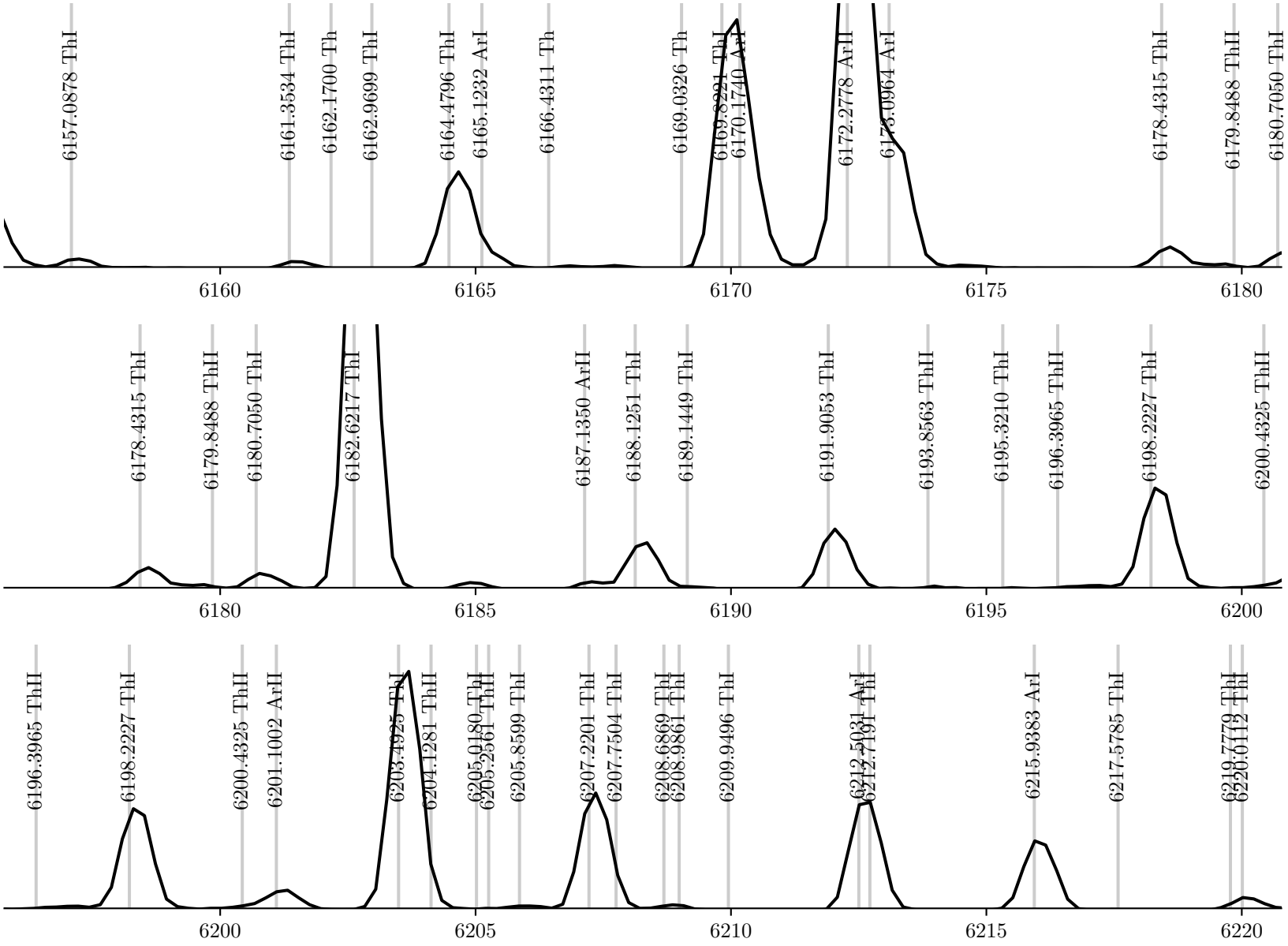
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6164.4796 ThI
6165.1232 ArI
6166.4311 Th
6169.0326 Th
6169.8221 ThI
6170.1740 ArI
6172.2778 ArII
6173.0964 ArI
6178.4315 ThI
6179.8488 ThII
6180.7050 ThI
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6180
6178.4315 ThI
6179.8488 ThII
6180.7050 ThI
6182.6217 ThI
6187.1350 ArII
6188.1251 ThI
6189.1449 ThI
6191.9053 ThI
6193.8563 ThII
6195.3210 ThI
6196.3965 ThII
6198.2227 ThI
6200.4325 ThII
6180
6185
6190
6195
6200
6196.3965 ThII
6198.2227 ThI
6200.4325 ThII
6201.1002 ArII
6203.4925 ThI
6204.1281 ThII
6205.0180 ThI
6205.2561 ThII
6205.8599 ThI
6207.2201 ThI
6207.7504 ThI
6208.6869 ThI
6208.9861 ThI
6209.9496 ThI
6212.5031 ArI
6212.7191 ThI
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6219.7779 ThI
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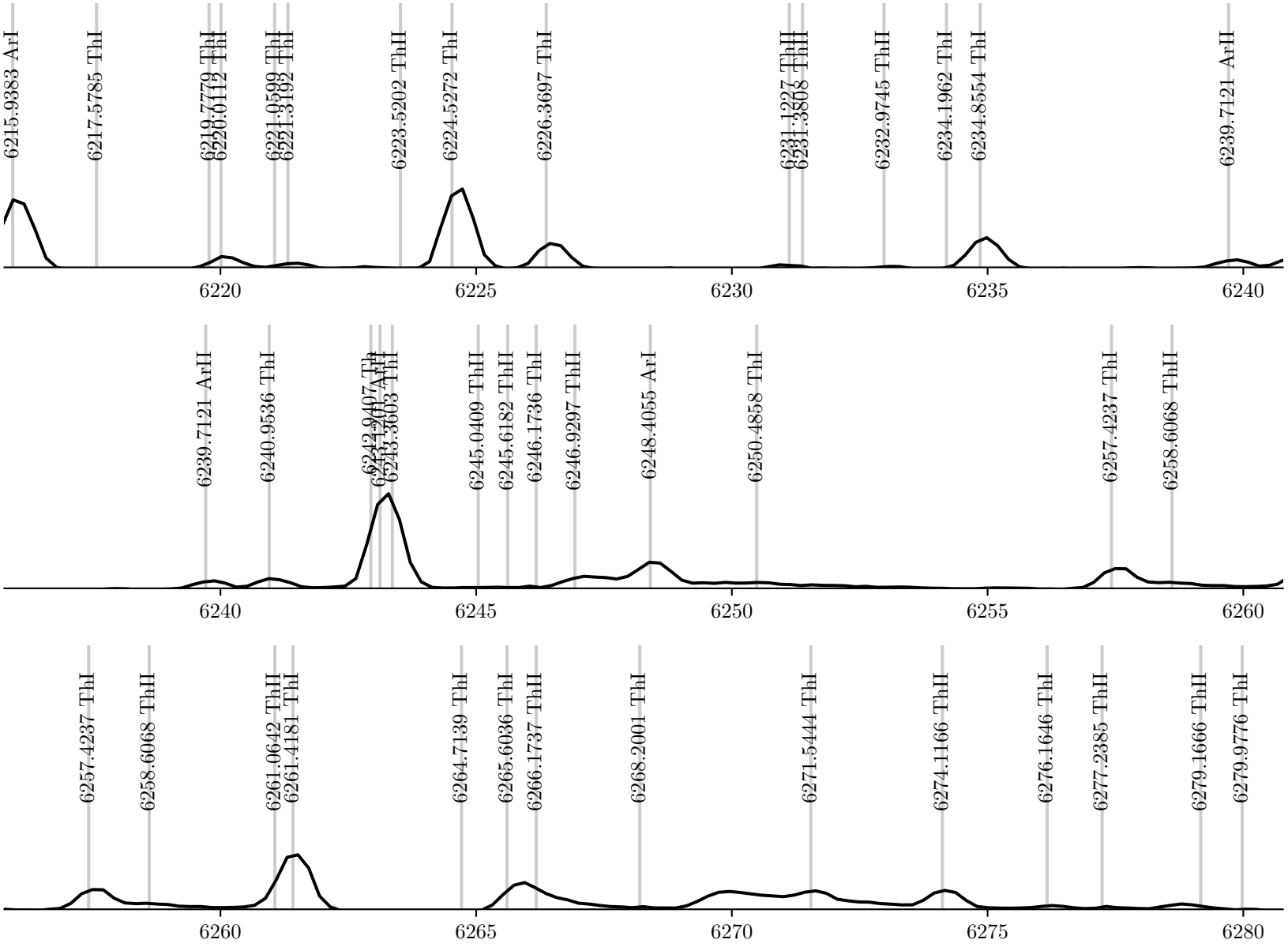
6215
6220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6224.5272 ThI
6226.3697 ThI
6231.1227 ThII
6231.3808 ThII
6232.9745 ThII
6234.1962 ThI
6234.8554 ThI
6239.7121 ArII
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6230
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6239.7121 ArII
6240.9536 ThI
6242.9407 ThI
6243.3603 ThI
6245.0409 ThII
6245.6182 ThII
6246.1736 ThI
6246.9297 ThII
6248.4055 ArI
6250.4858 ThI
6257.4237 ThI
6258.6068 ThII
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6257.4237 ThI
6258.6068 ThII
6261.0642 ThII
6261.4181 ThI
6264.7139 ThI
6265.6036 ThI
6266.1737 ThII
6268.2001 ThI
6271.5444 ThI
6274.1166 ThII
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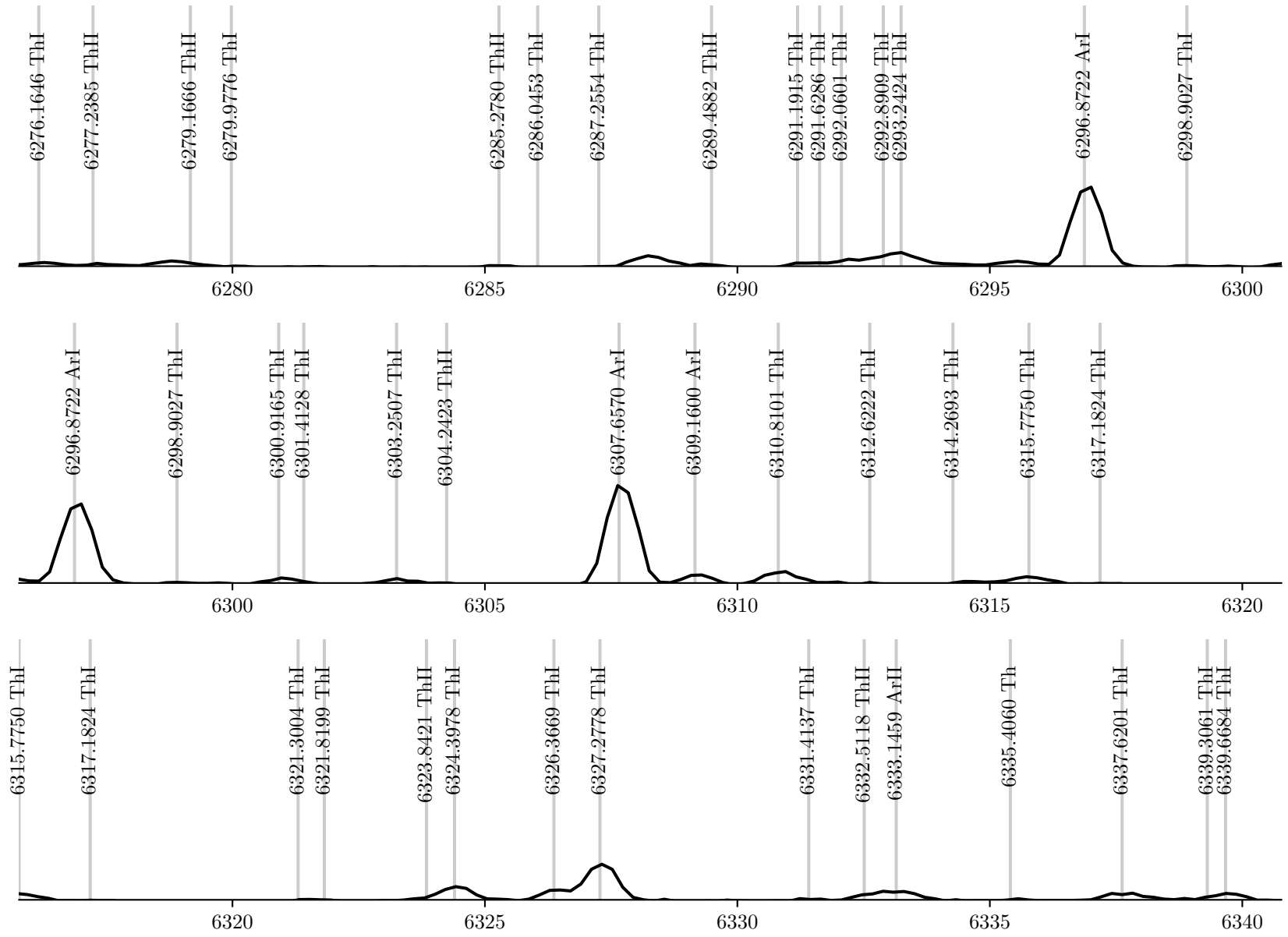
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6291.1915 ThI
6291.6286 ThI
6292.0601 ThI
6292.8909 ThI
6293.2424 ThI
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6298.9027 ThI
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6300
6296.8722 ArI
6298.9027 ThI
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6309.1600 ArI
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6323.8421 ThII
6324.3978 ThI
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6327.2778 ThI
6331.4137 ThI
6332.5118 ThII
6333.1459 ArII
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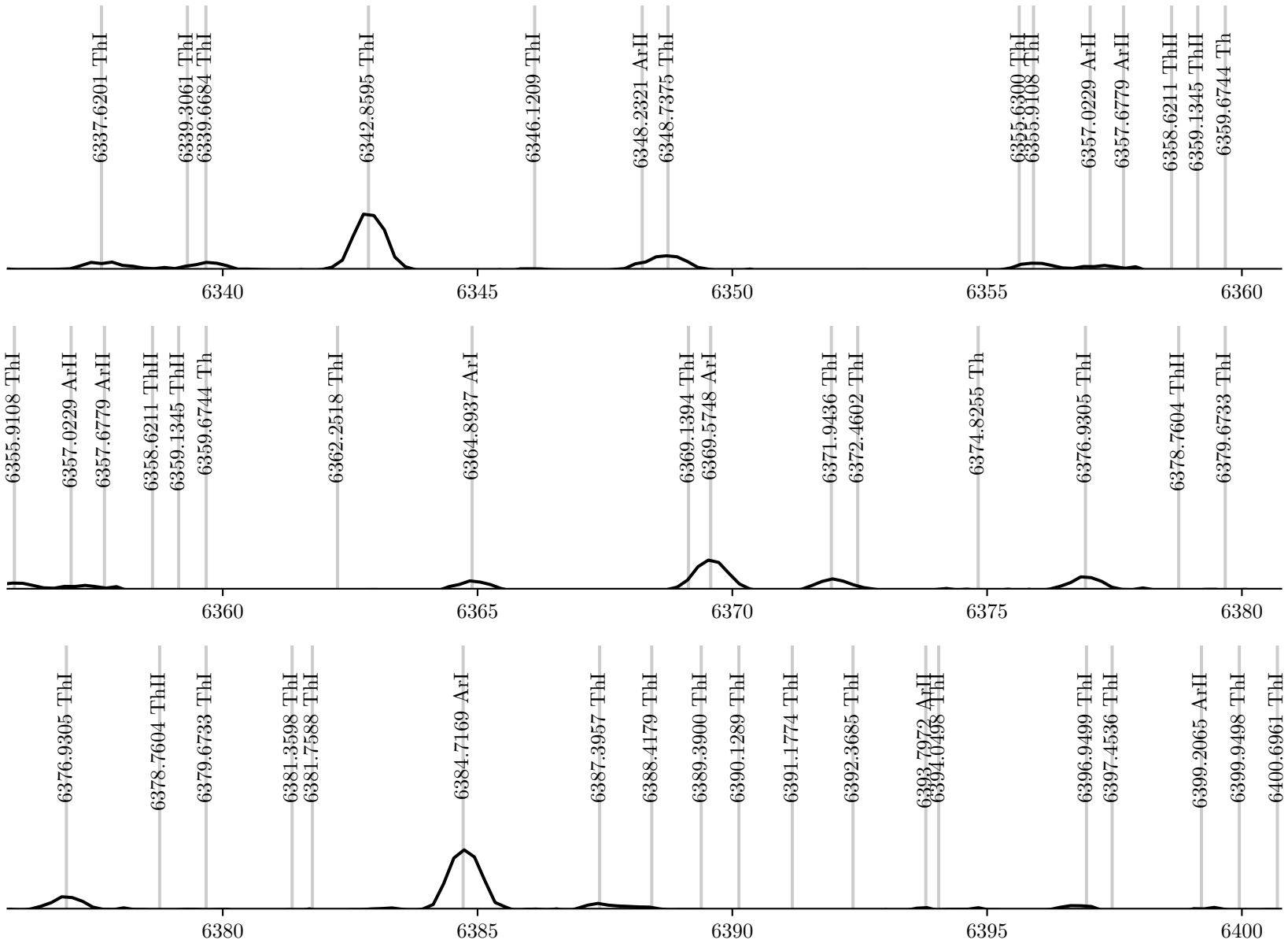
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6348.7375 ThI
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6355.9108 ThI
6357.0229 ArII
6357.6779 ArII
6358.6211 ThII
6359.1345 ThII
6359.6744 Th
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6345
6350
6355
6360
6355.9108 ThI
6357.0229 ArII
6357.6779 ArII
6358.6211 ThII
6359.1345 ThII
6359.6744 Th
6362.2518 ThI
6364.8937 ArI
6369.1394 ThI
6369.5748 ArI
6371.9436 ThI
6372.4602 ThI
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6376.9305 ThI
6378.7604 ThII
6379.6733 ThI
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6365
6370
6375
6380
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6378.7604 ThII
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6381.7588 ThI
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6387.3957 ThI
6388.4179 ThI
6389.3900 ThI
6390.1289 ThI
6391.1774 ThI
6392.3685 ThI
6393.7972 ArII
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6396.9499 ThI
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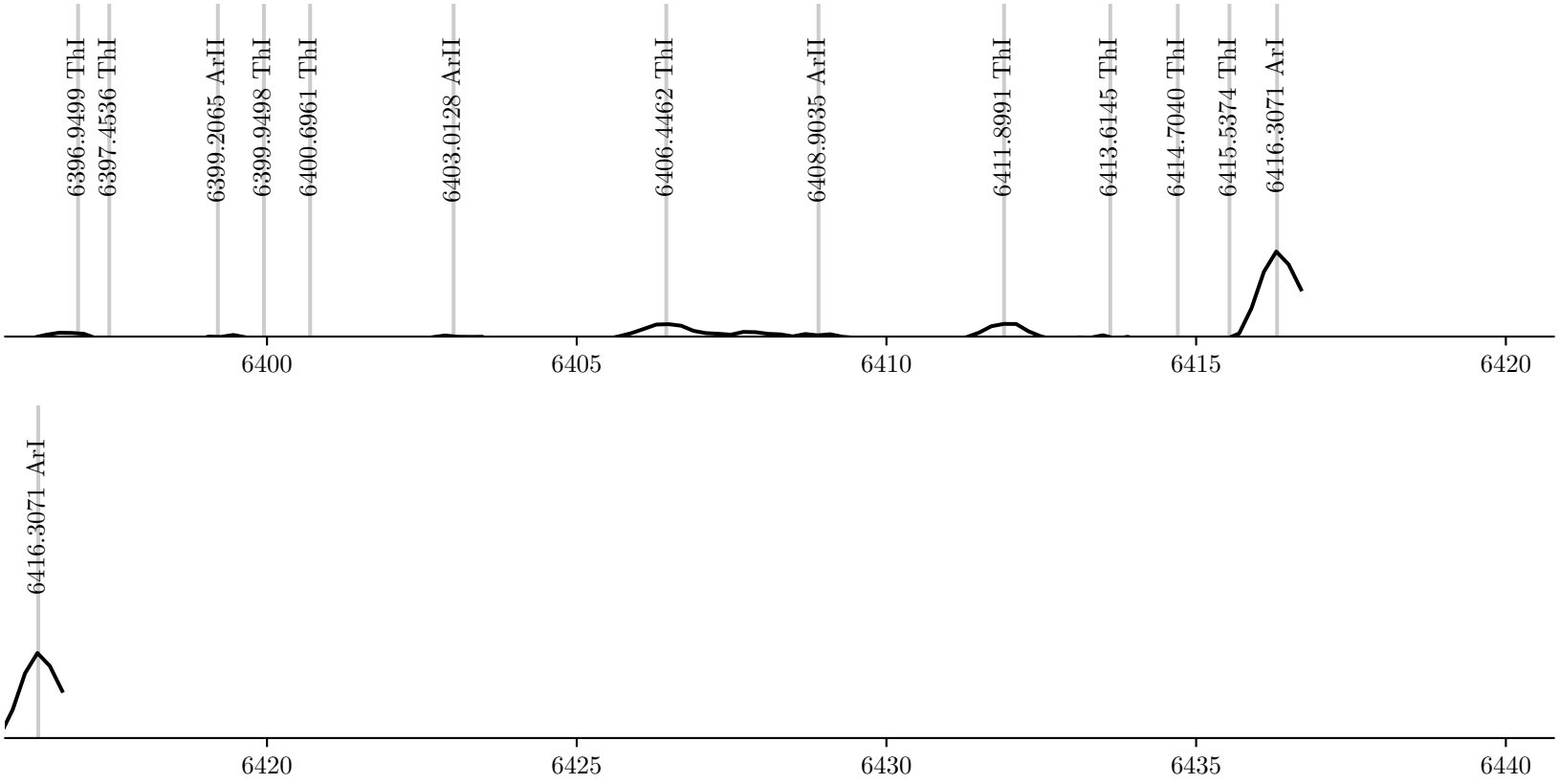
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6415.5374 ThI
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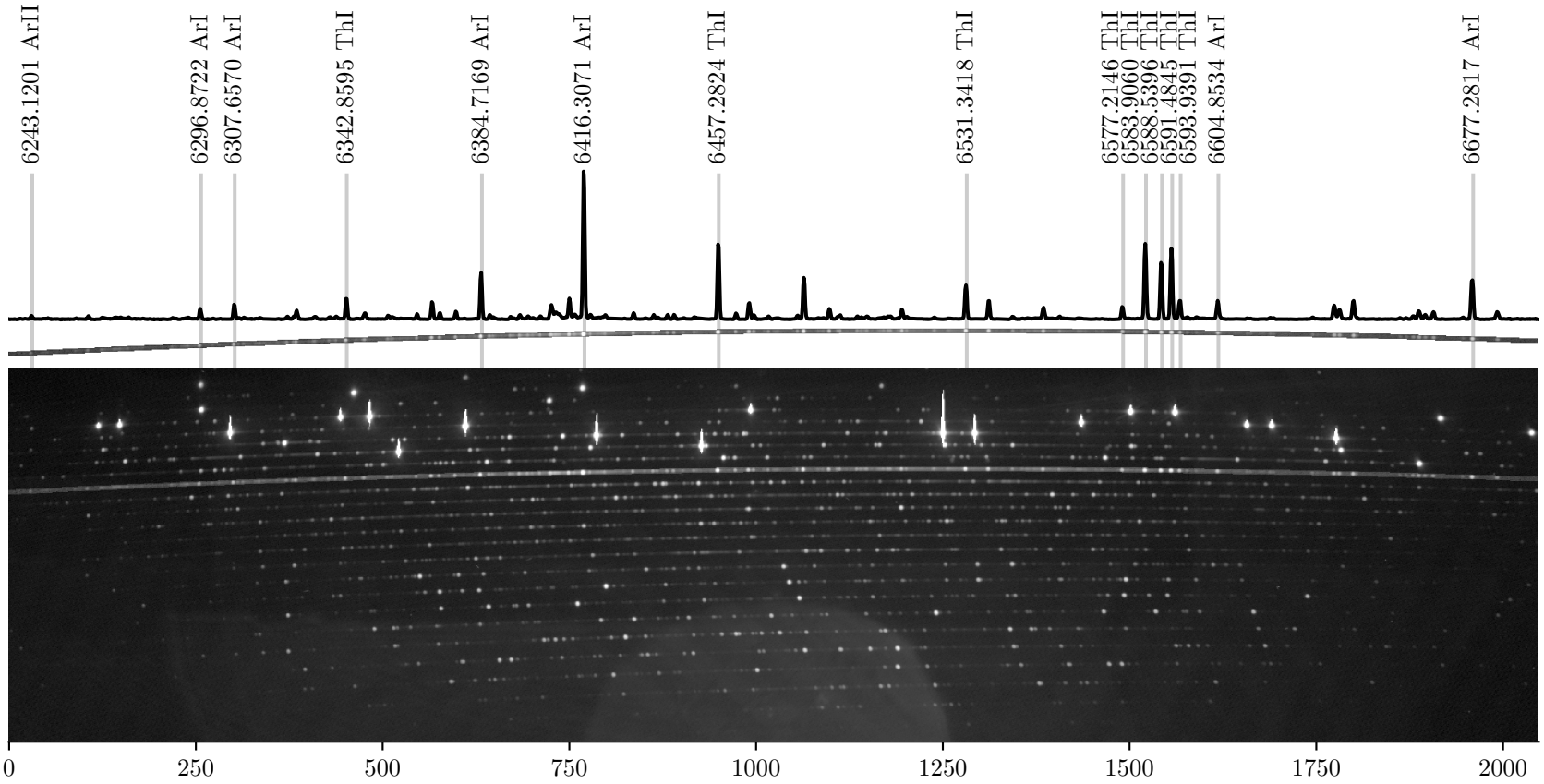
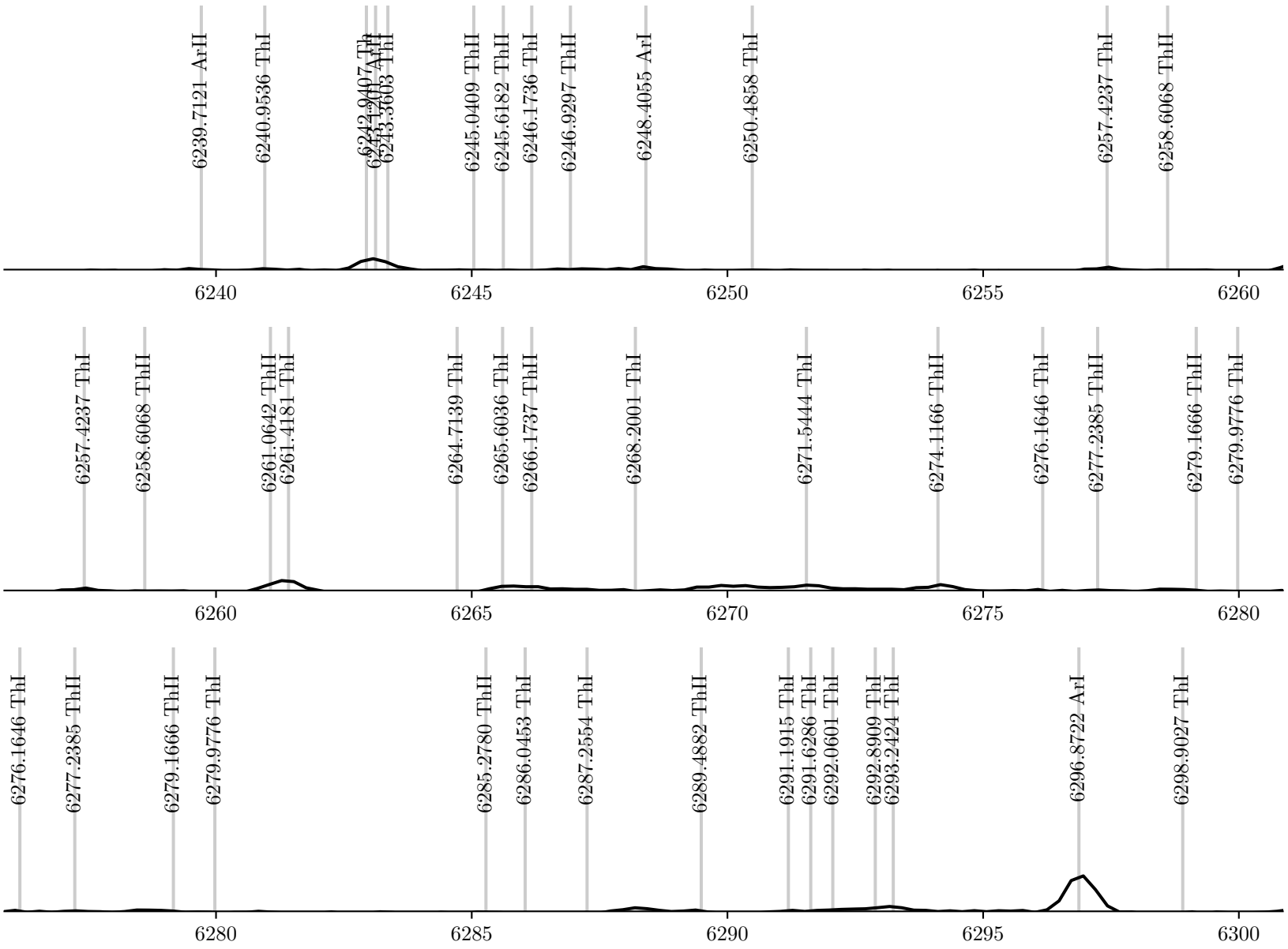
6440

Order 23



Dispersion:	0.225 Å pix^{-1}
Central wavelength:	6474 Å
Wavelength range:	6236 Å – 6696 Å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6257.4237 ThI
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6257.4237 ThI
6258.6068 ThII
6261.0642 ThII
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6266.1737 ThII
6268.2001 ThI
6271.5444 ThI
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6277.2385 ThII
6279.1666 ThII
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6285.2780 ThII
6286.0453 ThI
6287.2554 ThI
6289.4882 ThII
6291.1915 ThI
6291.6286 ThI
6292.0601 ThI
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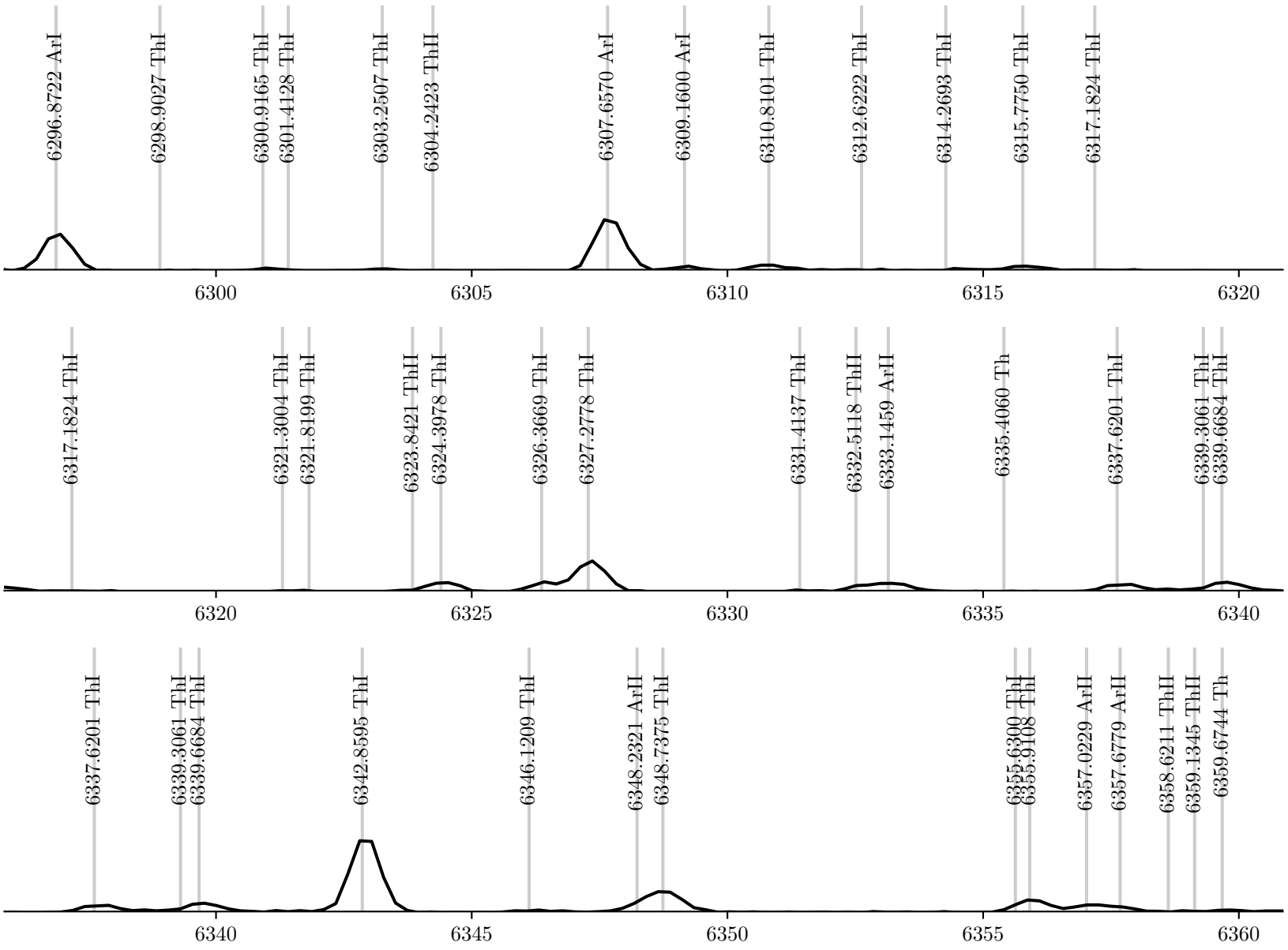
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6332.5118 ThII
6333.1459 ArII
6335.4060 Th
6337.6201 ThI
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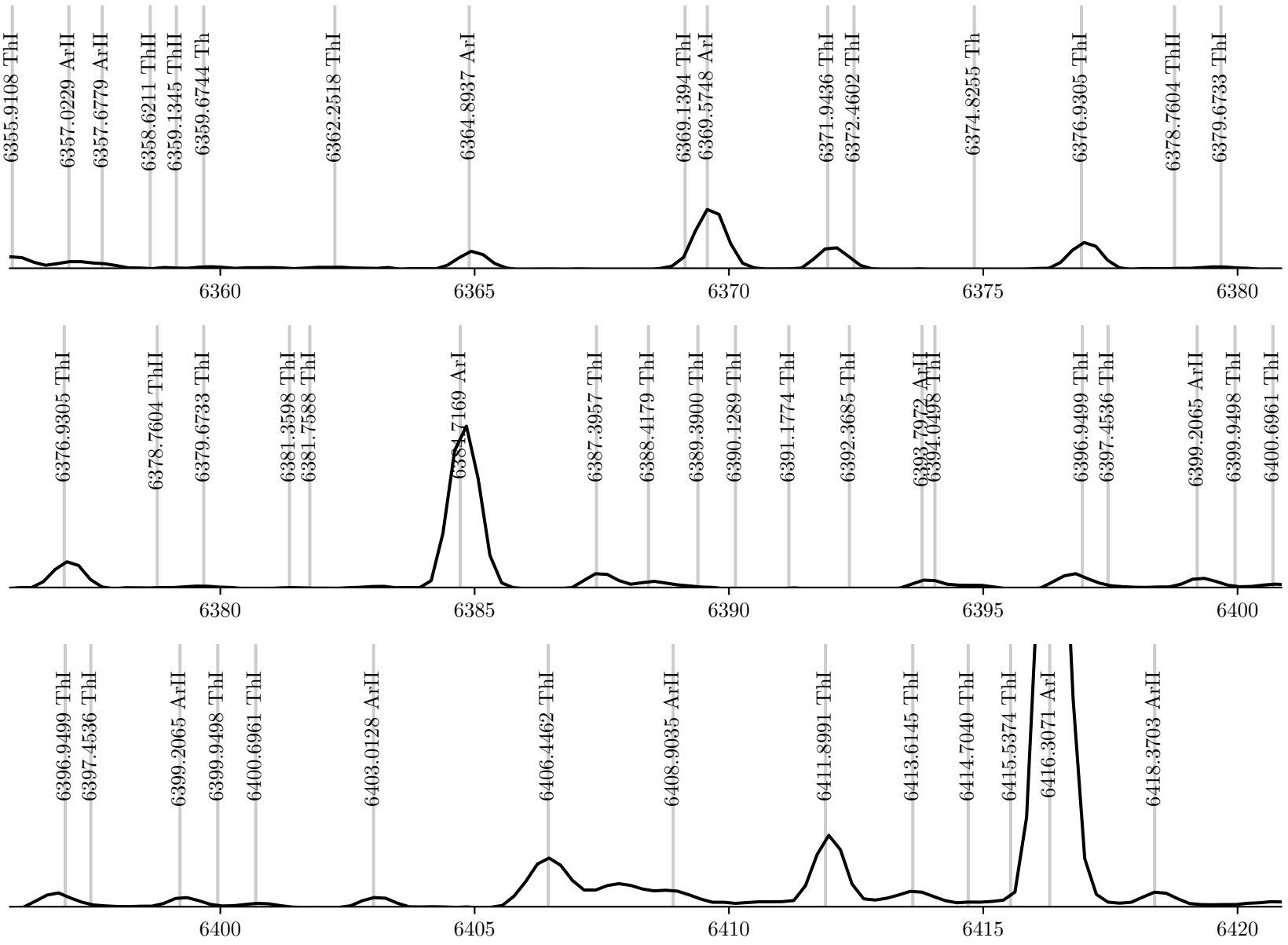
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6369.5748 ArI
6371.9436 ThI
6372.4602 ThI
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6406.4462 ThI
6408.9035 ArII
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6413.6145 ThI
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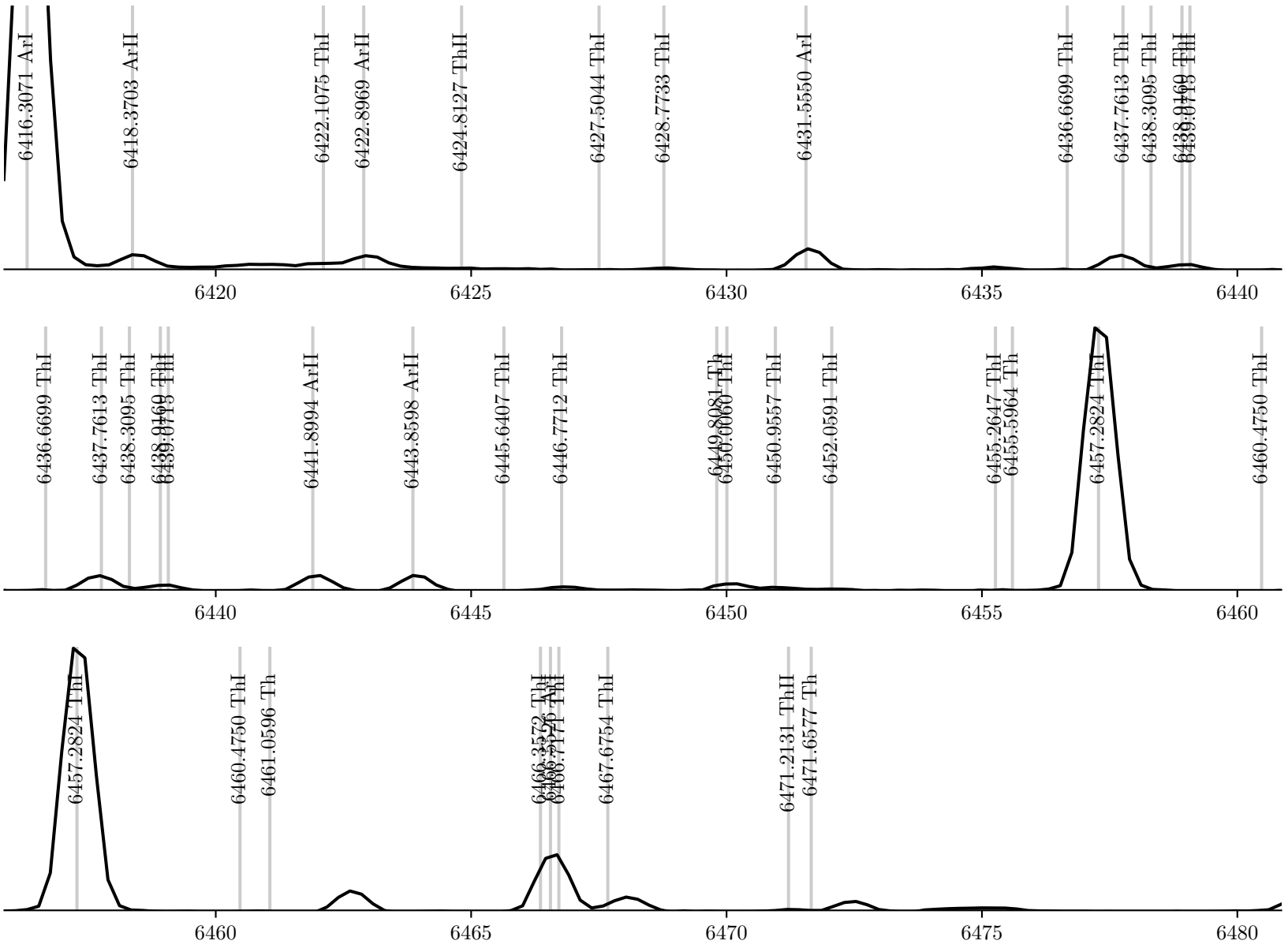
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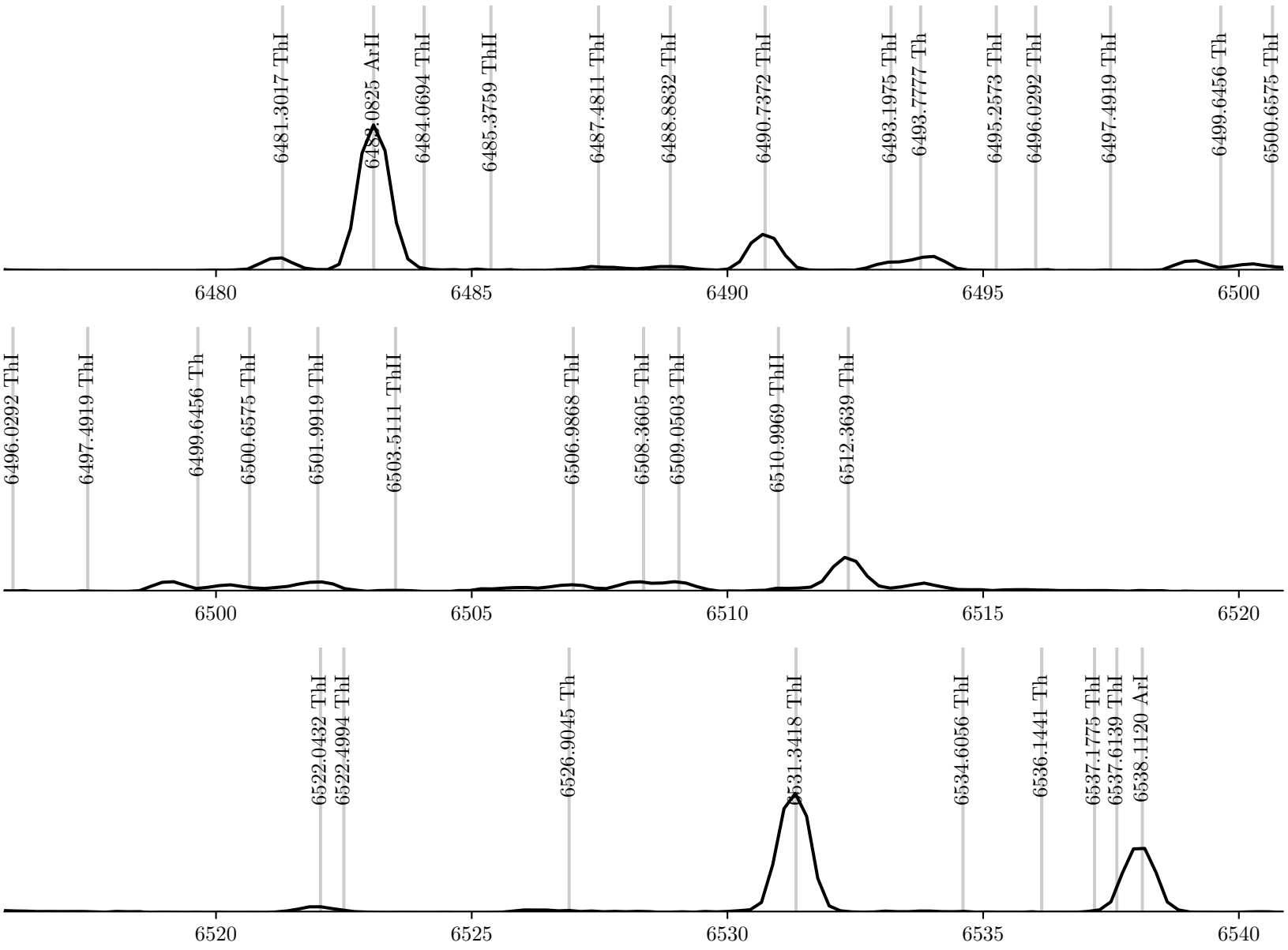
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6500
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6522.4994 ThI
6526.9045 Th
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6534.6056 ThI
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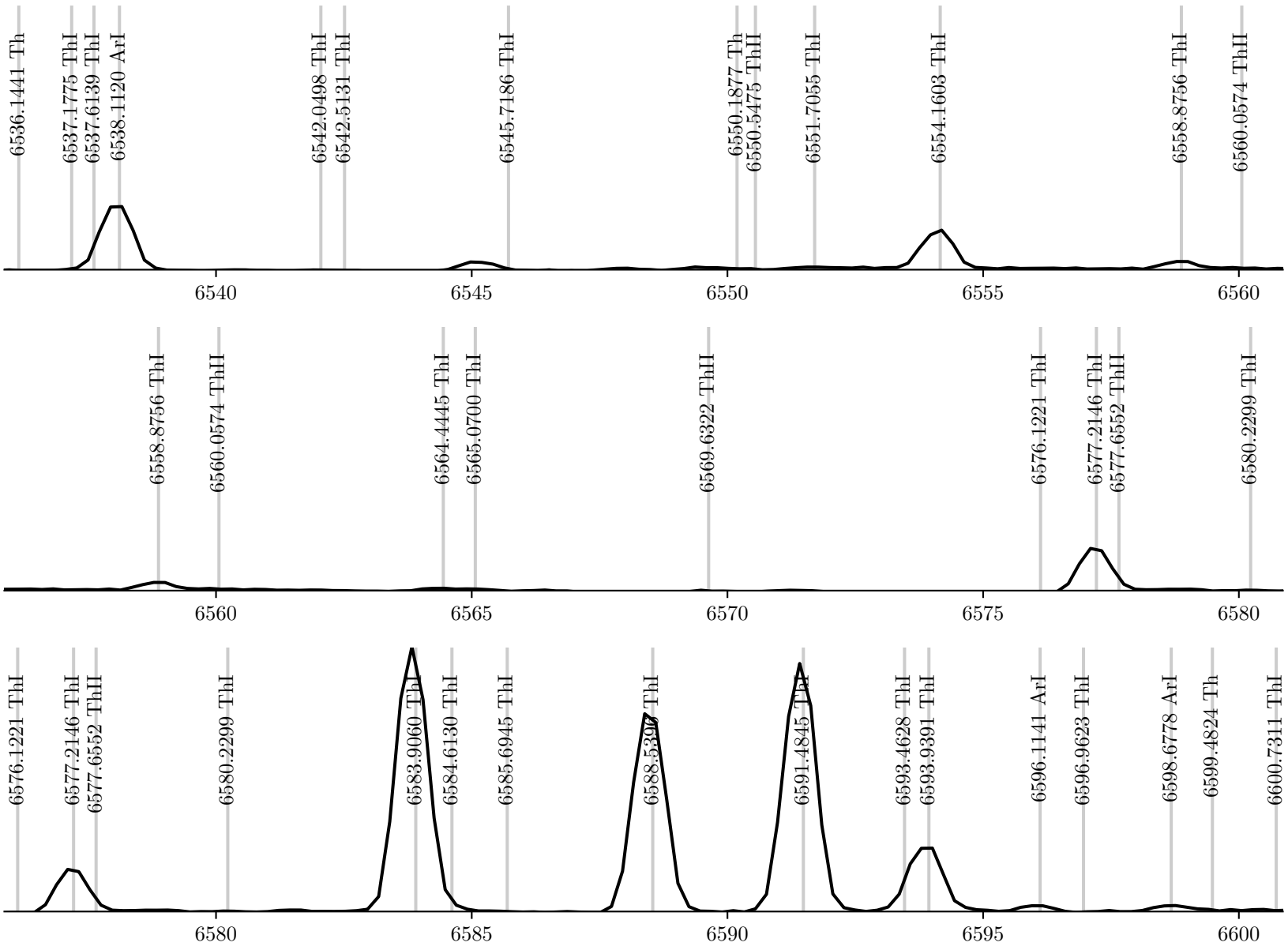
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6580
6576.1221 ThI
6577.2146 ThI
6577.6552 ThII
6580.2299 ThI
6583.9060 Th
6584.6130 ThI
6585.6945 ThI
6588.5300 ThI
6591.4845 ThI
6593.4628 ThI
6593.9391 ThI
6596.1141 ArI
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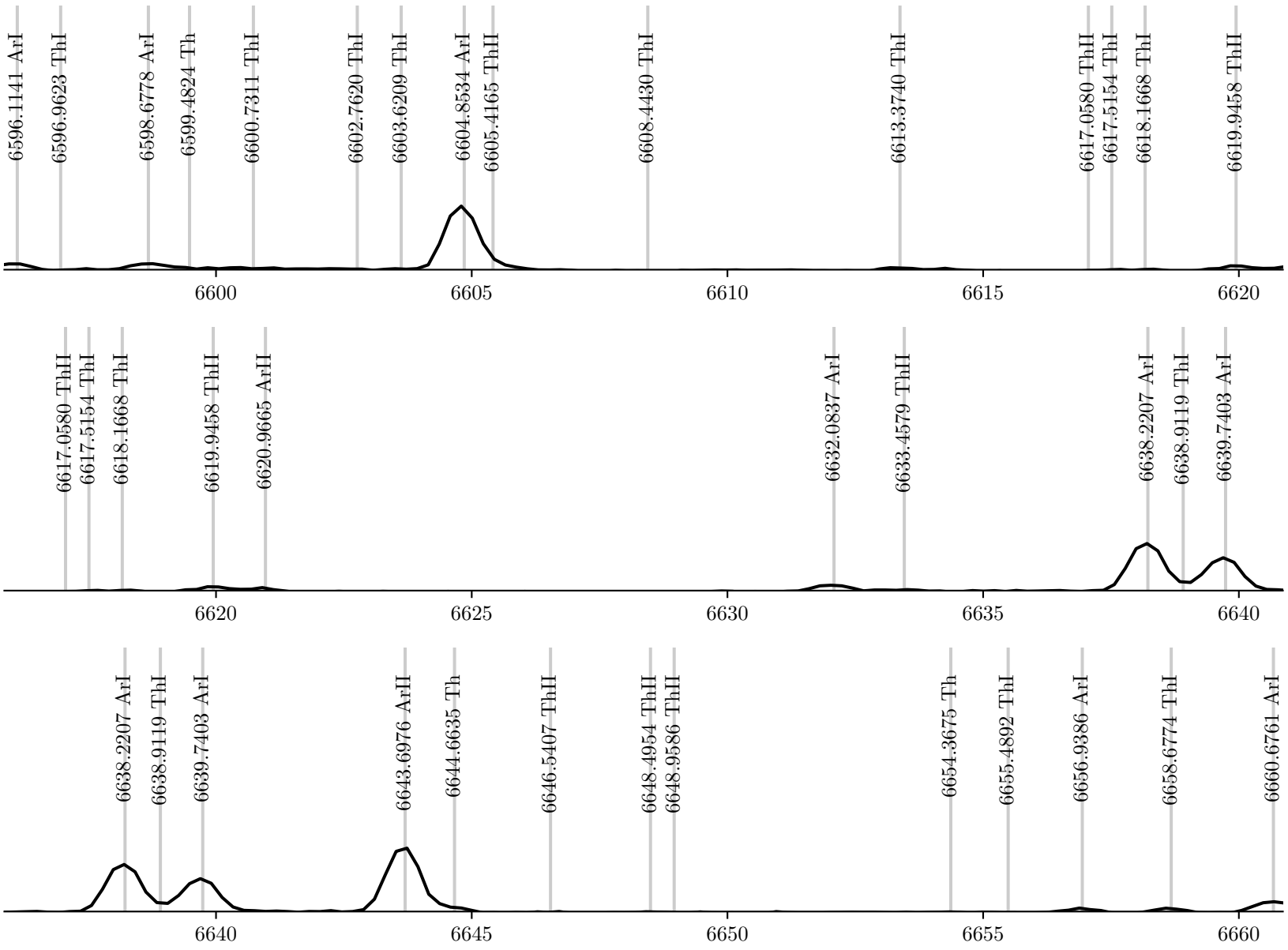
6600


6596.1141 ArI
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6600.7311 ThI
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6603.6209 ThI
6604.8534 ArI
6605.4165 ThII
6608.4430 ThI
6613.3740 ThI
6617.0580 ThII
6617.5154 ThI
6618.1668 ThI
6619.9458 ThII
6600
6605
6610
6615
6620
6617.0580 ThII
6617.5154 ThI
6618.1668 ThI
6619.9458 ThII
6620.9665 ArII
6632.0837 ArI
6633.4579 ThII
6638.2207 ArI
6638.9119 ThI
6639.7403 ArI
6620
6625
6630
6635
6640
6638.2207 ArI
6638.9119 ThI
6639.7403 ArI
6643.6976 ArII
6644.6635 Th
6646.5407 ThII
6648.4954 ThII
6648.9586 ThII
6654.3675 Th
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6658.6774 ThI
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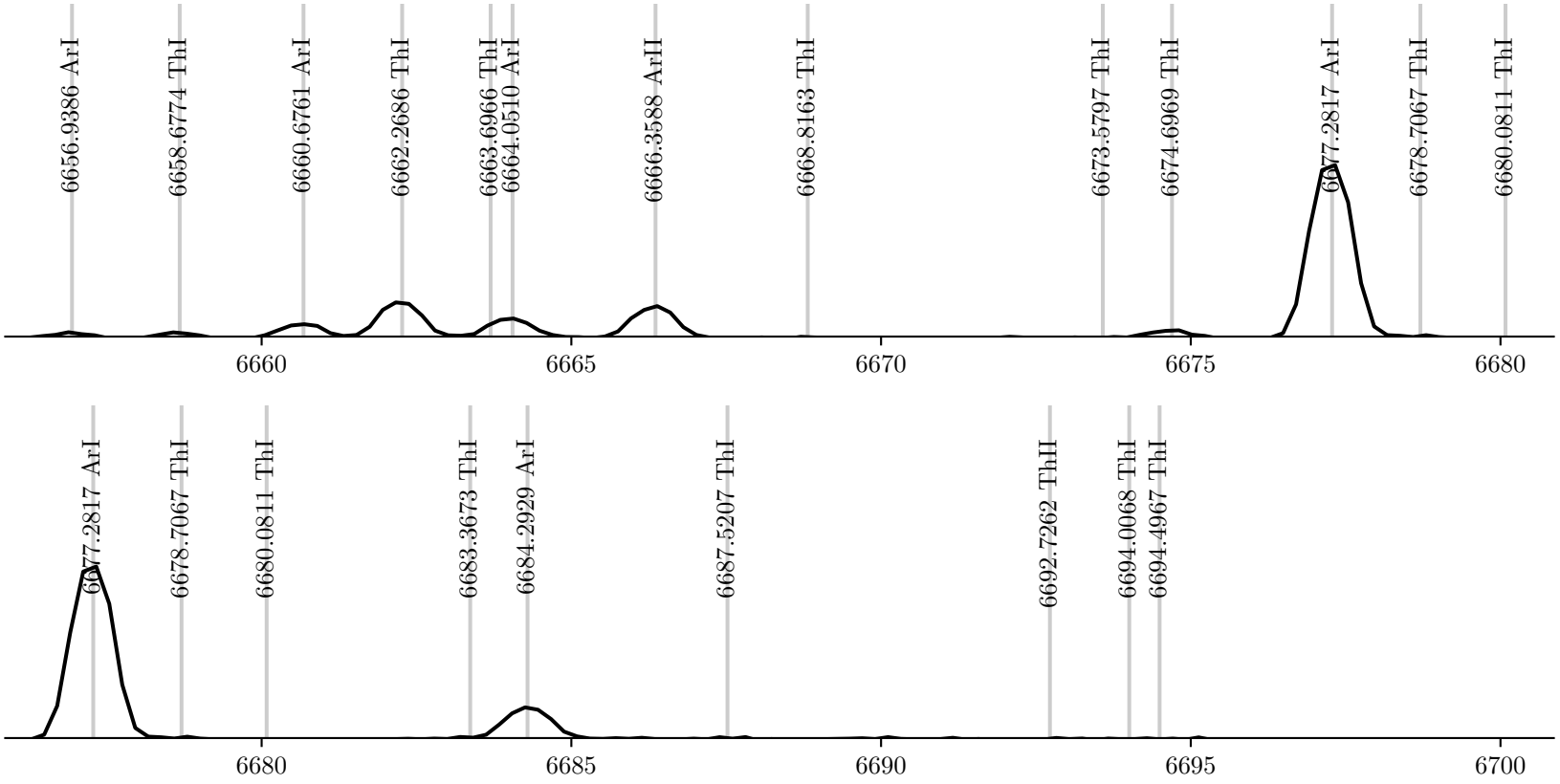
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6660.6761 ArI
6662.2686 ThI
6663.6966 ThI
6664.0510 ArI
6666.3588 ArII
6668.8163 ThI
6673.5797 ThI
6674.6969 ThI
6677.2817 ArI
6678.7067 ThI
6680.0811 ThI
6660
6665
6670
6675
6680
6677.2817 ArI
6678.7067 ThI
6680.0811 ThI
6683.3673 ThI
6684.2929 ArI
6687.5207 ThI
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6694.0068 ThI
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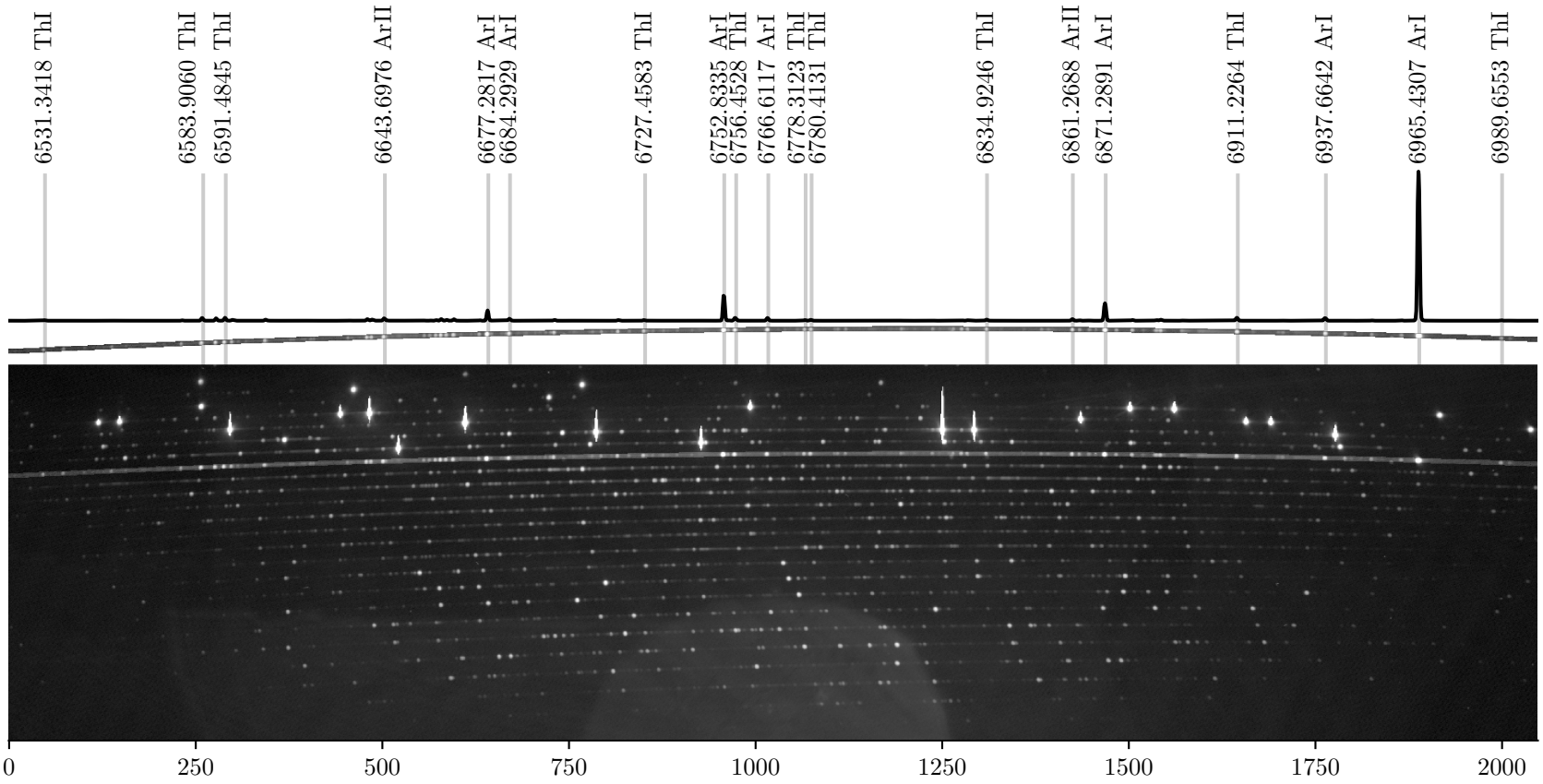
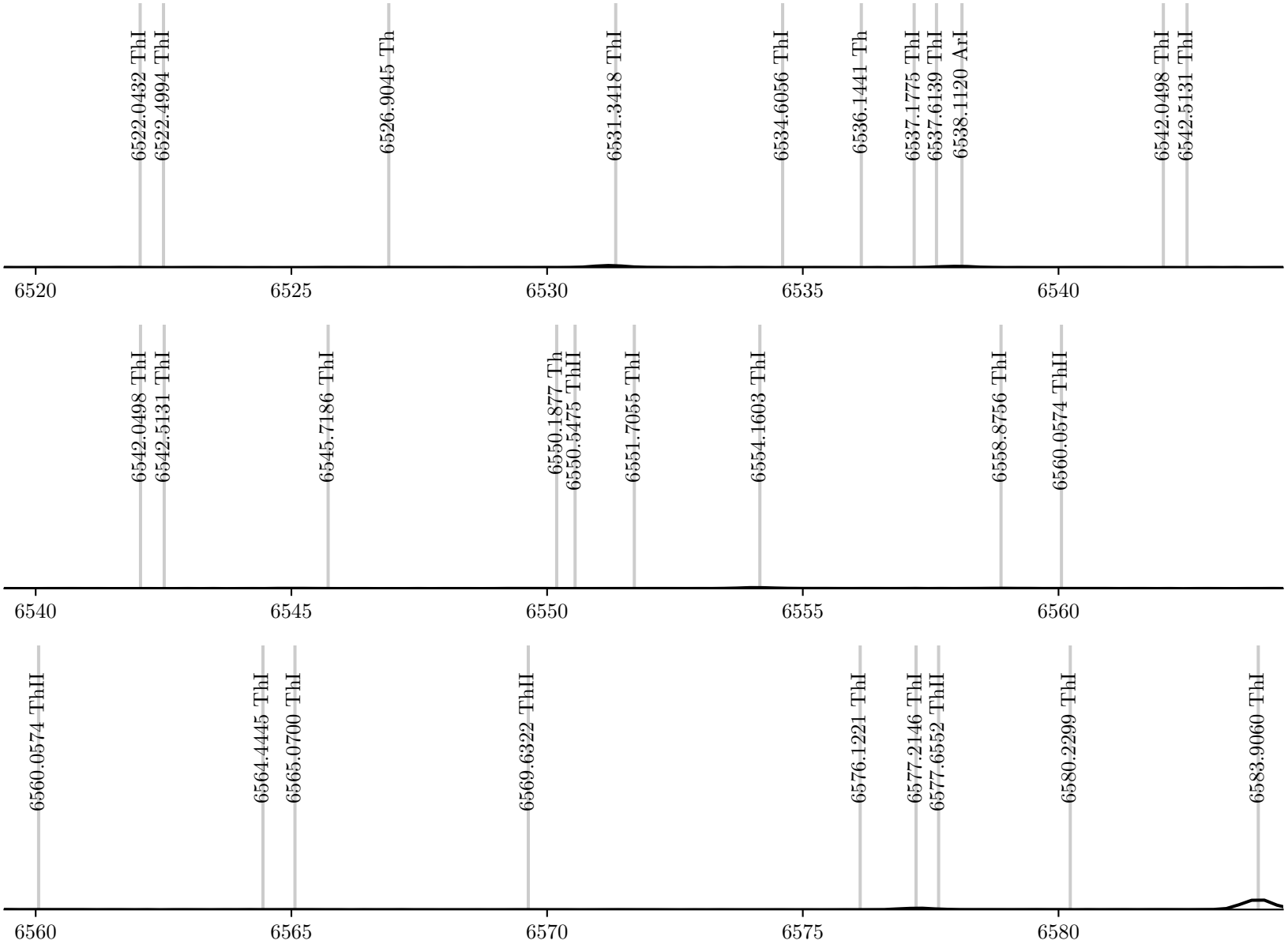
6695
6700

Order 22



Dispersion:	0.235 Å pix^{-1}
Central wavelength:	6768 Å
Wavelength range:	6519 Å – 7000 Å


6522.0432 ThI
6522.4994 ThI
6526.9045 Th
6531.3418 ThI
6534.6056 ThI
6536.1441 Th
6537.1775 ThI
6537.6139 ThI
6538.1120 ArI
6542.0498 ThI
6542.5131 ThI
6520
6525
6530
6535
6540
6542.0498 ThI
6542.5131 ThI
6545.7186 ThI
6550.1877 Th
6550.5475 ThII
6551.7055 ThI
6554.1603 ThI
6558.8756 ThI
6560.0574 ThII
6540
6545
6550
6555
6560
6560.0574 ThII
6564.4445 ThI
6565.0700 ThI
6569.6322 ThII
6576.1221 ThI
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6577.6552 ThII
6580.2299 ThI
6583.9060 ThI
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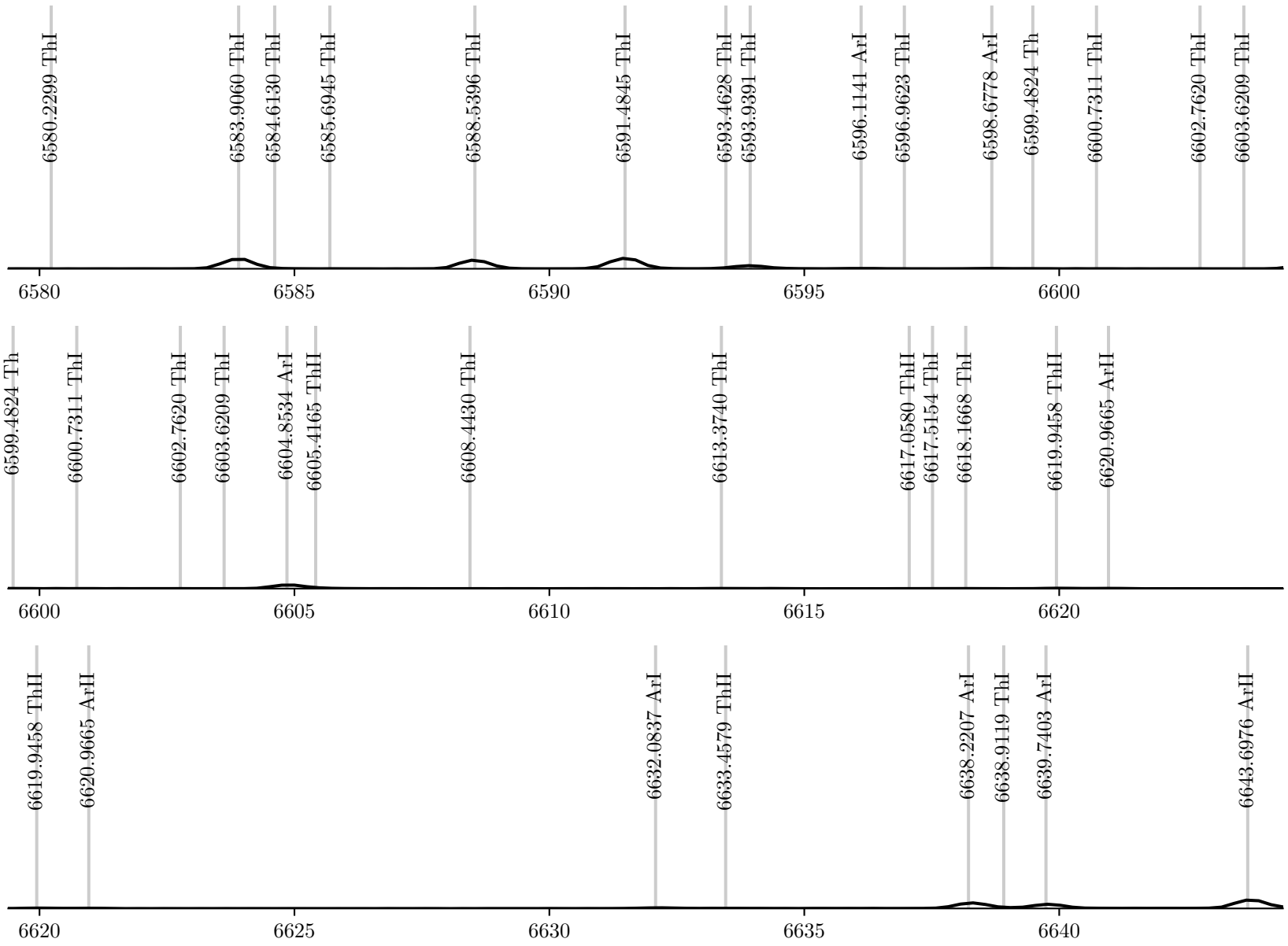
6570
6575
6580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6588.5396 ThI
6591.4845 ThI
6593.4628 ThI
6593.9391 ThI
6596.1141 ArI
6596.9623 ThI
6598.6778 ArI
6599.4824 Th
6600.7311 ThI
6602.7620 ThI
6603.6209 ThI
6580
6585
6590
6595
6600
6599.4824 Th
6600.7311 ThI
6602.7620 ThI
6603.6209 ThI
6604.8534 ArI
6605.4165 ThII
6608.4430 ThI
6613.3740 ThI
6617.0580 ThII
6617.5154 ThI
6618.1668 ThI
6619.9458 ThII
6620.9665 ArII
6600
6605
6610
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6620
6619.9458 ThII
6620.9665 ArII
6632.0837 ArI
6633.4579 ThII
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6638.9119 ThI
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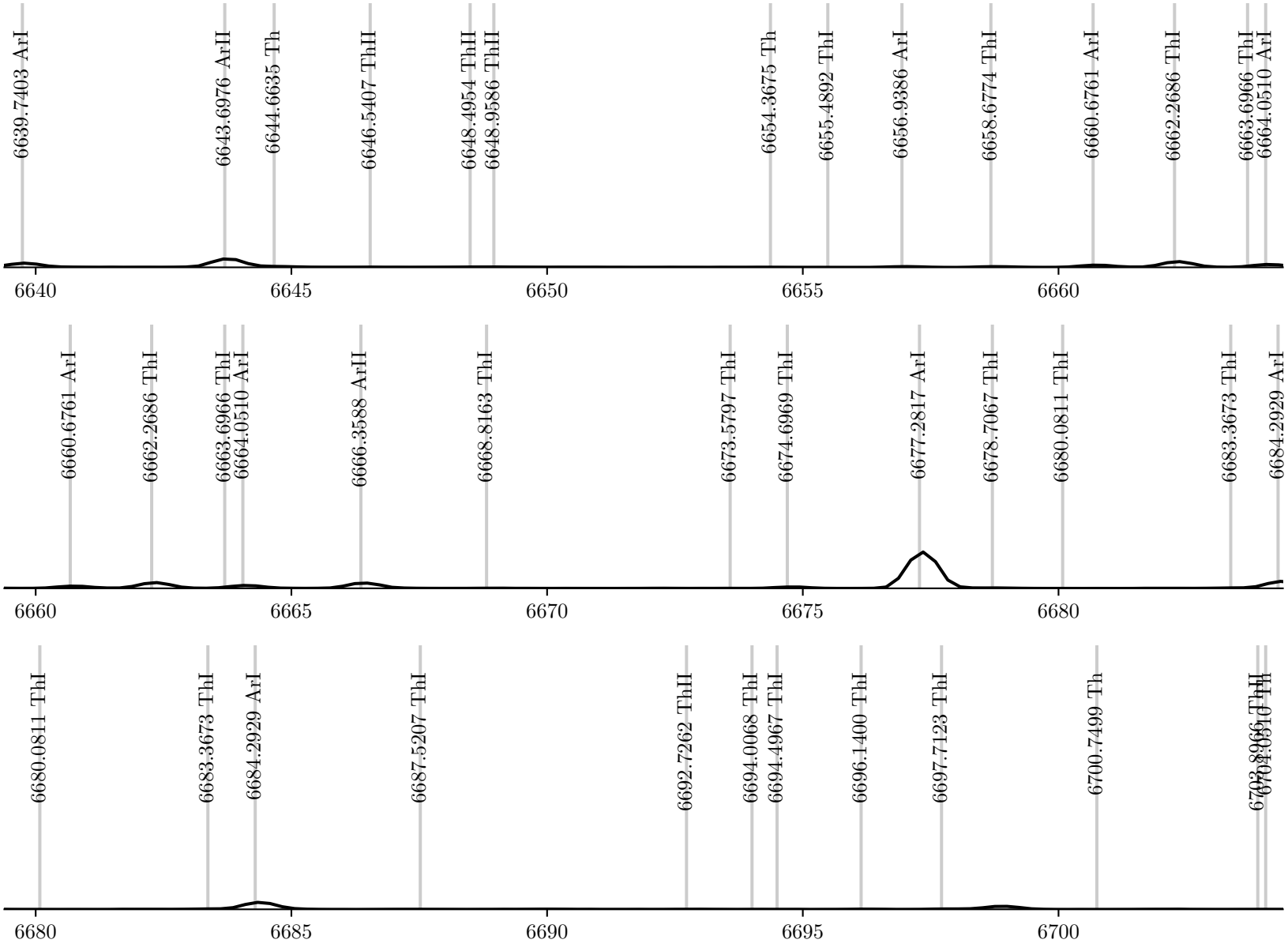
6639.7403 ArI
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6648.4954 ThII
6648.9586 ThII
6654.3675 Th
6655.4892 ThI
6656.9386 ArI
6658.6774 ThI
6660.6761 ArI
6662.2686 ThI
6663.6966 ThI
6664.0510 ArI
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6660
6660.6761 ArI
6662.2686 ThI
6663.6966 ThI
6664.0510 ArI
6666.3588 ArII
6668.8163 ThI
6673.5797 ThI
6674.6969 ThI
6677.2817 ArI
6678.7067 ThI
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6683.3673 ThI
6684.2929 ArI
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6665
6670
6675
6680
6680.0811 ThI
6683.3673 ThI
6684.2929 ArI
6687.5207 ThI
6692.7262 ThII
6694.0068 ThI
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6696.1400 ThI
6697.7123 ThI
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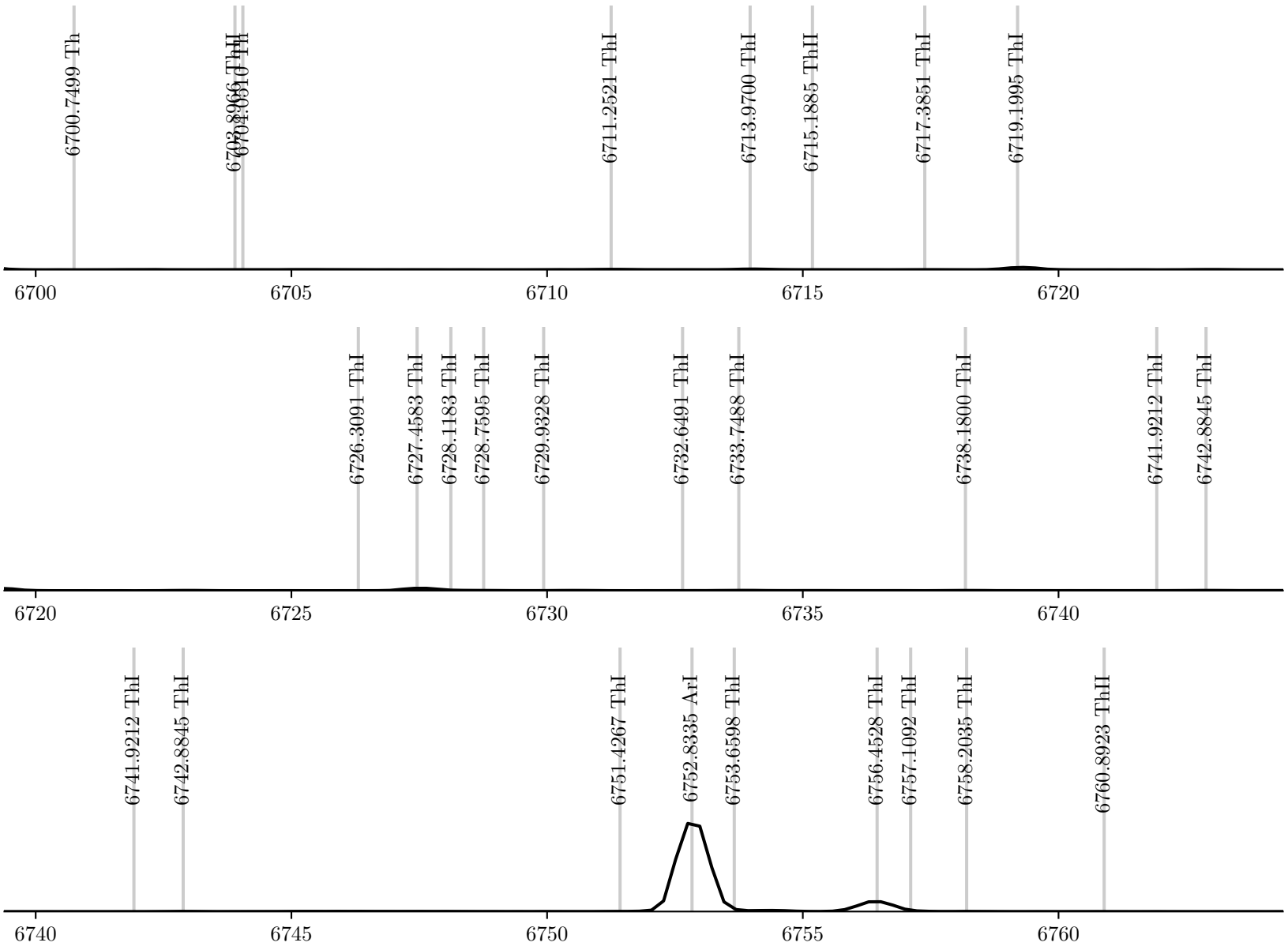
6690
6695
6700


6700.7499 Th
6711.2521 ThI
6713.9700 ThI
6715.1885 ThII
6717.3851 ThI
6719.1995 ThI
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6705
6710
6715
6720
6726.3091 ThI
6727.4583 ThI
6728.1183 ThI
6728.7595 ThI
6729.9328 ThI
6732.6491 ThI
6733.7488 ThI
6738.1800 ThI
6741.9212 ThI
6742.8845 ThI
6720
6725
6730
6735
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6741.9212 ThI
6742.8845 ThI
6751.4267 ThI
6752.8335 ArI
6753.6598 ThI
6756.4528 ThI
6757.1092 ThI
6758.2035 ThI
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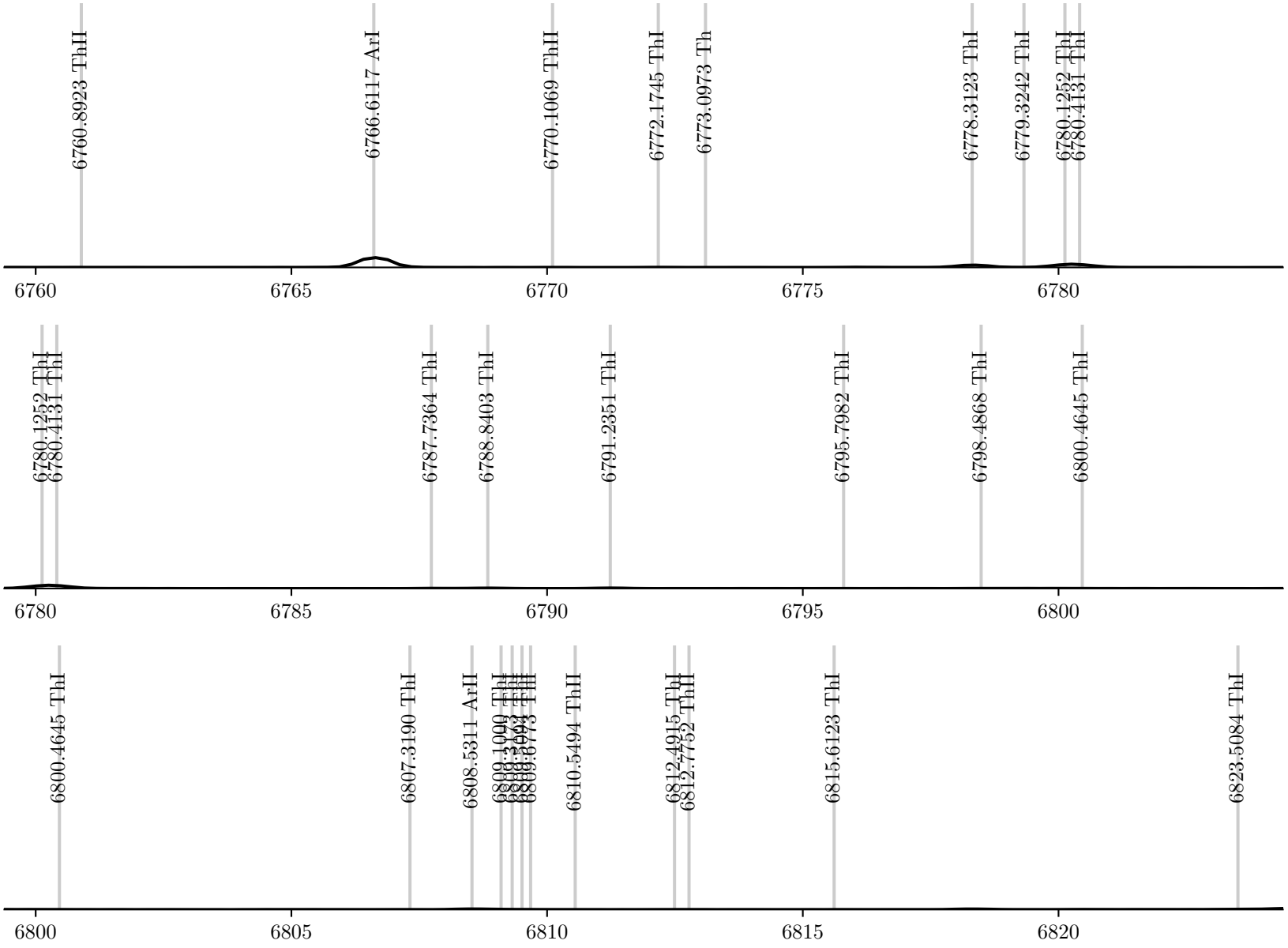
6750
6755
6760


6760.8923 ThII
6766.6117 ArI
6770.1069 ThII
6772.1745 ThI
6773.0973 Th
6778.3123 ThI
6779.3242 ThI
6780.1252 ThI
6780.4131 ThI
6760
6765
6770
6775
6780
6780.1252 ThI
6780.4131 ThI
6787.7364 ThI
6788.8403 ThI
6791.2351 ThI
6795.7982 ThI
6798.4868 ThI
6800.4645 ThI
6780
6785
6790
6795
6800
6800.4645 ThI
6807.3190 ThI
6808.5311 ArII
6810.5494 ThII
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6812.7752 ThII
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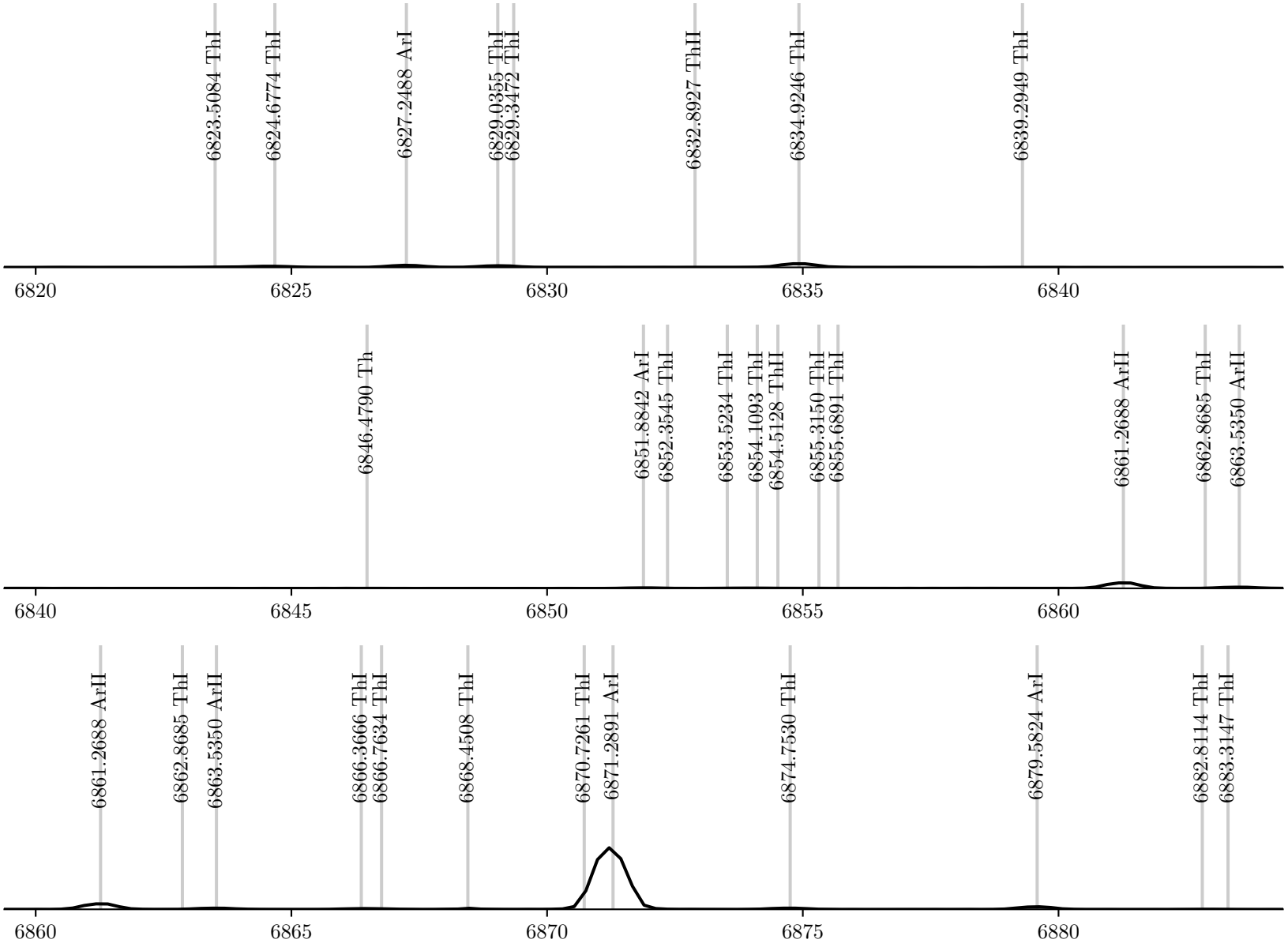
6823.5084 ThI
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6829.0355 ThI
6829.3472 ThI
6832.8927 ThII
6834.9246 ThI
6839.2949 ThI
6820
6825
6830
6835
6840
6846.4790 Th
6851.8842 ArI
6852.3545 ThI
6853.5234 ThI
6854.1093 ThI
6854.5128 ThII
6855.3150 ThI
6855.6891 ThI
6861.2688 ArII
6862.8685 ThI
6863.5350 ArII
6840
6845
6850
6855
6860
6861.2688 ArII
6862.8685 ThI
6863.5350 ArII
6866.3666 ThI
6866.7634 ThI
6868.4508 ThI
6870.7261 ThI
6871.2891 ArI
6874.7530 ThI
6879.5824 ArI
6882.8114 ThI
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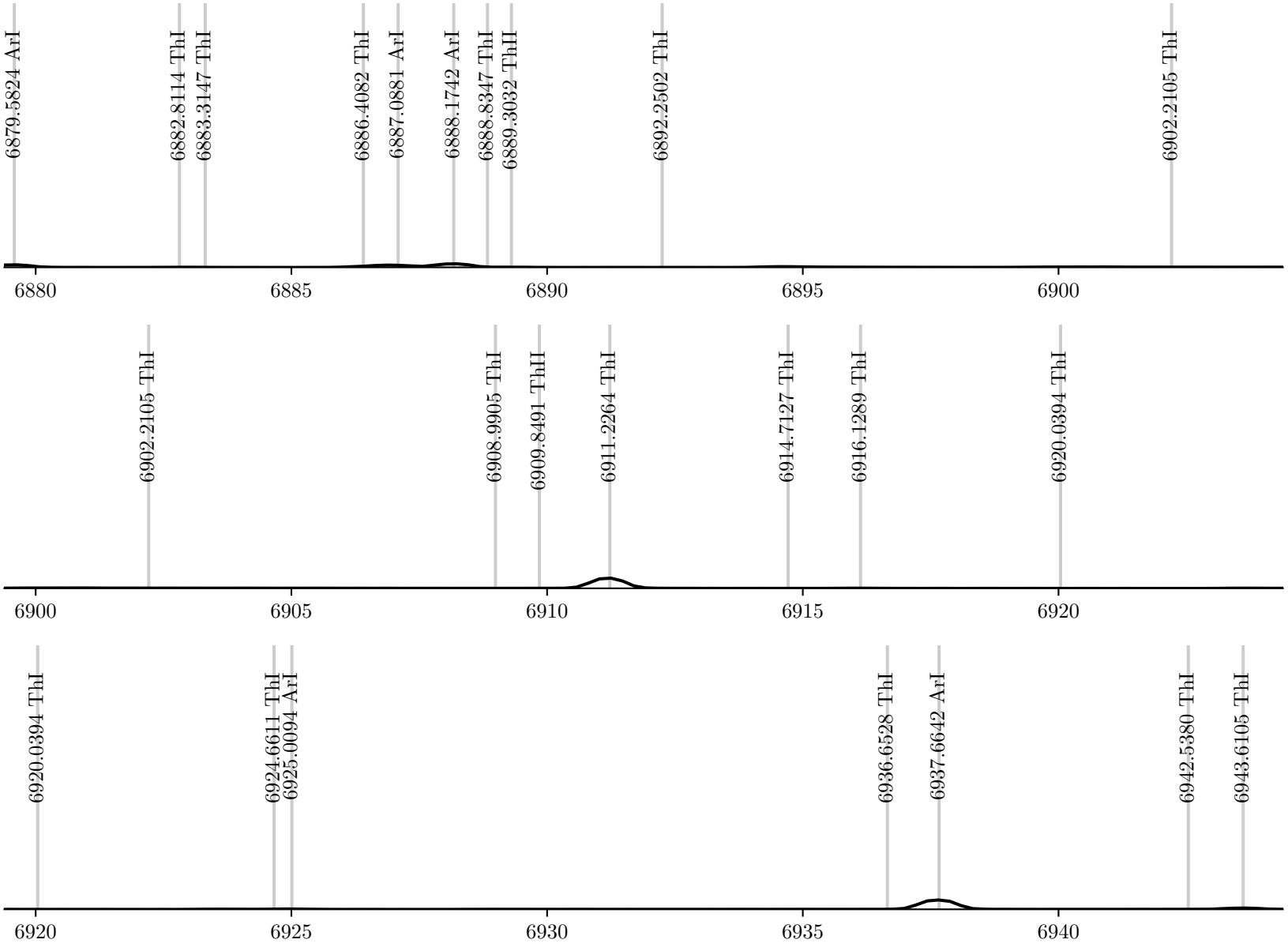
6879.5824 ArI
6882.8114 ThI
6883.3147 ThI
6886.4082 ThI
6887.0881 ArI
6888.1742 ArI
6888.8347 ThI
6889.3032 ThII
6892.2502 ThI
6902.2105 ThI
6880
6885
6890
6895
6900
6902.2105 ThI
6908.9905 ThI
6909.8491 ThII
6911.2264 ThI
6914.7127 ThI
6916.1289 ThI
6920.0394 ThI
6900
6905
6910
6915
6920
6920.0394 ThI
6924.6611 ThI
6925.0094 ArI
6936.6528 ThI
6937.6642 ArI
6942.5380 ThI
6943.6105 ThI
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6930
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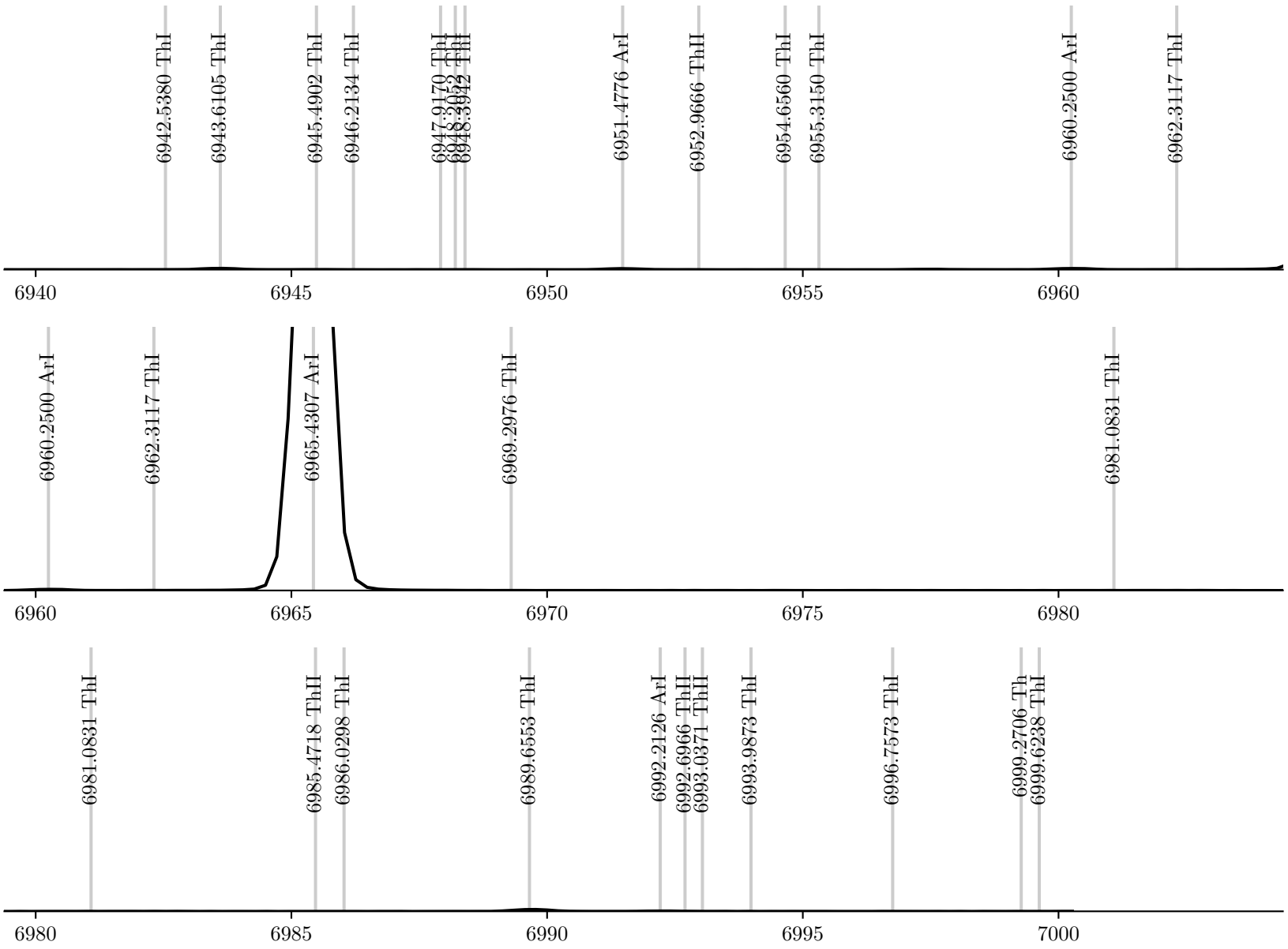
6940


6942.5380 ThI
6943.6105 ThI
6945.4902 ThI
6946.2134 ThI
6947.9170 ThI
6948.3052 ThI
6948.3942 ThI
6951.4776 ArI
6952.9666 ThII
6954.6560 ThI
6955.3150 ThI
6960.2500 ArI
6962.3117 ThI
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6950
6955
6960
6960.2500 ArI
6962.3117 ThI
6965.4307 ArI
6969.2976 ThI
6981.0831 ThI
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6965
6970
6975
6980
6981.0831 ThI
6985.4718 ThII
6986.0298 ThI
6989.6553 ThI
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6992.6966 ThII
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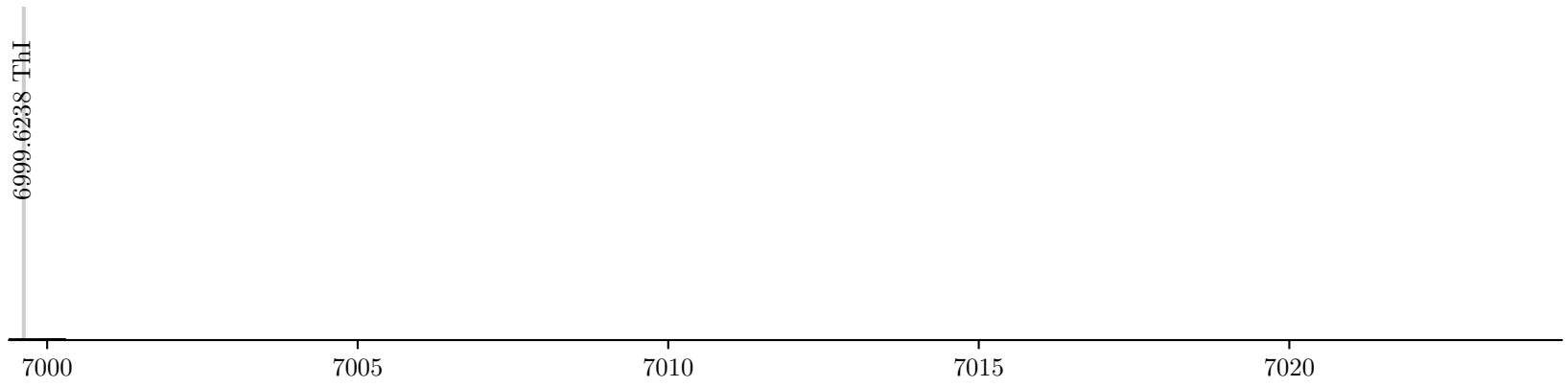
6995
7000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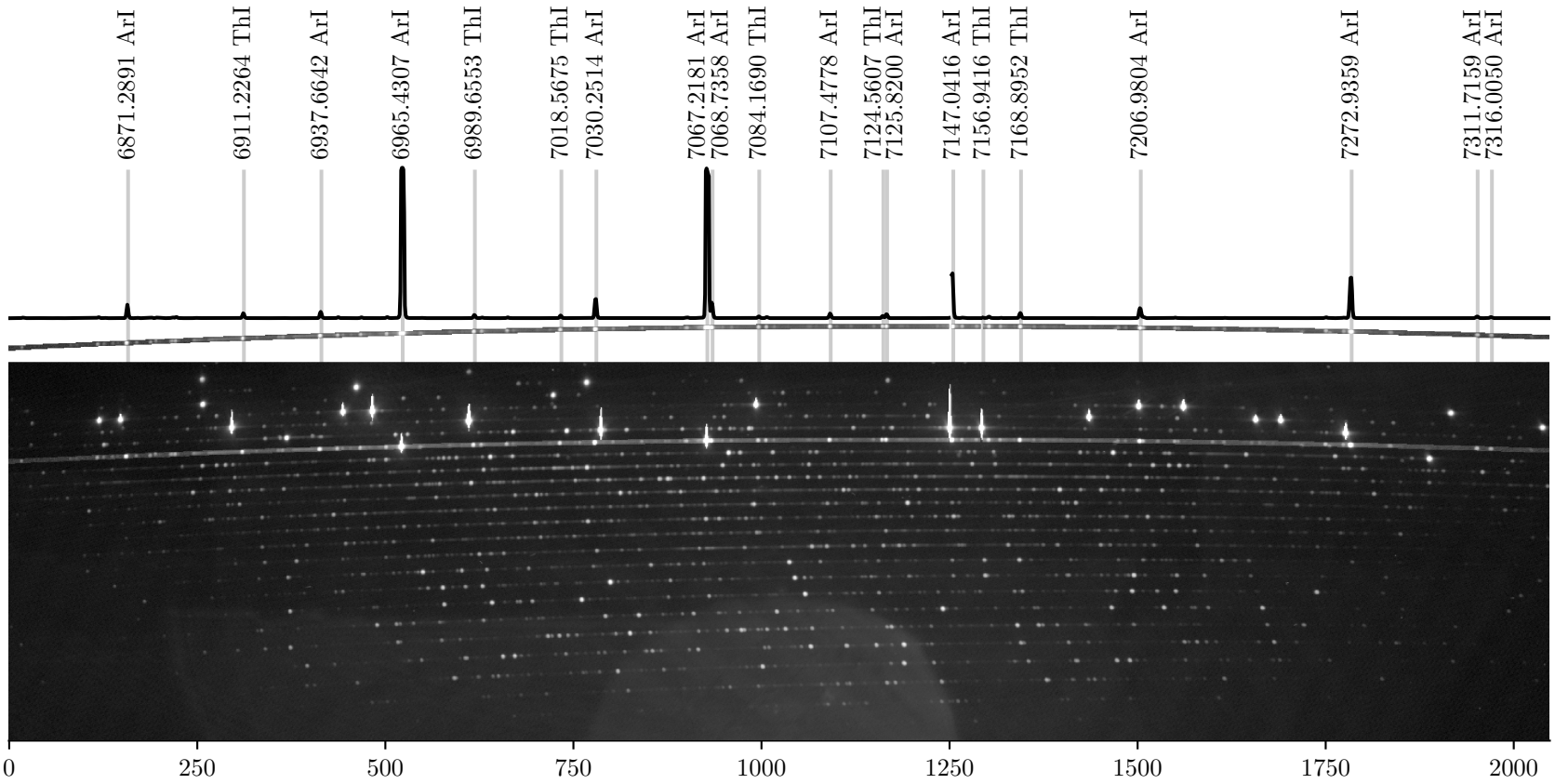
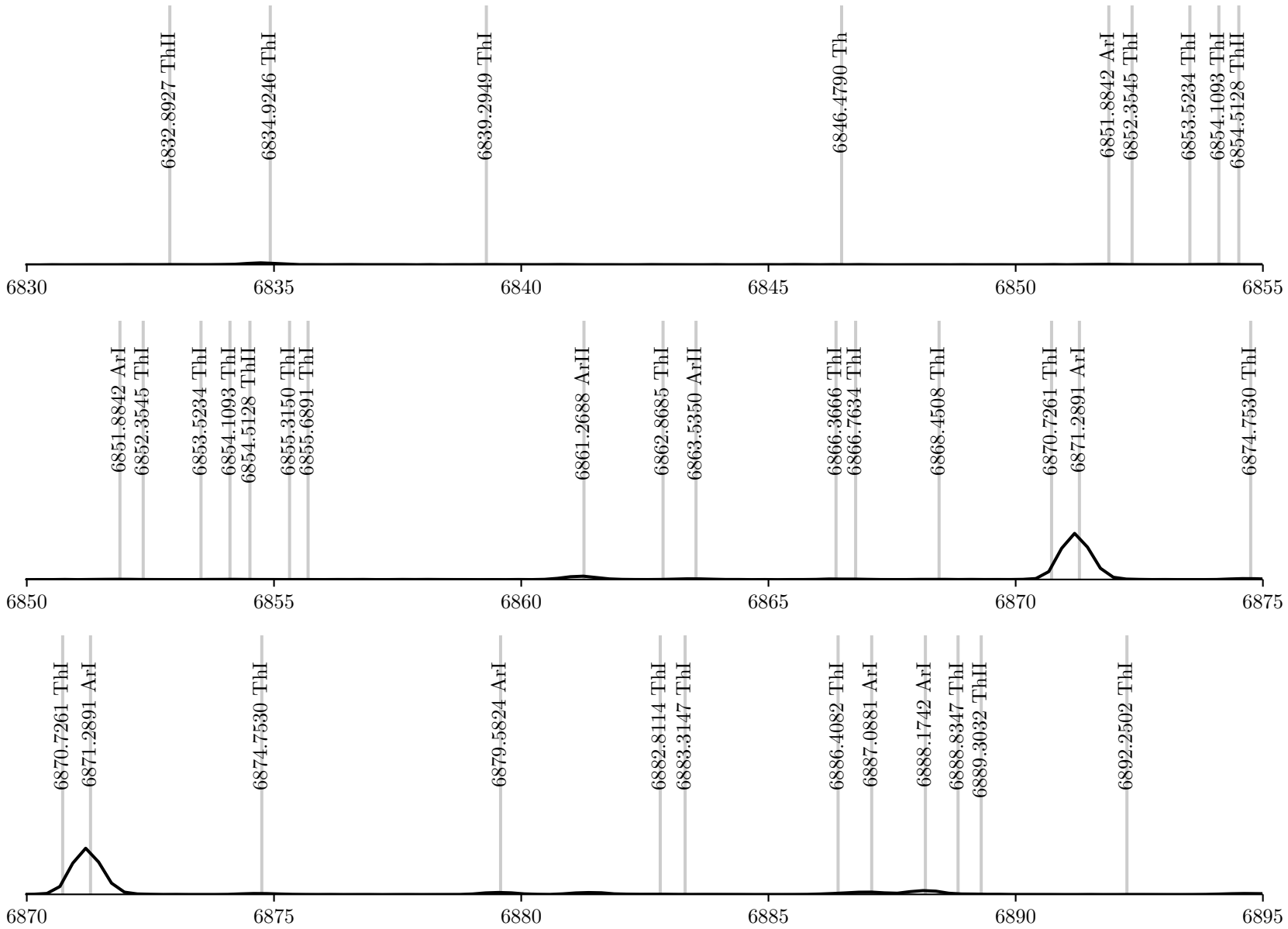
Order 21



Dispersion:	0.246 Å pix^{-1}
Central wavelength:	7091 Å
Wavelength range:	6830 Å – 7334 Å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6839.2949 ThI
6846.4790 Th
6851.8842 ArI
6852.3545 ThI
6853.5234 ThI
6854.1093 ThI
6854.5128 ThII
6830
6835
6840
6845
6850
6855
6851.8842 ArI
6852.3545 ThI
6853.5234 ThI
6854.1093 ThI
6854.5128 ThII
6855.3150 ThI
6855.6891 ThI
6861.2688 ArII
6862.8685 ThI
6863.5350 ArII
6866.3666 ThI
6866.7634 ThI
6868.4508 ThI
6870.7261 ThI
6871.2891 ArI
6874.7530 ThI
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6855
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6870
6875
6870.7261 ThI
6871.2891 ArI
6874.7530 ThI
6879.5824 ArI
6882.8114 ThI
6883.3147 ThI
6886.4082 ThI
6887.0881 ArI
6888.1742 ArI
6888.8347 ThI
6889.3032 ThII
6892.2502 ThI
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6885
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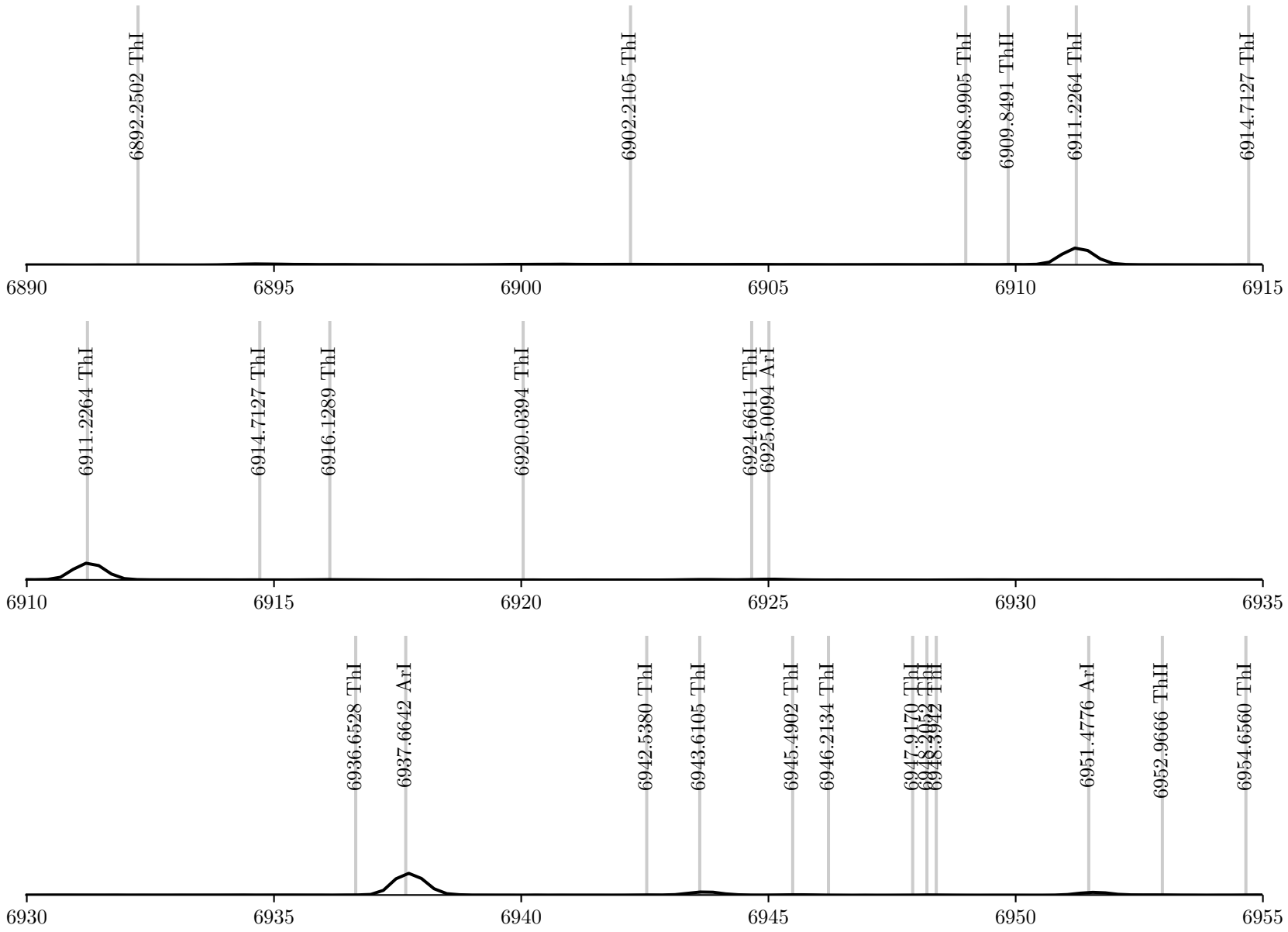
6895


6892.2502 ThI
6902.2105 ThI
6908.9905 ThI
6909.8491 ThII
6911.2264 ThI
6914.7127 ThI
6890
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6900
6905
6910
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6911.2264 ThI
6914.7127 ThI
6916.1289 ThI
6920.0394 ThI
6924.6611 ThI
6925.0094 ArI
6910
6915
6920
6925
6930
6935
6936.6528 ThI
6937.6642 ArI
6942.5380 ThI
6943.6105 ThI
6945.4902 ThI
6946.2134 ThI
6947.9170 ThI
6948.3052 ThI
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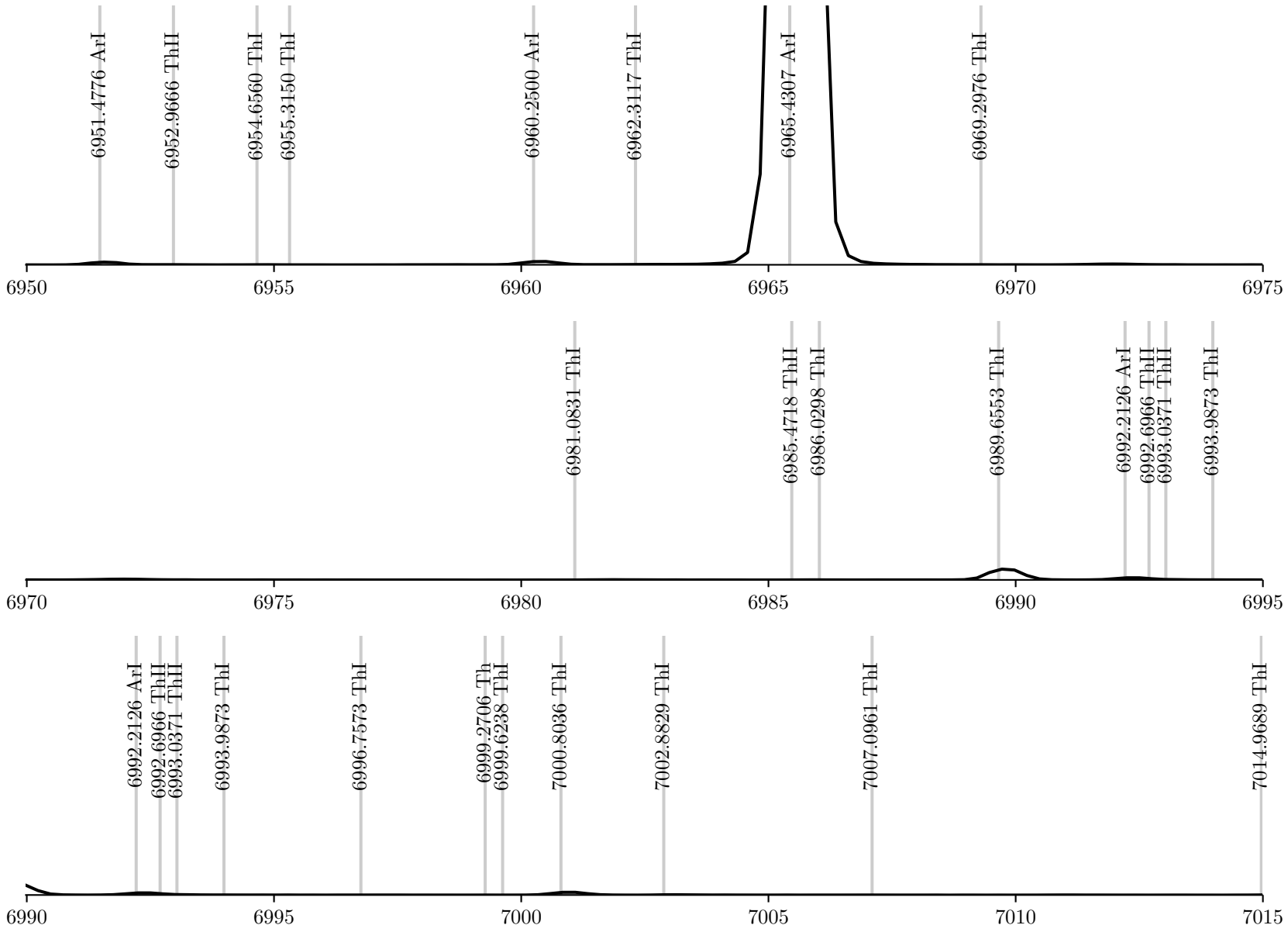
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6969.2976 ThI
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6981.0831 ThI
6985.4718 ThII
6986.0298 ThI
6989.6553 ThI
6992.2126 ArI
6992.6966 ThII
6993.0371 ThII
6993.9873 ThI
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6985
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6992.2126 ArI
6992.6966 ThII
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6993.9873 ThI
6996.7573 ThI
6999.2706 Th
6999.6238 ThI
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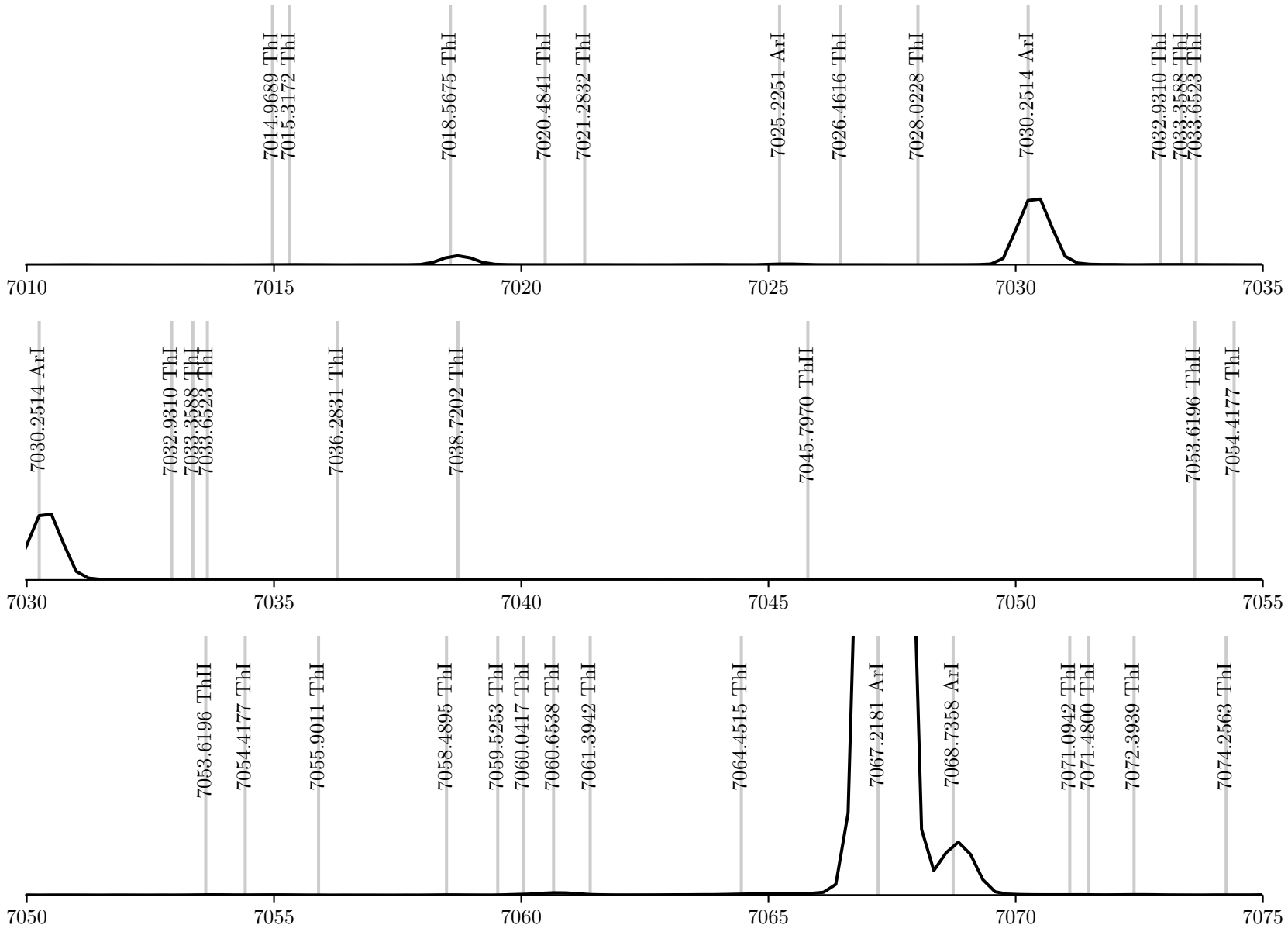
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7030
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7055.9011 ThI
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7060.0417 ThI
7060.6538 ThI
7061.3942 ThI
7064.4515 ThI
7067.2181 ArI
7068.7358 ArI
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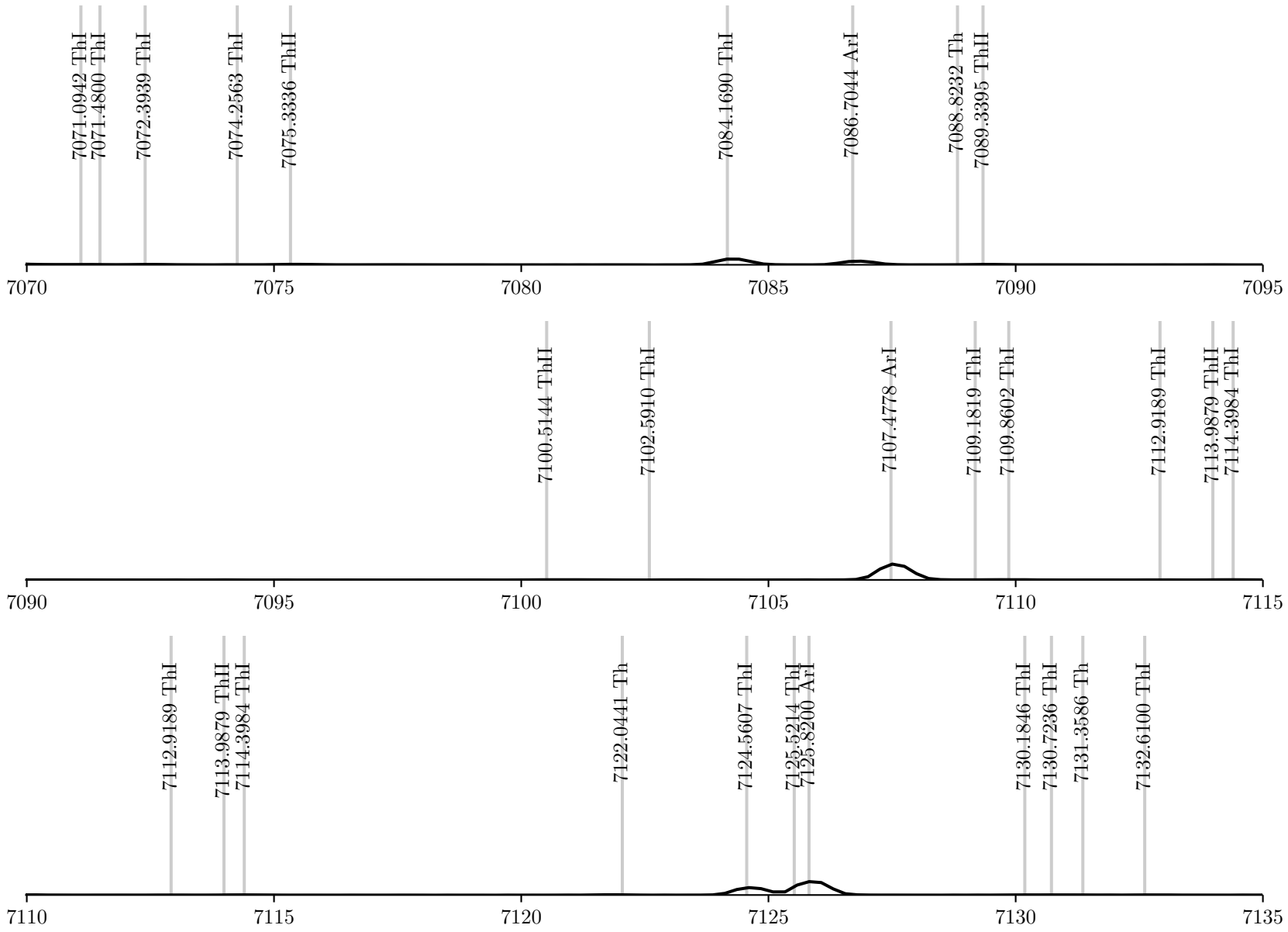
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7095
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7107.4778 ArI
7109.1819 ThI
7109.8602 ThI
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7113.9879 ThII
7114.3984 ThI
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7112.9189 ThI
7113.9879 ThII
7114.3984 ThI
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7124.5607 ThI
7125.5214 ThI
7125.8200 ArI
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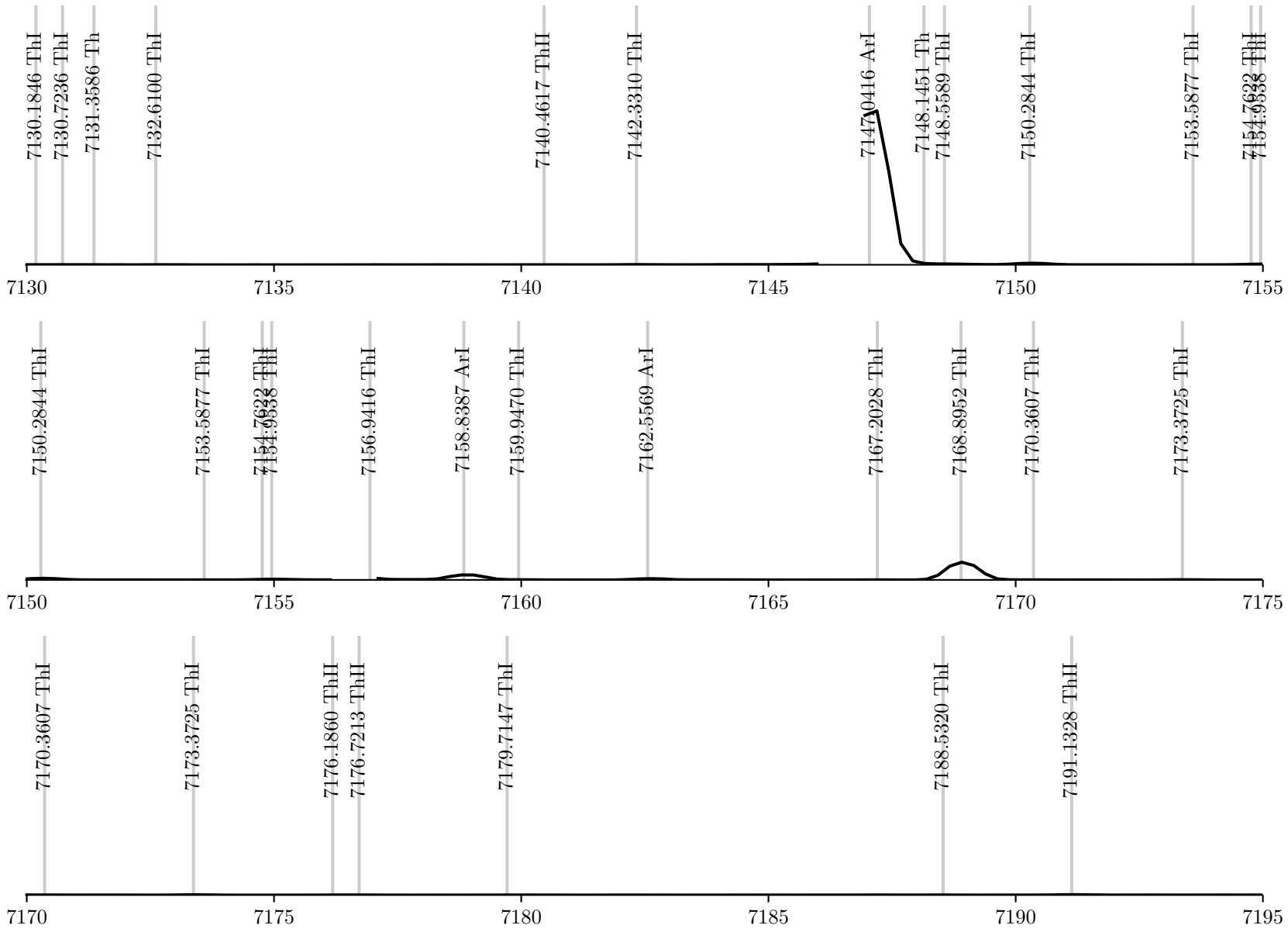
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7142.3310 ThI
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7148.1451 Th
7148.5589 ThI
7150.2844 ThI
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7154.7632 ThI
7154.9338 ThI
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7154.9338 ThI
7156.9416 ThI
7158.8387 ArI
7159.9470 ThI
7162.5569 ArI
7167.2028 ThI
7168.8952 ThI
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7170.3607 ThI
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7176.7213 ThII
7179.7147 ThI
7188.5320 ThI
7191.1328 ThII
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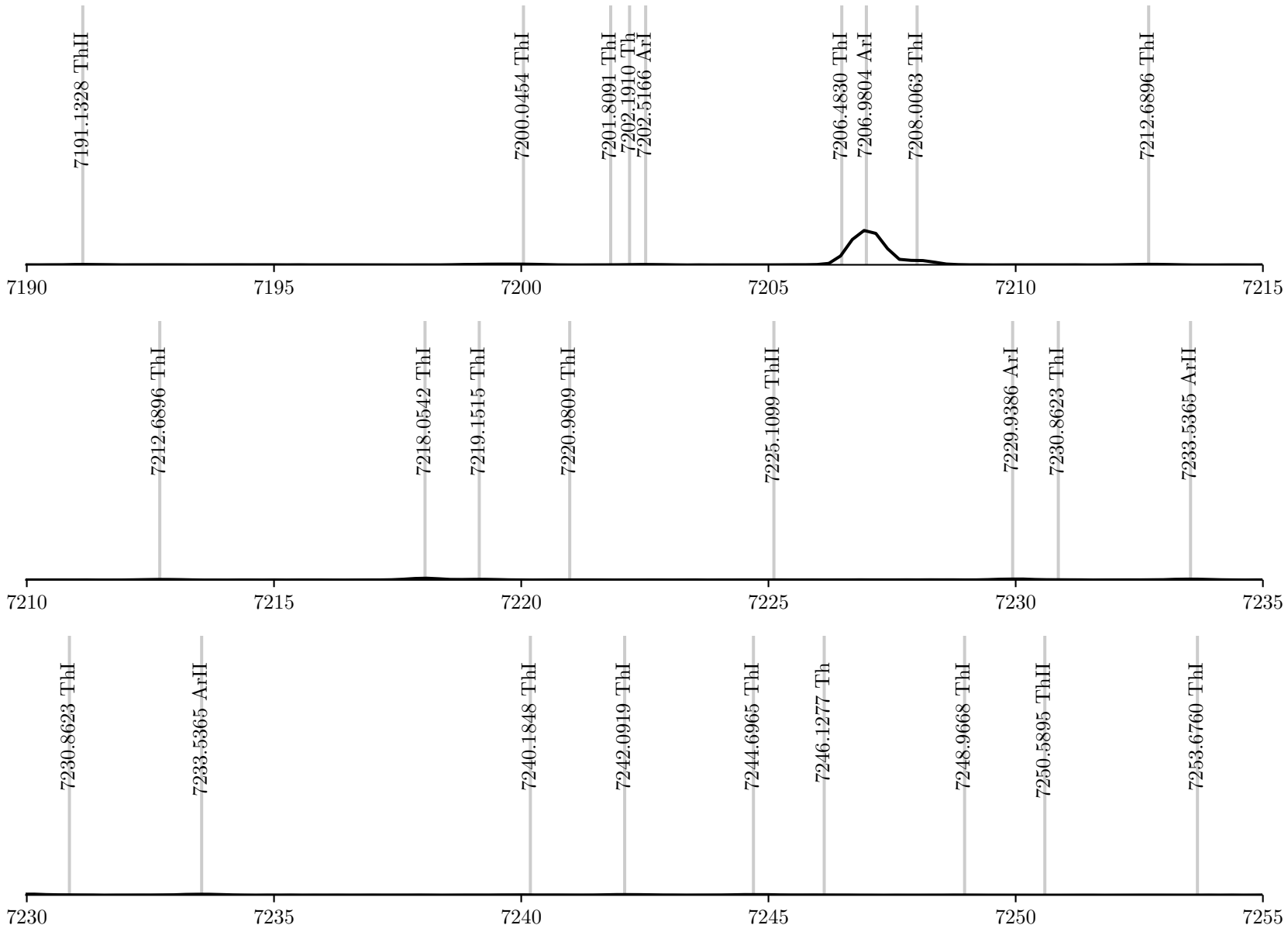
7191.1328 ThII
7200.0454 ThI
7201.8091 ThI
7202.1910 Th
7202.5166 ArI
7206.4830 ThI
7206.9804 ArI
7208.0063 ThI
7212.6896 ThI
7190
7195
7200
7205
7210
7215
7212.6896 ThI
7218.0542 ThI
7219.1515 ThI
7220.9809 ThI
7225.1099 ThII
7229.9386 ArI
7230.8623 ThI
7233.5365 ArII
7210
7215
7220
7225
7230
7235
7230.8623 ThI
7233.5365 ArII
7240.1848 ThI
7242.0919 ThI
7244.6965 ThI
7246.1277 Th
7248.9668 ThI
7250.5895 ThII
7253.6760 ThI
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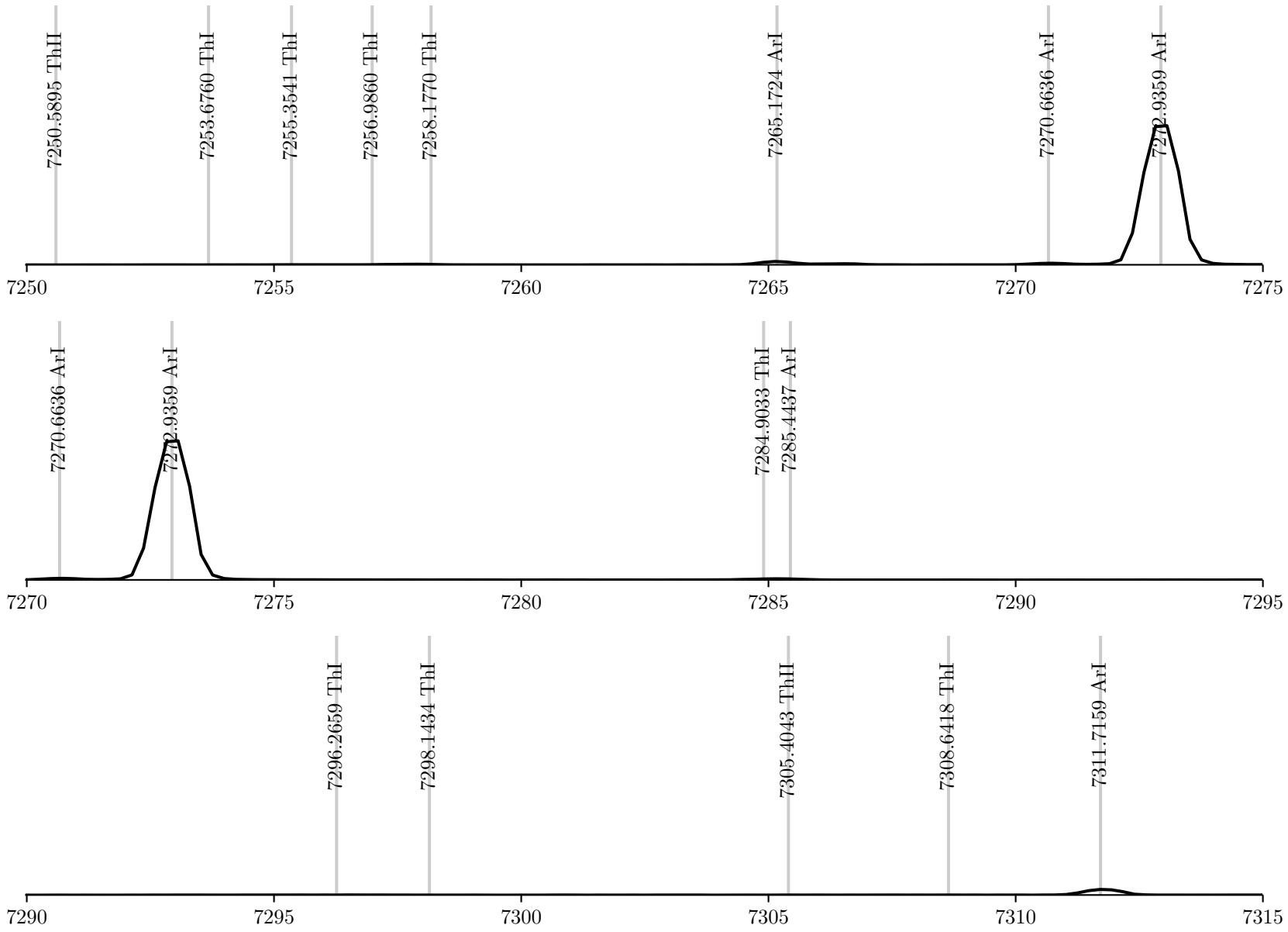
7250
7255


7250.5895 ThII
7253.6760 ThI
7255.3541 ThI
7256.9860 ThI
7258.1770 ThI
7265.1724 ArI
7270.6636 ArI
7272.9359 ArI
7250
7255
7260
7265
7270
7275
7270.6636 ArI
7272.9359 ArI
7284.9033 ThI
7285.4437 ArI
7270
7275
7280
7285
7290
7295
7296.2659 ThI
7298.1434 ThI
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7308.6418 ThI
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7305
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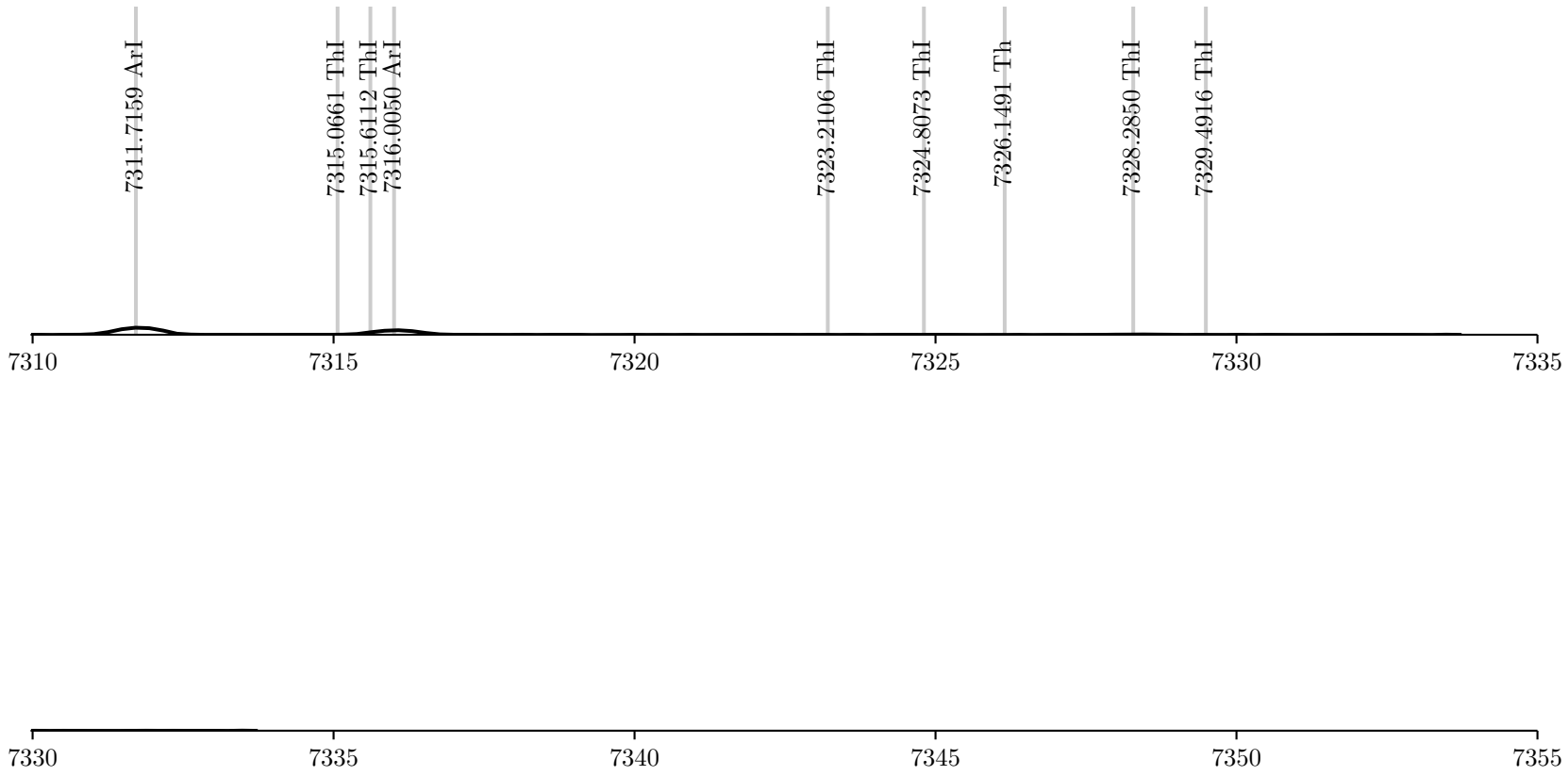
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7315.0661 ThI
7315.6112 ThI
7316.0050 ArI
7323.2106 ThI
7324.8073 ThI
7326.1491 Th
7328.2850 ThI
7329.4916 ThI
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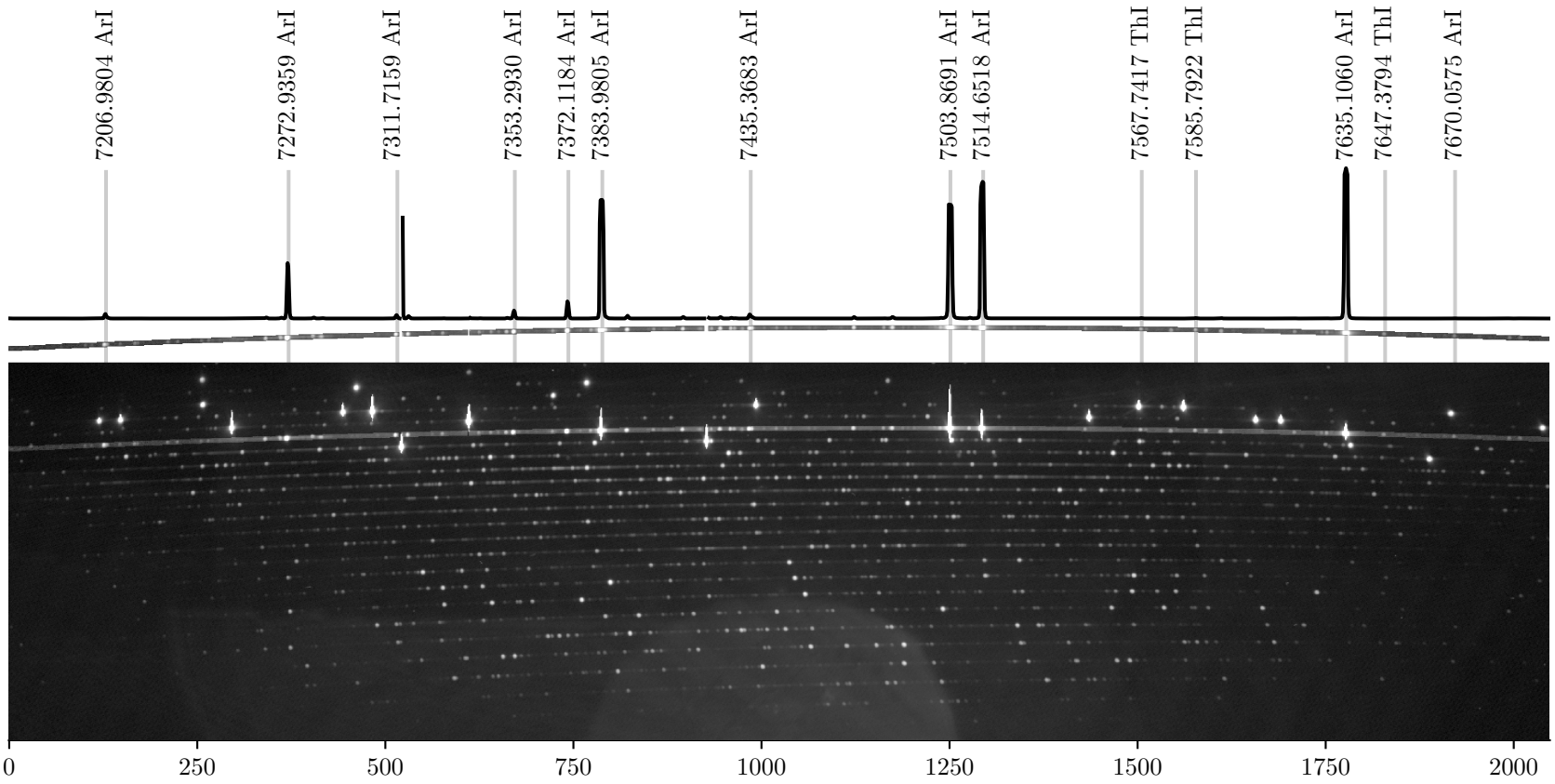
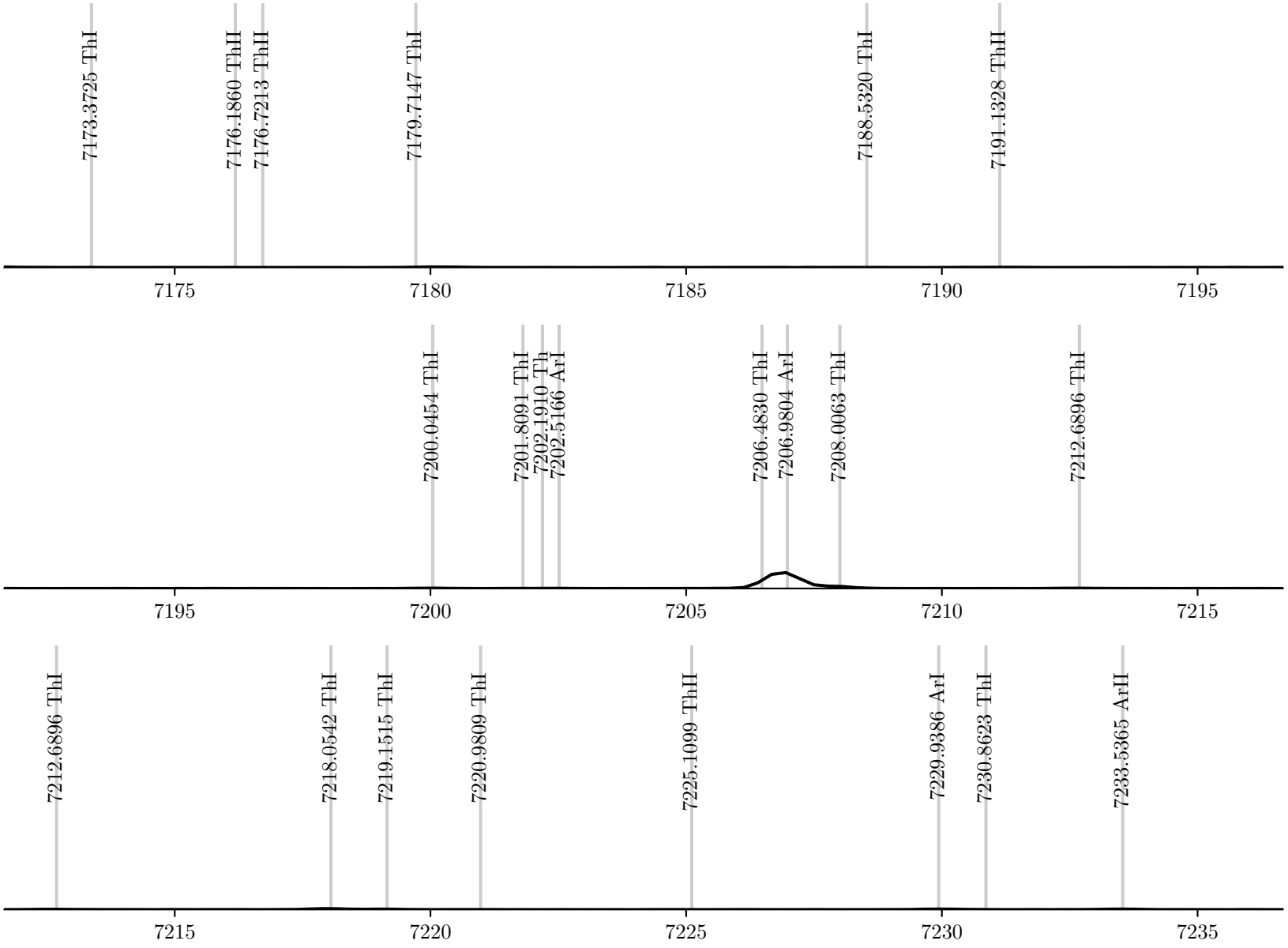
7355

Order 20



Dispersion:	0.258 Å pix^{-1}
Central wavelength:	7445 Å
Wavelength range:	7172 Å – 7701 Å


7173.3725 ThI
7176.1860 ThII
7176.7213 ThII
7179.7147 ThI
7188.5320 ThI
7191.1328 ThII
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7180
7185
7190
7195
7200.0454 ThI
7201.8091 ThI
7202.1910 Th
7202.5166 ArI
7206.4830 ThI
7206.9804 ArI
7208.0063 ThI
7212.6896 ThI
7195
7200
7205
7210
7215
7212.6896 ThI
7218.0542 ThI
7219.1515 ThI
7220.9809 ThI
7225.1099 ThII
7229.9386 ArI
7230.8623 ThI
7233.5365 ArII
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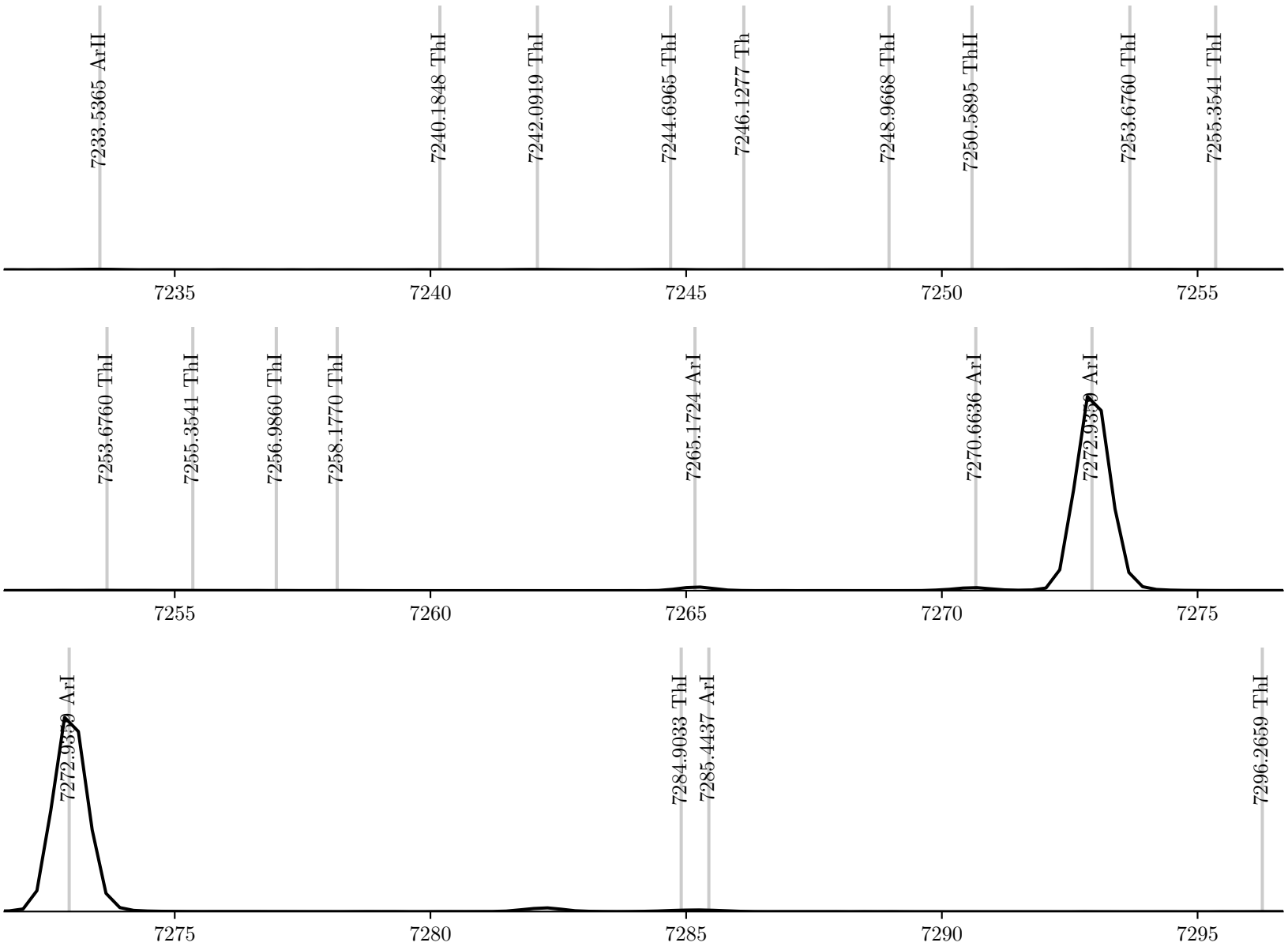
7233.5365 ArII
7240.1848 ThI
7242.0919 ThI
7244.6965 ThI
7246.1277 Th
7248.9668 ThI
7250.5895 ThII
7253.6760 ThI
7255.3541 ThI
7235
7240
7245
7250
7255
7253.6760 ThI
7255.3541 ThI
7256.9860 ThI
7258.1770 ThI
7265.1724 ArI
7270.6636 ArI
7272.9359 ArI
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7265
7270
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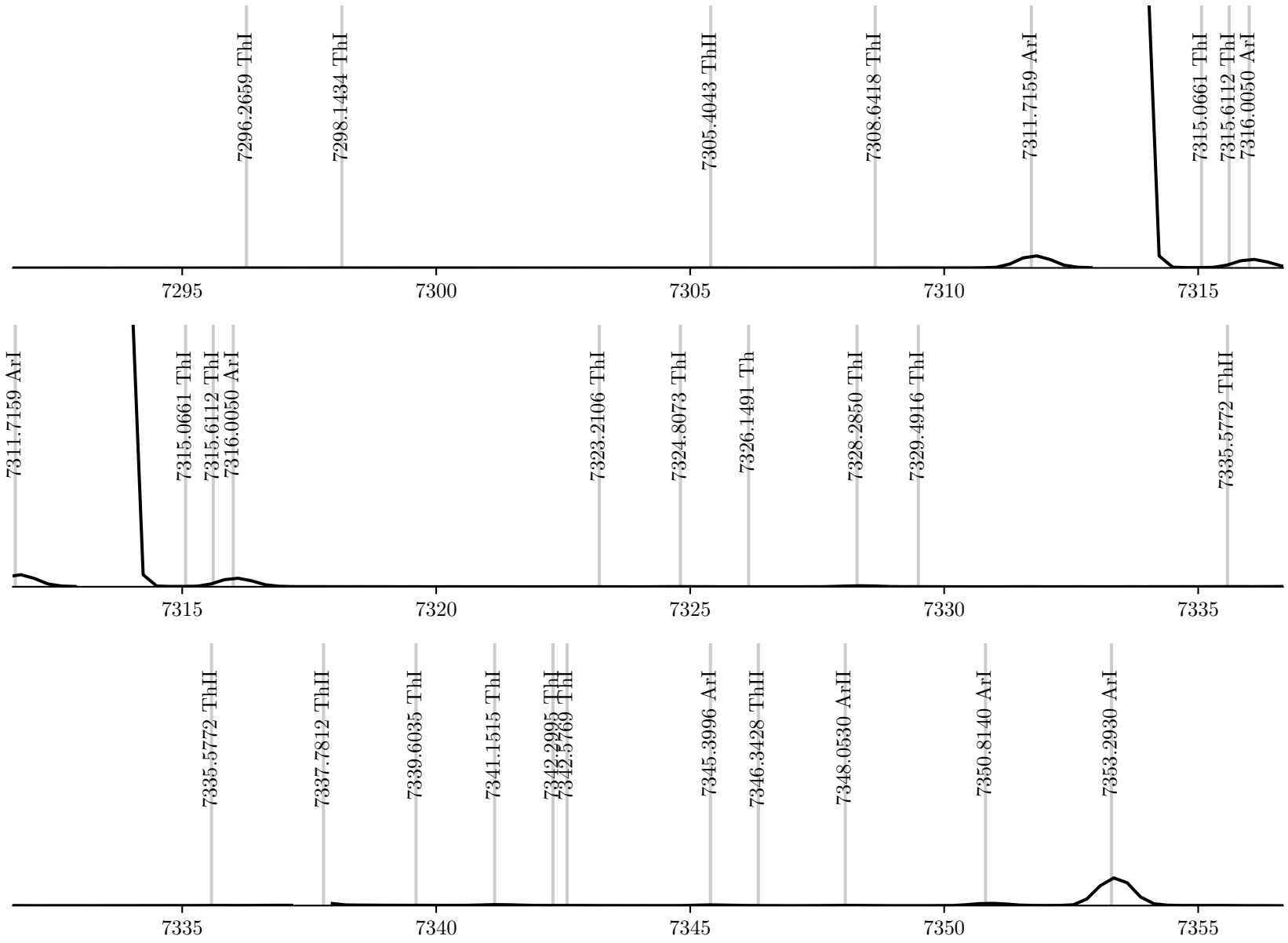
7296.2659 ThI
7298.1434 ThI
7305.4043 ThII
7308.6418 ThI
7311.7159 ArI
7315.0661 ThI
7315.6112 ThI
7316.0050 ArI
7295
7300
7305
7310
7315
7311.7159 ArI
7315.0661 ThI
7315.6112 ThI
7316.0050 ArI
7323.2106 ThI
7324.8073 ThI
7326.1491 Th
7328.2850 ThI
7329.4916 ThI
7335.5772 ThII
7315
7320
7325
7330
7335
7335.5772 ThII
7337.7812 ThII
7339.6035 ThI
7341.1515 ThI
7345.3996 ArI
7346.3428 ThII
7348.0530 ArII
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7345
7350
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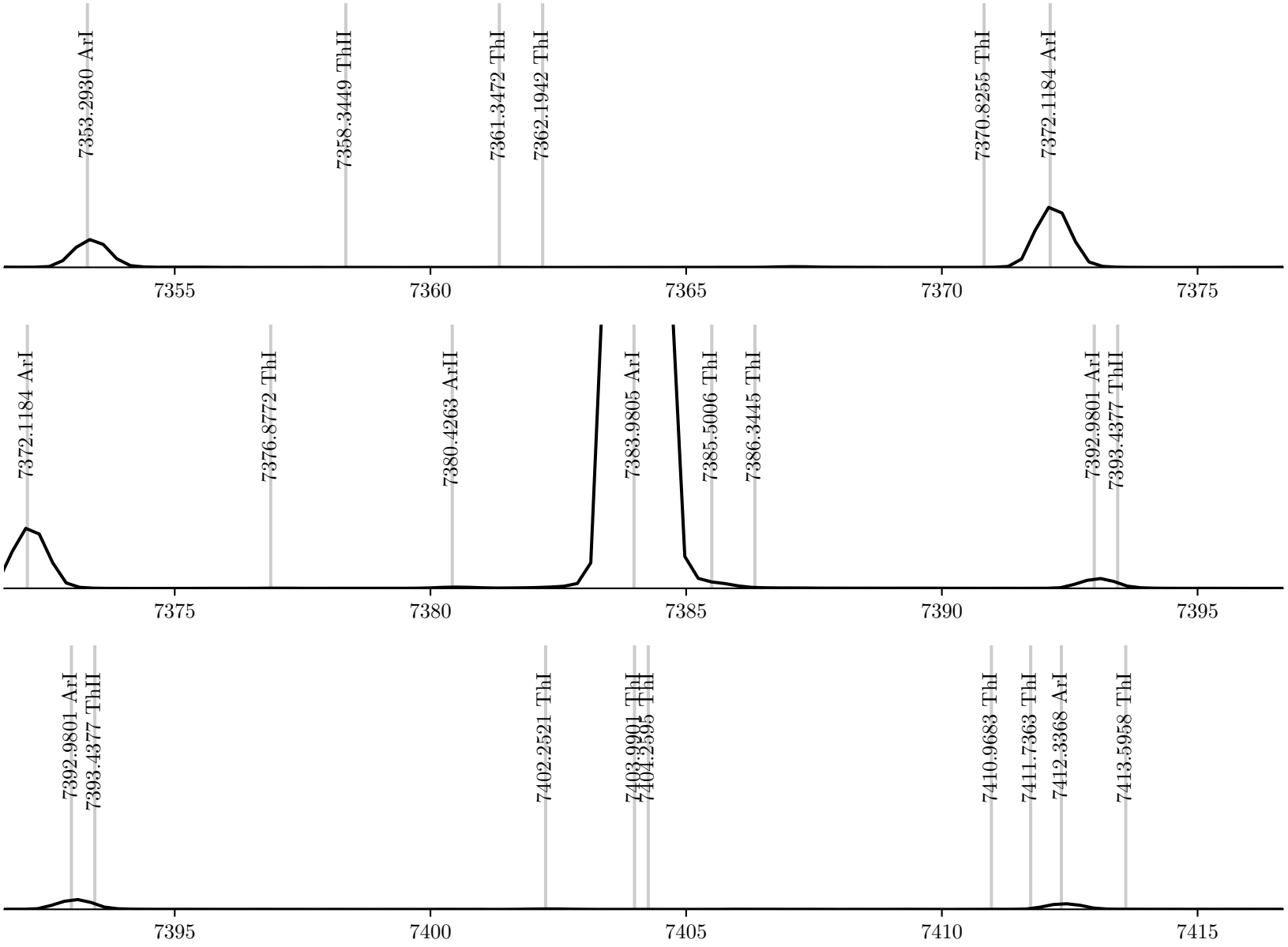
7353.2930 ArI
7358.3449 ThII
7361.3472 ThI
7362.1942 ThI
7370.8255 ThI
7372.1184 ArI
7355
7360
7365
7370
7375
7372.1184 ArI
7376.8772 ThI
7380.4263 ArII
7383.9805 ArI
7385.5006 ThI
7386.3445 ThI
7392.9801 ArI
7393.4377 ThII
7375
7380
7385
7390
7395
7392.9801 ArI
7393.4377 ThII
7402.2521 ThI
7403.9901 ThI
7404.2395 ThI
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7411.7363 ThI
7412.3368 ArI
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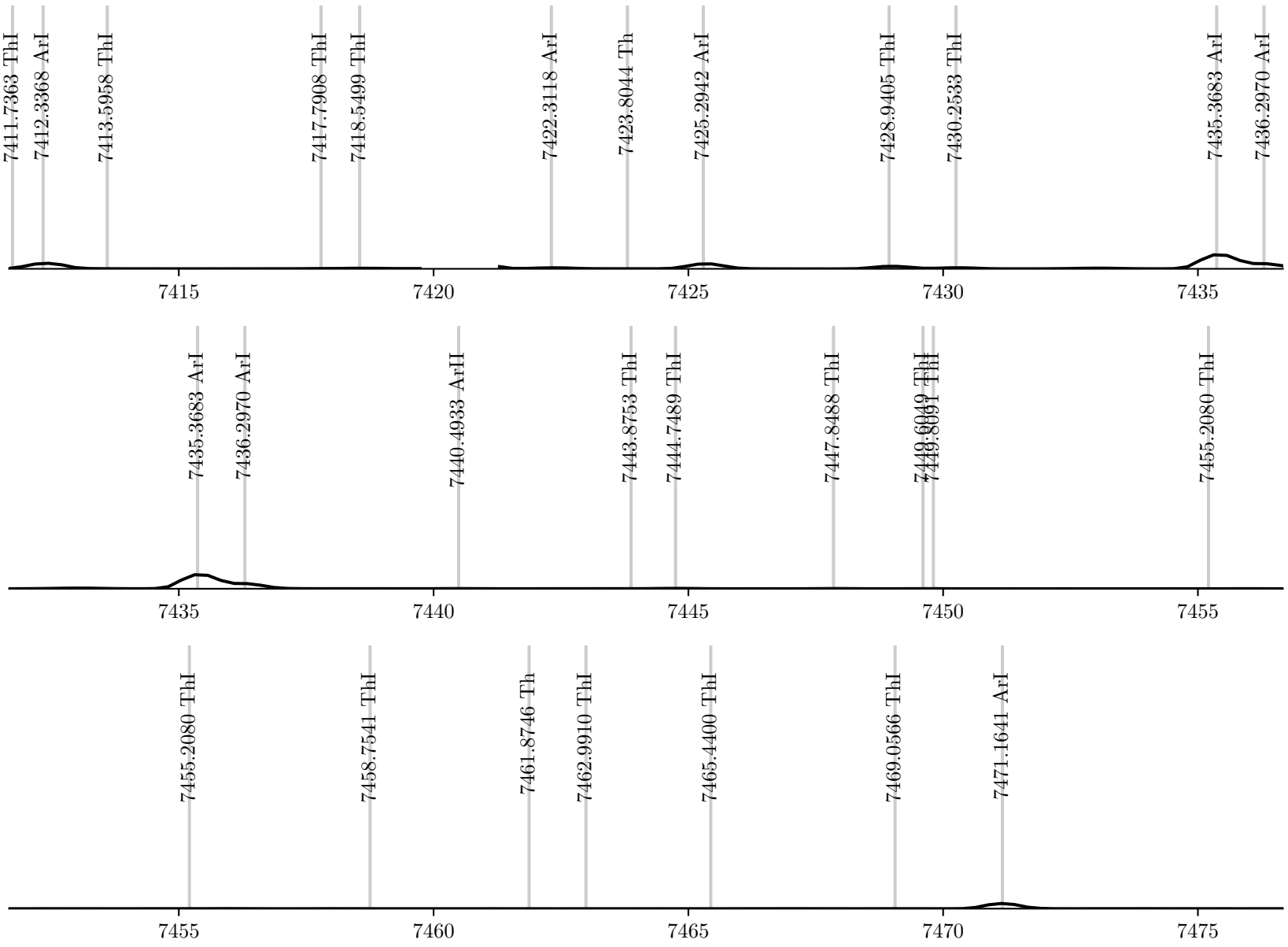
7405
7410
7415


7411.7363 ThI
7412.3368 ArI
7413.5958 ThI
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7418.5499 ThI
7422.3118 ArI
7423.8044 Th
7425.2942 ArI
7428.9405 ThI
7430.2533 ThI
7435.3683 ArI
7436.2970 ArI
7415
7420
7425
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7435
7435.3683 ArI
7436.2970 ArI
7440.4933 ArII
7443.8753 ThI
7444.7489 ThI
7447.8488 ThI
7455.2080 ThI
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7455.2080 ThI
7458.7541 ThI
7461.8746 Th
7462.9910 ThI
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7469.0566 ThI
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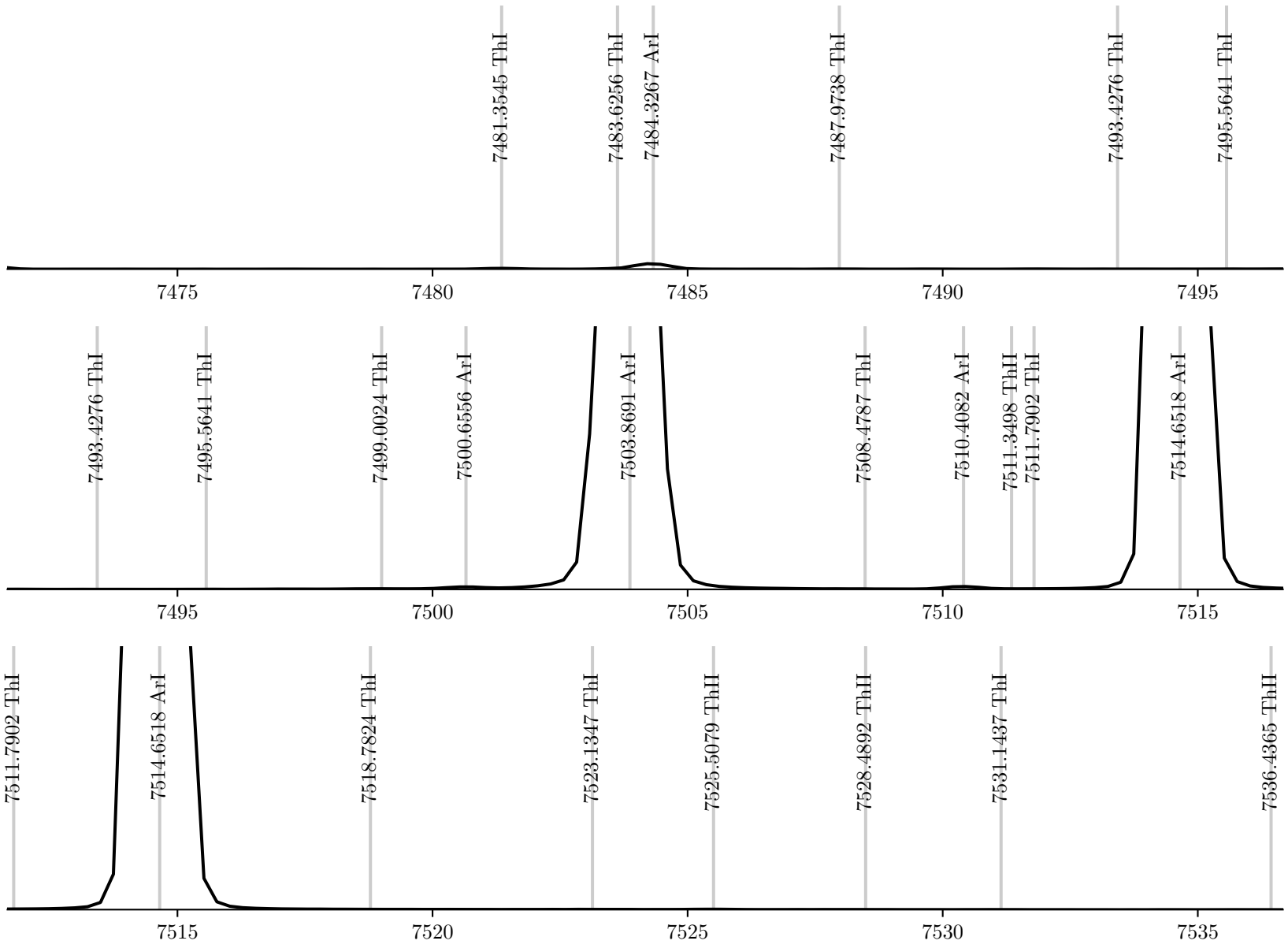
7481.3545 ThI
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7484.3267 ArI
7487.9738 ThI
7493.4276 ThI
7495.5641 ThI
7475
7480
7485
7490
7495
7493.4276 ThI
7495.5641 ThI
7499.0024 ThI
7500.6556 ArI
7503.8691 ArI
7508.4787 ThI
7510.4082 ArI
7511.3498 ThII
7511.7902 ThI
7514.6518 ArI
7495
7500
7505
7510
7515
7511.7902 ThI
7514.6518 ArI
7518.7824 ThI
7523.1347 ThI
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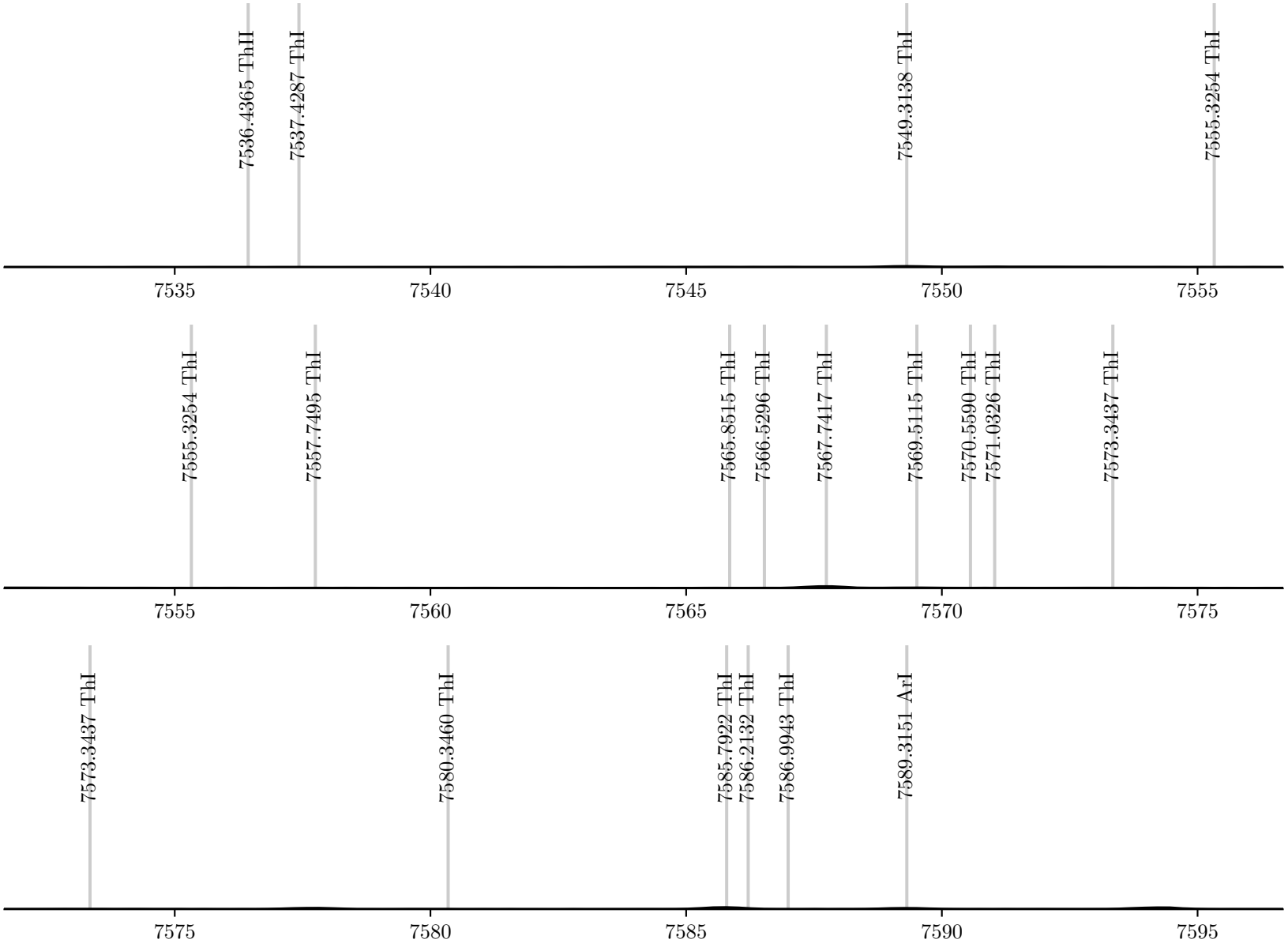
7536.4365 ThII
7537.4287 ThI
7549.3138 ThI
7555.3254 ThI
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7545
7550
7555
7555.3254 ThI
7557.7495 ThI
7565.8515 ThI
7566.5296 ThI
7567.7417 ThI
7569.5115 ThI
7570.5590 ThI
7571.0326 ThI
7573.3437 ThI
7555
7560
7565
7570
7575
7573.3437 ThI
7580.3460 ThI
7585.7922 ThI
7586.2132 ThI
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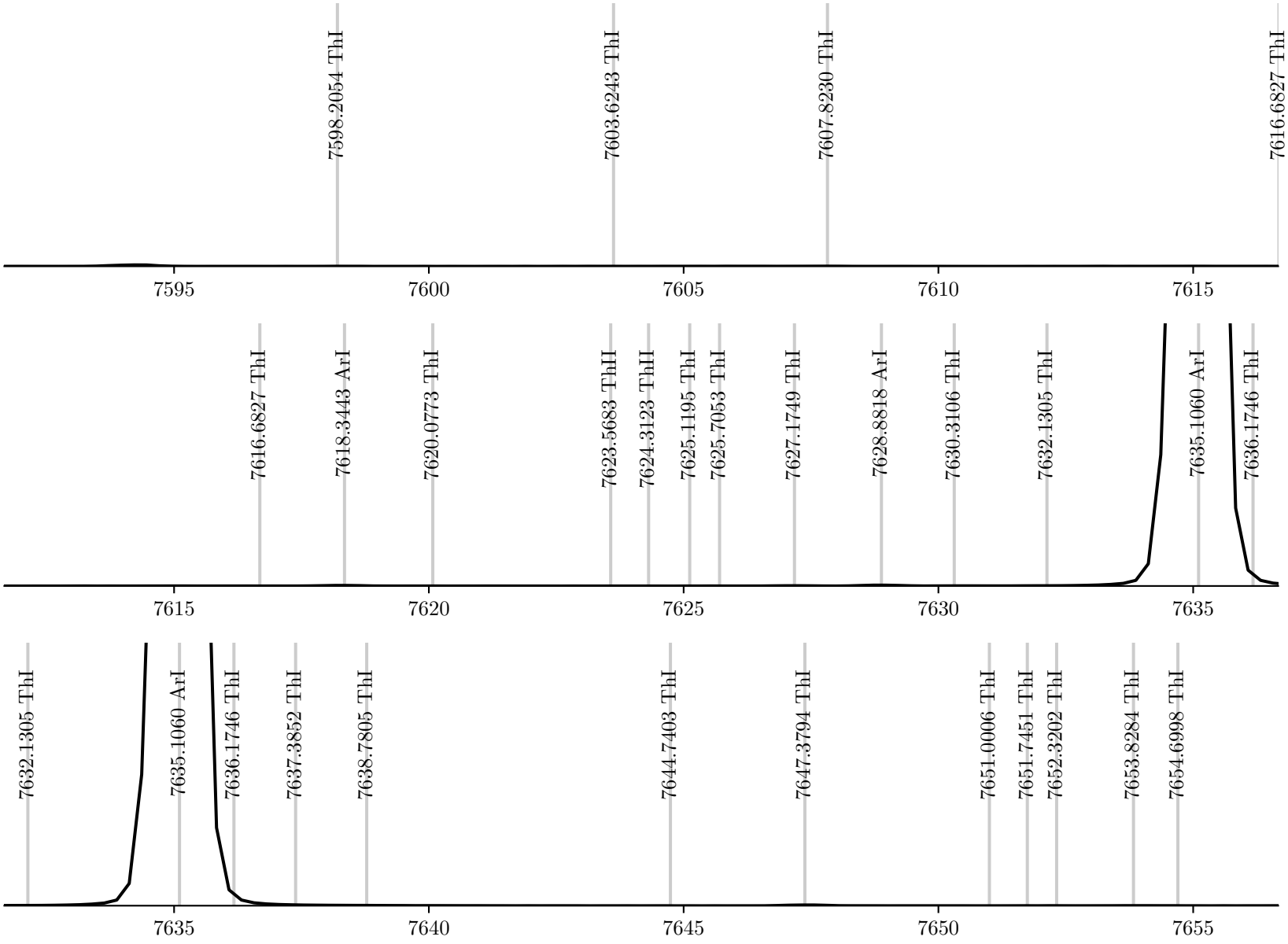
7595


7598.2054 ThI
7603.6243 ThI
7607.8230 ThI
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7595
7600
7605
7610
7615
7616.6827 ThI
7618.3443 ArI
7620.0773 ThI
7623.5683 ThII
7624.3123 ThII
7625.1195 ThI
7625.7053 ThI
7627.1749 ThI
7628.8818 ArI
7630.3106 ThI
7632.1305 ThI
7635.1060 ArI
7636.1746 ThI
7615
7620
7625
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7635
7632.1305 ThI
7635.1060 ArI
7636.1746 ThI
7637.3852 ThI
7638.7805 ThI
7644.7403 ThI
7647.3794 ThI
7651.0006 ThI
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7652.3202 ThI
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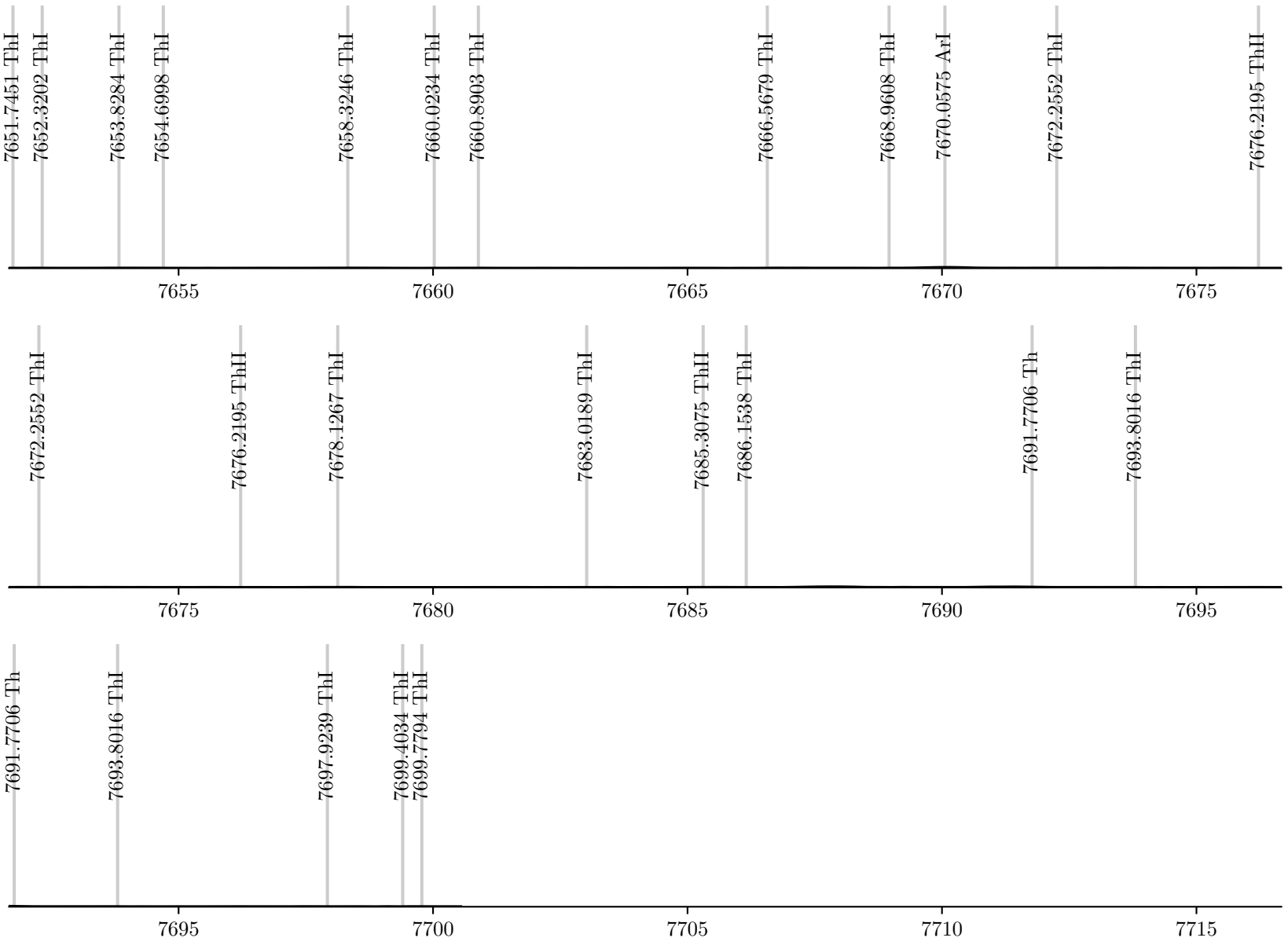
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7653.8284 ThI
7654.6998 ThI
7658.3246 ThI
7660.0234 ThI
7660.8903 ThI
7666.5679 ThI
7668.9608 ThI
7670.0575 ArI
7672.2552 ThI
7676.2195 ThII
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7675
7672.2552 ThI
7676.2195 ThII
7678.1267 ThI
7683.0189 ThI
7685.3075 ThII
7686.1538 ThI
7691.7706 Th
7693.8016 ThI
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7680
7685
7690
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7691.7706 Th
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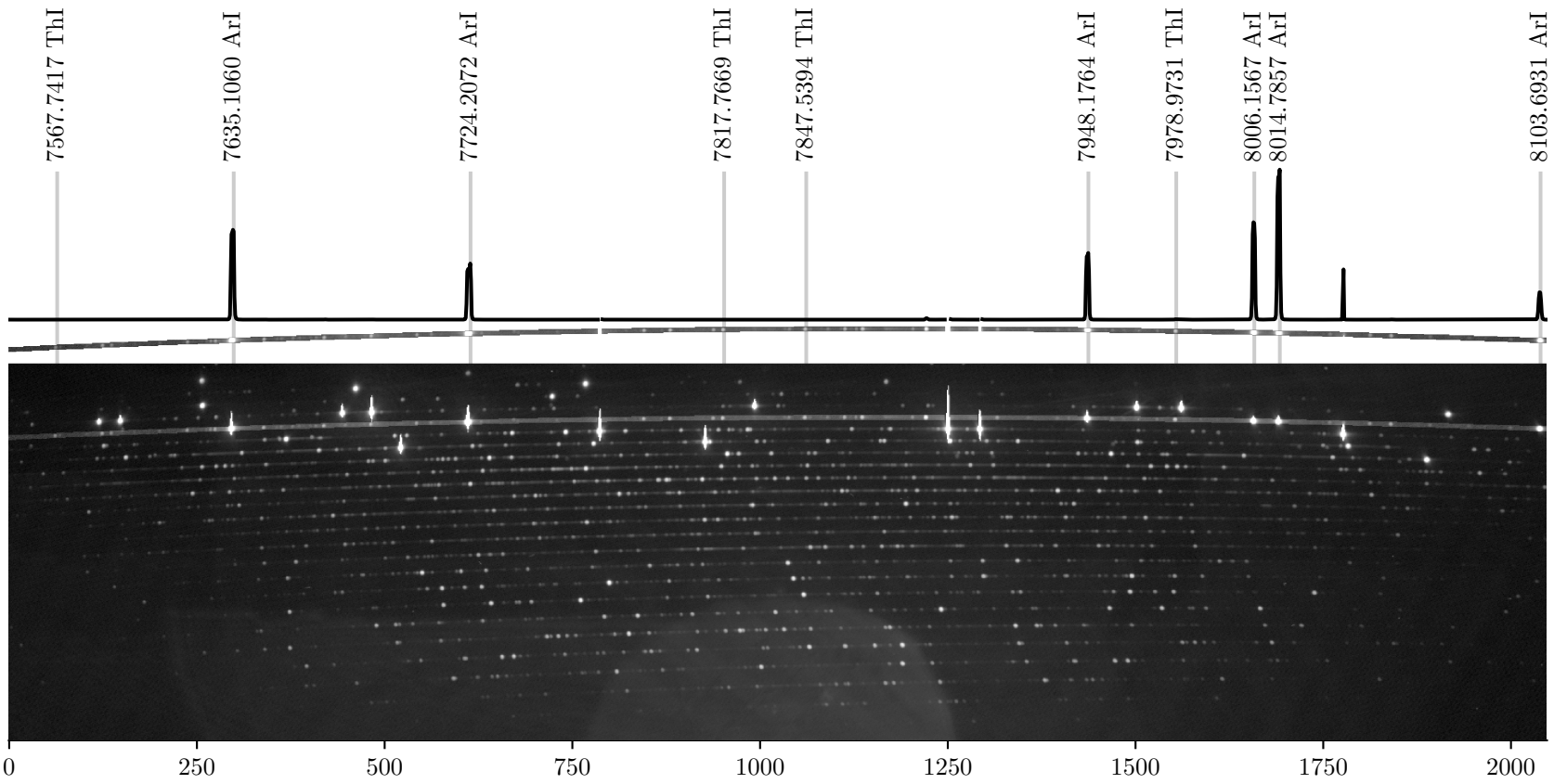
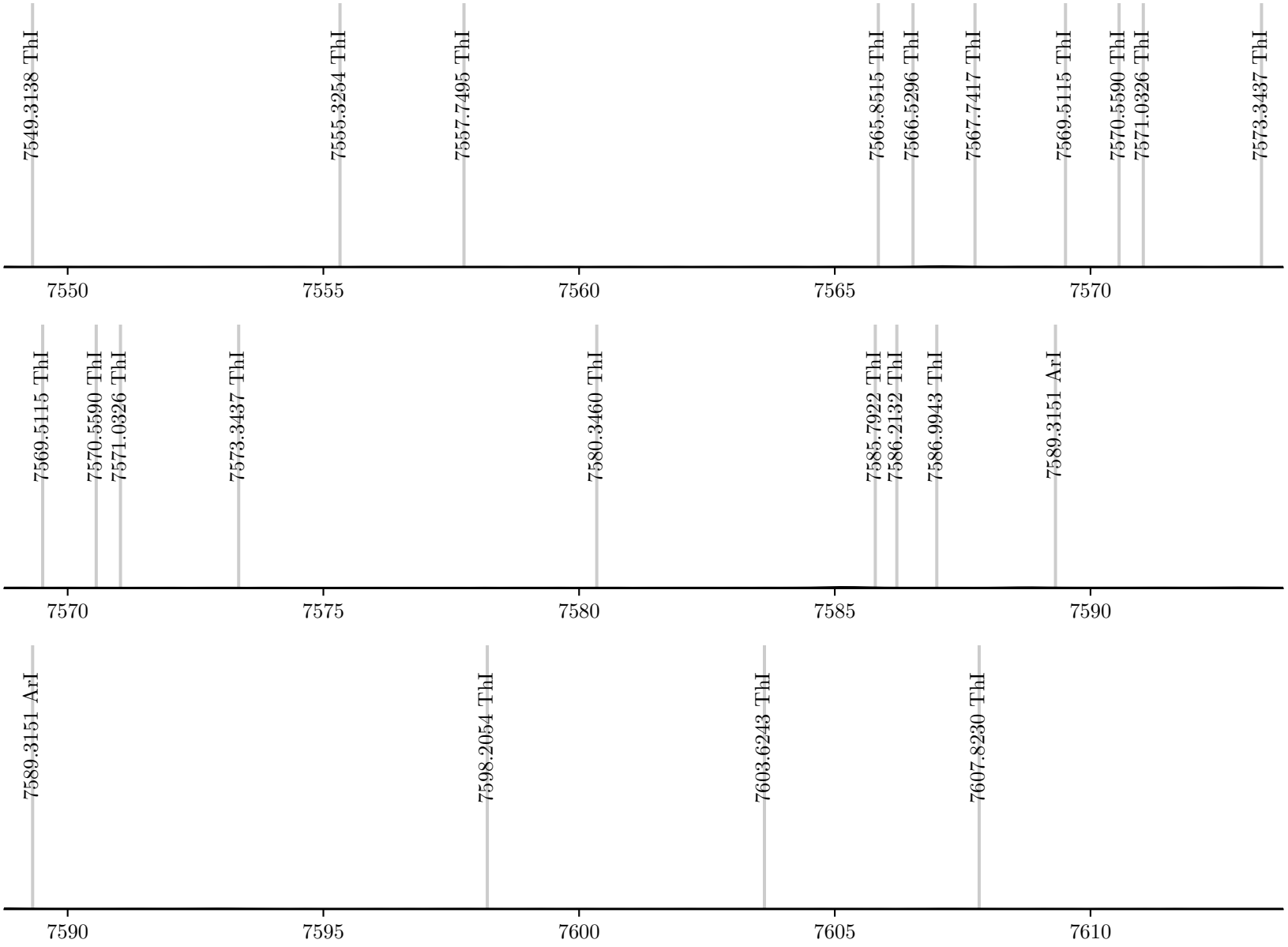
7715

Order 19



Dispersion:	0.272 Å pix^{-1}
Central wavelength:	7837 Å
Wavelength range:	7549 Å – 8106 Å


7549.3138 ThI
7555.3254 ThI
7557.7495 ThI
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7567.7417 ThI
7569.5115 ThI
7570.5590 ThI
7571.0326 ThI
7573.3437 ThI
7550
7555
7560
7565
7570
7569.5115 ThI
7570.5590 ThI
7571.0326 ThI
7573.3437 ThI
7580.3460 ThI
7585.7922 ThI
7586.2132 ThI
7586.9943 ThI
7589.3151 ArI
7570
7575
7580
7585
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7589.3151 ArI
7598.2054 ThI
7603.6243 ThI
7607.8230 ThI
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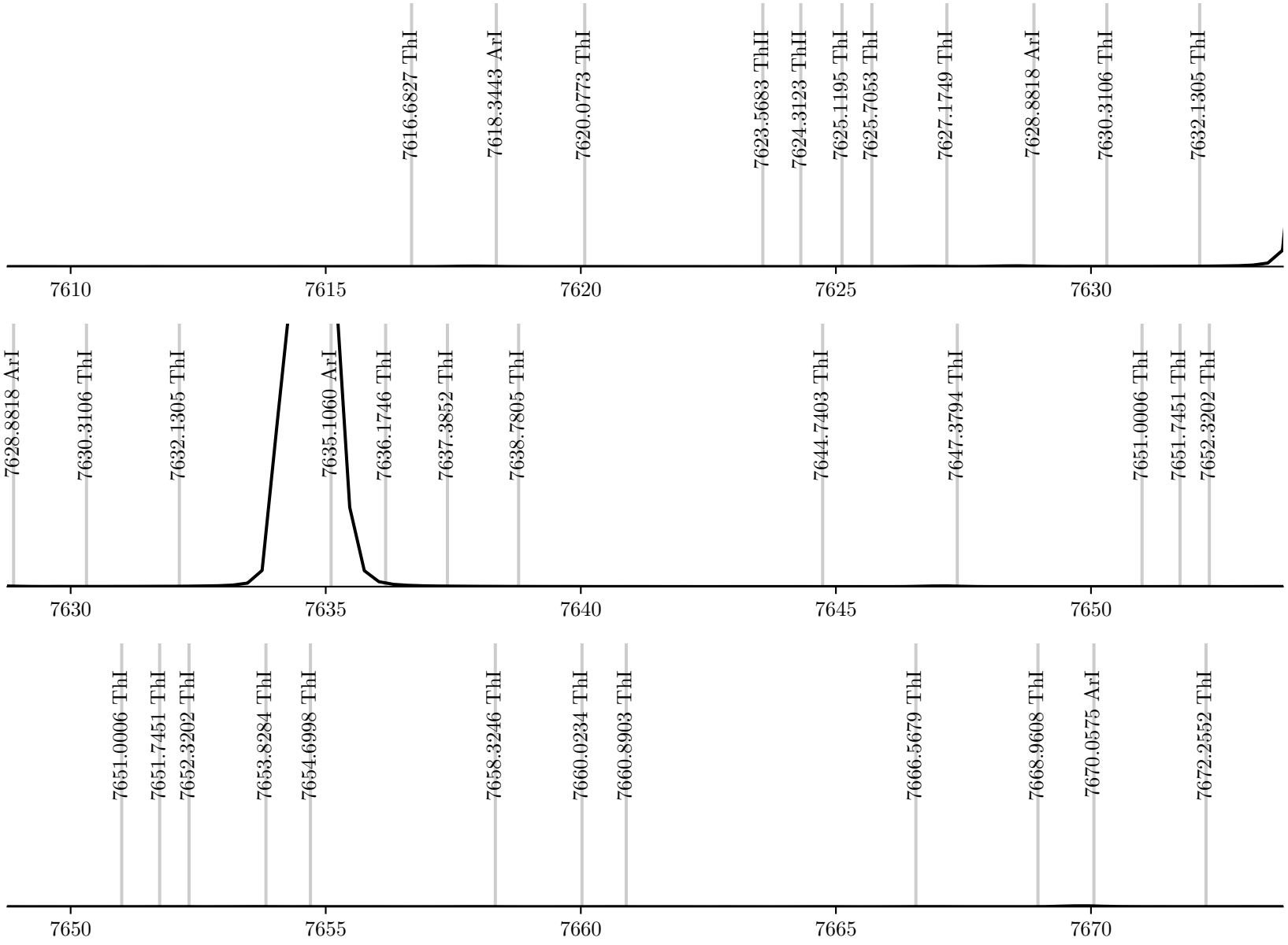
7600
7605
7610


7616.6827 ThI
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7623.5683 ThII
7624.3123 ThII
7625.1195 ThI
7625.7053 ThI
7627.1749 ThI
7628.8818 ArI
7630.3106 ThI
7632.1305 ThI
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7628.8818 ArI
7630.3106 ThI
7632.1305 ThI
7635.1060 ArI
7636.1746 ThI
7637.3852 ThI
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7647.3794 ThI
7651.0006 ThI
7651.7451 ThI
7652.3202 ThI
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7651.0006 ThI
7651.7451 ThI
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7653.8284 ThI
7654.6998 ThI
7658.3246 ThI
7660.0234 ThI
7660.8903 ThI
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7668.9608 ThI
7670.0575 ArI
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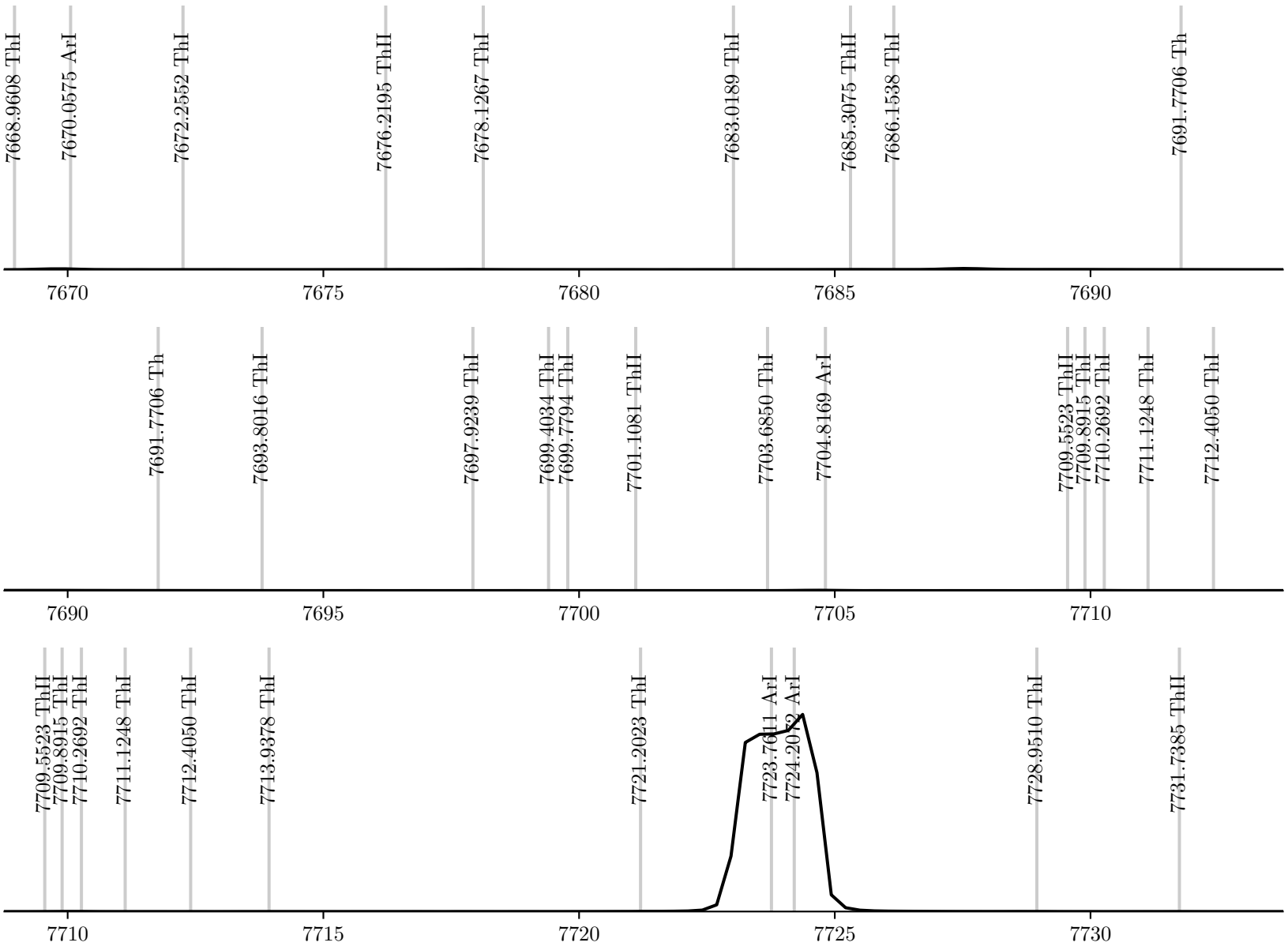
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7672.2552 ThI
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7685.3075 ThII
7686.1538 ThI
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7691.7706 Th
7693.8016 ThI
7697.9239 ThI
7699.4034 ThI
7699.7794 ThI
7701.1081 ThII
7703.6850 ThI
7704.8169 ArI
7709.5523 ThII
7709.8915 ThI
7710.2692 ThI
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7712.4050 ThI
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7710
7709.5523 ThII
7709.8915 ThI
7710.2692 ThI
7711.1248 ThI
7712.4050 ThI
7713.9378 ThI
7721.2023 ThI
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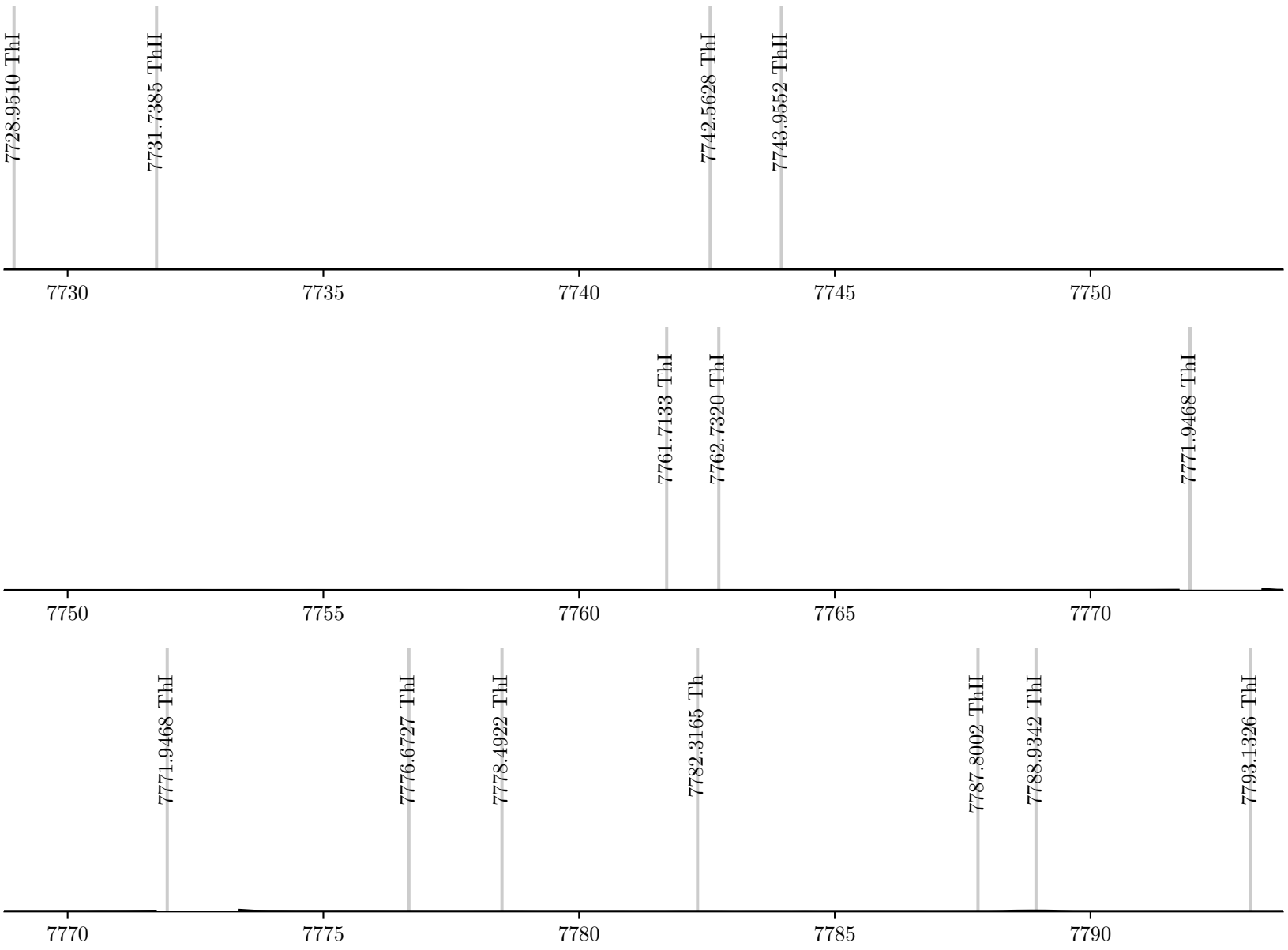
7728.9510 ThI
7731.7385 ThII
7742.5628 ThI
7743.9552 ThII
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7761.7133 ThI
7762.7320 ThI
7771.9468 ThI
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7771.9468 ThI
7776.6727 ThI
7778.4922 ThI
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7787.8002 ThII
7788.9342 ThI
7793.1326 ThI
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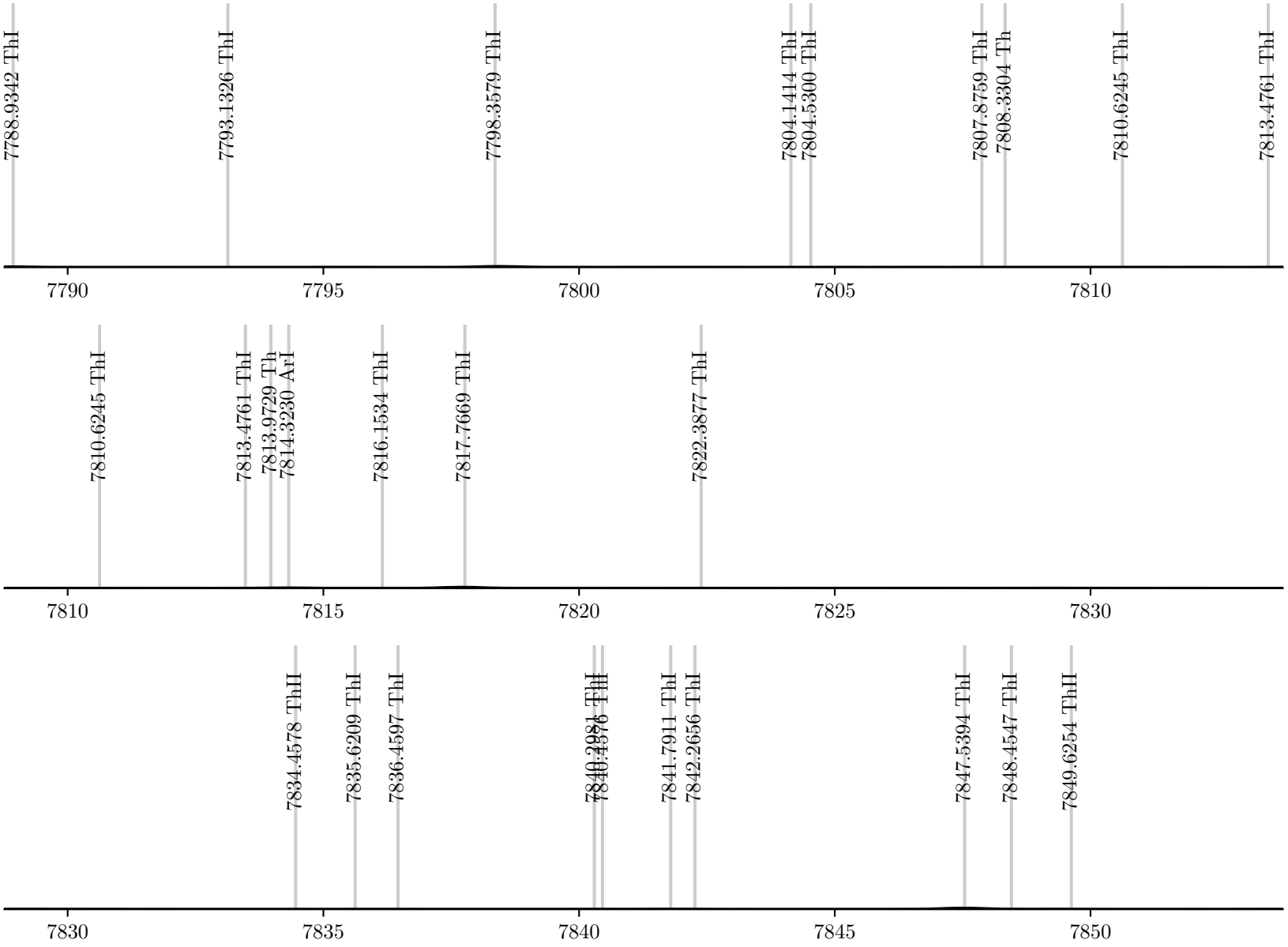
7788.9342 ThI
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7798.3579 ThI
7804.1414 ThI
7804.5300 ThI
7807.8759 ThI
7808.3304 Th
7810.6245 ThI
7813.4761 ThI
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7800
7805
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7810.6245 ThI
7813.4761 ThI
7813.9729 Th
7814.3230 ArI
7816.1534 ThI
7817.7669 ThI
7822.3877 ThI
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7834.4578 ThII
7835.6209 ThI
7836.4597 ThI
7841.7911 ThI
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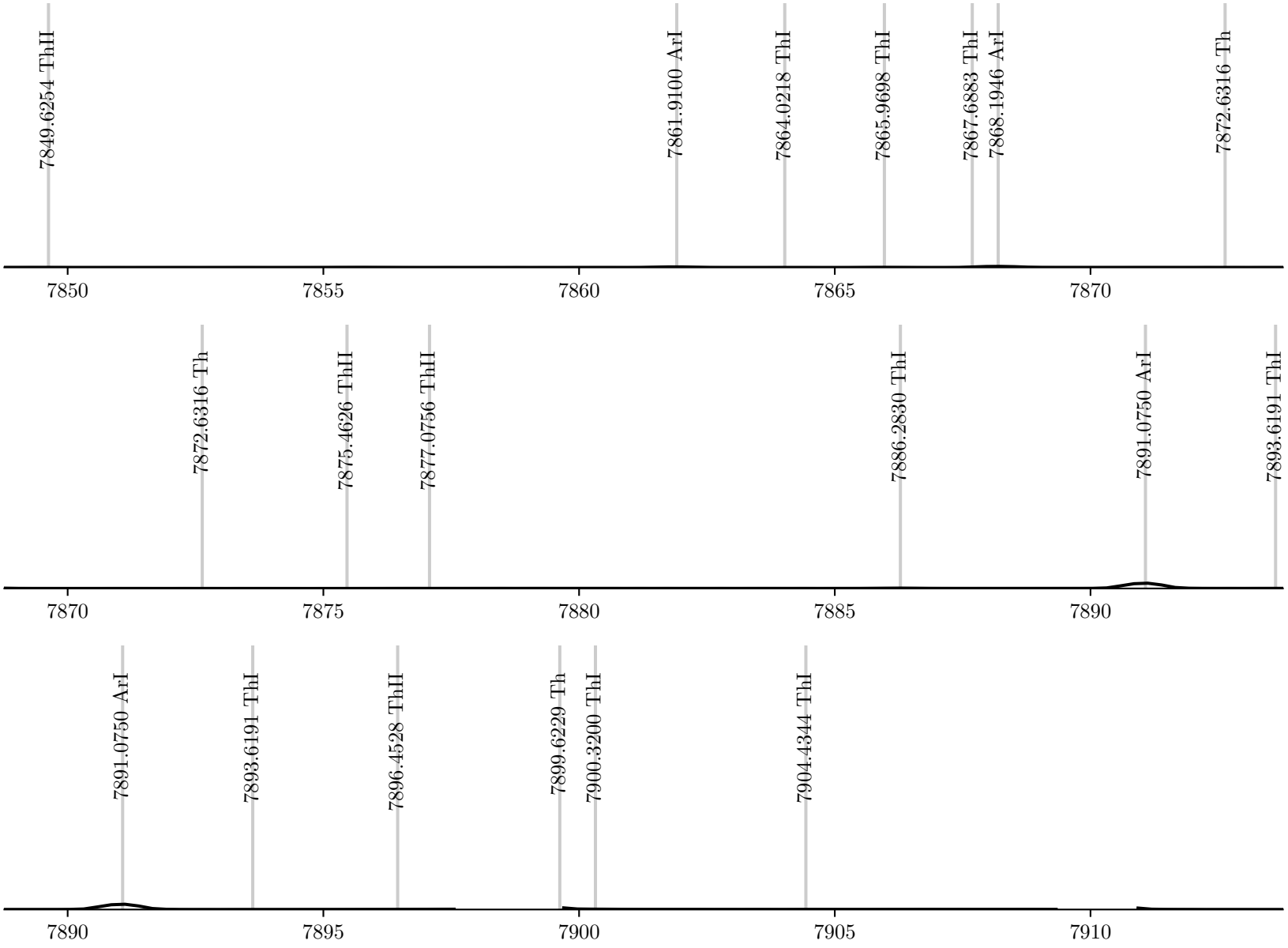
7849.6254 ThII
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7864.0218 ThI
7865.9698 ThI
7867.6883 ThI
7868.1946 ArI
7872.6316 Th
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7870
7872.6316 Th
7875.4626 ThII
7877.0756 ThII
7886.2830 ThI
7891.0750 ArI
7893.6191 ThI
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7880
7885
7890
7891.0750 ArI
7893.6191 ThI
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7900.3200 ThI
7904.4344 ThI
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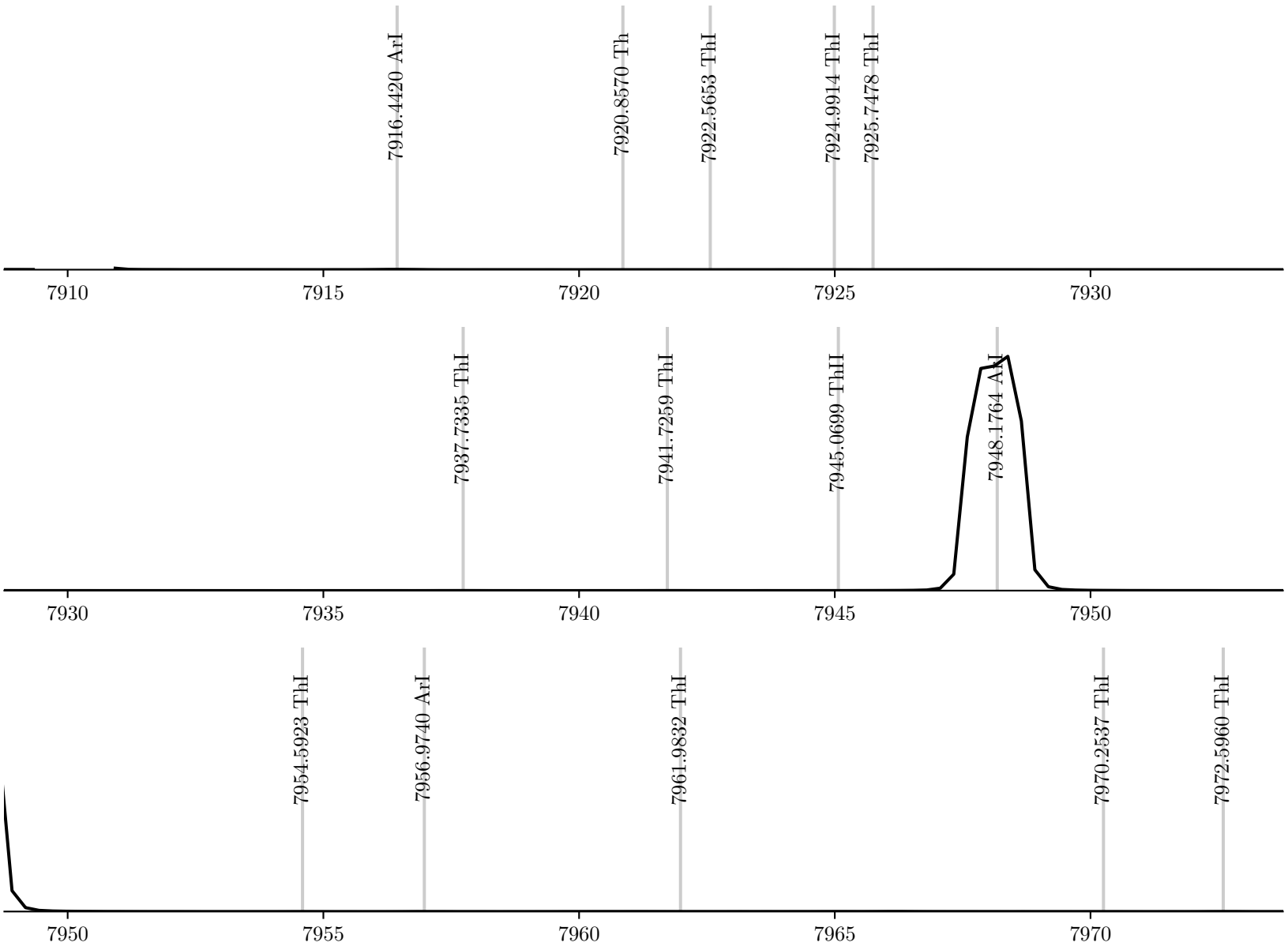
7900
7905
7910


7916.4420 ArI
7920.8570 Th
7922.5653 ThI
7924.9914 ThI
7925.7478 ThI
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7925
7930
7937.7335 ThI
7941.7259 ThI
7945.0699 ThII
7948.1764 AlI
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7954.5923 ThI
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7970.2537 ThI
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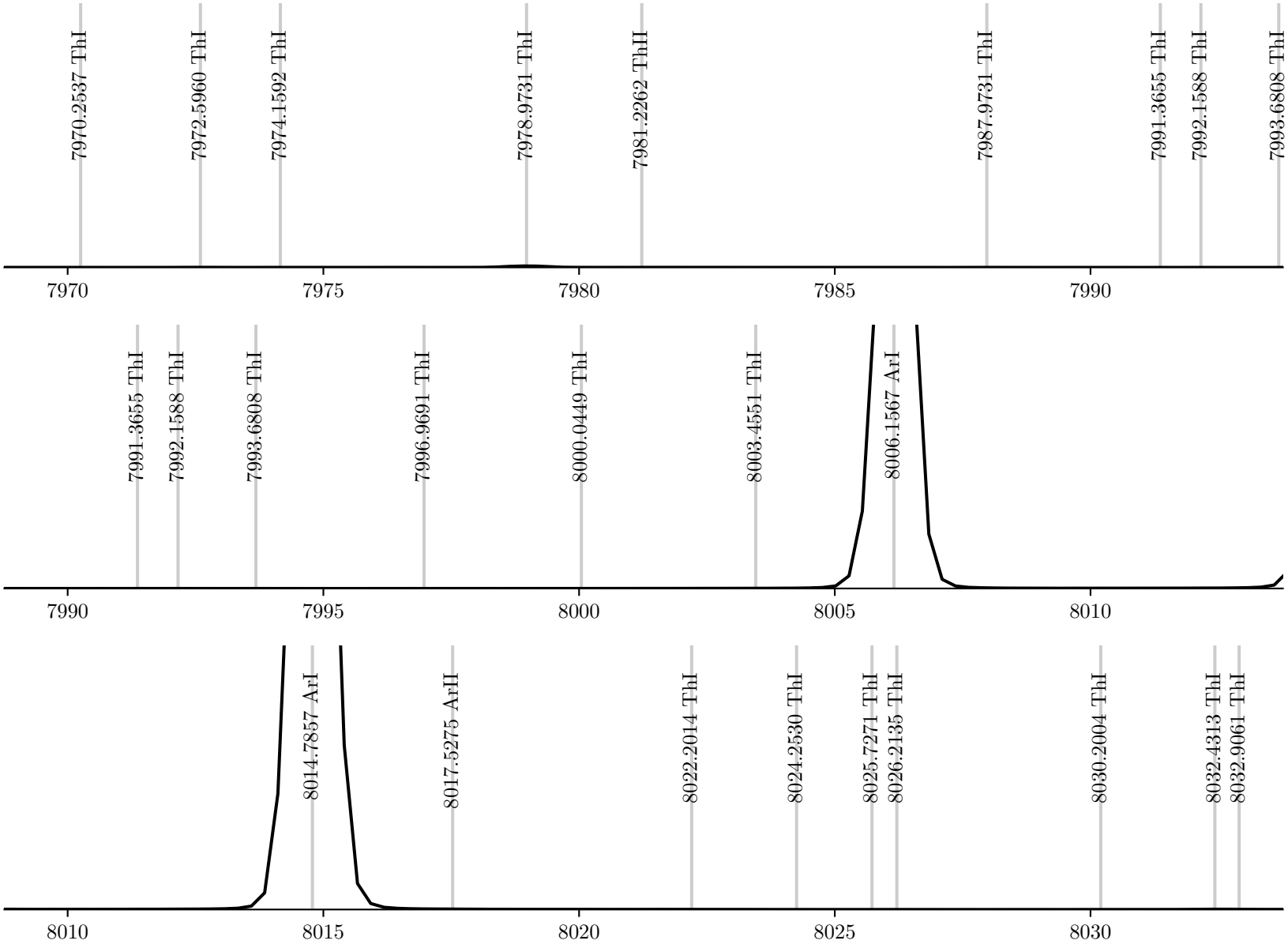
7970


7970.2537 ThI
7972.5960 ThI
7974.1592 ThI
7978.9731 ThI
7981.2262 ThII
7987.9731 ThI
7991.3655 ThI
7992.1588 ThI
7993.6808 ThI
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7992.1588 ThI
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7996.9691 ThI
8000.0449 ThI
8003.4551 ThI
8006.1567 ArI
7990
7995
8000
8005
8010
8014.7857 ArI
8017.5275 ArII
8022.2014 ThI
8024.2530 ThI
8025.7271 ThI
8026.2135 ThI
8030.2004 ThI
8032.4313 ThI
8032.9061 ThI
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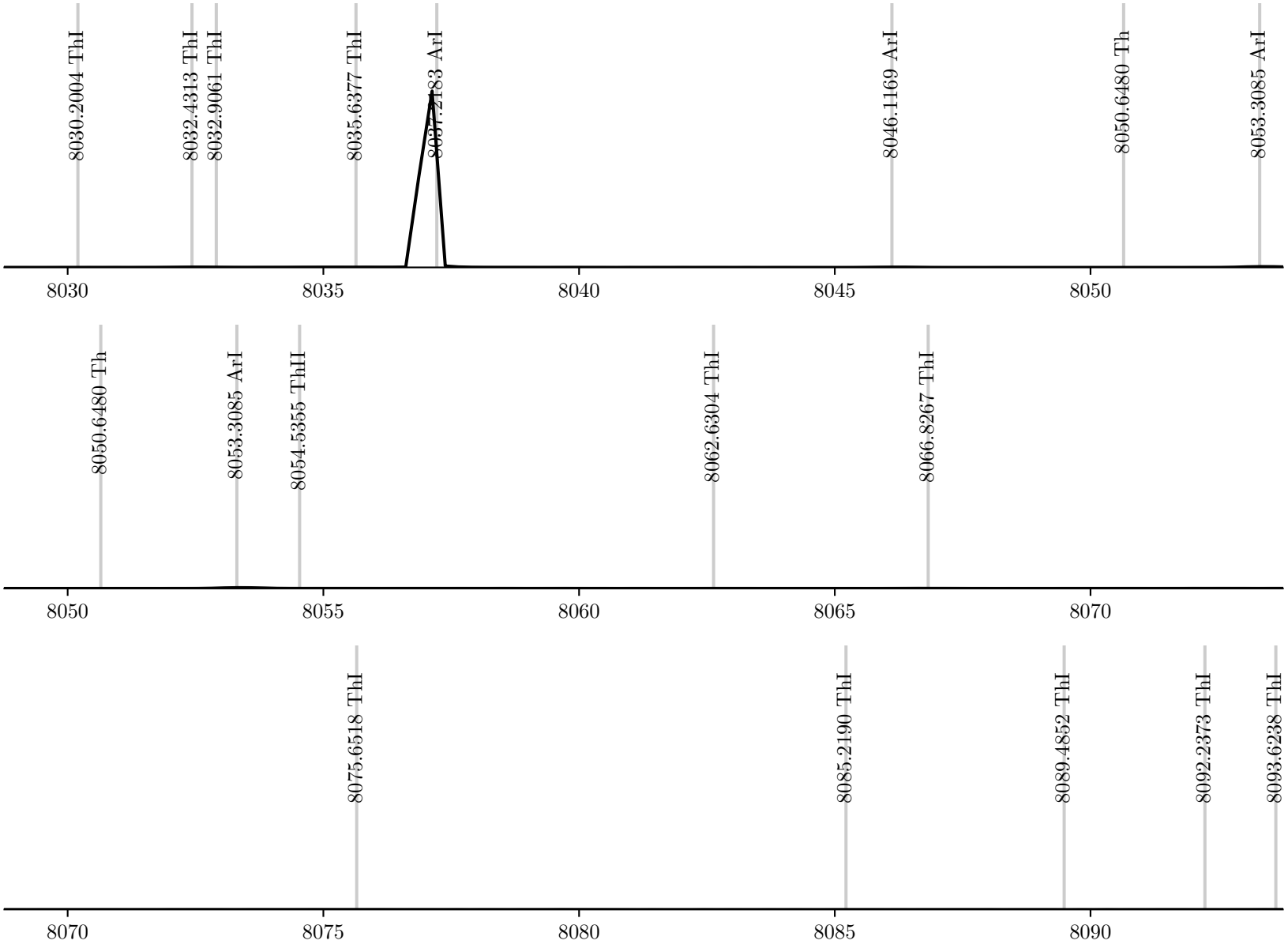
8030


8030.2004 ThI
8032.4313 ThI
8032.9061 ThI
8035.6377 ThI
8037.2183 ArI
8046.1169 ArI
8050.6480 Th
8053.3085 ArI
8030
8035
8040
8045
8050
8050.6480 Th
8053.3085 ArI
8054.5355 ThII
8062.6304 ThI
8066.8267 ThI
8050
8055
8060
8065
8070
8075.6518 ThI
8085.2190 ThI
8089.4852 ThI
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8093.6238 ThI
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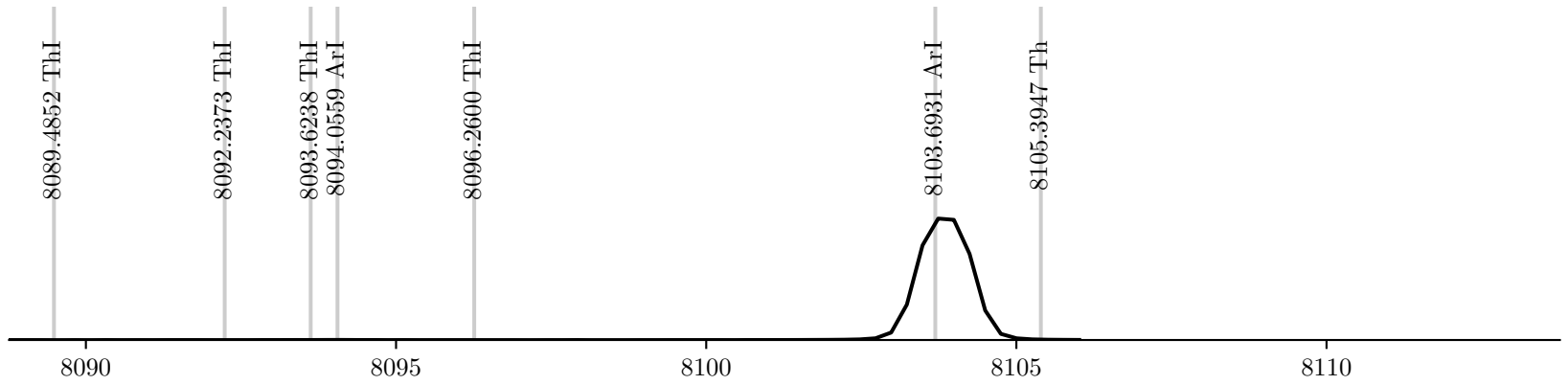
8089.4852 ThI
8092.2373 ThI
8093.6238 ThI
8094.0559 ArI
8096.2600 ThI
8103.6931 ArI
8105.3947 Th
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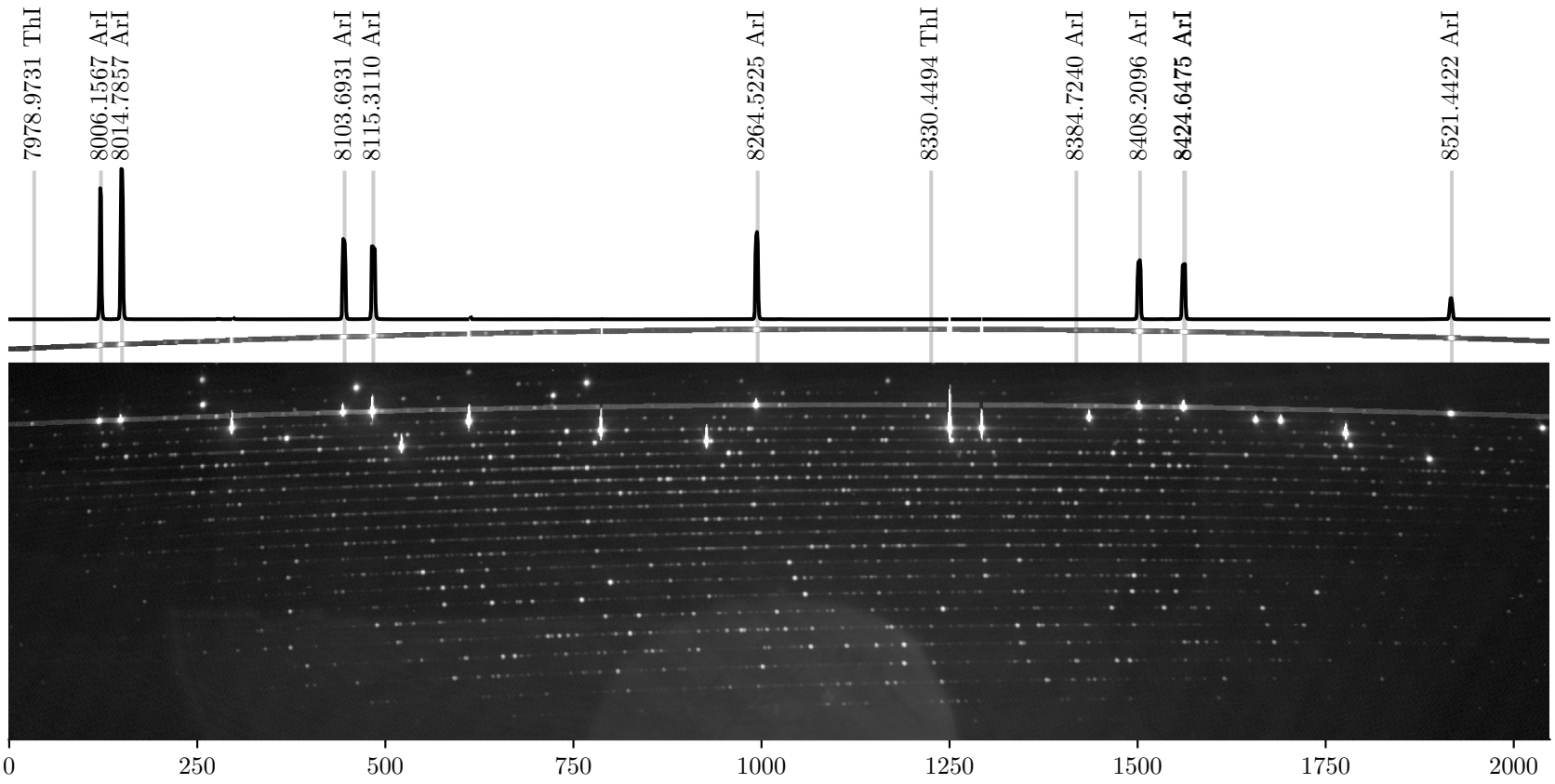
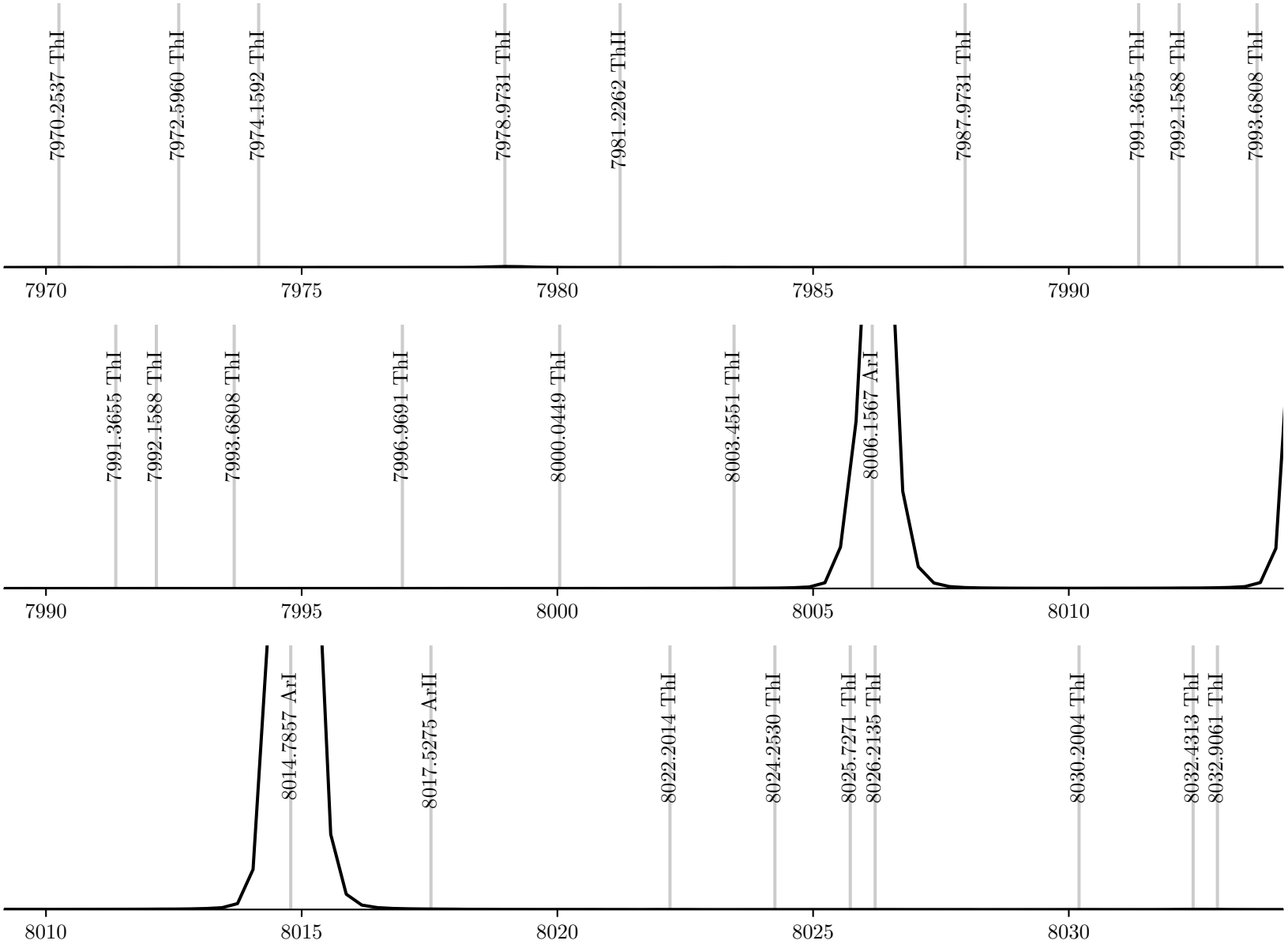
8105
8110

Order 18



Dispersion:	0.287 Å pix^{-1}
Central wavelength:	8273 Å
Wavelength range:	7969 Å – 8556 Å


7970.2537 ThI
7972.5960 ThI
7974.1592 ThI
7978.9731 ThI
7981.2262 ThII
7987.9731 ThI
7991.3655 ThI
7992.1588 ThI
7993.6808 ThI
7970
7975
7980
7985
7990
7991.3655 ThI
7992.1588 ThI
7993.6808 ThI
7996.9691 ThI
8000.0449 ThI
8003.4551 ThI
8006.1567 ArI
7990
7995
8000
8005
8010
8014.7857 ArI
8017.5275 ArII
8022.2014 ThI
8024.2530 ThI
8025.7271 ThI
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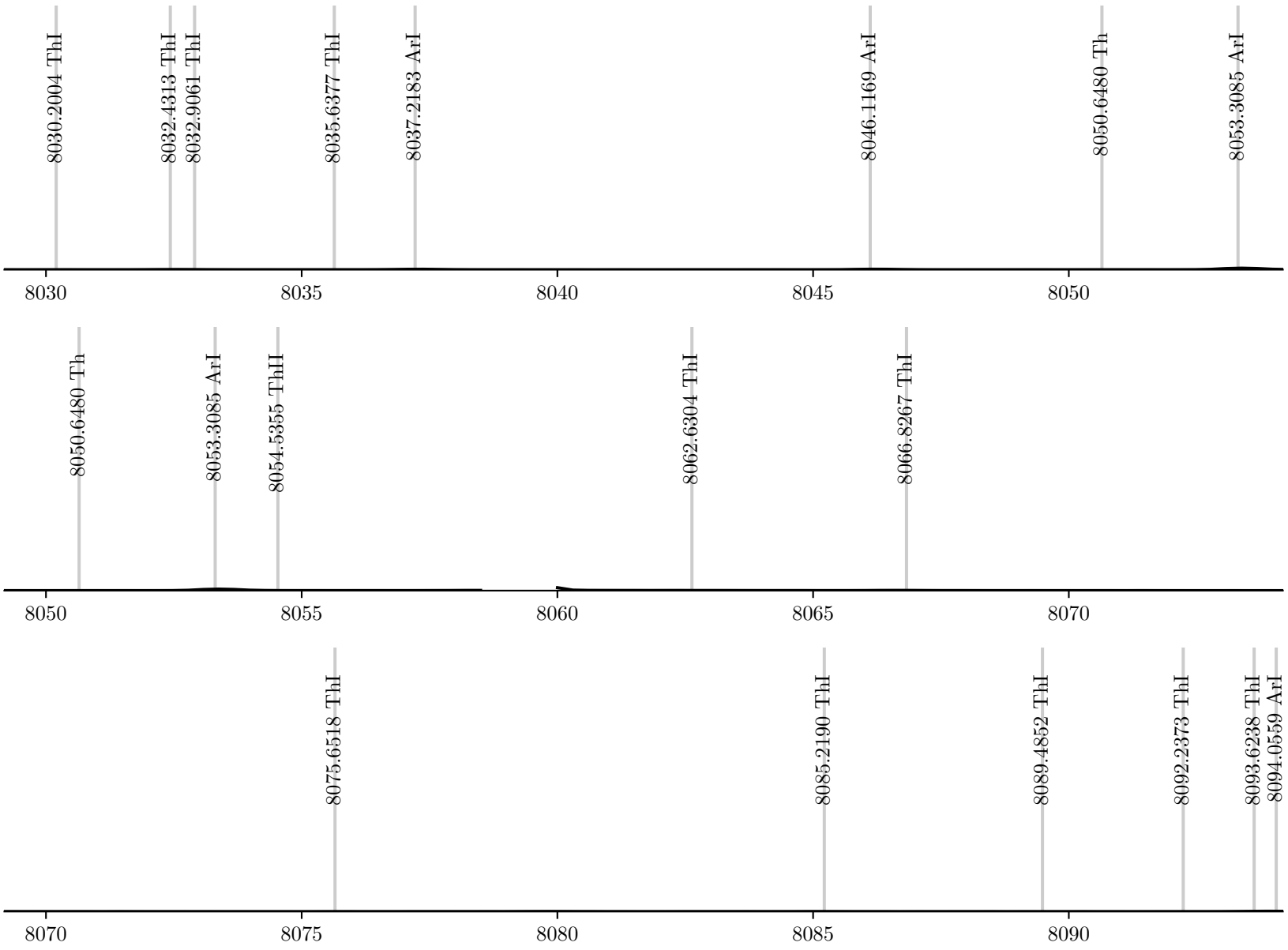
8030


8030.2004 ThI
8032.4313 ThI
8032.9061 ThI
8035.6377 ThI
8037.2183 ArI
8046.1169 ArI
8050.6480 Th
8053.3085 ArI
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8035
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8045
8050
8050.6480 Th
8053.3085 ArI
8054.5355 ThII
8062.6304 ThI
8066.8267 ThI
8050
8055
8060
8065
8070
8075.6518 ThI
8085.2190 ThI
8089.4852 ThI
8092.2373 ThI
8093.6238 ThI
8094.0559 ArI
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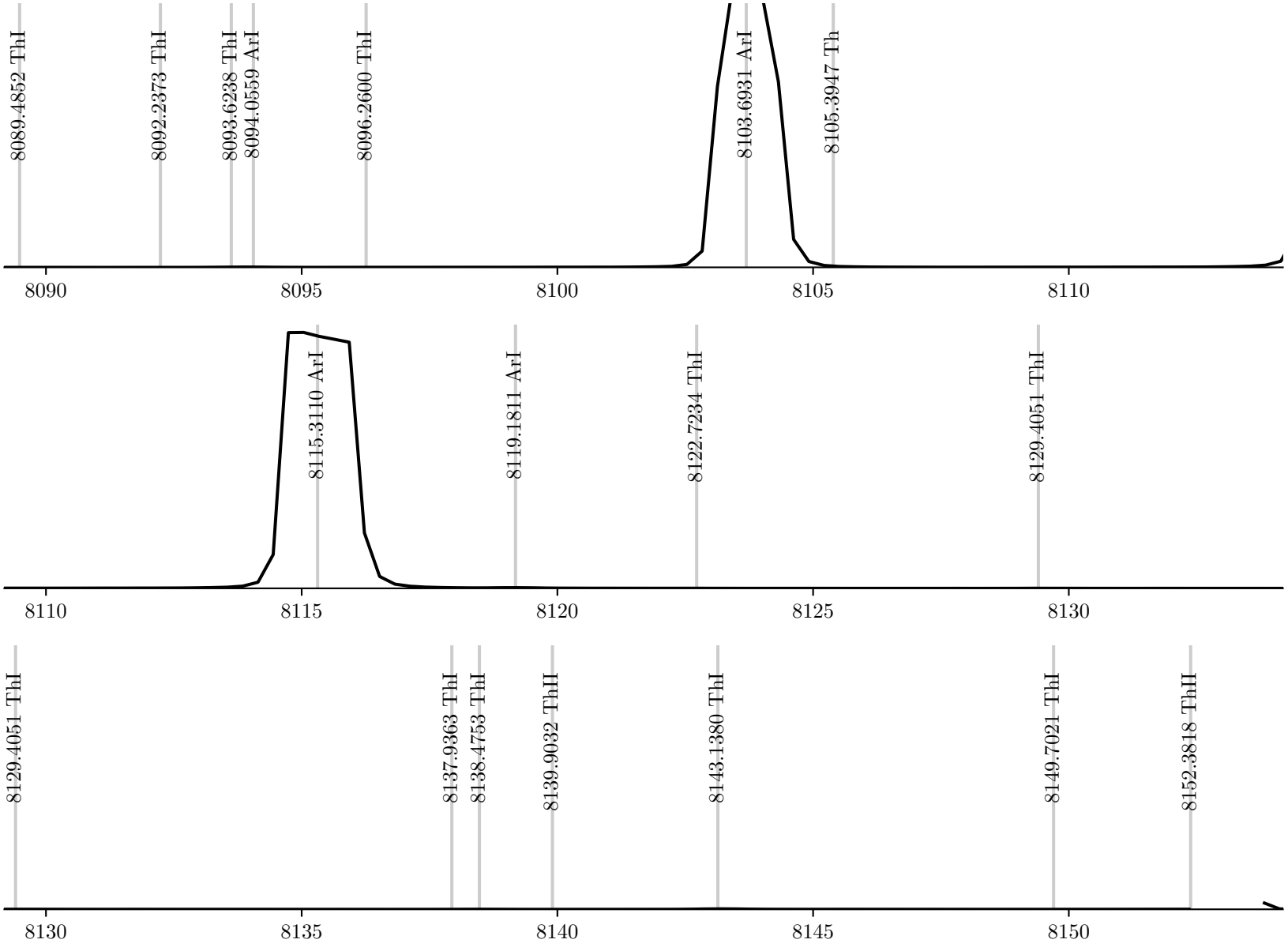
8075
8080
8085
8090


8089.4852 ThI
8092.2373 ThI
8093.6238 ThI
8094.0559 ArI
8096.2600 ThI
8103.6931 ArI
8105.3947 Th
8090
8095
8100
8105
8110
8115.3110 ArI
8119.1811 ArI
8122.7234 ThI
8129.4051 ThI
8110
8115
8120
8125
8130
8129.4051 ThI
8137.9363 ThI
8138.4753 ThI
8139.9032 ThII
8143.1380 ThI
8149.7021 ThI
8152.3818 ThII
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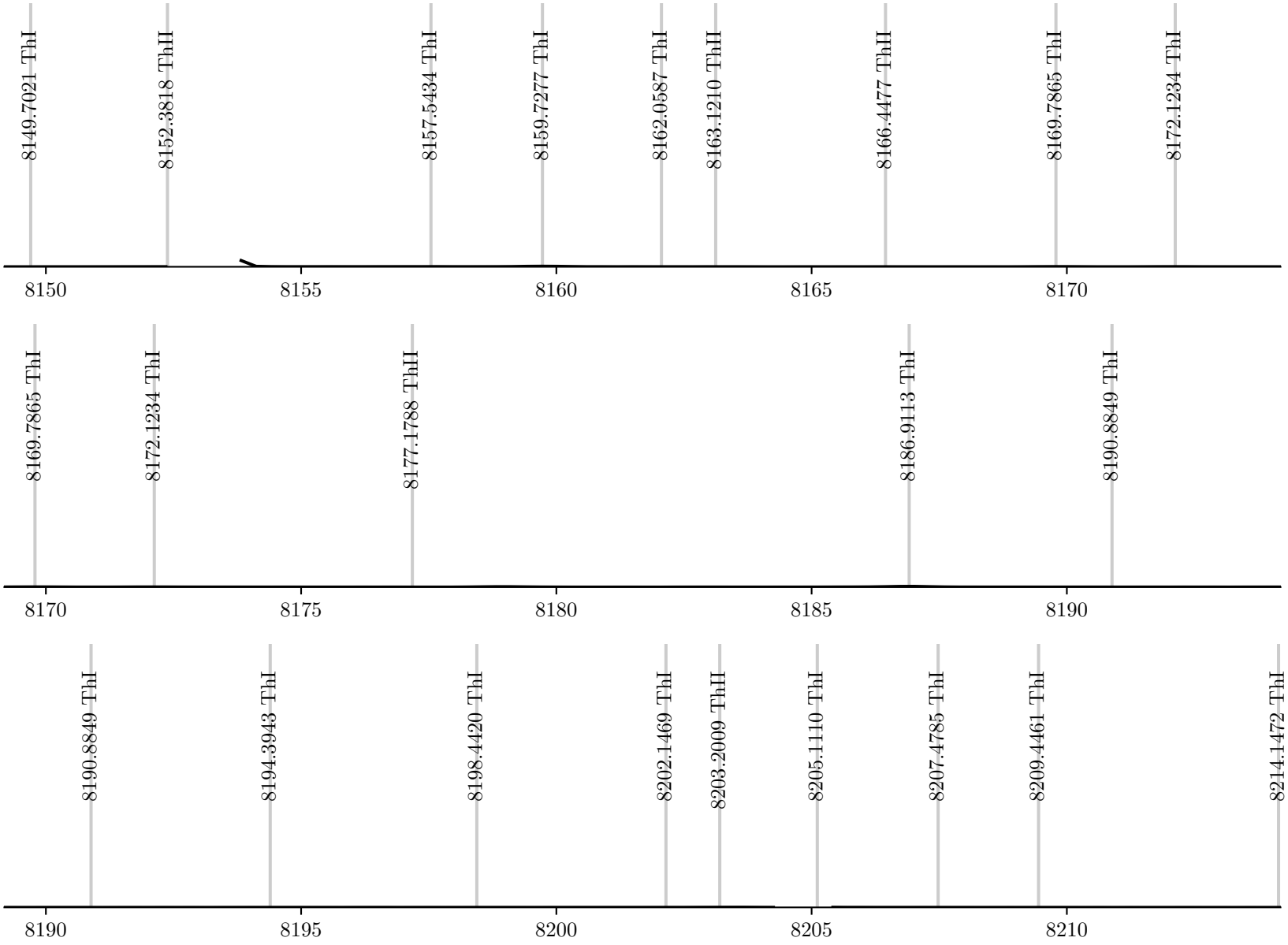
8150


8149.7021 ThI
8152.3818 ThII
8157.5434 ThI
8159.7277 ThI
8162.0587 ThI
8163.1210 ThII
8166.4477 ThII
8169.7865 ThI
8172.1234 ThI
8150
8155
8160
8165
8170
8169.7865 ThI
8172.1234 ThI
8177.1788 ThII
8186.9113 ThI
8190.8849 ThI
8170
8175
8180
8185
8190
8190.8849 ThI
8194.3943 ThI
8198.4420 ThI
8202.1469 ThI
8203.2009 ThII
8205.1110 ThI
8207.4785 ThI
8209.4461 ThI
8214.1472 ThI
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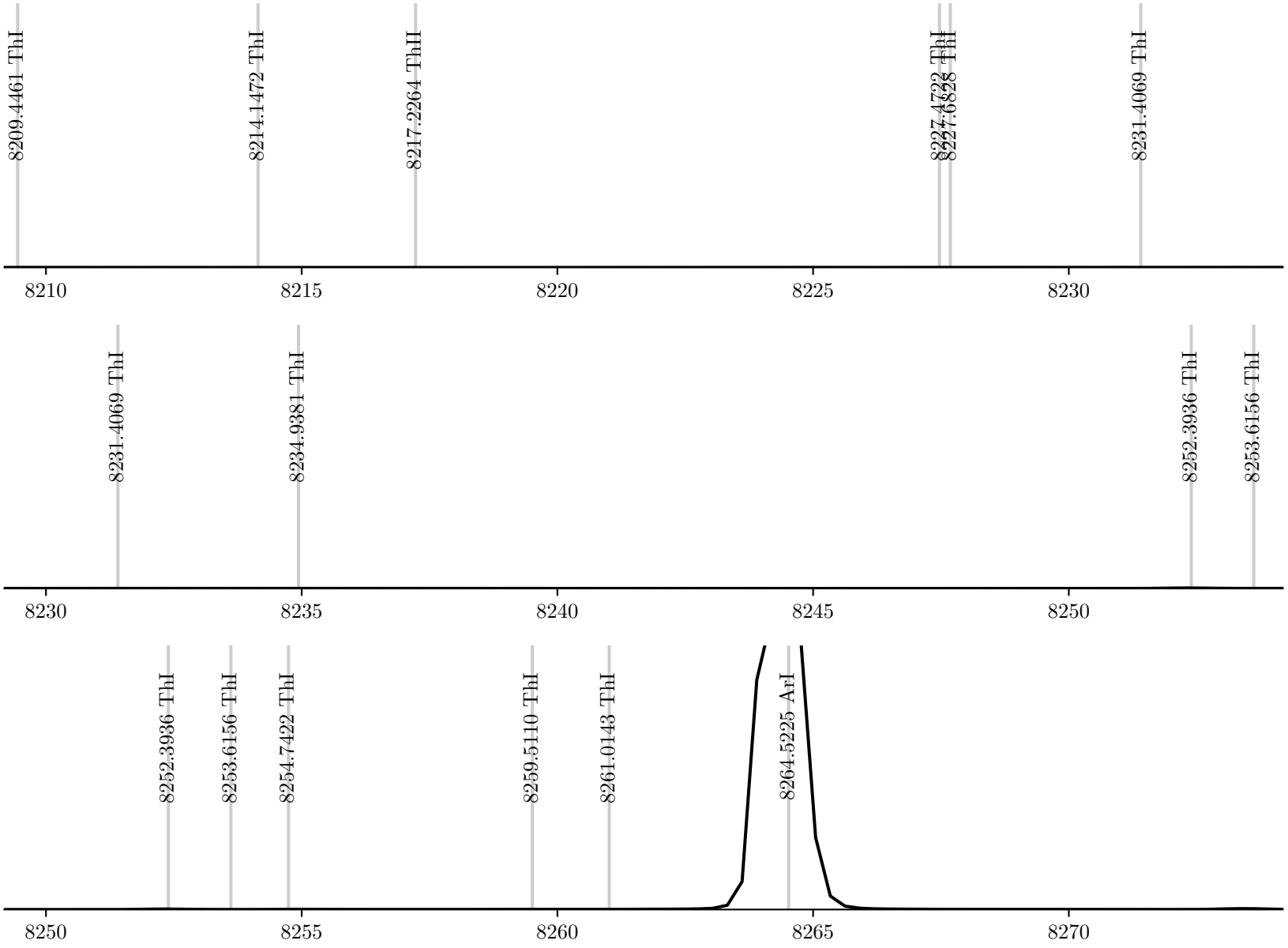
8195
8200
8205
8210


8209.4461 ThI
8214.1472 ThI
8217.2264 ThII
8227.4722 ThI
8227.6828 ThI
8231.4069 ThI
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8215
8220
8225
8230
8231.4069 ThI
8234.9381 ThI
8252.3936 ThI
8253.6156 ThI
8230
8235
8240
8245
8250
8252.3936 ThI
8253.6156 ThI
8254.7422 ThI
8259.5110 ThI
8261.0143 ThI
8264.5225 ArI
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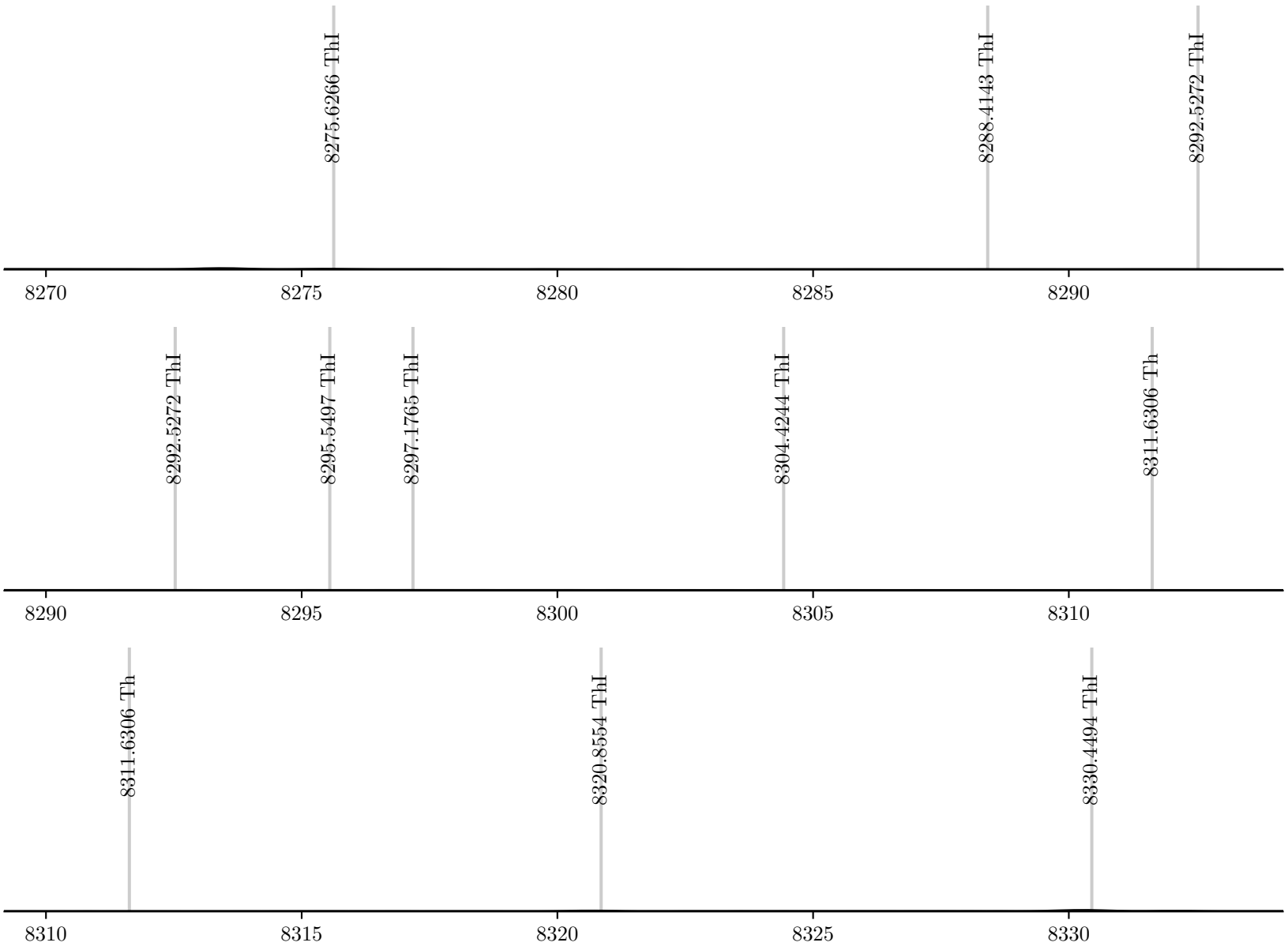
8260
8265
8270


8275.6266 ThI
8288.4143 ThI
8292.5272 ThI
8270
8275
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8292.5272 ThI
8295.5497 ThI
8297.1765 ThI
8304.4244 ThI
8311.6306 Th
8290
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8311.6306 Th
8320.8554 ThI
8330.4494 ThI
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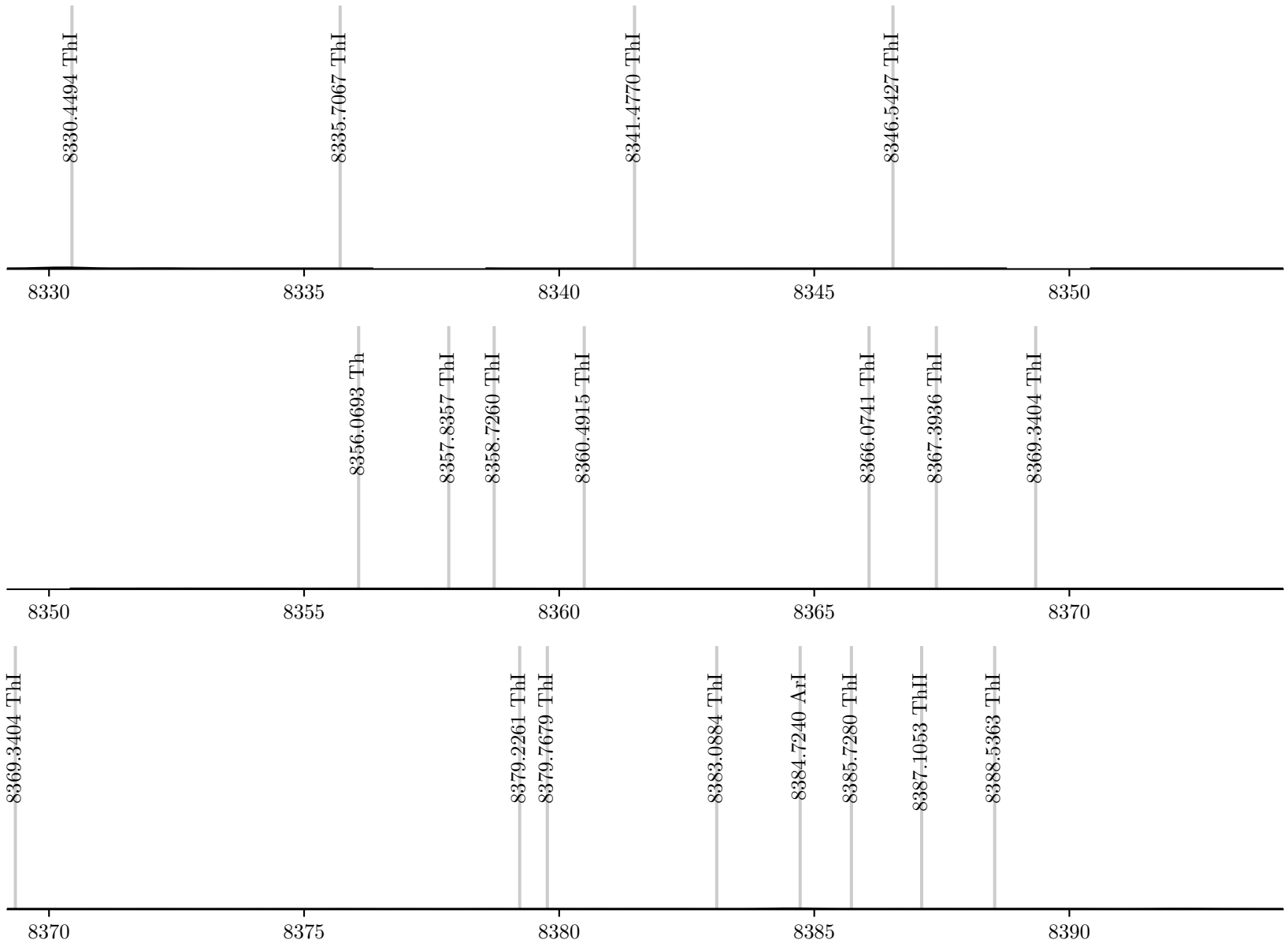
8320
8325
8330


8330.4494 ThI
8335.7067 ThI
8341.4770 ThI
8346.5427 ThI
8330
8335
8340
8345
8350
8356.0693 Th
8357.8357 ThI
8358.7260 ThI
8360.4915 ThI
8366.0741 ThI
8367.3936 ThI
8369.3404 ThI
8350
8355
8360
8365
8370
8369.3404 ThI
8379.2261 ThI
8379.7679 ThI
8383.0884 ThI
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8385.7280 ThI
8387.1053 ThII
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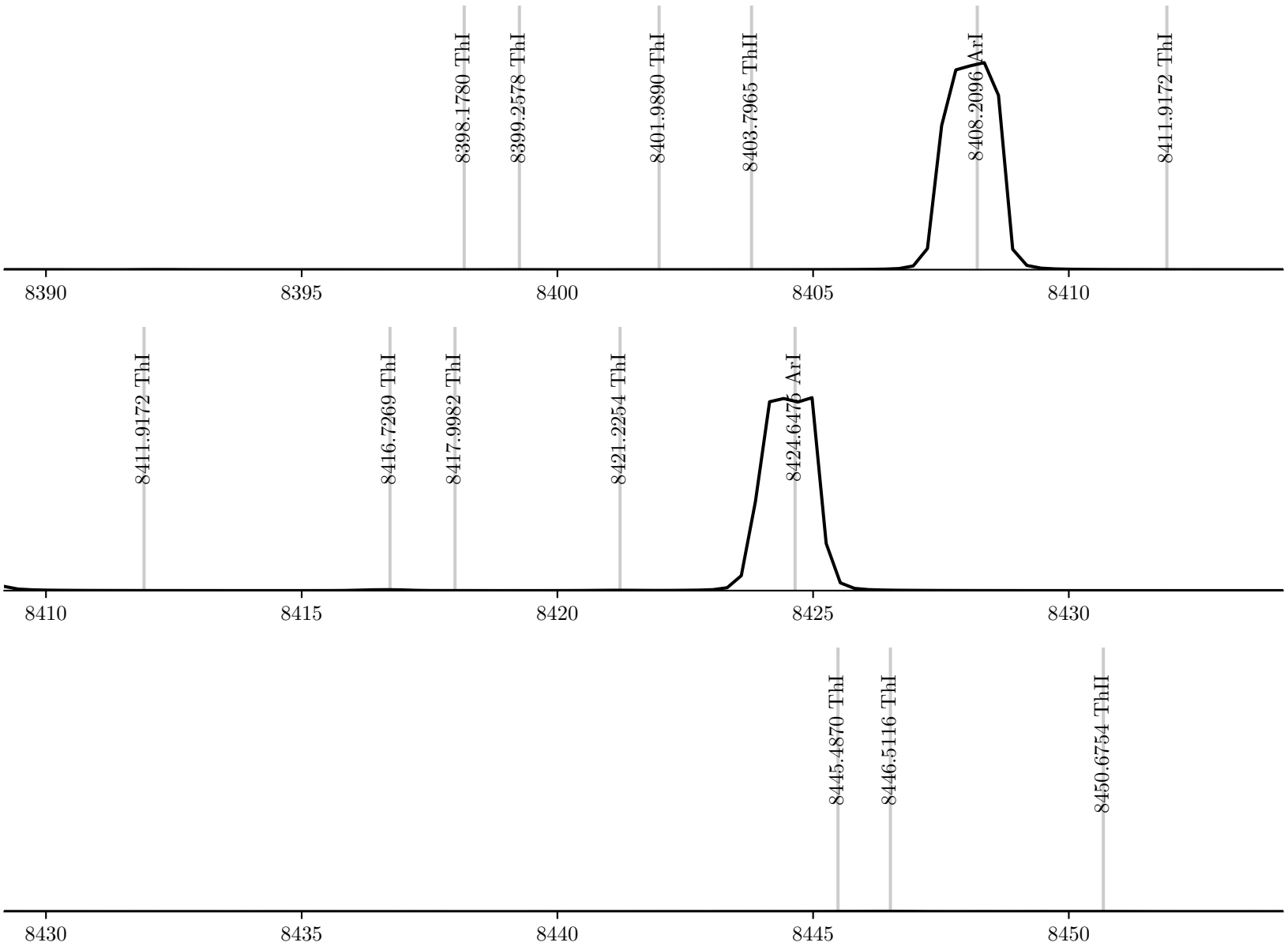
8385
8390


8398.1780 ThI
8399.2578 ThI
8401.9890 ThI
8403.7965 ThII
8408.2096 ArI
8411.9172 ThI
8390
8395
8400
8405
8410
8411.9172 ThI
8416.7269 ThI
8417.9982 ThI
8421.2254 ThI
8424.6476 ArI
8410
8415
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8445.4870 ThI
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8450.6754 ThII
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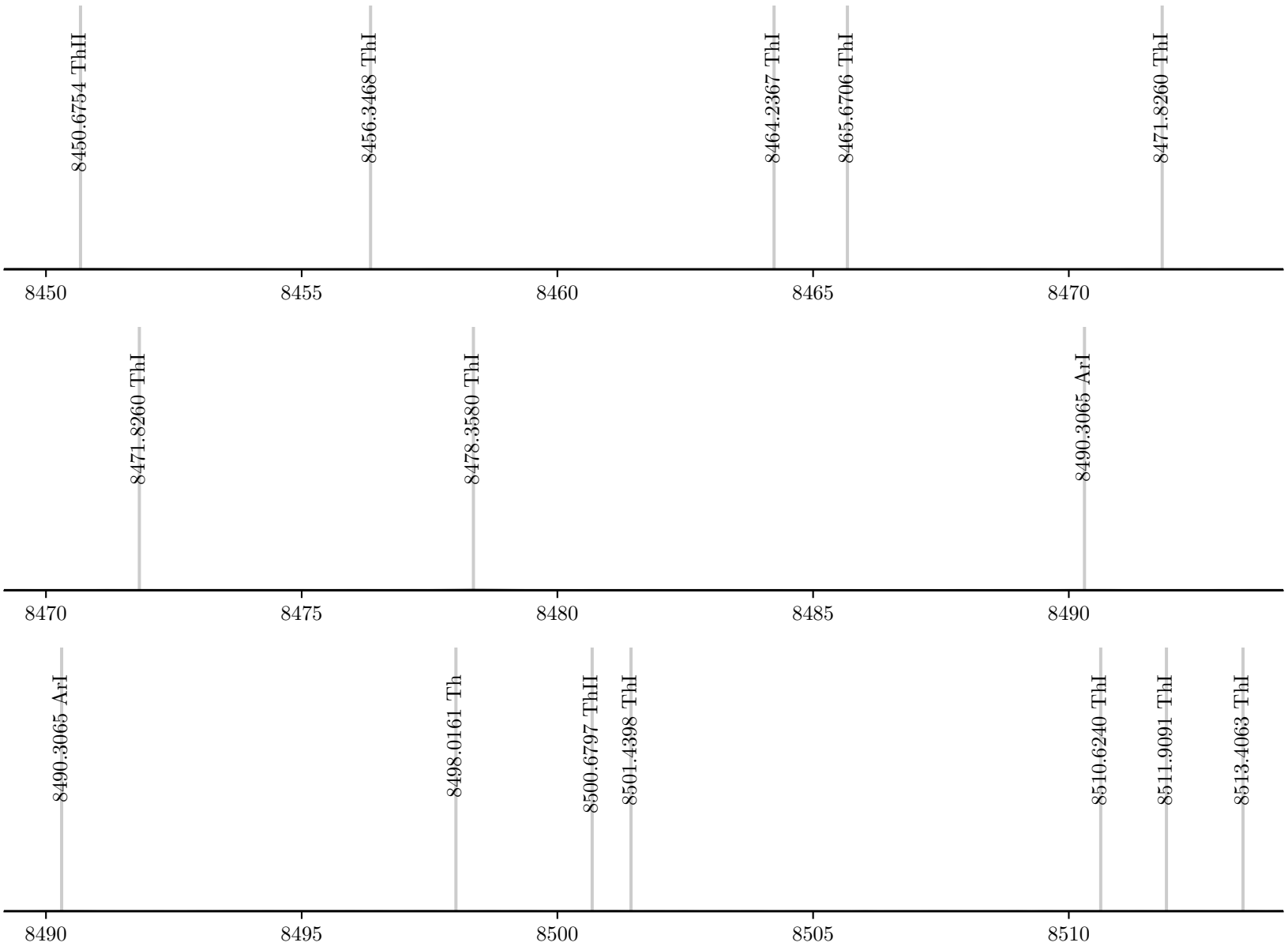
8450


8450.6754 ThII
8456.3468 ThI
8464.2367 ThI
8465.6706 ThI
8471.8260 ThI
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8455
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8465
8470
8471.8260 ThI
8478.3580 ThI
8490.3065 ArI
8470
8475
8480
8485
8490
8490.3065 ArI
8498.0161 Th
8500.6797 ThII
8501.4398 ThI
8510.6240 ThI
8511.9091 ThI
8513.4063 ThI
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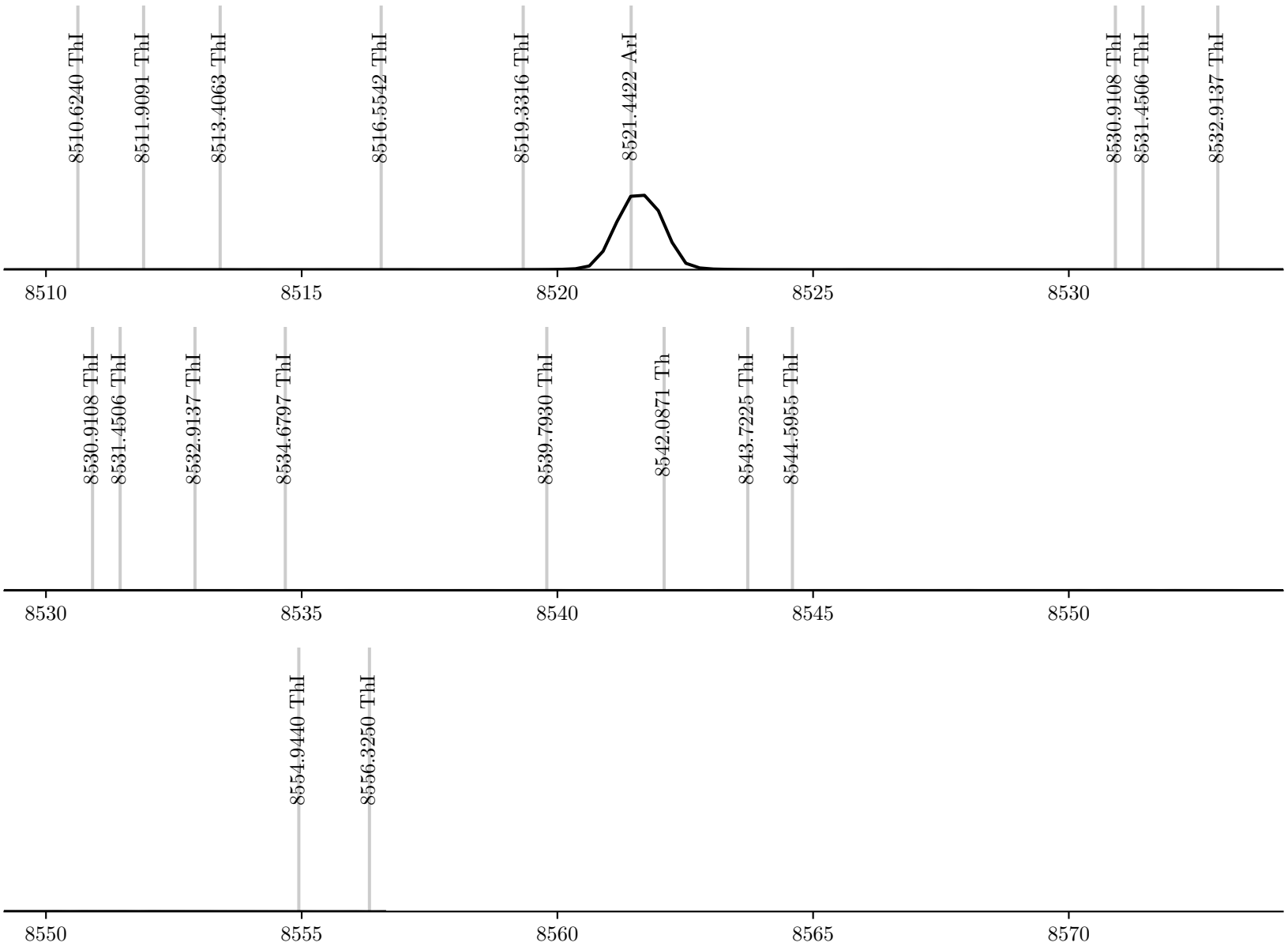
8500
8505
8510


8510.6240 ThI
8511.9091 ThI
8513.4063 ThI
8516.5542 ThI
8519.3316 ThI
8521.4422 ArI
8530.9108 ThI
8531.4506 ThI
8532.9137 ThI
8510
8515
8520
8525
8530
8530.9108 ThI
8531.4506 ThI
8532.9137 ThI
8534.6797 ThI
8539.7930 ThI
8542.0871 Th
8543.7225 ThI
8544.5955 ThI
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8554.9440 ThI
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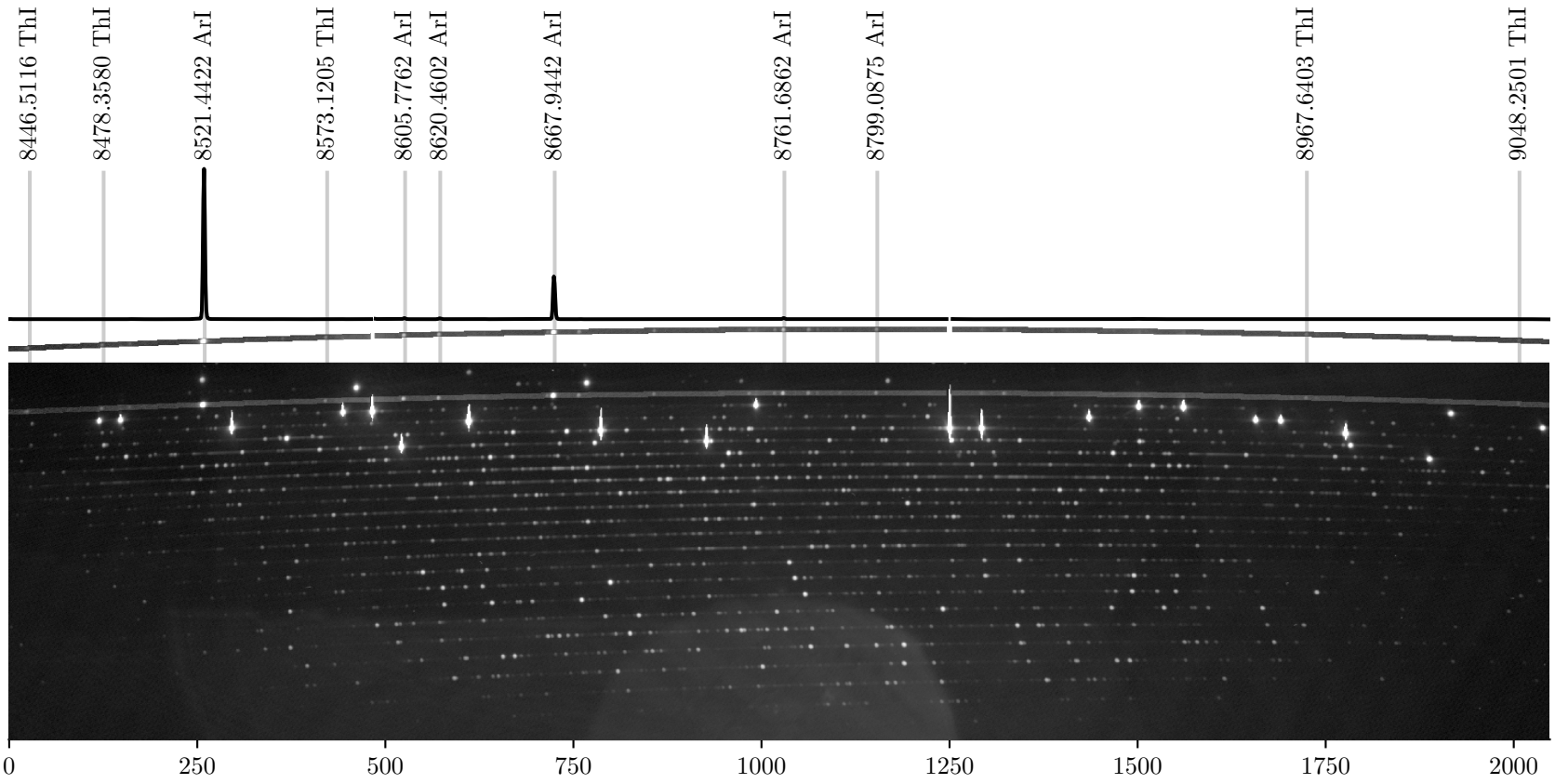
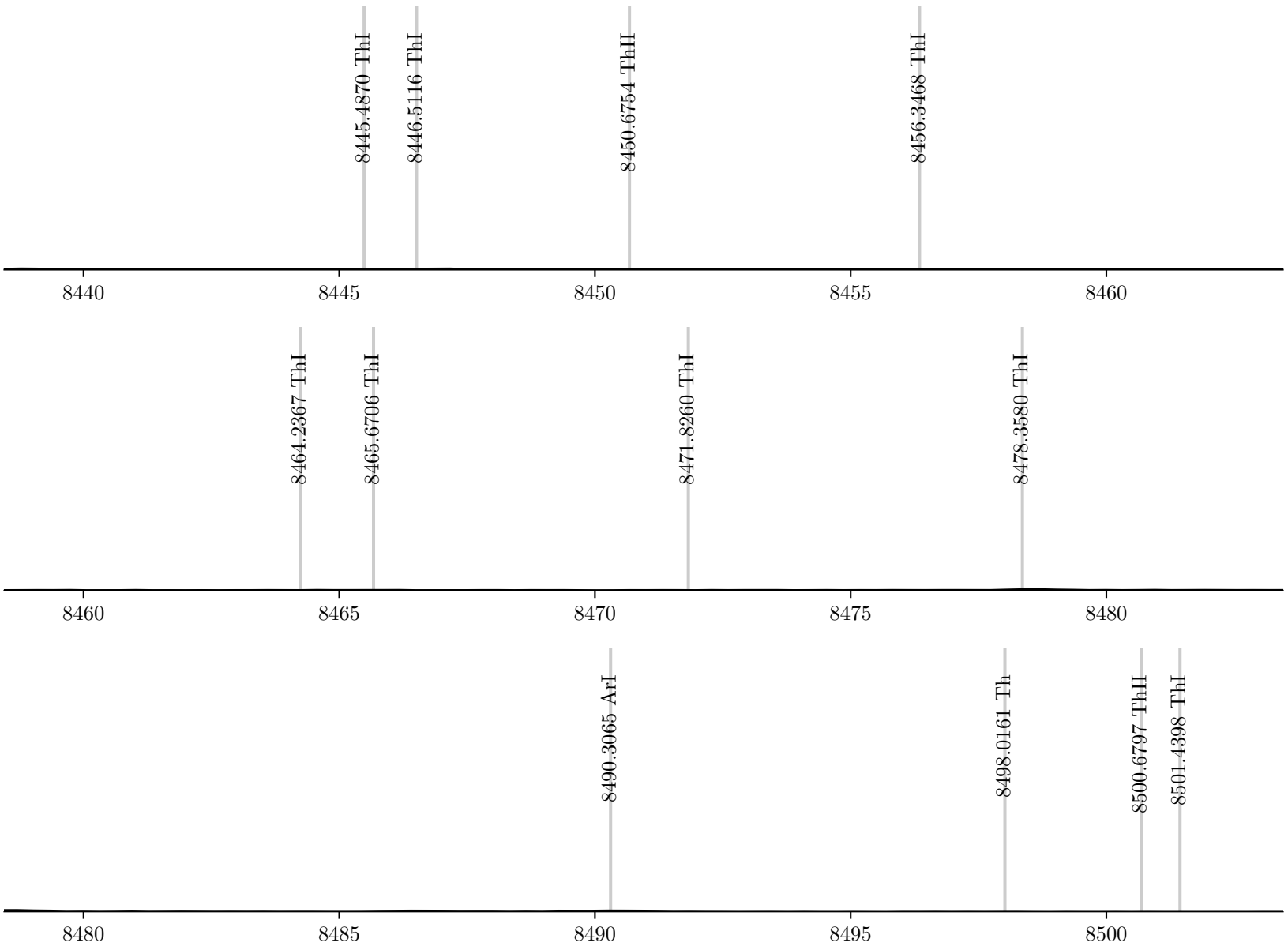
8560
8565
8570

Order 17



Dispersion:	0.304 Å pix^{-1}
Central wavelength:	8760 Å
Wavelength range:	8438 Å – 9060 Å


8445.4870 ThI
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8456.3468 ThI
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8455
8460
8464.2367 ThI
8465.6706 ThI
8471.8260 ThI
8478.3580 ThI
8460
8465
8470
8475
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8490.3065 ArI
8498.0161 Th
8500.6797 ThII
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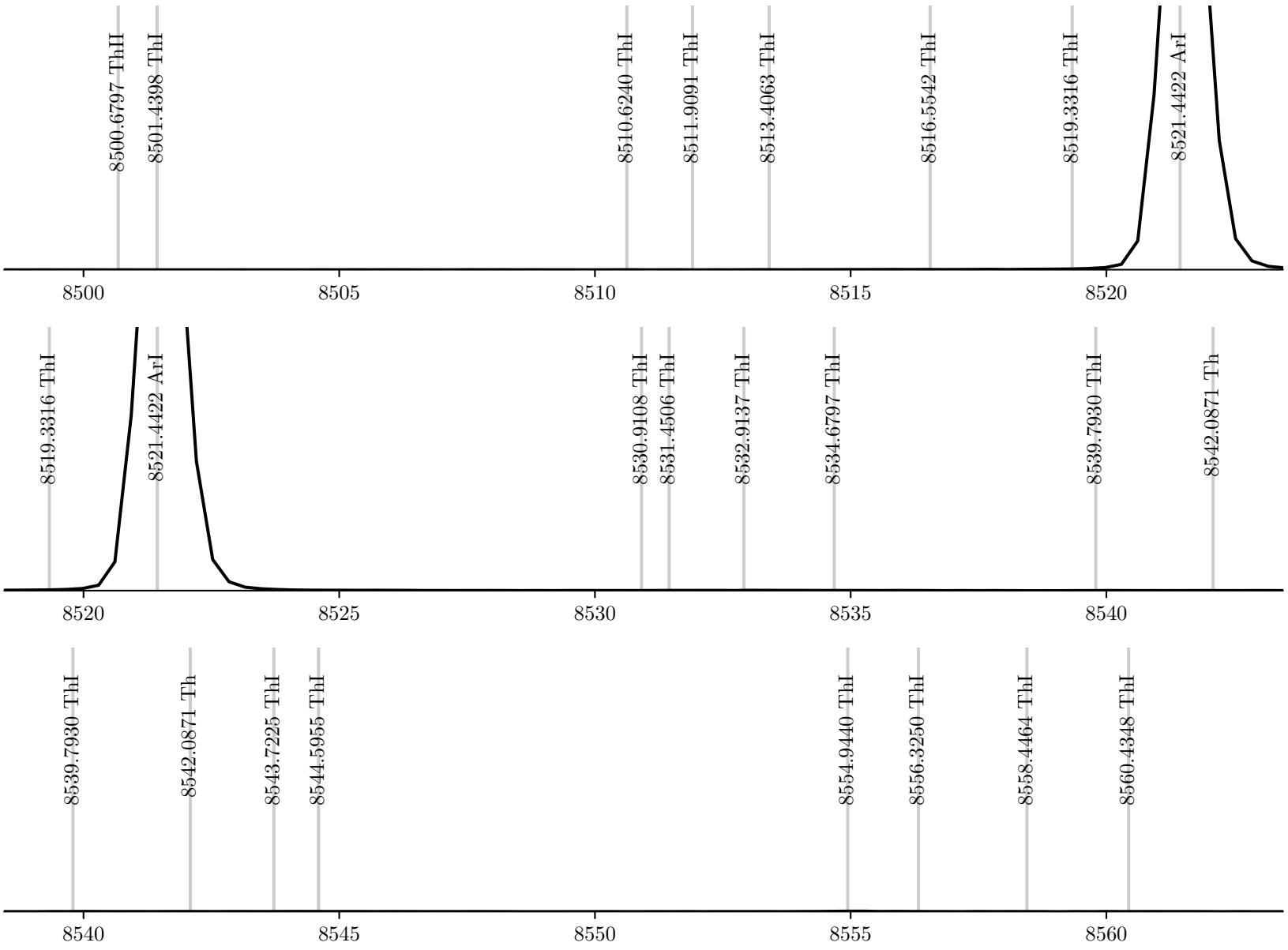
8490
8495
8500


8500.6797 ThII
8501.4398 ThI
8510.6240 ThI
8511.9091 ThI
8513.4063 ThI
8516.5542 ThI
8519.3316 ThI
8521.4422 ArI
8500
8505
8510
8515
8520
8519.3316 ThI
8521.4422 ArI
8530.9108 ThI
8531.4506 ThI
8532.9137 ThI
8534.6797 ThI
8539.7930 ThI
8542.0871 Th
8520
8525
8530
8535
8540
8539.7930 ThI
8542.0871 Th
8543.7225 ThI
8544.5955 ThI
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8556.3250 ThI
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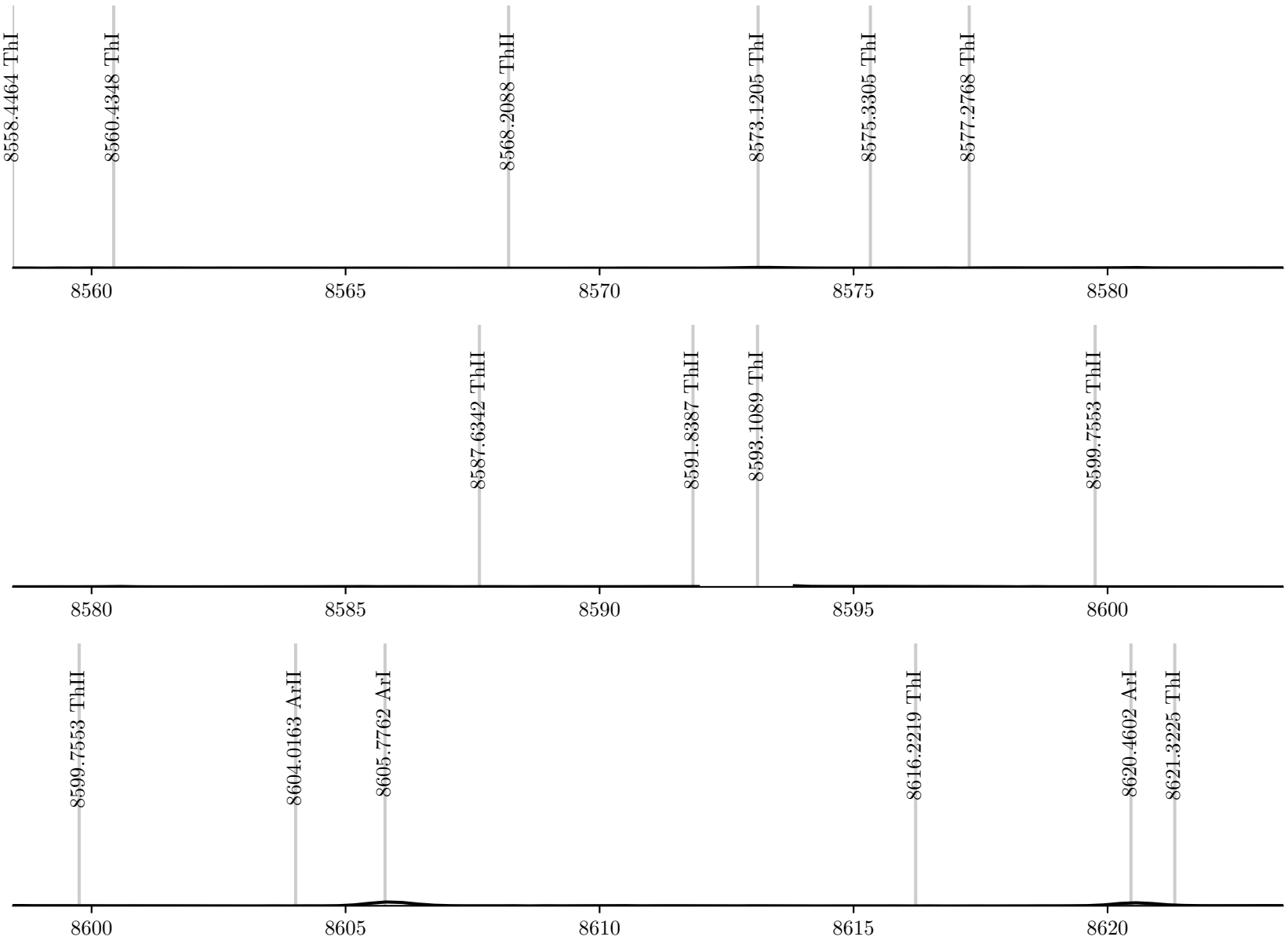
8560


8558.4464 ThI
8560.4348 ThI
8568.2088 ThII
8573.1205 ThI
8575.3305 ThI
8577.2768 ThI
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8570
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8580
8587.6342 ThII
8591.8387 ThII
8593.1089 ThI
8599.7553 ThII
8580
8585
8590
8595
8600
8599.7553 ThII
8604.0163 ArII
8605.7762 ArI
8616.2219 ThI
8620.4602 ArI
8621.3225 ThI
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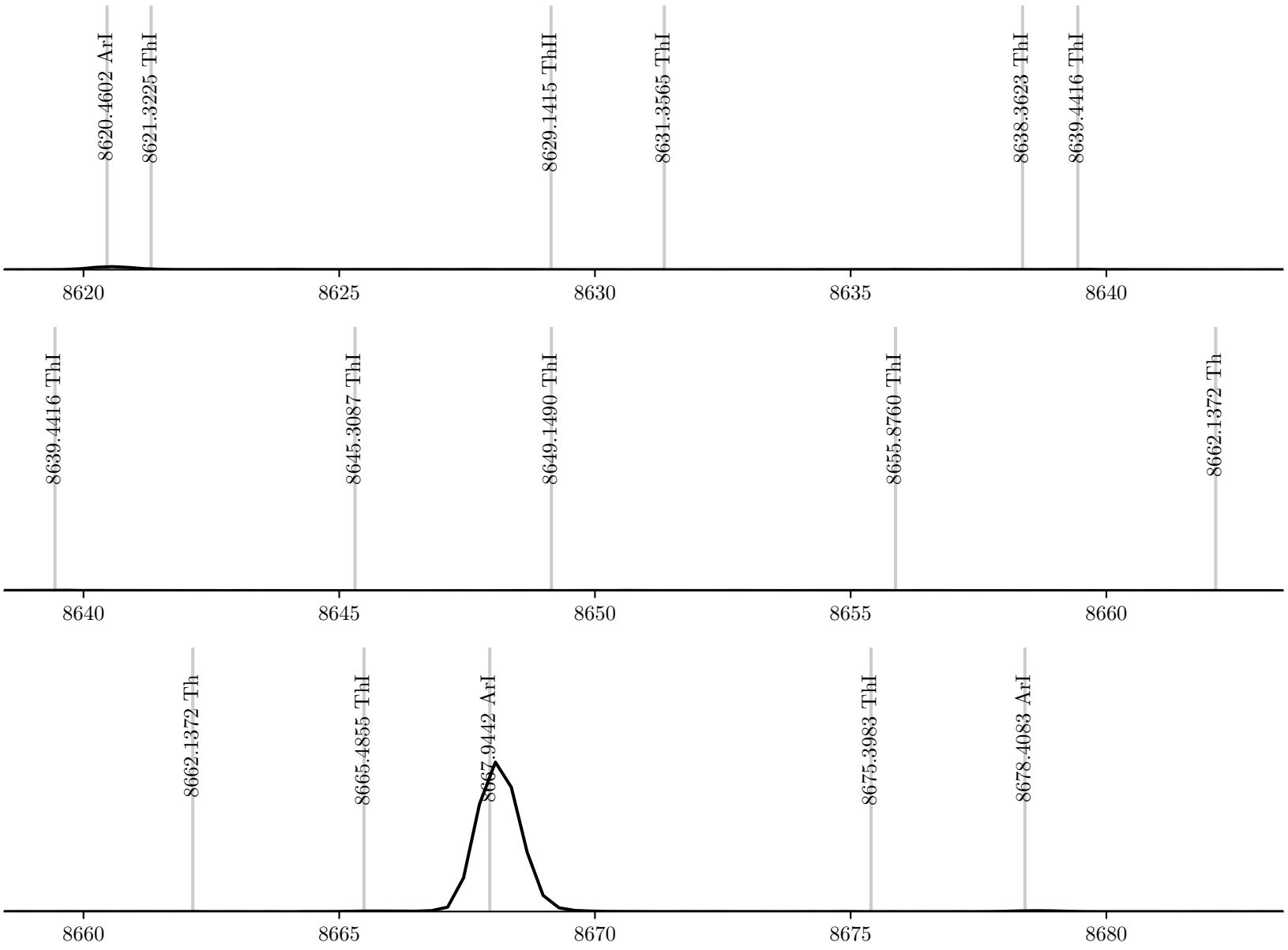
8615
8620


8620.4602 ArI
8621.3225 ThI
8629.1415 ThII
8631.3565 ThI
8638.3623 ThI
8639.4416 ThI
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8625
8630
8635
8640
8639.4416 ThI
8645.3087 ThI
8649.1490 ThI
8655.8760 ThI
8662.1372 Th
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8645
8650
8655
8660
8662.1372 Th
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8667.9442 ArI
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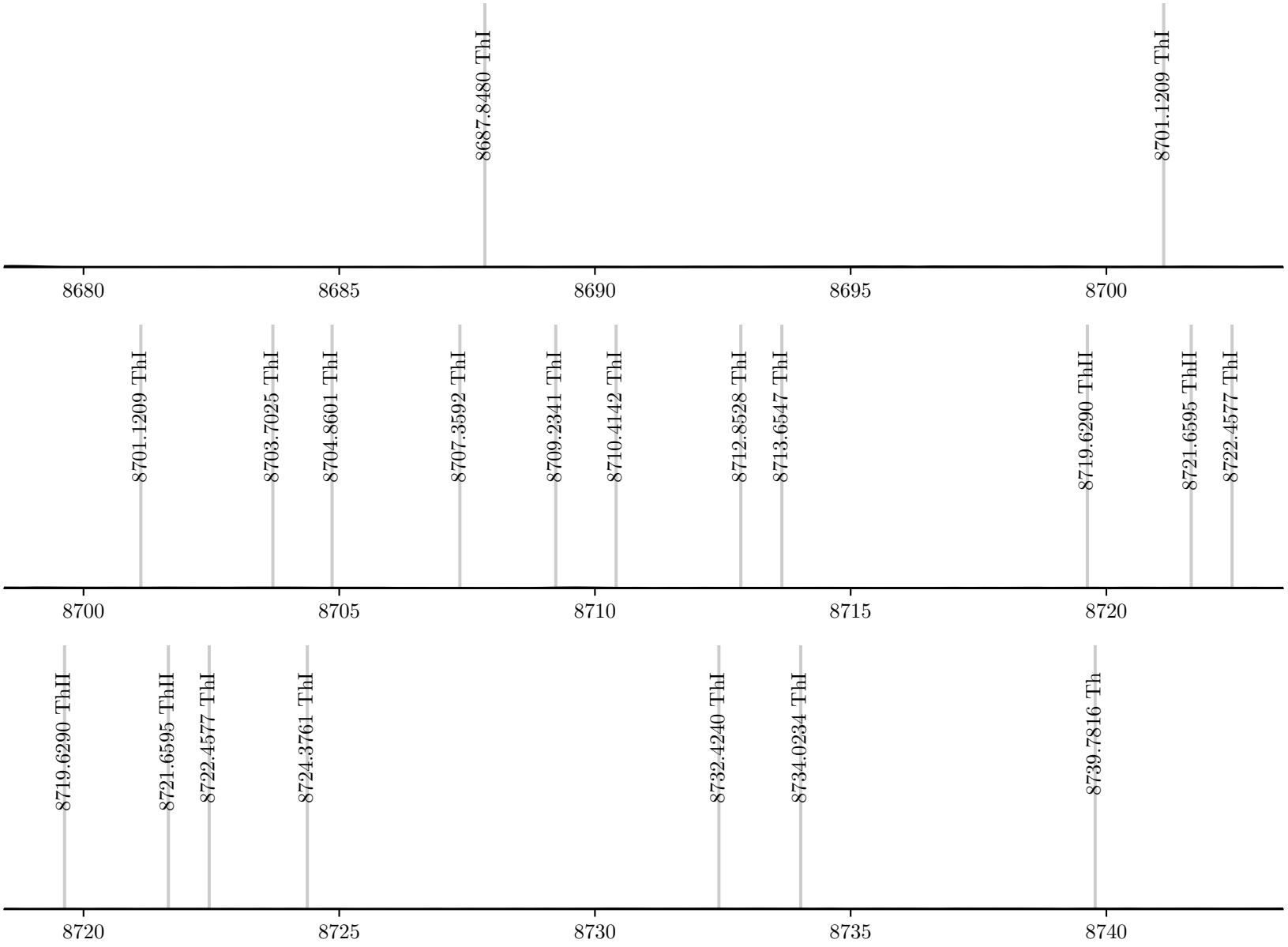
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8701.1209 ThI
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8707.3592 ThI
8709.2341 ThI
8710.4142 ThI
8712.8528 ThI
8713.6547 ThI
8719.6290 ThII
8721.6595 ThII
8722.4577 ThI
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8705
8710
8715
8720
8719.6290 ThII
8721.6595 ThII
8722.4577 ThI
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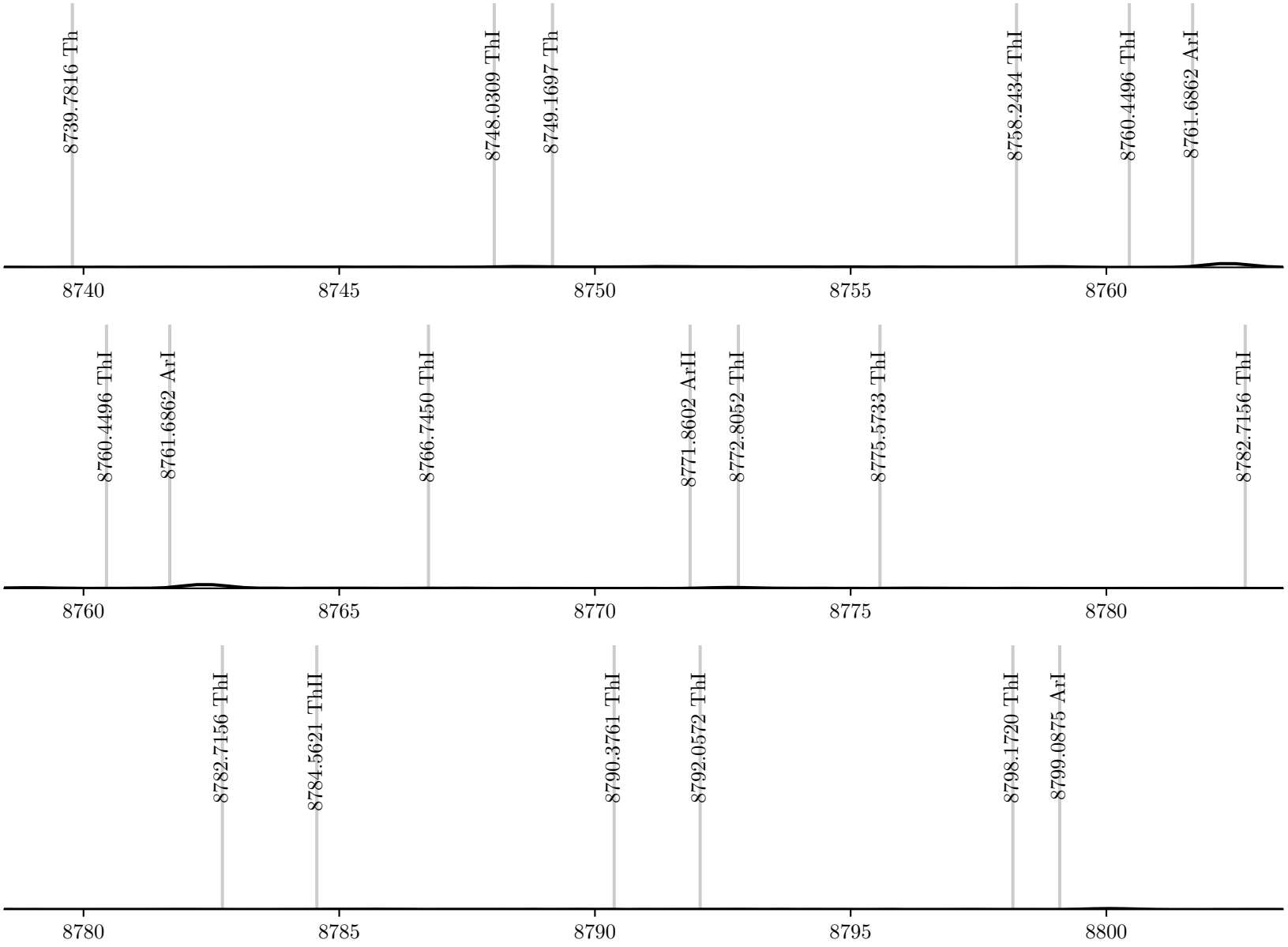
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8749.1697 Th
8758.2434 ThI
8760.4496 ThI
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8745
8750
8755
8760
8760.4496 ThI
8761.6862 ArI
8766.7450 ThI
8771.8602 ArII
8772.8052 ThI
8775.5733 ThI
8782.7156 ThI
8760
8765
8770
8775
8780
8782.7156 ThI
8784.5621 ThII
8790.3761 ThI
8792.0572 ThI
8798.1720 ThI
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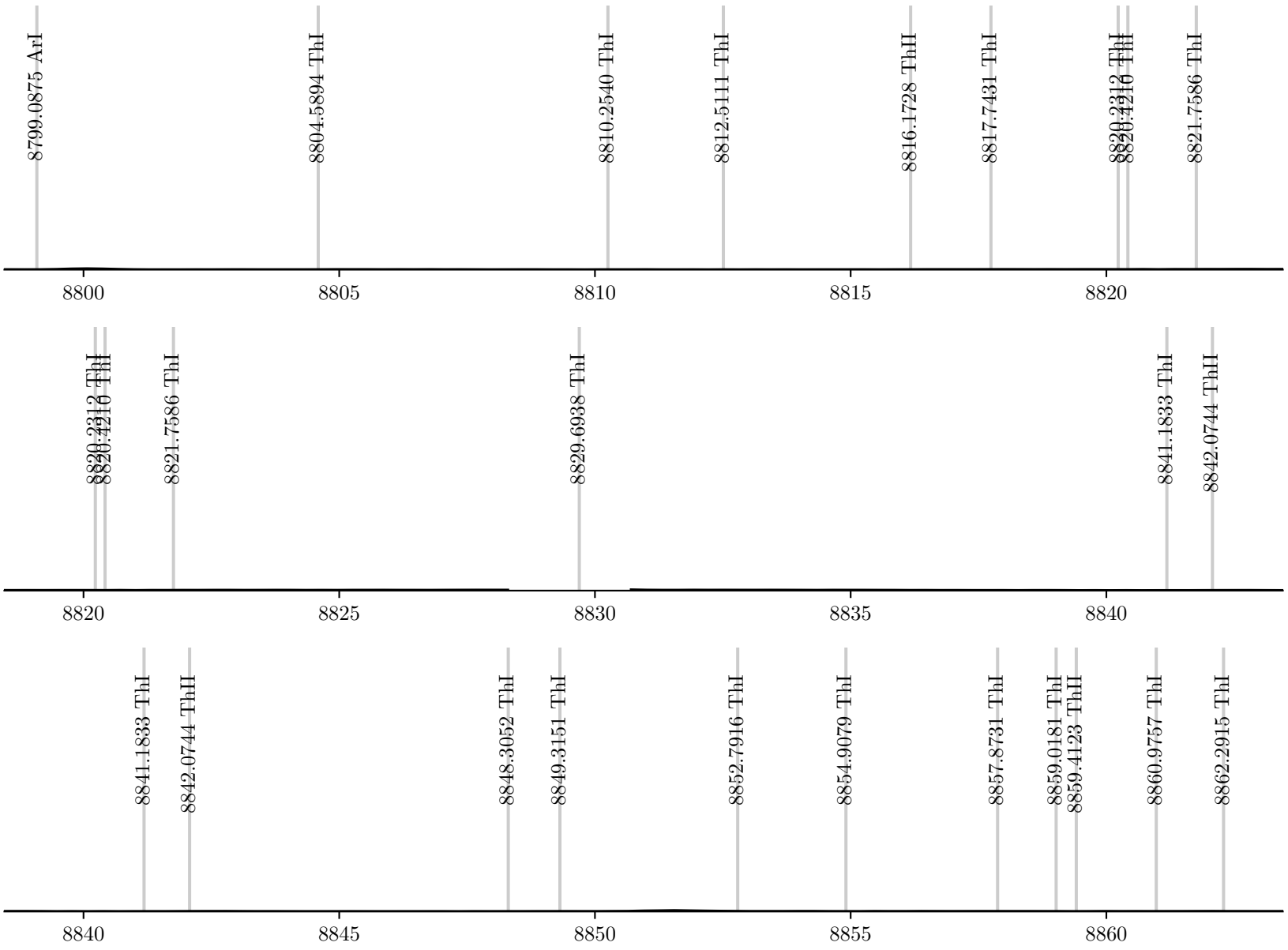
8790
8795
8800


8799.0875 ArI
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8812.5111 ThI
8816.1728 ThII
8817.7431 ThI
8820.2312 ThI
8820.4210 ThI
8821.7586 ThI
8800
8805
8810
8815
8820
8820.2312 ThI
8820.4210 ThI
8821.7586 ThI
8829.6938 ThI
8841.1833 ThI
8842.0744 ThII
8820
8825
8830
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8841.1833 ThI
8842.0744 ThII
8848.3052 ThI
8849.3151 ThI
8852.7916 ThI
8854.9079 ThI
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8859.4123 ThII
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8862.2915 ThI
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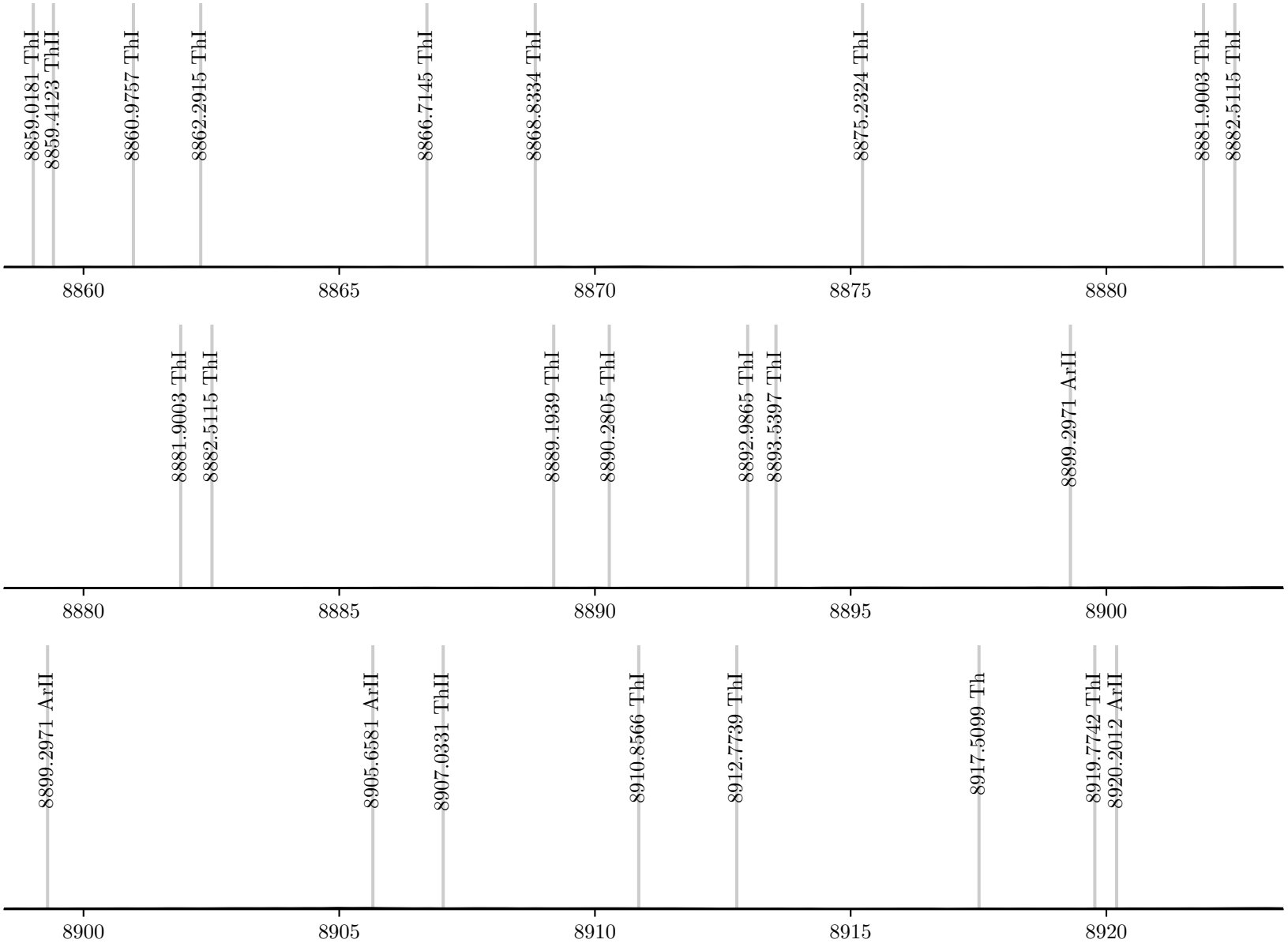
8860


8859.0181 ThI
8859.4123 ThII
8860.9757 ThI
8862.2915 ThI
8866.7145 ThI
8868.8334 ThI
8875.2324 ThI
8881.9003 ThI
8882.5115 ThI
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8870
8875
8880
8881.9003 ThI
8882.5115 ThI
8889.1939 ThI
8890.2805 ThI
8892.9865 ThI
8893.5397 ThI
8899.2971 ArII
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8890
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8900
8899.2971 ArII
8905.6581 ArII
8907.0331 ThII
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8919.7742 ThI
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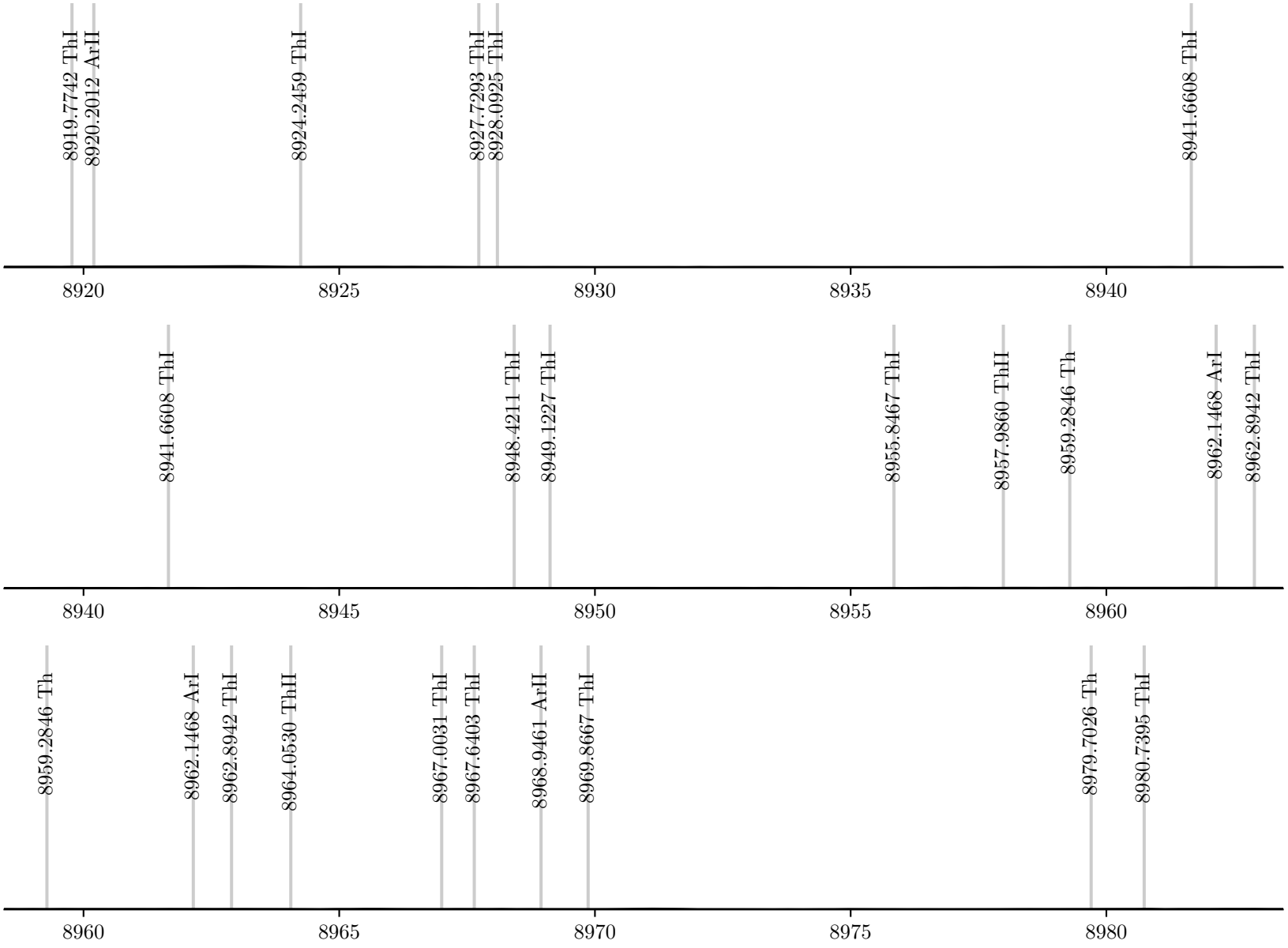
8920


8919.7742 ThI
8920.2012 ArII
8924.2459 ThI
8927.7293 ThI
8928.0925 ThI
8941.6608 ThI
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8941.6608 ThI
8948.4211 ThI
8949.1227 ThI
8955.8467 ThI
8957.9860 ThII
8959.2846 Th
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8962.1468 ArI
8962.8942 ThI
8964.0530 ThII
8967.0031 ThI
8967.6403 ThI
8968.9461 ArII
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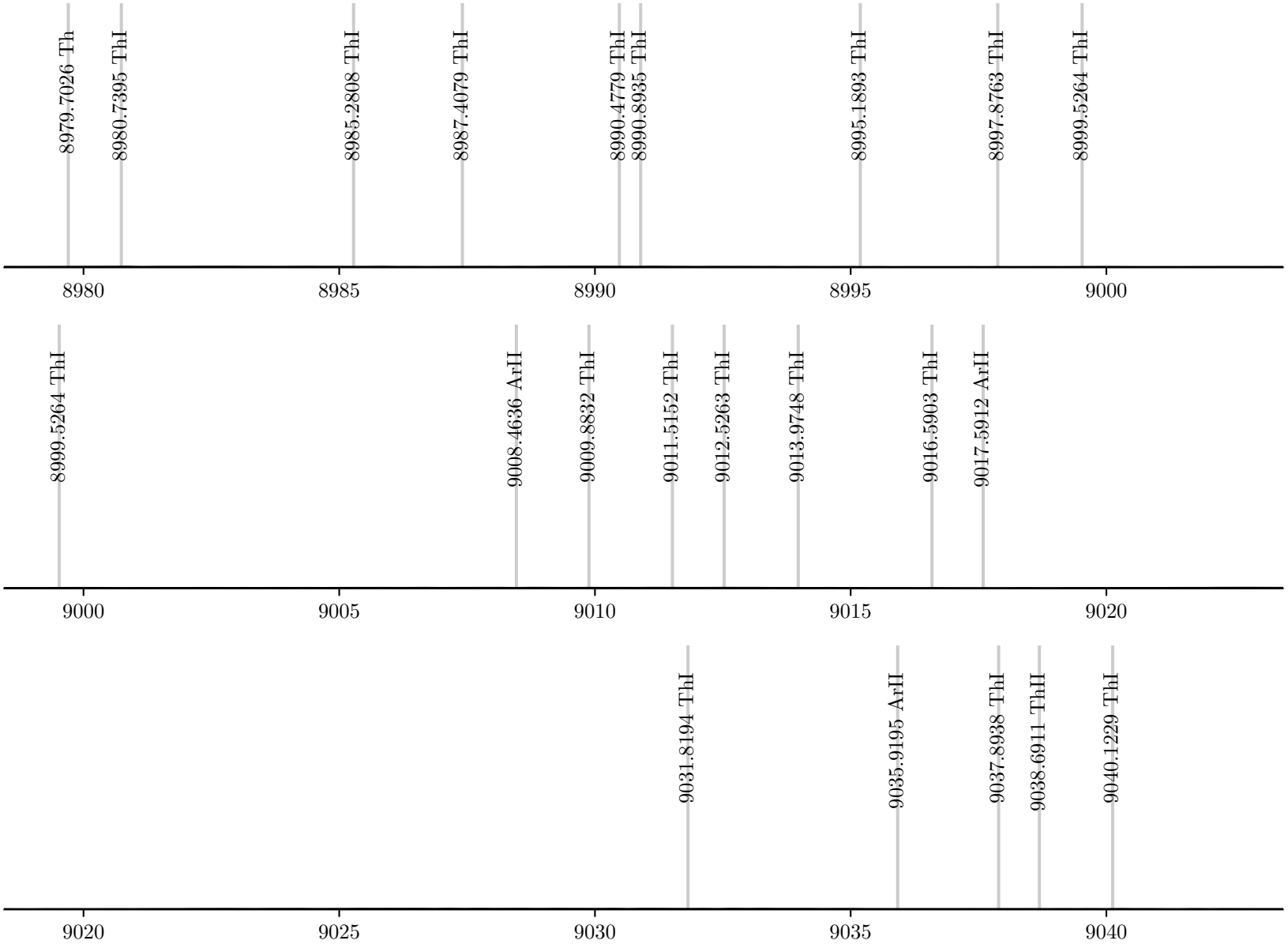
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8985.2808 ThI
8987.4079 ThI
8990.4779 ThI
8990.8935 ThI
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9016.5903 ThI
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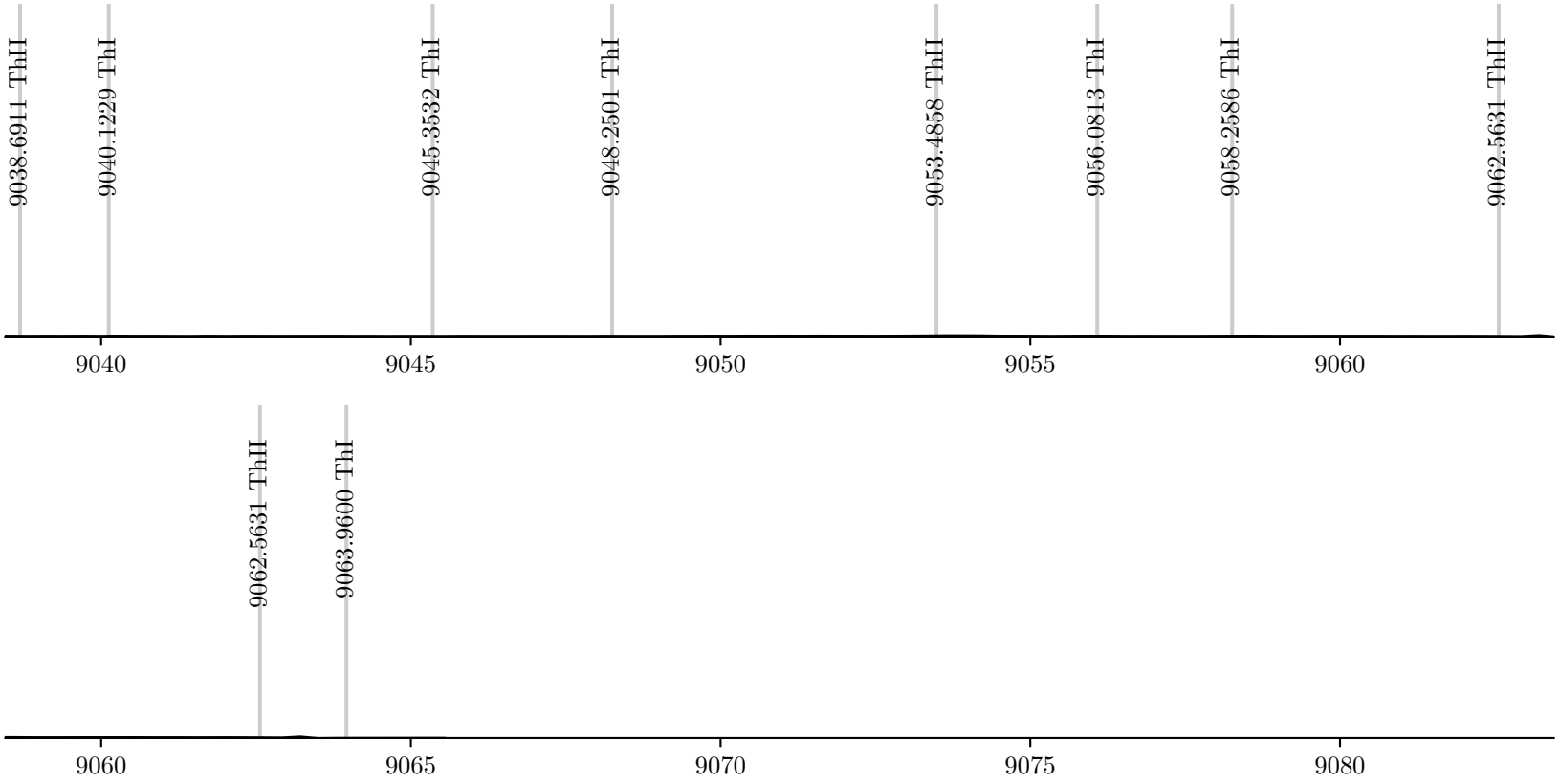
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9050
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9060
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9080